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Geological Survey of India

केंद्र शासित प्रदेश: जम्मू एवं कश्मीर के डोडा-लावा, डोडा और किश्तवाड़ जिलों में
एन.एच. 244 के साथ मेसो स्केल (1:10000) भूस्खलन संवेदनशीलता गलियारा मानचित्रण

Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir.

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(Field Season: 2024-25)

अरुण देव, निदेशक
मुहम्मद इब्राहिम वरिष्ठ भूविज्ञानी

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महानिदेशक, भारतीय भूवैज्ञानिक सर्वेक्षण, कोलकाता की पूर्व अनुमति के बिना आंशिक या पूर्ण रूप से प्रकाशित नहीं किया जाये

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By

Arun Dev, Director and Mohd. Ibrahim, Senior. Geologist

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**केंद्र शासित प्रदेश: जम्मू एवं कश्मीर के डोडा-लावा, डोडा और किश्तवाड़ जिलों में
एन.एच.-244 के साथ मेसो स्केल (1:10000) भूस्खलन संवेदनशीलता गलियारा मानचित्रण**

(एफएसपी मद कोड: एम4एलएसएसएम/एनसी/एनआर/एसयू-जेके/2024/50064)

अरुण देव, निदेशक तथा मोहम्मद इब्राहिम, वरिष्ठ भूवैज्ञानिक

सारांश

डोडा-लावा खंड राष्ट्रीय राजमार्ग एन.एच.-244 का एक महत्वपूर्ण हिस्सा है, जो डोडा और किश्तवाड़ जिलों के लिए प्राथमिक जीवन रेखा के रूप में कार्य करता है। राष्ट्रीय राजमार्ग (एन.एच.-244) के डोडा और किश्तवाड़ खंड के बीच भूस्खलन की बार-बार सक्रियता सभी मौसमों में निरंतर यातायात के लिए एक बड़ी बाधा है। डोडा और किश्तवाड़ के बीच 4 लेन राजमार्ग का निर्माण कार्य प्रगति पर है, जहां ऊबड़-खाबड़ स्थलाकृति के कारण कई स्थानों पर खड़ी चट्टानें मौजूद हैं। डोडा और किश्तवाड़ के बीच इस राष्ट्रीय राजमार्ग (एन.एच.-244) के चौड़ीकरण के दौरान, सड़क चट्टान के ढेरों और उपरिभार सामग्री के बड़े पैमाने पर स्खलन से प्रभावित हुई है। सड़क चौड़ीकरण के कारण कई पुराने भूस्खलन पुनः सक्रिय हो गए हैं, जो जीवन और संपत्ति की सुरक्षा को ध्यान में रखते हुए अत्यंत चिंता का विषय है। इस रणनीतिक राजमार्ग पर बार-बार होने वाले भूस्खलन से यातायात में गंभीर व्यवधान उत्पन्न होता है, जिससे क्षेत्र के निवासियों की समग्र आर्थिक गतिविधियों और अन्य गतिविधियों पर भी प्रभाव पड़ता है।

डोडा पुल और लावा के बीच भूस्खलन की बार-बार होने वाली घटनाओं के कारण, लगभग 45 लाइन-किलोमीटर की कुल लंबाई को कवर करते हुए, पहाड़ी ढलानों के स्खलन के प्रति संवेदनशीलता का स्तर और उनकी संवेदनशीलता के आधार पर उन्हें वर्गीकृत करने के लिए एक मेसो-स्केल (1:10,000) भूस्खलन संवेदनशीलता मानचित्रण किया गया है। अध्ययन क्षेत्र लघु हिमालयी क्षेत्र का एक हिस्सा है, जो भारतीय सर्वेक्षण विभाग की टोपोशीट संख्या 43ओ/12, 15 और 16 के अंतर्गत आता है। यह अध्ययन क्षेत्र को विभिन्न भूस्खलन संवेदनशीलता वर्गों में वर्गीकृत करने के लिए किया गया है, जो कि अनुमानी-सह-निर्धारक मॉडल पर आधारित है, जहां आंतरिक कारकों के विभिन्न विषयगत मानचित्र तैयार किए गए और भूस्खलन संवेदनशीलता अनुमानित रेटिंग के रूप में मूल्यांकित किए गए।

संचयी संवेदनशीलता मान को भूस्खलन संवेदनशीलता अनुमानित रेटिंग को एकीकृत करके गणना किया जाता है, जिससे क्षेत्र के विभिन्न क्षेत्रों के लिए अलग-अलग कुल अनुमानित संवेदनशीलता मान प्राप्त होते हैं। कट ढलान, उपरिभार और चट्टानी क्षेत्रों के लिए डोमेन आधारित कुल अनुमानित संवेदनशीलता मान की गणना की गई है। डोमेन आधारित डिजिटल एलिवेशन मॉडल व्युत्पन्न मानचित्रों के निर्माण के लिए ए.एल.ओ.एस. डिजिटल एलिवेशन मॉडल डेटा (10 मीटर) का उपयोग करके अध्ययन किया गया है जिसमें बेस मैप (पहलू मानचित्र), ढलान पहलू, ढलान कोण, सापेक्ष ऊंचाई, जल निकासी, स्ट्रीम पावर इंडेक्स, भू-आकृति और भूमि उपयोग-भूमि आवरण मानचित्र शामिल हैं। इन मानचित्रों को आर्क जीआईएस 10.8, ग्रास जीआईएस 7.6 और गूगल अर्थ प्रो का उपयोग करके तैयार किया गया है। पहलू के आधार पर तैयार किए गए विभिन्न क्षेत्र आधारित मानचित्र हैं - लिथोलॉजिकल मानचित्र, संरचनात्मक मानचित्र, ढलान निर्माण सामग्री, ढलान निर्माण सामग्री मोटाई, जल विज्ञान, भूमि उपयोग भूमि आवरण, भू-तकनीकी और गतिज विफलता मानचित्र।

वर्तमान अध्ययन क्षेत्र चिनाब घाटी के साथ-साथ स्थित है जो पुराजीव चट्टानों और चतुर्थक निक्षेपों को उजागर करता है। अत्यंत नूतन कल्प से नूतन कल्प के चतुर्थक निक्षेप जलोढ़, स्क्री और हिमोढ़ द्वारा दर्शाए गए हैं। डोडा पुल से लावा तक का अध्ययन क्षेत्र सलखला संरचना की चट्टानों को उजागर करता है, जिनमें अभ्रक-शिस्ट, कार्टजाइट, नाइस, संगमरमर और चतुर्थक निक्षेप शामिल हैं। कुछ स्थानों पर सलखला संरचना में प्राग्जीव कल्प के कपलास ग्रेनाइट का अंतर्वेधन है। ये शैल इकाइयां डोडा पुल-लावा खंड, एन.एच.-244 पर उजागर हैं। गडसू से हीरपोरा तक का मार्ग अधिकांशतः नदी तलछट सामग्री और मिश्रोढ सामग्रियों द्वारा आच्छादित है, जहां आकस्मिक चट्टानें उजागर होती हैं।

शैल संहति में लगातार तीन से चार जोड़ होते हैं और इसलिए, शैल संहति वर्ग VI-VIII (आई.एस.आर.एम., 1981) के अंतर्गत आते हैं। शैल हल्के से मध्यम रूप से अपक्षयित हैं। क्षेत्र आधारित एक-अक्षीय दाब सहनशीलता मानों के आधार पर, शैल संहति कमज़ोर से मध्यम रूप से मजबूत वर्गों के अंतर्गत आते हैं। क्षेत्र अवलोकनों के आधार पर, शैल संहति प्रमुख रूप से RMRb मानों के साथ सामान्य से अच्छे वर्ग के अंतर्गत आते हैं, जिनका मान औसत 33.6-79.3 के बीच होता है। कुछ शैल उद्भेदन भी लगभग 33.6 RMRb के साथ खराब श्रेणी के अंतर्गत आते हैं। क्षेत्र आधारित शैल उद्भेदन श्रेणी IV खराब से श्रेणी II अच्छी श्रेणी तक भिन्न होते हैं। श्रेणी IV खराब श्रेणी के अंतर्गत आने वाले कुछ शैल संहति के अतिरिक्त, अध्ययन क्षेत्र में अधिकांश शैल संहति सामान्य और अच्छे श्रेणियों के अंतर्गत आते हैं। डोडा-लावा गलियारे में

कुल 26 भूस्खलनों की पहचान कर उन्हें विभिन्न प्रकार की विफलताओं में व्यवस्थित रूप से वर्गीकृत किया गया है। इनमें 12 मलबा स्खलन, 6 समतलीय विफलताएं, 3 वेज विफलताएं, 2 टॉपलिंग विफलताएं और 3 कट ढलान विफलताएं शामिल हैं। डोडा पुल से प्रेमनगर तक का खंड मुख्य रूप से मलबा स्खलन और मलबा पात से प्रभावित है, जिसकी विशेषता अक्सर खड़ी मानवजनित कट ढलानों के कारण कंकड़ के आकार के टुकड़ों का ढलान से नीचे की ओर खिसकना है। मिश्रोढ़ और जलोढ़ अनुक्रमों के भीतर अंतर्संयोजित कंकड़ संस्तरों की उपस्थिति गुरुत्वाकर्षण और जलवैज्ञानिक दबाव के तहत अपरदन और ढलान अस्थिरता को और बढ़ा देती है। गडसू में ढीले मलबे के भीतर एक महत्वपूर्ण सक्रिय मलबा स्खलन दर्ज किया गया है।

इसके विपरीत, प्रेमनगर से लावा तक का क्षेत्र शैल स्खलन के प्रति अधिक संवेदनशील है, जिसमें ठाठरी, द्रबशला और लावा के आसपास प्रमुख घटनाएं दर्ज की गई हैं। ये विफलताएं मुख्य रूप से आधारशिला में संरचनात्मक असंततताओं द्वारा नियंत्रित होती हैं, जिनमें संयुक्त क्वार्ट्जाइट और शिस्ट क्रम शामिल हैं, जो सड़क कटाव के साथ-साथ कटाव और अपक्षय के कारण और भी बढ़ जाती हैं।

गतिज विश्लेषण से पता चलता है कि क्षेत्र में 75% ढलान दिशा उत्तरी गोलार्ध की ओर है, जिसमें पश्चिमी भाग (49%) की तुलना में पूर्वी भाग (51%) की अधिक प्रधानता है। संयुक्त आंकड़ों से यह पता चलता है कि व्यापक रूप से फैले जे1 और जे2 मुख्यतः उत्तर-पश्चिम दिशा की ओर झुके हुए हैं, मध्यम रूप से फैले जे3 मुख्यतः दक्षिण-पश्चिमी दिशा की ओर झुके हुए हैं, और व्यापक रूप से फैले जे4 समान रूप से उत्तर-पश्चिमी और दक्षिण-पूर्वी दिशाओं में झुके हुए हैं। ढलान का जे1 के साथ विपरीत संबंध और जे2 के साथ सकारात्मक संबंध से यह स्पष्ट है कि क्षेत्र की वर्तमान स्थलाकृति डी2 विकृति द्वारा नियंत्रित है जो कि एक परिपक्व स्थलाकृति है। इसके अतिरिक्त, कील निर्माण में जे3 फैब्रिक की भागीदारी भी इसके पहले के फैब्रिक के साथ क्रॉस-कट संबंध को पुष्ट करती है।

क्षेत्र के एल.एस.एम. के प्रमाणीकरण के दौरान यह पाया गया कि उच्च भूस्खलन संभाव्यता वाला क्षेत्र, जो कुल क्षेत्रफल का केवल 26.87% है, उसमें 62.96% भूस्खलन दर्ज किए गए हैं। मध्यम संभाव्यता क्षेत्र, जो 34.45% क्षेत्र को आच्छादित करता है, में 22.22% भूस्खलन दर्ज किए गए हैं, जबकि निम्न संभाव्यता क्षेत्र, जो 38.66% क्षेत्र को आच्छादित करता है, में केवल 14.81% भूस्खलन दर्ज किए गए हैं। इस प्रकार, मेसोस्केल एल.एस.एम. मानचित्र में उच्च और मध्यम संभाव्यता वर्गों के अंतर्गत भूस्खलनों का संचयी प्रतिशत 85.18% है।

Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir.

(FSP Item Code: M4LSSM/NC/NR/SU-JK/2024/50064)

Arun Dev, Director and Mohd. Ibrahim, Senior Geologist

Abstract

The Doda-Lawa section, forming a vital part of National Highway NH-244, serves as the primary lifeline for the Doda and Kishtwar districts. Repeated activation of landslides between Doda and Kishtwar section of National Highway (NH-244) is a major obstruction for the free flow of traffic on it in all weathers. The construction of 4 Lane highway between Doda and Kishtwar is under progress through rugged topography forming steep scarps at places. During widening of this National Highway (NH-244) between Doda and Kishtwar, the road has been affected by number of mega slides both in the rock masses as well as in the overburden material. Number of old slides have also been reactivated due to road widening, which is a cause of great concern considering the safety of the life and properties. The frequent land sliding on this strategic highway causes serious disruption of traffic thereby, affecting the overall economic development and other activities of inhabitants in the area.

Due to the recurrent occurrence of landslides along the stretch between Doda Bridge and Lawa, a meso-scale (1:10,000) landslide susceptibility mapping was undertaken, covering a total length of approximately 45 line-kilometres and to assess the degree of susceptibility of hill slopes and to categorize them depending upon the vulnerability. The area of investigation is a part of Lesser Himalayan zone and falls in Survey of India (SOI) toposheet nos. 43O/12, 15 & 16. The study was carried out to categorise the area into different landslide susceptibility classes based on heuristic cum deterministic model where different thematic maps of the intrinsic factors are prepared and rated as Landslide Susceptibility Estimated Rating (LSER). The cumulative susceptibility values are calculated by integrating the LSER which gives different Total Estimated Susceptibility Values (TESV) for various domains of the area. The domain based TESP was calculated for cut slope, overburden and rock domains. The study was carried out using ALOS Digital Elevation Model (DEM) data (10m) for the generation of domain based DEM derivative maps which include Base Map (Facet Map), Slope Aspect, Slope Angle, Relative Relief, Drainage,

Stream Power Index (SPI), Landform (Geomorphon) and Land use Land cover (LULC) maps using Arc GIS 10.8, GRASS GIS 7.6 and Google Earth Pro. The various field based maps prepared based on facet are Lithological Map, Structural Map, Slope Forming Material (SFM), SFM thickness, Hydrological, Land use land cover (LULC), Geotechnical and kinematic failure maps.

The present study area falls along the Chenab valley and exposes Palaeozoic rocks and Quaternary deposits. Quaternary deposits of Pleistocene to Holocene age are represented by alluvium, scree and moraine. The study area from Doda Pul to Lawa exposes rocks of Salkhala Formation which comprises of mica-schist, quartzite, gneiss, marble and Quaternary deposits. At places Kaplas Granite of Proterozoic age is intrusive within Salkhala Formation. These rock units are exposed along the Doda Pul-Lawa section, NH-244. The route stretch from Gadsu to Heerpora is mostly covered by RBM and colluvium materials with occasional rock exposures.

The rock mass has consistent three to four set of joints and, therefore, fall between VI-VIII in rock mass class (ISRM, 1981). The rocks are slight to moderately weathered. Based on the field based UCS values rock masses fall in weak to moderately strong classes. On the basis of field observations, the rock masses dominantly fall in fair to good category with RMR_b values ranging between avg. 33.6-79.3. Some out crops also fall in poor category with approximately 33.6 RMR_b . Field based RMR varies from Class IV Poor to Class II Good category. Most of the rock masses in the study area falls into Fair and Good categories, except some falling in Class IV Poor category.

A total of 26 landslides were identified and systematically classified into various failure types along the Doda–Lawa corridor. These include 12 debris slides, 6 planar failures, 3 wedge failures, 2 toppling failures, and 3 cut slope failures. The segment from Doda Bridge to Premnagar is predominantly affected by debris slides and debris falls, characterized by the downslope movement of pebble-sized fragments often triggered by steep anthropogenic cut slopes. The presence of interbedded pebble beds within colluvial and alluvial sequences further contributes to erosion and slope instability under gravitational and hydrological stress. A significant active debris slide was recorded at Gadsu, confined within loose debris material.

In contrast, the stretch from Premnagar to Lawa exhibits increased susceptibility to rock slides, with notable occurrences documented in the vicinity of Thatri, Drabshalla, and Lawa. These

failures are primarily controlled by structural discontinuities in the bedrock, including jointed quartzites and schist sequences, exacerbated by undercutting and weathering along road cuts.

Kinematic analysis reveals that the area shows 75% slope directions towards northern hemisphere with slight dominance towards eastern quarter (51%) than western (49%). Whereas the joint data shows pervasive J1 and J2 predominantly dip towards NW direction, moderately spaced J3 fabric predominantly dipping towards SW direction, widely spaced J4 uniformly dipping towards NW and SE directions. The opposite relation of slope with J1 and positive relation of slope with J2 strongly support the view that present topography of the area is controlled by D2 deformation and is a mature topography. Further, involvement of J3 fabric in wedge formation again supports its cross-cut relation with earlier fabric.

The LSM of the area on validation shows 62.96% landslides in high susceptibility zone which covers only 26.87% area, 22.22% landslides in moderate susceptibility zone which covers 34.45% area and 14.81% landslides in low susceptibility zone which covers only 38.66% area. The cumulative percentage of landslides under high and moderate susceptibility classes in mesoscale LSM map is 85.18%.

Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir.

(Field Season 2024-25)

M4LSSM/NC/NR/SU-JK/2024/50064

By

**Arun Dev, Director
Mohd Ibrahim Sr. Geologist**

CHAPTER 1

INTRODUCTION

1.1: General

As per the MoM-approved Landslide Vision and Strategy Document 2020-2030, the Geological Survey of India (GSI) has planned meso scale (1: 10,000) landslide susceptibility mapping as one of its major programme on landslide investigations in different Regions. This up scaled task is prioritized, as, i) the landslide susceptibility mapping on 1:50,000 scale for the entire landslide prone terrain in Northern Western Himalaya, has already been completed under National Landslide Susceptibility Mapping (NLSM) program, ii) a substantial area 15.49% has been identified as highly susceptible to landslides in the area and iii) an up scaled landslide susceptibility information will provide apt geo-information for any stakeholder for use in local infrastructure and mitigation planning.

The UT of Jammu & Kashmir has aspecial important in the country from defense, hydro-electric projects, religious places and tourism. Being mountainous state, communication routes play a vital role in its all-round development. NH 244 starts at near Khanabal,Achabal, Kokernag, Daksum, via Sinthan pass (Elevation: 3748 m), Kishtwar via Thathri, Doda and terminates at NH-44 near Batote. It constitutes the life line for the vast interior areas of twin districts of UT: J&K (Doda and Kishtwar). The Sinthan Pass (Elevation: 3748 m) is famous Tourist Circuit linked through this highway, besides an alternate vital linkto the Kashmir valley with Jammu Region. Several old landslides and debris flows exists along NH-244, from Batote, Doda, Kishtwar, Sinthan Pass area, Doda, Kishtwar and Anantnag districts, which reoccurs every year. Besides,

new landslides also take place in every rainy season. These slides disrupt the free flow of the pilgrims of famous Machail Mata Yatra during rainy period and general public at large and face lot of inconvenience. Apart from this, the whole area remains under constant threat of slope failure owing to its topography and geological condition.

The Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir., is also listed in minutes of meeting held via video conferencing between GSI, NDMA and NHAI on 15th July, 2020. In the minute it was mentioned that GSI will prepare 1:10,000 scale landslide susceptibility map along the susceptible highways like NH-244. It was also mentioned in the minutes that the Government is going to develop fresh highway along Jammu-Ramban Doda – Sinthan Top Pass - Anantnag as alternate route to the Kashmir valley also necessitates doing such studies.

The Jammu and Kashmir region is quite vulnerable to landslide hazards considering that the region is affected frequently by the monsoon rainfall as well as western disturbances winter rainfall and particularly, both Jammu the Kashmir division witness the prolonged winter and rainy seasons. The region is also susceptible to major seismic activities/earth quakes as it falls under the Seismic Zone IV. The overall economic development of the region is highly dependable on the NH44 and NH-244 from Batote to Kishtwar that provides all-weather-connectivity to the Kashmir Division from Jammu. Being a mountainous region, these communication routes play a vital role in its all-round development.

Repeated activation of landslides between Doda and Kishtwar section of National Highway (NH-244) is a major obstruction for the free flow of traffic on it in all weathers. The construction of 4 Lane highway between Doda and Kishtwar is under progress through rugged topography forming steep scarps at places. During widening of this National Highway (NH-244) between Doda and Kishtwar, the road has been affected by number of mega slides both in the rock masses as well as in the overburden material. Number of old slides have also been reactivated due to road widening, which is a cause of great concern considering the safety of the life and properties. The frequent land sliding on this strategic highway causes serious disruption of traffic thereby, affecting the overall economic development and other activities of inhabitants in the area.

The UT of Jammu & Kashmir has a special important in the country from defence, hydro-electric projects, religious places and tourism. Being a mountainous state, communication routes play a vital role in its all round development. NH 244 starts at near Khanabal, Achabal, Kokernag, Daksum, via Sinthan pass (Elevation: 3748 m), Kishtwar via Thathri, Doda and terminates at NH-44 near Batote. It constitutes the life line for the vast interior areas of twin districts of UT: J&K (Doda and Kishtwar). The Sinthan Pass (Elevation: 3748 m) is famous Tourist Circuit linked through this highway, besides an alternate vital link to the Kashmir valley with Jammu Region. Several old landslides and debris flows exists along NH-244, from Batote, Doda, Kishtwar, Sinthan Pass area, Doda, Kishtwar and Anantnag districts, which reoccurs every year. Besides, new landslides also take place in every rainy season. These slides disrupt the free flow of the pilgrims during rainy period and general public at large and face lot of inconvenience. Apart from this, the whole area remains under constant threat of slope failure owing to its topography and geological condition.

Thus, keeping the above reasons in view, the meso-scale (1:10,000) Landslide susceptibility Mapping (LSM), along Doda and Kishtwar road corridor was proposed by GSI under FSP: 2024-25, vide item Code No. M4LSSM/NC/NR/SU-JK/2024/50064, and to access the degree of susceptibility of hill slopes and to categorize them depending upon the vulnerability. LSM is a major and effective tool which divides the landslide prone hilly terrain into different zones according to the relative degree of susceptibility to landslides. Landslide susceptibility maps generated after the mapping are of a big use for the assessment of damage potential, and are pre-requisite for the quantification of landslide hazards and risk. The susceptibility analysis carried on larger scale are very supportive for selecting sites that are less prone to ground instability while delineating precise locations of unstable/landslide prone areas, where suitable environmental programmes can be initiated by adopting suitable mitigation measures. The area of investigation is a part of Lesser Himalayan zone and falls in Survey of India Toposheet No. 43O/12, 15 and 16

1.2. Objective

The general objective of present study was landslide susceptibility mapping along the Doda –Lawa NH-244 Road corridor. The specific objectives are:

- I. To identify slope-failure domains and to prepare domain-specific geofactor maps in GIS.
- II. To rate and weigh relevant geofactor maps for different failure domains.
- III. To integrate the rated and weighted geofactor maps as per identified failure domain in GIS.
- IV. To combine the domain-specific landslide susceptibility models into a combined and classified landslide susceptibility map on a 10k scale.
- V. To qualitatively/ quantitatively assess the landslide risk and to frame a generic mitigation /management plan on a 10k scale for relevant value addition and direct use by the stakeholders.

1.3. Location and Accessibility

The area of investigation is a part of Lesser Himalayan zone and falls in Survey of India (SOI) toposheet nos. 43O/12 ,15 & 16. (Fig.1.1).

The study area accessible through metalled/or non-metalled roads and mule tracks. The area is approachable from Jammu through NH-44 and NH-244 The rail link is also operative upto Udhampur. At few places the area can be approached through the link metalled/un-metalled roads. The nearest railway station, Udhampur is 20 km away from the study area. The nearest Airport is Jammu Airport. It is located at a distance of 80 kms from Udhampur.

1.4 Physiography and Drainage

The study area lies within the higher Himalayan ranges and the relief varies predominantly from 875m to 4668m above mean sea level (MSL). It is characterized by rugged topography comprising high ranges and deeply dissected valleys, escarpments and cliff faces, 4668m, 4557m, 3755m and Selo Paliri etc are some of the highest peaks lies in the northeastern corner of the study area whereas the lowest point (875m) is near Barhut over the River bed of Chenab towards western part of the study area.

Deep gorges and V-shaped valleys have been carved out by the tributaries (Gahan Nadi, Kuli Gad, Kal Nai Gad, Kagune Gad, Jai Gad, Ladu Gad and Chili Nadi) of Chenab River in the area. Chenab forms the 7th order major draining river of the area which flows in North-South

direction from Jarwan to Thathri upto 1155 point and in east -west direction from 1155 point to Barhut. The drainage Pattern is dendritic to trellis (Fig.1.3). Almost all the peaks receive snowfall in the winter season.

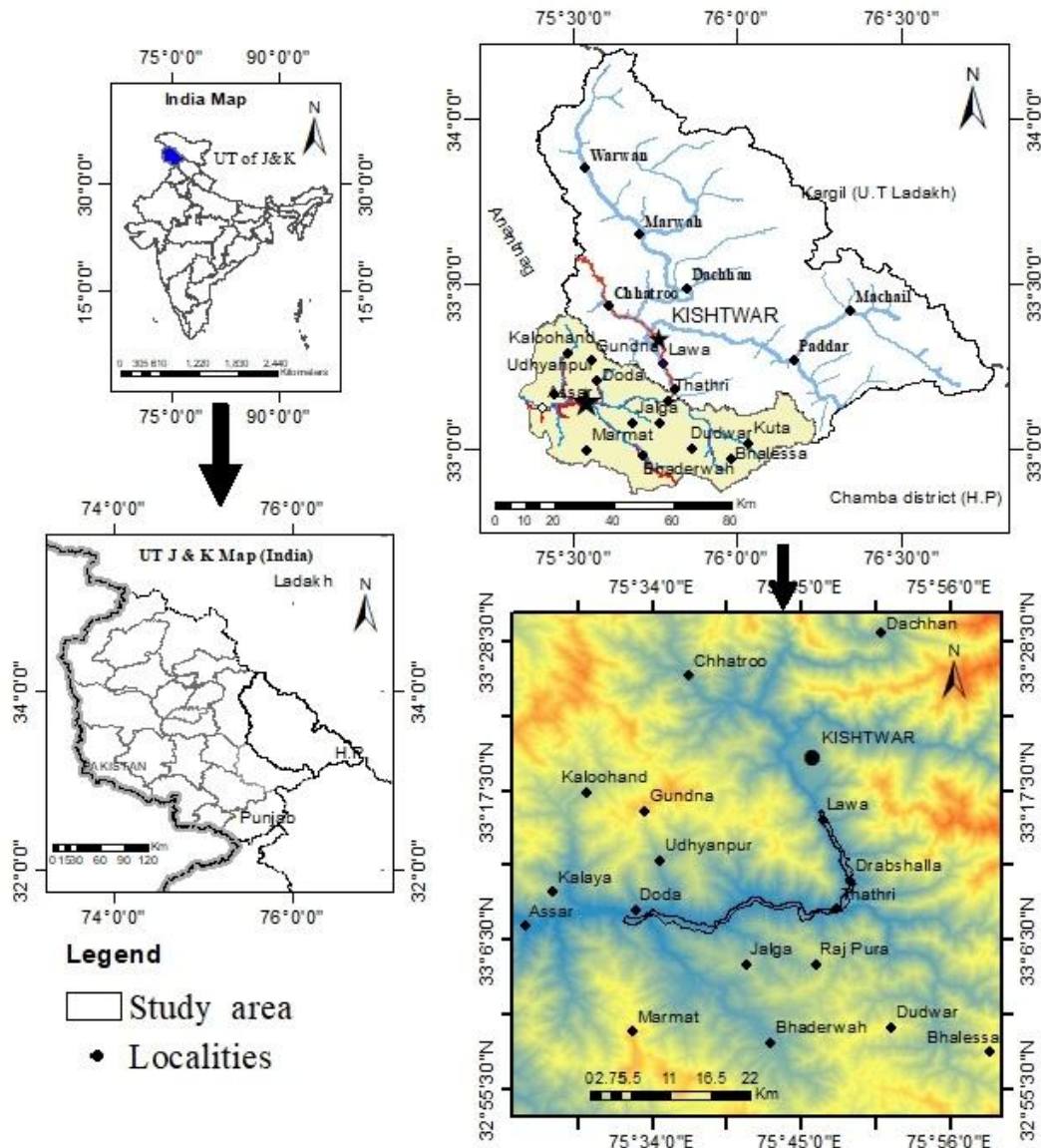


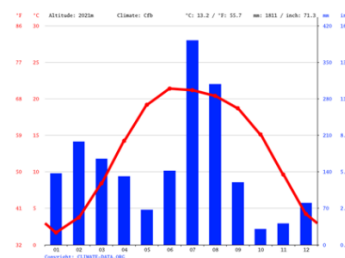
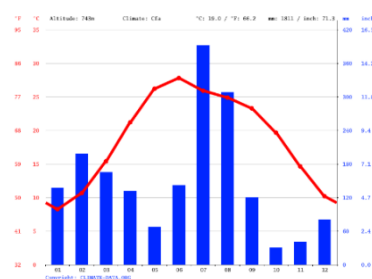
Figure 1.1: Location map of the Study area

1.5. Climate

The study area covers the southern part of Kishtwar district and Northern part of Doda District. The data of meteorological observatory at Bhadarwah (Climate of Jammu & Kashmir, Climatological Summaries of States Series No. 20 India Meteorological Department, July 2014),

SW of the area, would represents true picture of the climate. The climate of the area is temperate. The summers are long, warm and mostly clear and winters are short, very cold and partly cloudy. Over the year, the summer temperature can reach up to 40°C whereas the winter temperatures are usually between -3.8°C to 25°C. Rainfall, spatially, is highly variable depending upon the altitude but often medium to heavy during monsoon. The average annual rainfall in the Doda district is 1334.5 mm whereas of the Kishtwar is 887.8 mm. The rainfall in southwest monsoon season (June-September) is about 33% of the annual normal rainfall, while the rainfall in pre-monsoon months (March to May) accounts for 33% of the annual rainfall, March being the month of the highest rainfall with an average of 213.5 mm. The winter months (December to February) account for about 27% of the annual rainfall. At the higher elevations, considerable amount of precipitation is received in the form of snow. About 64 cm of snowfall occurs annually in the district. Snowfall mainly occurs in the winter months November to March. About 75% of snowfall occurs in December & February.

Considerable amount of precipitation is received in the form of snowfall in winter months in the elevated areas in the northeastern part of the area and remain covered for almost 6-8 months (October to June) apart from the permanent snow patches in the North Eastern part, Lower areas remain snow covered for a period of 4 to 5 months (November to February). The area gets precipitation in the form of rain and snow. Rainy season generally starts from June end and continue till early September followed by the winter season, which continues till February.



. (Fig. 1.2). Annual Rainfall data of Kishtwar and Doda District (Source- Climatic- Data org)

1.6 Flora and Fauna

The study area is partly covered with dense forests and partly by sparse forests. Natural vegetation in the area extends up to about 3500 m elevation beyond which only shrubs, grass and

some low plants are found. The lower altitudes support luxuriant forests dominated by conifers. The higher hills are covered by different varieties of trees including Kail (*Pinus Wallichiana*), Deodar (*Cedrus Deodara*), Chir (*Pinus Roxburghii*), ReiaBudul (*Abiesspectabilis*) Chir (*Pinuslongifolia*), Fir (*Abies Pindrow*), Spruce (*PinusSmithina*) etc. Kuth, a medicinal plant of high value, and Dhupgrow at higher elevations in some of the areas. Some others medicinal herbs shrubs and mushrooms are also common. The low-lying areas exhibit mostly deciduous species including Chinar, Poplar, Willow, Keekar, Almond, Pear, Cherry, Apple and Walnut (*Juglanaregia*). The main crops are maize, paddy and barley. Vegetables and fruits are rare. Paddy and vegetable growing are restricted to the low-lying irrigated areas.

The dominant fauna in the forests include Monkey, Black Bear (*Selenarctos Thibetanus*), Leopard (*FelisPardus*), Kashmir Stag (*Cervus Cashmirianus*), Musk Deer (*Moschus Moschiferus*), Markhor (*Capra FalconeriCashmiriensis*), Wolf (*Canislupus*), Fox (*Vulpes Montana*), Jackal (*Canis Aurcus*) etc. Game birds like hill partridges, fowl, rock pigeons and the pretty 'Neel' are found in relative abundance. During summer there is a regular movement of huge flocks of sheep to these areas because of the abundance of grassy meadows.

1.7 Previous Work

The GSI has received the attention regarding the problems of sliding along NH-44, NH-1A erstwhile, since 1960s. Raju (1961) carried out field study and survey to study the areas prone to severe landslides along National Highway between Batote and Srinagar and suggested certain remedial and control measures for the numerous slides (e.g. Phagla Passi and Makerkot slide). Subsequently, Ashraf et al., (1966) of the Geological Survey also visited certain sections of the Highway and tendered advice such construction of retaining walls. Tiku A.K. (1965) carried out experimental blasting conducted near the Khuni Nala on the Jammu-Srinagar National Highway, Jammu and Kashmir. Verma (1967) also studied the slides along National Highway between Banihal and Peerah, a majority of them Banihal and Peerah, Jawahar Tunnel. He also studied the widening of the road at a few points and the foundations for the Panthal and the Makerkot bridges.

Ajay Kumar (2007) carried out preliminary geotechnical investigation of landslide along Jammu-Srinagar National Highway (NH-1A) between Ramban-Banihal sections, Doda district,

J&K. National Landslide Susceptibility Mapping on 1:50000 scales is among new programs towards landslides studies. During 2015-16 (Pradeep Pal and Sharma Sanjeev Kumar, 2016 and 2018-19 (Sharma and Akram, 2019) macro-scale study on NLSM covering (43O/4, 43O/8 and 43O/3, 43O/7) Doda, Reasi, Ramban and Udhampur districts has been carried out. As per this study, The Landslide susceptibility map for 1293 Sq Km of area (2015-16) reveals that around 10% (129 Sq Km) of area comes under high susceptibility zone and is located mostly along the rim of the Bagliyar reservoir and along the main drainage course/network of Chenab River, parts of Chanderkot and Ramban and is vulnerable to landslides. During 2018-19 out of 1290 sq km mapped area about 31.77 percent falls under “High” susceptible zone (HSZ) while 20.08 percent falls under “Moderate” susceptible zone (MSZ) and the rest 48.05 percent falls under “Low” susceptible zone (LSZ) respectively. The areas/localities like Pogal, Shagan and Ramsoo etc. come under high susceptibility zone.

The GSI has carried out Systematic Geological mapping on 1:63,360 scale in parts of the area under investigation. The Systematic geological mapping on 1:50,000 scales have also been carried out in this area (GSI 2012).

1.8 Scope of the Work

The study area has been covered during Macro Scale (1:50000) LSM during National Landslide Susceptibility Mapping (NLSM) programme (Pal and Sharma, 2016, Sharma and Akram, 2019). The seamless Macroscale LSM categorizes the study area into three classes ‘high’, ‘moderate’ and ‘low’ based on cumulative distribution of landslides. Much of the area under study falls in low category, owing to the scale constraints of Macro Scale LSM.

The Meso-scale LSM studies being carried out on large scale (1:10000) have its utility in ensuring adequate and proper landuse zoning to reduce landslide risks, and also to provide a basis for detailed studies for the design and construction of engineering structures for long-term remediation. The mitigation measures given based on this study will help the stakeholders to carry out relevant landslide remedies. Further, the necessary slope stability measures suggested would help in undertaking relevant engineering solutions to protect the slopes. The landslide susceptibility map prepared is based on the study of pre-disposing factors of a slope. The effect of

external triggering factors, like rainfall and earthquake, although discussed, has not been used for the landslide susceptibility modelling. It is to further state that the generated mesoscale susceptibility map is suitable for preliminary planning purposes only. Appropriate detailed geotechnical studies may be presented before actual developmental work in the area.

1.9 Nature and Quantum of Work

The details of the nature and quantum of work, time schedule etc., as per the annual programme of GSI, NR along with the achievements for Field Seasons 2024-25 is given in (Table 1.1). This programme envisages the preparation of a LSZ map on meso scale (1:10,000) covering an area of 45-line km from Doda to Lava along NH-244. The final Landslide Susceptibility Zonation (LSZ) map thus generated will help the stakeholders in identification of suitable areas for future developmental works; take appropriate mitigation measures, environmental regeneration, construction and maintenance of good road network. The details of number of field days spent by each officer are given in (Table 2.2).

Table 1.1. Physical Targets and Achievements as per the NQT

Nature of Work	Work Proposed for FS 2024-25	Total Achievements
A. Quantitative Target:		
Field Coverage: Landslide Susceptibility Mapping (Sq. Km) (1:10,000 scale)	45 L Km	45 L km
Field sample collection		
Soil	10 Nos	10
Rock	10 Nos	10
Qualitative Task: Detailed pre-field landslide inventory map including ground deformation, creeps etc. and identification of failure domains ,Facet map (if selected as mapping unit), Landforms, Slope, Forming Material, engineering geological, landslide domain, Overburden thickness, geotechnical maps etc., Preparation of Landslide Management Map, Preparation of prefield thematic geofactor maps on 10k based on failure domains (e.g., slope morphometry, Drainage, Stream Power Index, Land use/land cover map, Rock: UCS/ Point Load Soil: qualitative: grain size analysis, specific gravity, Density (dry), Density (wet), Porosity, NMC, C and ϕ .	45 L Km	45 L Km

Table 1.2. Number of field days of the Officers

S. No.	Name of Officer	No. of field days
1	Sh. Arun Dev, Director	86
2	Sh. Mohd Ibrahim, Sr. Geologist	83

1.10. Project workflow scheme- Sequence of Stages of Investigation:

The baseline data acquisition for relevant causative geofactors involves an integrated approach using Remote Sensing and field-based data collection. This process encompasses multiple stages aimed at acquiring and processing geofactor data for landslide susceptibility mapping (LSM) studies at the Meso Scale. The workflow includes pre-field studies utilizing satellite data, followed by field studies, laboratory analytical studies, and post-field data processing involving rating, overlay analysis, and modelling using Geographic Information System (GIS) techniques. Headquarters-based data processing focuses on selecting and segregating failure domains based on intrinsic factors. In this study, specific causative geofactors were identified and utilized to develop susceptibility models for distinct domains such as debris slides, cut slope failures, rock slides, and rock falls. Field data from rocky terrains revealed various types of rock failures including planar, wedge, and toppling rock failures, which informed the creation of separate susceptibility maps for rock slide and rock fall domains. These individual maps were later combined (mosaiced) to produce a unified landslide susceptibility map. Finally, a comprehensive Meso Scale LSM was developed by merging all four domain-based susceptibility maps. The entire process, including the various studies conducted at each stage, is illustrated in a detailed flow chart (Fig.3.5).

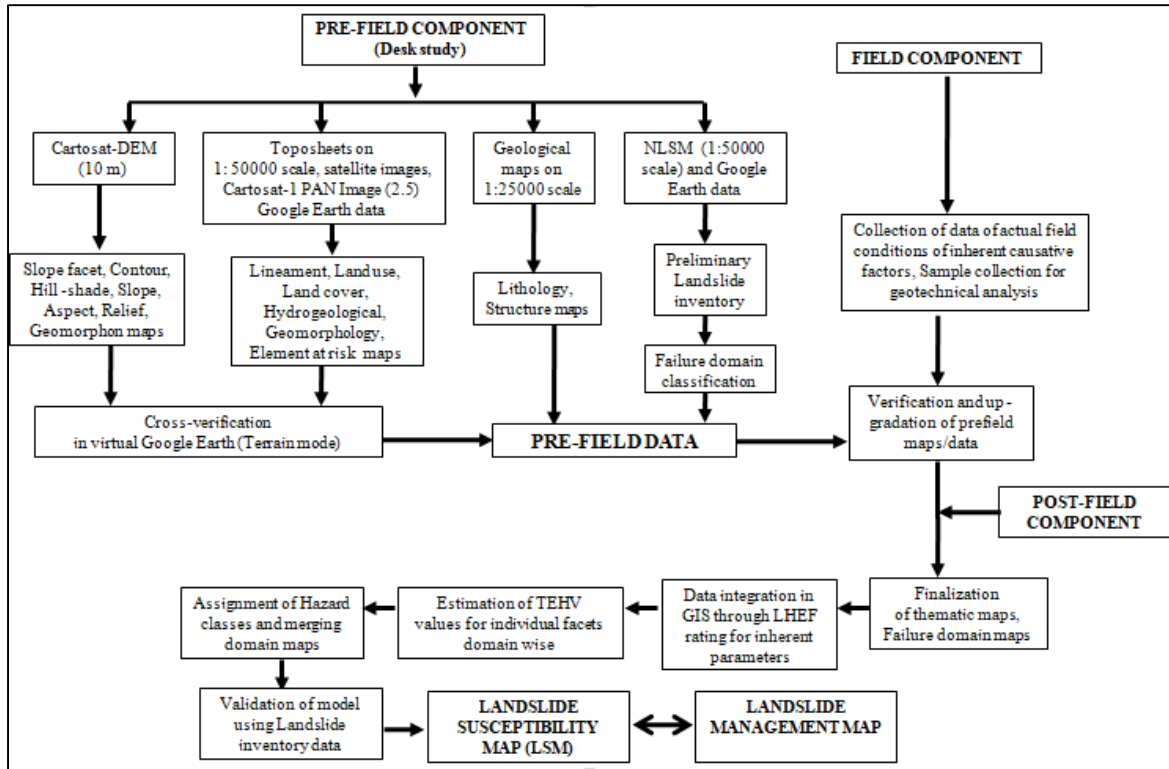


Fig.3.7 Flow chart showing workflow in mesoscale landslide susceptibility mapping studies.

1.10 Acknowledgements

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CHAPTER-2

STUDY AREA

2.1 Detailed Description of the Study Area

The Study area of investigation is a part of Lesser Himalayan zone and falls in Survey of India (SOI) toposheet nos. 43O/12 ,15 & 16. It lies within the higher Himalayan ranges and the relief varies predominantly from 875m to 4668m above mean sea level (MSL). It is characterized by rugged topography comprising high ranges and deeply dissected valleys, escarpments and cliff faces, 4668m, 4557m, 3755m and Selo Paliri etc are some of the highest peaks lies in the northeastern corner of the study area whereas the lowest point (875m) is near Barhut over the River bed of Chenab towards western part of the study area.

2.2 Regional Geology of the Study Area

The study area exposes Palaeo Proterozoic rocks of undifferentiated central crystallines (=Nanga Parbat Crystallines). Lopara and Dul Formation (Meso-Proterozoic) of Kishtwar window zone, Undifferentiated Salkhala Formation (Proterozoic), Bhaderwah Formation of Neo-Proterozoic age and Undifferentiated Quaternary deposits of Pleistocene to Holocene age.

Central Crystallines: The Undifferentiated Central Crystallines (=Nanga Parbat Crystallines=Giambal Group), well exposed in the central part of the area are the oldest rocks. In the adjoining toposheet, 52C/4, these rocks have been divided into Padar and Liond Formation. The Central Crystallines comprise gneissic suite of biotite gneiss, garnetiferous biotite gneiss, kyanite gneiss, migmatite gneiss, leucocratic gneiss, porphyroblastic granite gneiss, porphyritic biotite gneiss and subordinate high grade pelitic schist with thin bands of quartzite, phyllite and marble.

Lopara Formation: The Lopara Formation of Kishtwar Window Zone of Meso-Proterozoic age is well exposed in the northeastern corner of the toposheet. It consists of a sequence of inter-bedded slate, phyllite and quartzite.

Dul Formation: The Dul Formation comprising quartzite with thin beds of phyllite is exposed in the south within a tectonic window referred to as Kishtwar window.

Table 2.1: General Geological set-up of the study area (after compiled geological map, GSI)

Group/Age	Formation	Lithology
Quaternary (Pleistocene to Holocene)	Undifferentiated Quaternary deposit	Alluvium, moraines and scree.
(Palaeozoic)	Kaplas (=Piparan Granite)	Porphyritic biotite granite with quartzite and slate bands
(Neoproterozoic)	Bhaderwah	Slate, phyllite and quartzite and quartzose slates.
(Mesoproterozoic)	Pias	Hornblende granite gneiss characterized by bluish quartz and chlorite
Kishtwar Window zone (Mesoproterozoic)	Dul	Quartzite with thin beds of phyllite and quartzite
	Lopara	Interbedded slate, phyllite and quartzite
(Undifferentiated proterozoic)	Salkhala	Carbonaceous slate, quartzite, dolomite with stromatolitic cherty limestone.
		Stromatolitic limestone and slate
		Phyllite, carbonaceous phyllite, Pebbly phyllite, schist, Limestone, Marble (M) and quartzite band with basic enclaves.
Undifferentiated Gambal group(=Central Crystalline) Palaeo-proterozoic)		Undifferentiated gneisses and schist with thin bands of ferruginous Schist and marble.

Salkhala Formation:

The Undifferentiated Salkhala Formation of Proterozoic age shows thrust contact (?) with Central Crystallines, marked by MCT. This formation is well exposed in Champal, Tante Ki Gali, Bachhara, Dadi, Bhandera and Duri area. It comprises phyllite, carbonaceous phyllite, garnetiferous mica schist with bands of quartz, dunite schist, staurolite schist, sillimanite schist and calc-silicate rocks with marble band and basic enclaves. Garnet mica schist remains the host rock even for the migmatized assemblage.

Bhaderwah Formation:

The Undifferentiated Salkhala Formation is overlain by the Bhaderwah Formation of Neo Proterozoic age which is well exposed in the southernmost part of the toposheet. Its contact with underlying Salkhala is gradational one. These rocks are light to dark grey slates and silver grey to buff coloured phyllites inter bedded with dirty grey to grey bands of quartzite and quartzose slates. These slates show incipient development of brownish crystals of garnet in many localities to have

closer affinity with garnetiferous phyllite. In contrast to the thin impersistent bands of quartzite and green coloured chlorite phyllite, phyllite, quartzite and slates are the dominant constituents.

Undifferentiated Quaternary deposit:

Small patches of Undifferentiated Quaternary sediments represented by alluvium, scree and moraines are present in the area.

Pias Granite: The grey to greenish coloured granite gneiss containing bluish quartz, feldspar, hornblende, blue quartz and chlorite and chlorite has tentatively been assigned Neo Proterozoic age. This intrusive granite occurs within Kishtwar window Zone.

Kaplas Granite: Leucocratic, coarse grained, porphyritic, biotite granite (=Piparan Granite) of Palaeozoic age in contrast is seen intruded within the rocks of Central Crystallines and Salkhalas.

2.3 Geology of the Study Area

The area of investigation is a part of Lesser Himalayan zone and falls in Survey of India Toposheet No. 43O/12 and geologically exposes Palaeozoic rocks and Quaternary deposits. The study area from Doda Pul to Lawa exposes rocks of Salkhala Formation which comprises of mica-schist, quartzite, gneiss, marble and overlain by Quaternary deposits. Quaternary deposits of Pleistocene to Holocene age are represented by alluvium, scree and moraine at places Kaplas Granite of Proterozoic age is intrusive within Salkhala Formation. The Salkhala Formation is tectonically overlain by the high grade metamorphites (Padar and Liond formations of Central Crystallines) along the trace of Atholi Thrust (Main Central Thrust). The Kaplas Granite of Proterozoic age is intrusive into Salkhala Formation of rocks.

2.2.1 Salkhala Formation: Salkhala Formation which comprises of quartzite, mica-schist, gneiss and marble. The brief description of different litho-units is given below:

2.2.1. Quartzite Fine to medium grained. Its light brown to buff in color with some reddish-brown staining. Massive and compact appearance with a blocky structure. Some patches exhibit siliceous luster on fresh surfaces. The visible fractures and joints may be due to tectonic stress. It mainly exposed all along NH-244 from Pul Doda to Lawa, near premnagar at places it is massive having band of 5-6m thick.



Figure 2.1 Field photographs showing a.Quartzite exposure near Prem Nagar b. Near Thattri along NH-244..

Table 2.2: Generalised Stratigraphy of the area, J&K (Source: Compiled Geological Map, GSI).

Zone	Group	Formation	Lithology	Age (as per GSI compiled map)
	Pir Panjal	Panjal Volcanic	Andesitic to basic lava flows with arenite and chert band	Early Permian
		Agglomeratic Slate	Slate, phyllite, quartzite with conglomerate, tuff and ash bed	Late Carboniferous to Early Permian
	Intrusive	Kaplas Granite	Leucocratic to Mesocratic biotite granite	Palaeozoic
Parautochthonous Zone		Sincha	Phyllite, limestone, quartzite and basal conglomerate	Neo Proterozoic
		disconformity represented by Basal Conglomerate unit.....		
		Ramban	Arenite, Slate, carbonaceous phyllite, carbonate and quartzite bands	Neo Proterozoic
.....Vaikrita Thrust/ Panjal Thrust.....				
Nappe Zone		Dumgali	Schists and quartzites	Neo Proterozoic
		Salkhala	Chlorite phyllite, grey phyllite, quartzite, slate, Garnet schist, quartz biotite schist, quartzite	Mesoproterozoic

Mica schist: It's a medium- to high-grade metamorphic rock characterized by a foliated texture. The rock displays a mix of gray, dark bluish-black, and rusty orange hues. Rusty staining (likely from weathered iron-bearing minerals) at some places suggests oxidation of biotite or garnet. Well-developed foliation is visible, with schistosity planes running obliquely. Fine- to medium-grained

with visible alignment of minerals. The rock is mainly composed of quartz, muscovite, biotite, chlorite, and possibly garnet at some places.

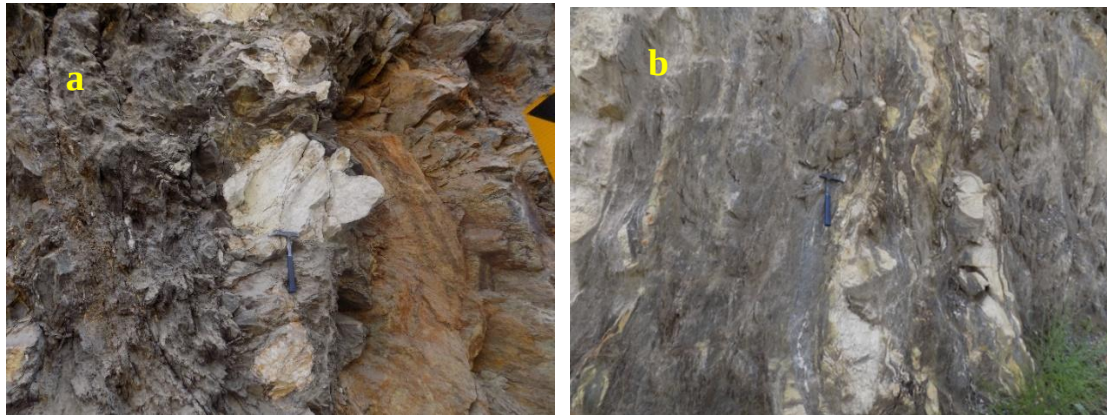


Figure 2.2 Field photographs showing a. Mica schist.b. Mica schist with Quartzite band exposures along NH-244

Marble band: Its light gray to off-white; fine- to medium-grained crystals. Smooth, polished-like exposure in some areas, possibly due to weathering and erosion. The beds appear thick-bedded to massive, with a distinct dip towards the road. Foliation is weak or absent, but some banding or layering may be present due to mineral alignment. It is well exposed near Drabshalla having width of 3m -5m and Thatri along NH-244.



Fig.2.4 Field Photograph showing a.Exposure of marble band near Darabshalla.b.exposure of marble band at Thatri

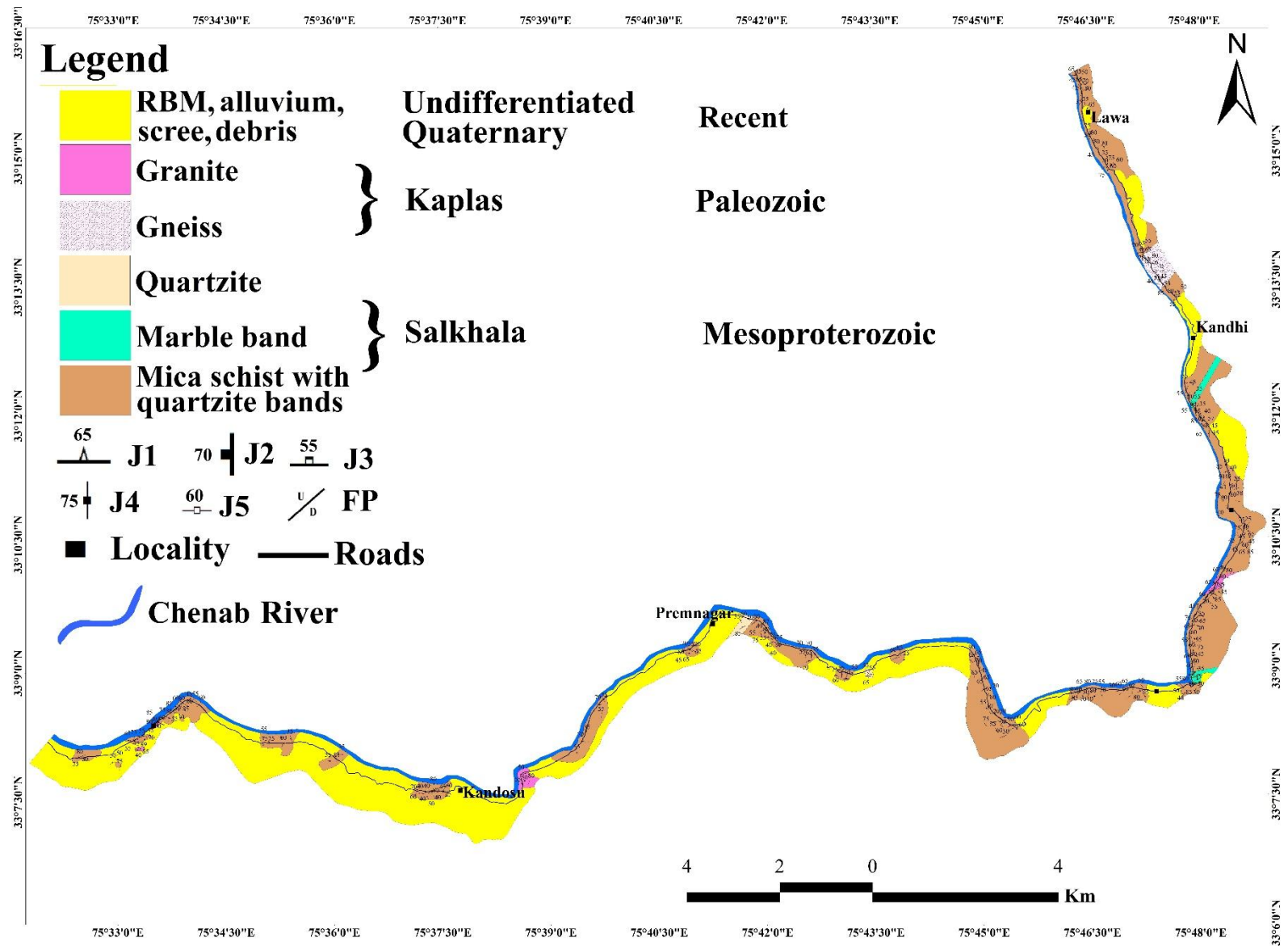


Fig.2.3. Geological Map of the Study area

Gneiss: It's a medium- to coarse-grained; distinctly foliated with a gneissic banding pattern. Alternating light and dark mineral bands. Light bands are mostly composed of quartz and feldspar and dark bands: mainly biotite, hornblende, and possibly garnet. Strongly developed gneissic banding — parallel to subparallel light and dark mineral layers. Band thickness varies between ~0.5–3 cm (2.5a & b). This unit is exposed mainly near Darabshalla and Kandhani along NH-244.

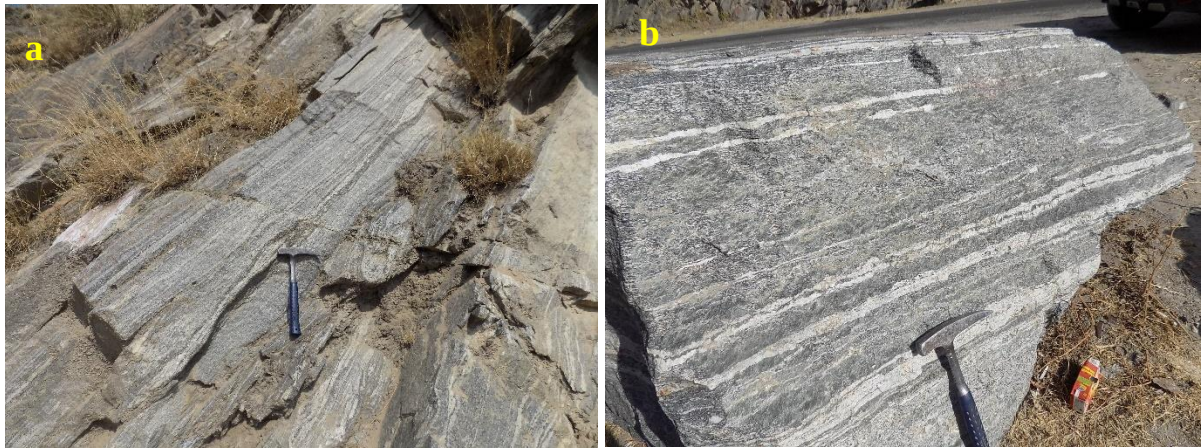


Fig.2.5. Field Photograph showing a. Exposure of gneiss near Darabshalla, b. banding in gneiss near Kandholi

Kaplas Granite Leucocratic, coarse grained, porphyritic, biotite granite (=Piparan Granite) of Palaeozoic age in contrast is seen intruded within the rocks of Salkhalas well exposed near Pul Doda along NH-244. (Fig.2.6)



Fig.2.6. Field Photograph showing Kaplas Granite intrusive exposure near Doda bridge along NH-244

2.2.3. Undifferentiated Quaternary: The undifferentiated Quaternary deposits of Holocene age are represented by alluvium, colluvium, moraine and scree, slope wash. Low-lying areas are occupied by alluvial deposits formed by the several major tributaries of Chenab River and Tawi rivers. The quaternary deposits are observed in the form of terrace deposits, channel deposits and as overburden material the gentle slope faces in the area. These deposits are

generally poorly sorted consisting an assemblage of rounded to well-rounded boulder/cobble/pebble/gravel size clasts of basalt, siltstone/sandstone, limestone/dolomite and quartzites etc and fine to medium grained sand, silt and clay in the area. These deposits cover the older formations in the area.



Fig.2.7. Field Photograph showing Quaternary exposure a. near Thattri, b. near Prem Nagar along NH-244

2.4 Structural deformation and Joints

The rocks are highly deformed and shows three prominent joint planes (J1, J2 & J3) with fourth inconsistent one (J4) randomly developed. In most of the cases J1 representing the first deformation stage (D1), is showing parallelism with S0 (J0) possibly indicating the isoclinal nature of F1 folding. The J1//J0 trends in NW-SE with northerly dip. The J2 representing the D2 cuts the J1 in opposite direction and trends in NW-SE with southerly steep dip. The J3 cuts across the strike of both the J1 & J2 and trends in NE-SW with steep either way dip.

CHAPTER- 3

LANSLSLIDE INVENTORY

3.1. Landslide Inventory data:

The landslide inventory map was prepared on the basis of landslide data from old workers, demarcated from satellite data and during field work. For collecting the landslide data of the study area, old GSI reports including that of Macroscale LSM (Pradeep Pal and Sharma Sanjeev Kumar, 2016 and 2018-19 & Sharma and Akram, 2019), published data were consulted. Further, high resolution Google Earth Pro imagery was used to identify and digitise the landslides. During pre-field work, the landslides were classified into debris and rock slides, wherever possible. Further, cut slope failures were also demarcated. The landslide data prepared during prefield studies were verified and updated during the fieldwork. Additional landslides were identified and classified during field work. The morphological parameters were observed in the field for all the slides and the 42 point proforma was recorded.

A total of 26 landslides were identified and systematically classified into various failure types along the Doda–Lawa corridor. These include 12 debris slides, 6 planar failures, 3 wedge failures, 2 toppling failures, and 3 cut slope failures (Figs 3.1& 3.2). The segment from Doda Bridge to Premnagar is predominantly affected by debris slides and debris falls, characterized by the downslope movement of pebble-sized fragments often triggered by steep anthropogenic cut slopes. The presence of interbedded pebble beds within colluvial and alluvial sequences further contributes to erosion and slope instability under gravitational and hydrological stress. A significant active debris slide was recorded at Gadsu, confined within loose debris material.

In contrast, the stretch from Premnagar to Lawa exhibits increased susceptibility to rock slides, with notable occurrences documented in the vicinities of Thattri, Drabshalla, and Lawa. These failures are primarily controlled by structural discontinuities in the bedrock, including jointed quartzites and schist sequences, exacerbated by undercutting and weathering along road cuts.

The landslide types were categorized into five classes based on their nature and scale. These classes include debris slides, planar failure, wedge failure, toppling debris fall, and cut slope instabilities. The collection of such data was very much essential to assess the impact of slope failure and instabilities in the area and to cross-verify the accuracy of the Mesoscale LSM (Fig.5.4).

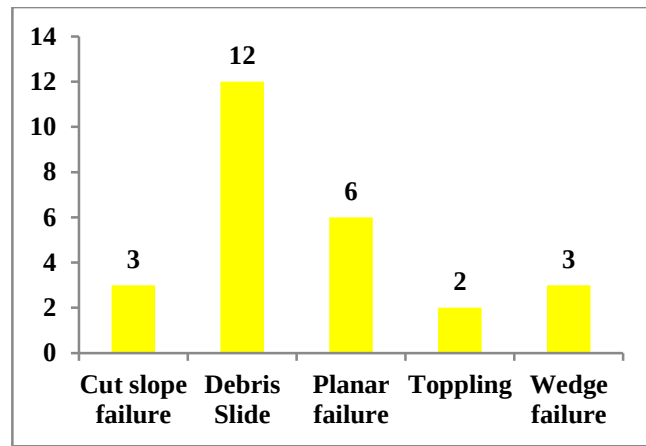
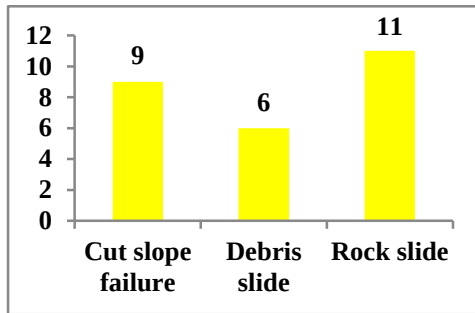


Fig. 3.1 Bar graphs showing Landslide Inventory (a) Prefield Google earth-based Landslides (b) Field based landslides of the study area.

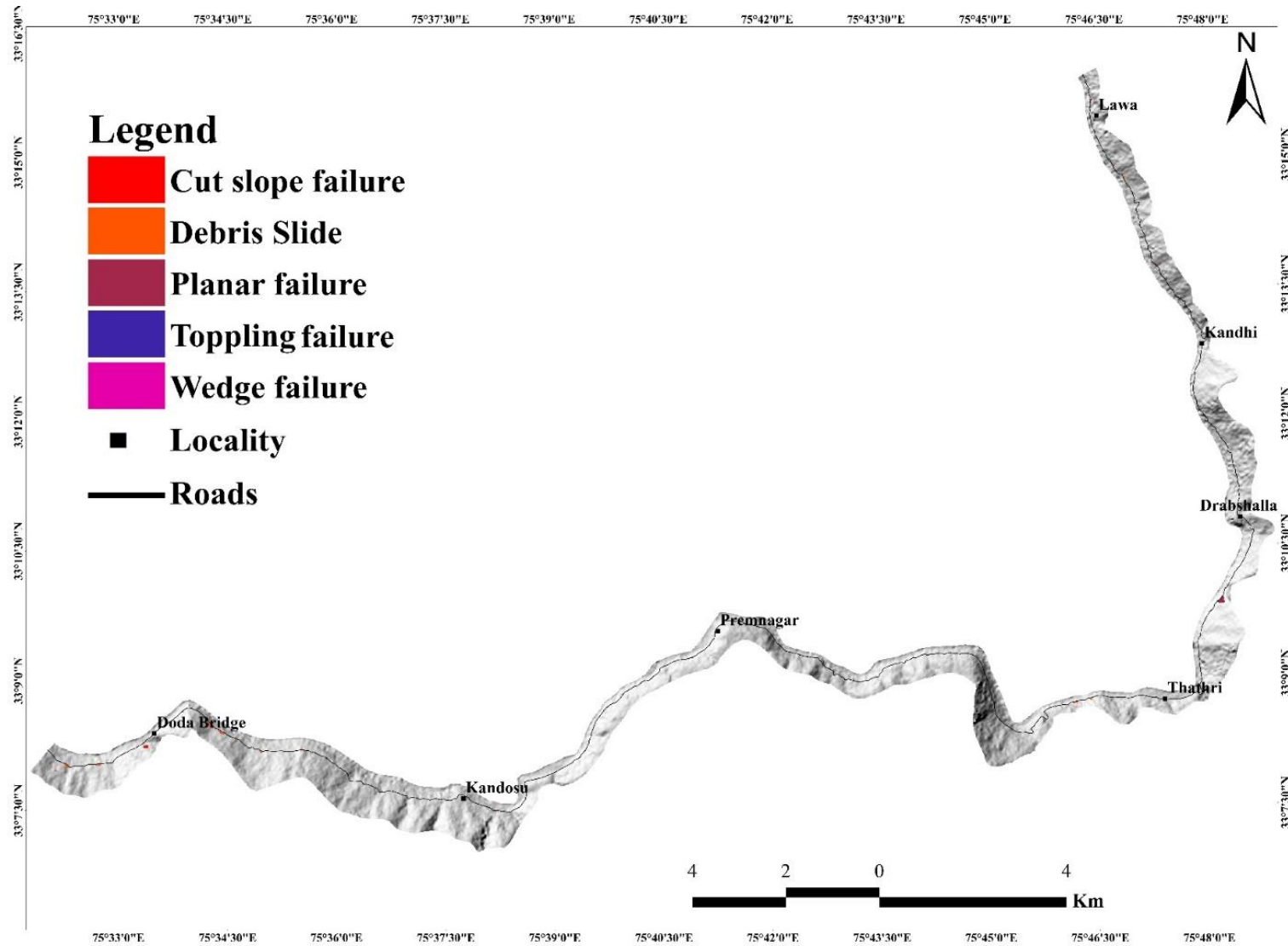


Fig. 3.2 Landslide incidence map of study area generated using Google Pro imagery and field study.



Fig.3.3 Field photographs showing landslides (a) Debris slide near Ganpath bridge (b) Debris slide near Premnagar.



Fig 3.4 Field photographs showing landslides (c) Toppling failure near Kandni (d)Toppling failure in quartzite and interbedded schist near Thatri.



Fig 3.5 Field photographs showing landslides (e) Wedge failure near Lawa (f) Planar in mica schist



Fig 3.6 Field photographs showing landslides (g) Major landslide at Gadsu (h) Rock slide near Darabshalla.

3.2. Selection of geo-factors: Based on the landslide types, numbers of geo-factors were selected (Table: 3.1).

Table: 3.1: Domain based geofactors.

Geofactors	Debris slide	Rock slide	Remarks
Slope angle	Yes	Yes	Both debris and rock slides are more prevalent in the NE and SW part, where the NH-244 passes along steeper slopes.
LULC	Yes	Yes	Debris and rock slides are dominant in the NE part, where slopes are covered by thin grass or are barren. Dense and moderate vegetation in the central part results in fewer slides.
Geomorphon	Yes	No	Debris slides and flows are more common in hollows and slopes in the NE part.
SFM	Yes	No	Debris slides are more dominant in Colluvium and RBM with a thin colluvium cover.
Hydrogeology	Yes	No	The SW part is more wet & dripping than NE part.
Relative Relief	Yes	No	The relative relief is more along the cut slopes.
SMR	No	Yes	Rocky slopes consist of lithologies such as quartzite, schist, marble, granite and gneiss, with varying foliation intensity.
SPI	Yes	Yes	Rill and gully erosion are more common in the SW and central parts.
SFM Thickness	Only for cutslope failure		The SFM with higher thickness experiences more debris slides along the NH-244 around Ganpath bridge and around Darabshalla.

The area was divided into distinct zones based on terrain characteristics: overburden covered slopes, rocky slopes, and cut slope sections. The overburden covered slopes were identified as susceptible to debris slides and debris flows due to the presence of thick, unconsolidated material. The NH-244 road cut through this terrain steeply exposed substantial unconsolidated overburden material and created higher relative relief along the cut slope. Consequently, debris slides, rock slides, and rock falls were observed on the back slope of the cut slope. This led to the delineation of a specific domain for cut slope failures. Along the rocky slopes, which are steep, barren, and characterized by multiple joints, rock slides and rock falls occur along different joint planes. To assess stability within the cut slope failure domain, Slope Mass Rating (SMR) was used at rock exposures, evaluating susceptibility to planar slides, wedge slides, and toppling failures.

CHAPTER–4

THEMATIC MAPS

4.1. Digital Elevation Model (DEM) and Image derivative thematic maps:

Various morphometric maps derived from the ALOS DEM include facet maps, slope maps, aspect maps, drainage maps, stream power index (SPI) maps, and geomorphon maps. Additionally, land use/land cover (LULC) maps, element at risk maps, and active landslide inventory maps were created using the latest Google Earth imagery and subsequently verified in the field.

4.1.1 Slope Facet Map: The slope facet map was created using a computer algorithm applied to a DEM-derived Triangular Irregular Network (TIN) continuous surface (Paul and Kumar, 2022; Paul and Ibrahim, 2022, 2023). The slope gradient classification was conducted with the following intervals: less than 15° for gentle slopes, 15-25° for moderate slopes, 25-35° for moderately steep slopes, 35-45° for steep slopes, and greater than 45° for very steep slopes. The slope aspect classification was divided into 30° intervals, resulting in the following classes: 0-29°, 30-59°, 60-89°, 90-119°, 120-149°, 150-179°, 180-209°, 210-239°, 240-269°, 270-299°, 300-329°, and 330-360°, arranged clockwise from NNE to NNW. In total, 1,726 facets were generated (Fig. 4.1). The minimum facet size chosen for this study was 3,250 square meters. Most facets were limited to an area of 10,000 square meters, providing a better representation of the terrain for most of the study area.

4.1.2. Slope gradient, aspect and relative relief map: The slope gradient map was created from the ALOS DEM using ArcGIS software. The area distribution is as follows: 12.06% of the area has very gentle slopes (0-15°), 19.06% has gentle slopes (15-25°), 35.8% has moderately steep slopes (25-35°), 26.39% has steep slopes (35-45°), and 6.64% has very steep slopes (>45°) (Figs. 5.2). The NW part of the study area exhibits a steep and very steep slopes. Nearly all landslides were observed along cut slopes where over-steepening occurs due to road or track construction. Most debris slides (>90%) are found on moderately steep slopes.

The aspect map was prepared by classifying the area into 12 directions varying from 0° to 360° with each class representing 30°. 68% of the area shows a southerly slope direction (Fig 4.3).

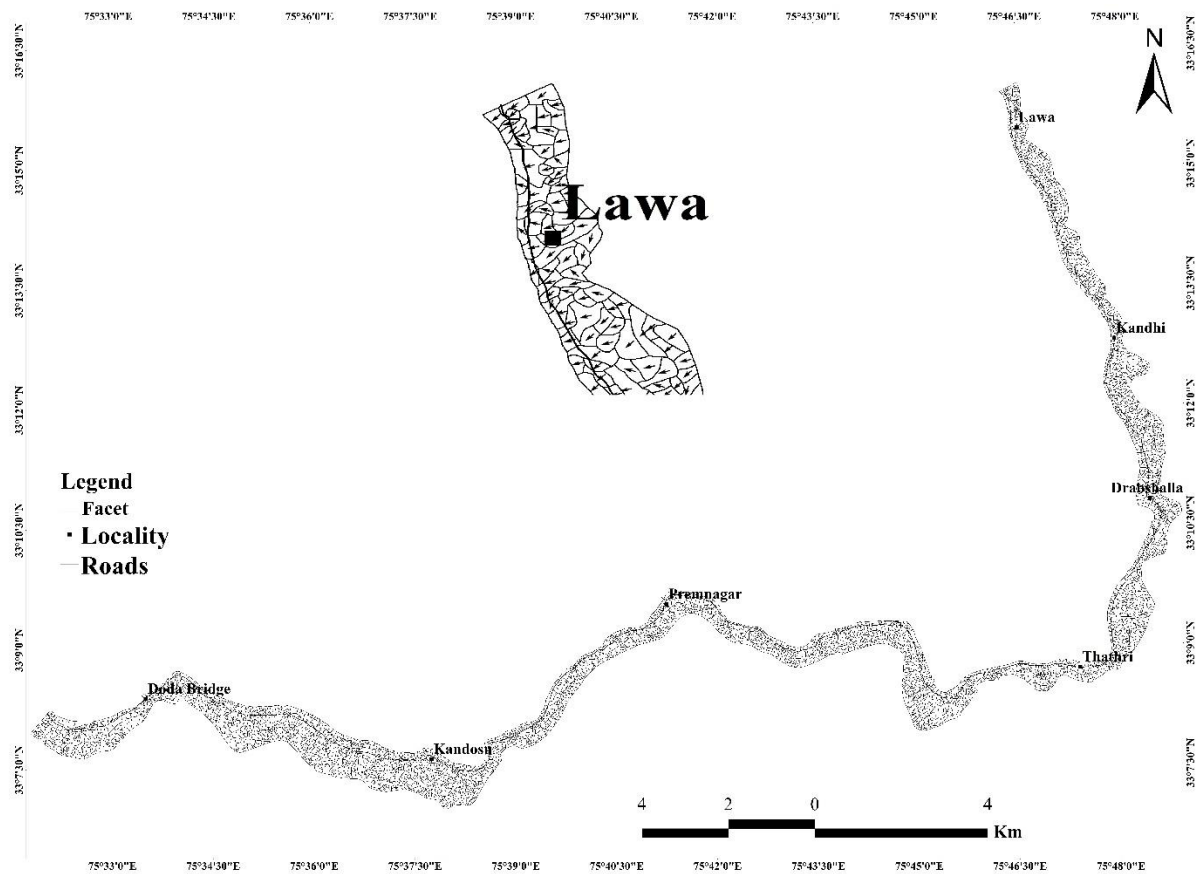


Fig.4.1. Facet map prepared using TIN model showing geometrical configuration of slope.

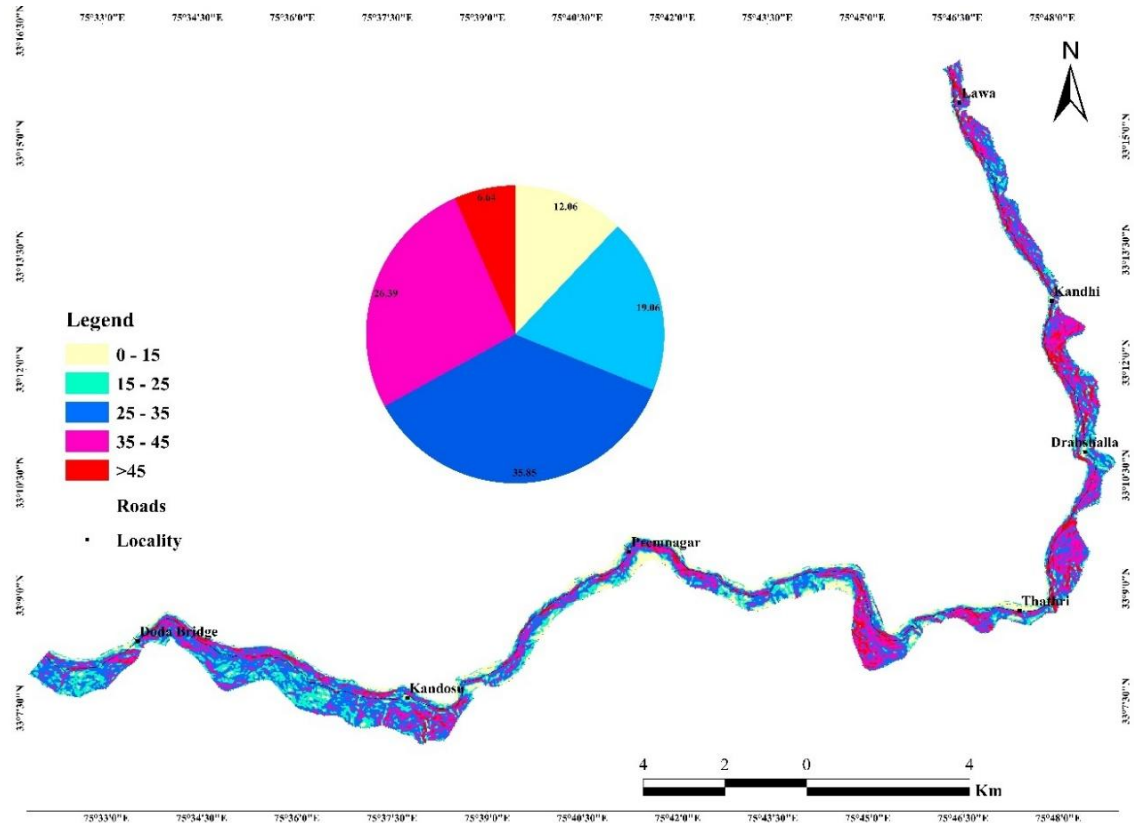


Fig.4.2. Slope map of the study area

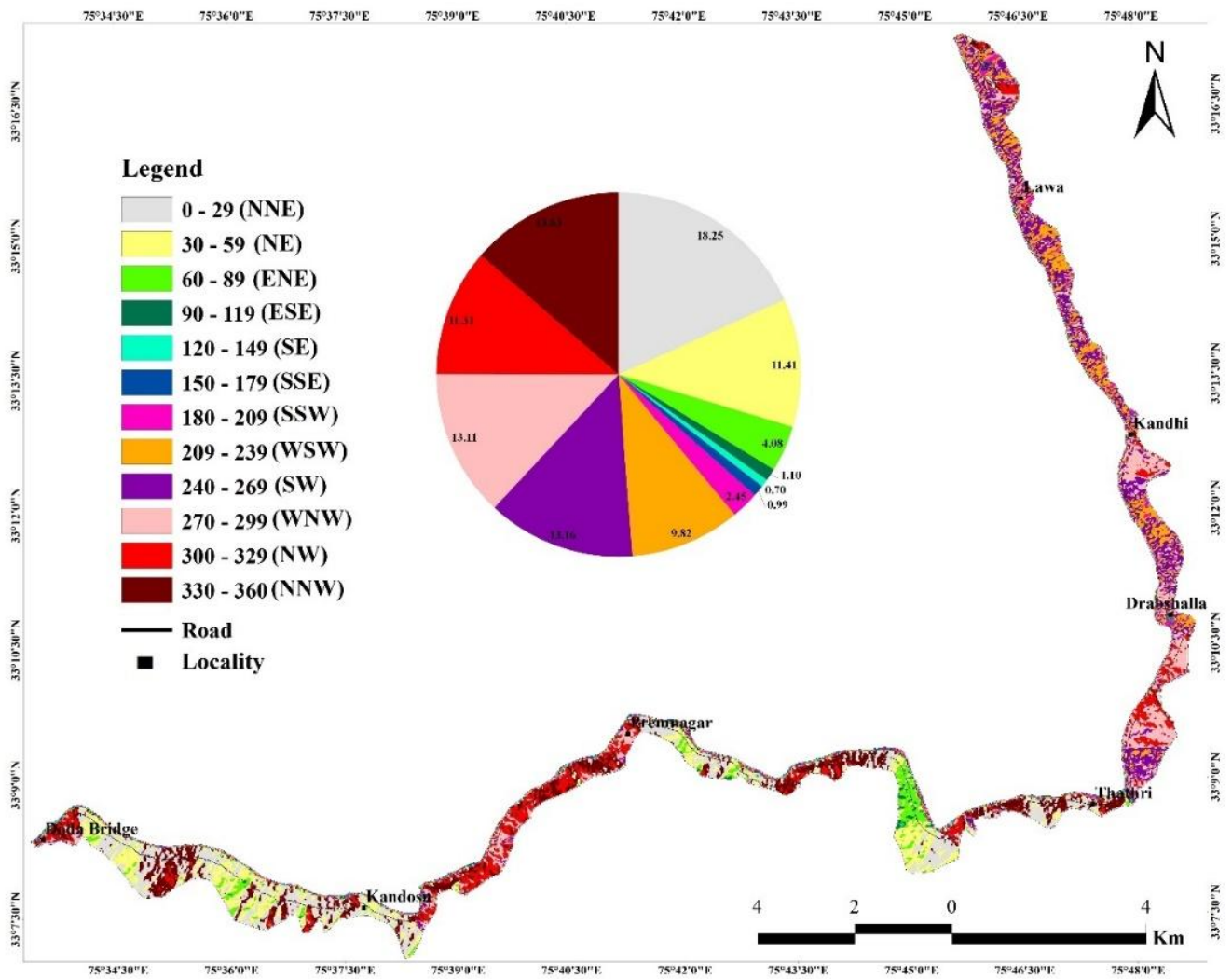


Fig. 4.3. Aspect map of the study area generated from ALOS DEM. The analysis revealed that 19% of the facets have a relative relief of less than 50m, 45% have a relative relief between 50-100m, 24% have a relative relief between 100-150m, 9% have a relative relief between 150-200m, and the remaining 3% of facets exhibit a relative relief greater than 200m (Fig. 4.4).

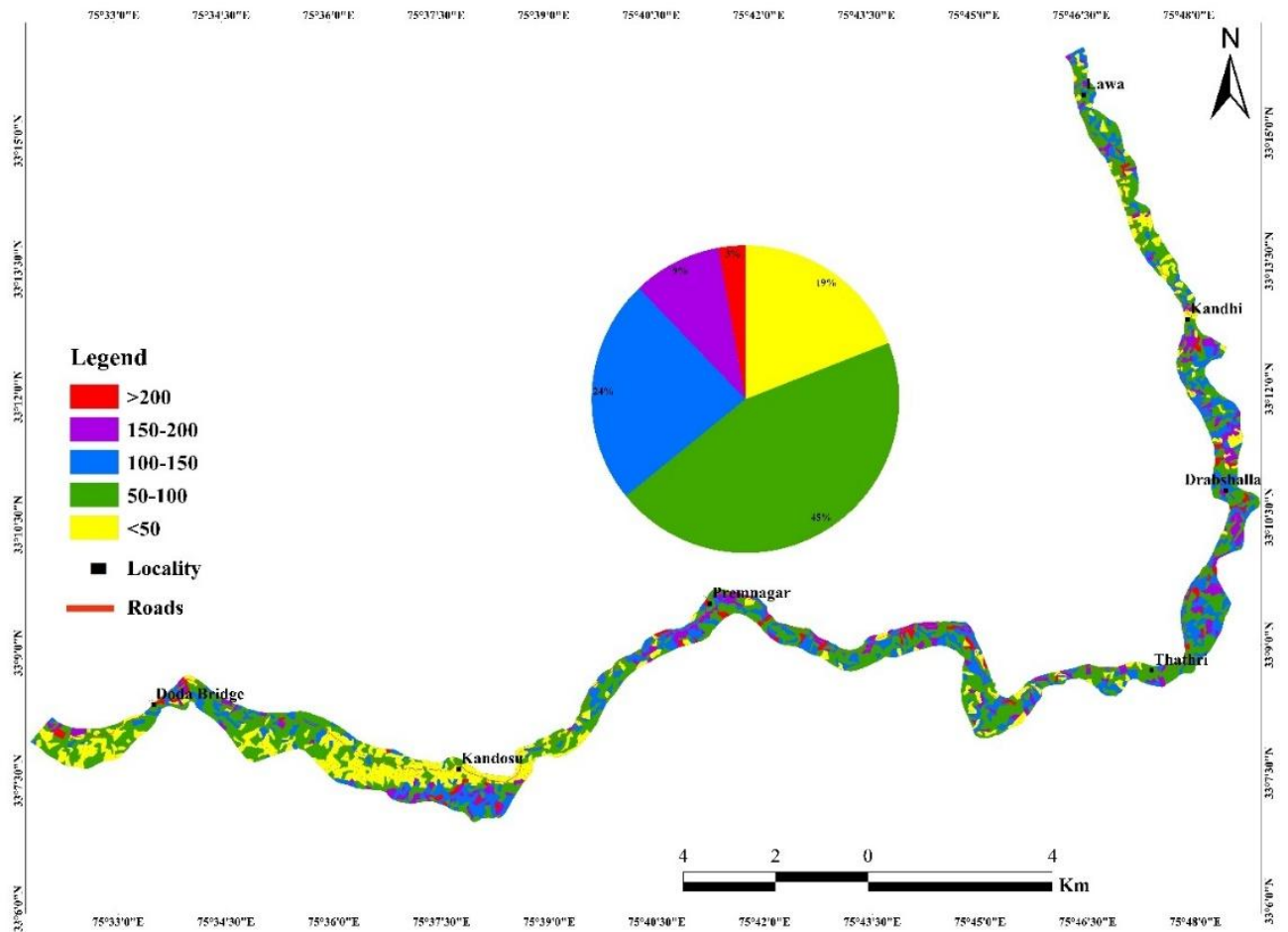


Fig. 4.4. Relative relief map of the study area generated from ALOS DEM.

4.1.3 Stream power index map (SPI):

The area is drained by the Chenab River and its tributaries, exhibiting an overall sub-parallel drainage pattern. The Stream Power Index (SPI), which measures the erosive power of flowing water at specific points on the topographic surface, was calculated based on the slope gradient and contributing area. The SPI identifies potential locations for gully formation in the landscape. Using ArcGIS software, the SPI map was generated from ALOS DEM data and is calculated with the following equation:

$$SPI = \ln(CA * \tan G) \text{ -----Eq-6}$$

where CA is the upstream catchment area and G is the slope gradient at a cell.

The SPI ranges from -14 to 6 and has been classified into least erosion (<0), very low erosion (0-1.5), low erosion (1.5-3), moderate erosion (3-4.5), and high erosion (4.5-6) classes. The ridges, spurs and summits show the least erosion with <0.0 SPI value, all the

1st order and adequate 2nd order streams fall in SPI class of 0-2, most of the 2nd order and few 3rd order streams fall in SPI class 2-4, most of the 3rd order streams fall in SPI class 4-6, the remaining 4th and higher order streams fall in highest SPI category of >6 value (Figs.5.6). Most landslides are found located in areas with moderate erosion SPI. This suggests that higher-order streams significantly contribute to landslides.

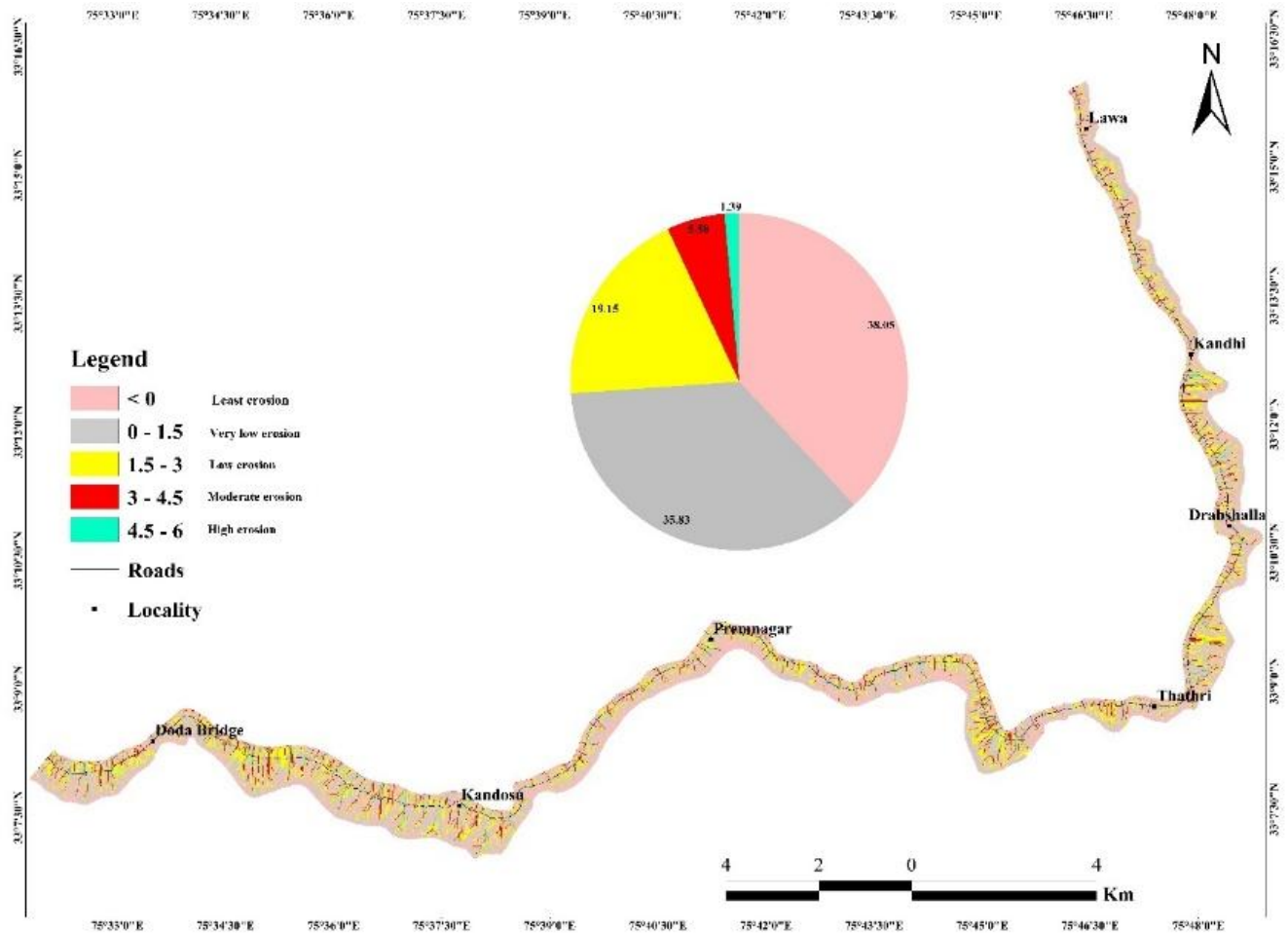


Fig. 4.5. Stream power index (SPI) map of the study area generated from ALOS DEM.

4.1.4 Geomorphon map: The geomorphic properties of the land surface (e.g., slope angle, aspect, curvature) give rise to diverse terrain morphology and cause the natural slopes to manifest a variety of complex landforms. The Geomorphon map was prepared using the r.geomorphon extension module of GRASSGIS7.6. The geomorphon map shows the following landform types in the area: slope (58% area), hollow (15.5%), spur (13%), shoulder (4.6%), valley (4%) and ridge (2%) and the remaining landforms generated are foot slope, depression and summit, comprises <3% of the area. (Fig.4.6).

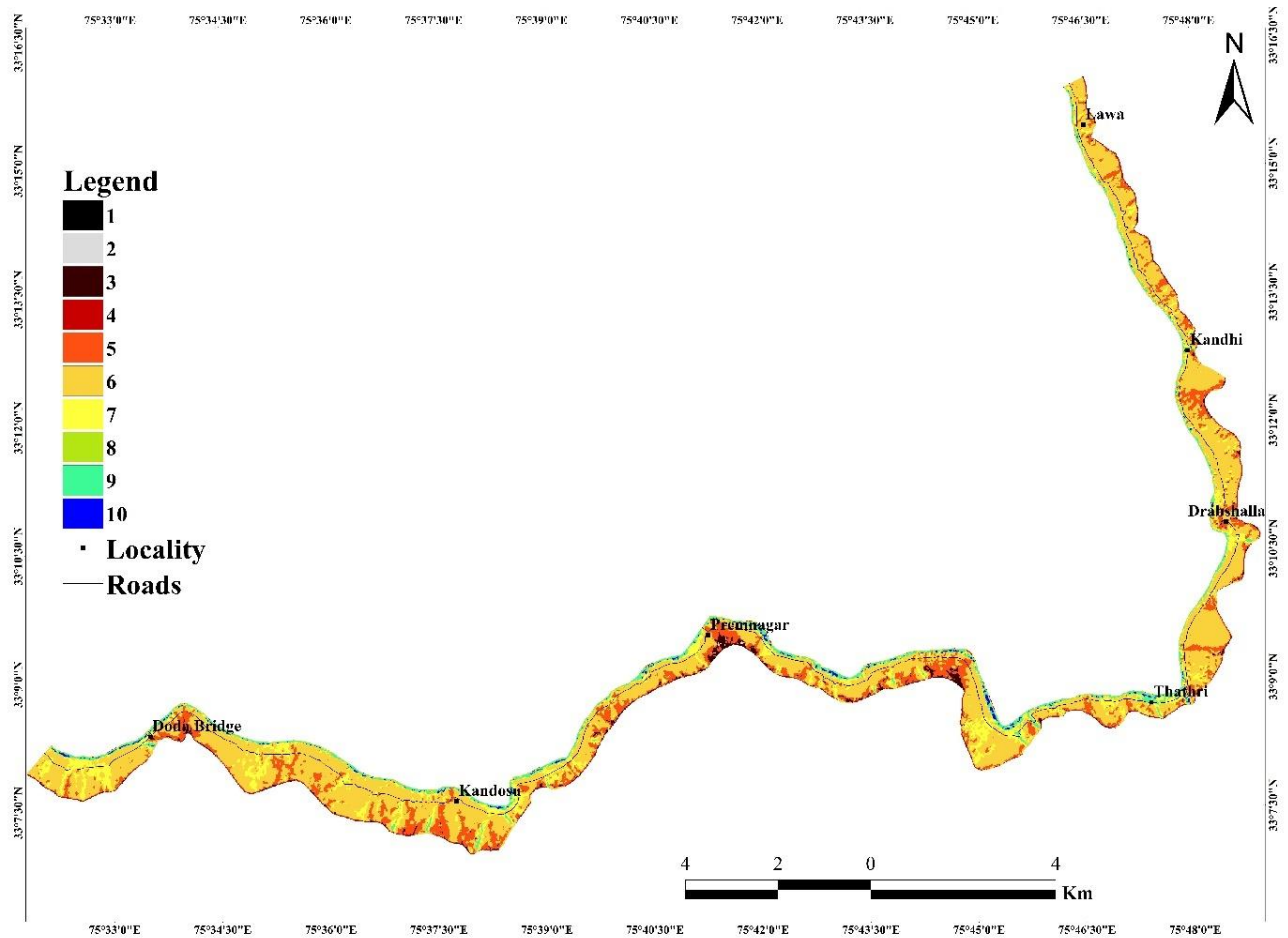


Fig. 4.6. Geomorphon map of study area generated from ALOSDEM.

4.1.5 Land use land cover (LULC) map:

The Land Use and Land Cover (LULC) map was initially created using Google Earth Pro through image interpretation during the pre-field phase. It was subsequently verified and updated based on findings from field studies. The final LULC map incorporates these field inputs for accuracy and comprehensiveness. In the study area, moderate forest dominates, accounting for 23% of the total area. This is followed by cultivated land and barren land, which covers 21% each. Sparse forest makes up 16% of the area, primarily located from Pul Doda to Thatri. Grass cover 7 % of the study area, with dense forests accounting for 5%. Settlements also occupy 3% of the area, with most of these found from Premnagar to Darabshalla. The remaining 4% of the area is covered by river, petrol pump, roads, security establishment and transmission towers (Fig. 4.7).

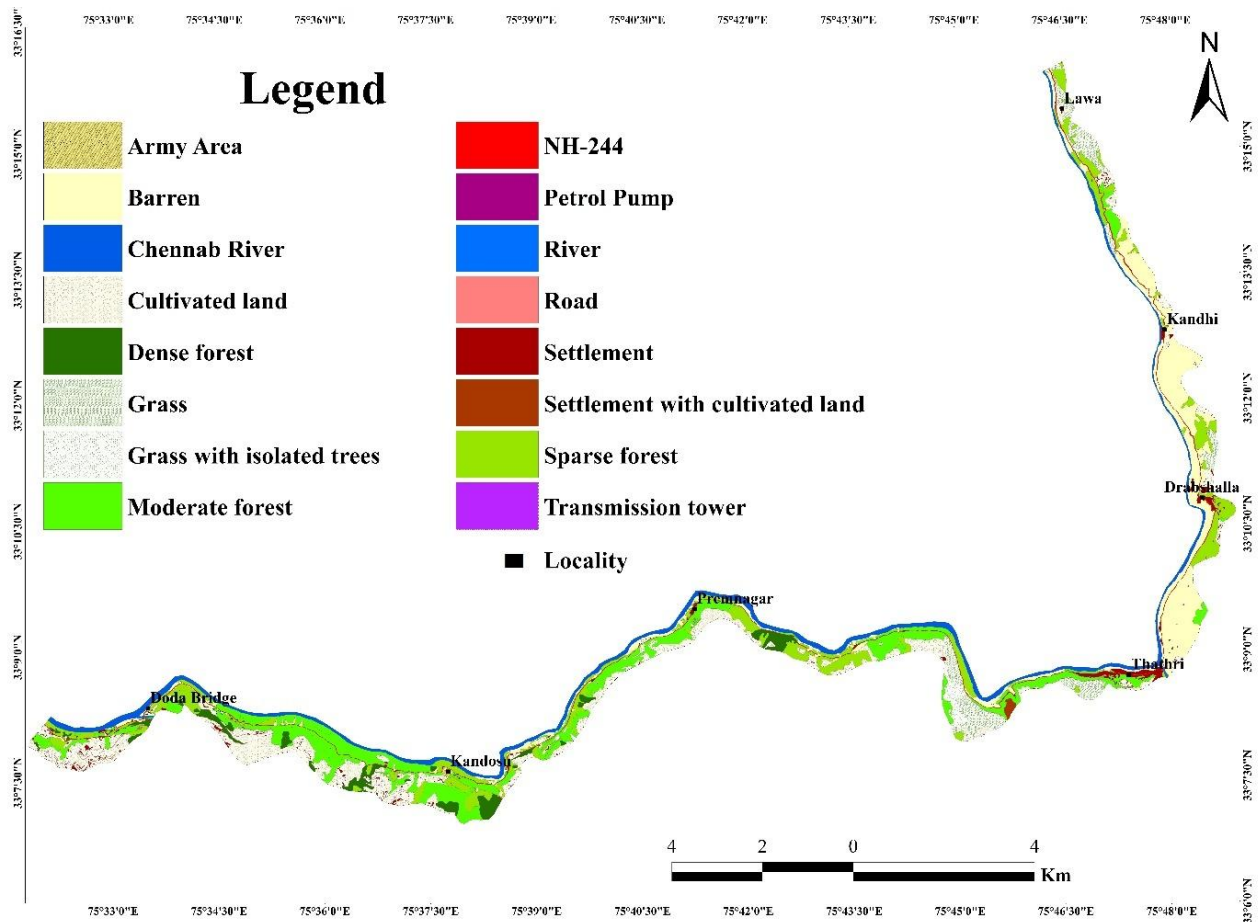


Fig. 4.7. LULC map of study area.

4.2 Field derivate thematic maps: During the field work various inherent causative factors for landslides were identified and mapped. These geofactor maps include SFM map, SFM thickness map, hydrogeological map, SMR map, structural map for kinematic failure, geotechnical map of rocks and overburden (on basis of sample analysis), landslide inventory map, updated lulc map and element at risk map.

4.2.1. Lithological map: Quartzite, mica schist with quartzite bands and marble of Salkhala Formation and gneiss, granite of Kaplas and Quaternary deposits consist of RBM, Alluvium, scree, debris exposed in the study area. The geological map of the area shows 48.5% mica schist with quartzite bands, 37.7 % of quaternary deposits, 9% quartzite, 4.5% of area is represented granite mostly towards the north-western extent of the study area. The remaining area is covered by marble band (0.11%) and gneiss (0.05%) (Fig.4.8).

The rock exposure in the area is found all along the road section from Ganpath bridge to Lawa. The rock exposure covers 45% of the total study area.

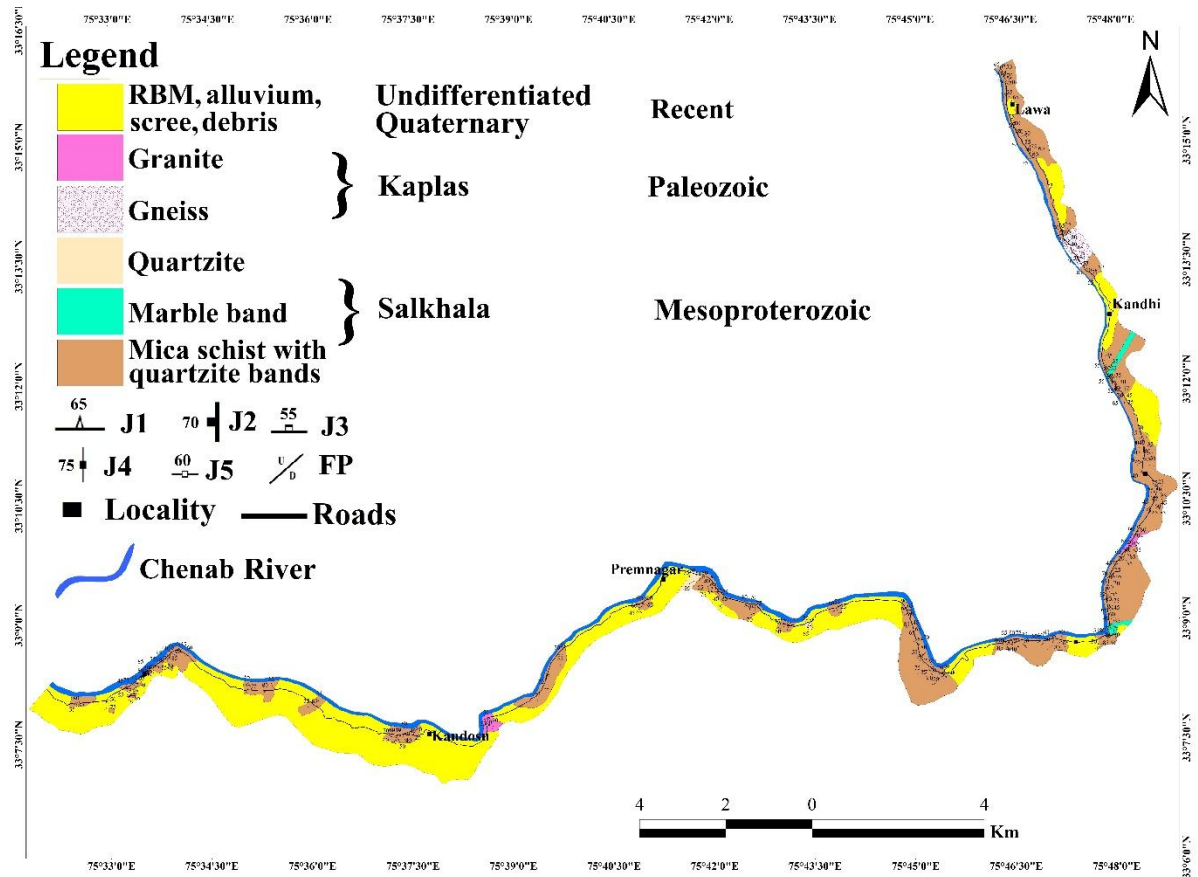


Fig:4.8 Geological map of the study area.

4.2.2 Structural details: The rocks are highly deformed and shows three prominent joint planes (J1, J2 & J3) with fourth inconsistent one (J4) randomly developed. In most of the cases J1 representing the first deformation stage (D1), is showing parallelism with S0 (J0) possibly indicating the isoclinal nature of F1 folding. The J1//J0 trends in NW-SE with northerly dip. The J2 representing the D2 cuts the J1 in opposite direction and trends in NW-SE with southerly steep dip. The J3 cuts across the strike of both the J1 & J2 and trends in NE-SW with steep either way dip.

4.2.3. Slope forming material (SFM) map: The north-western part of the study area is predominantly rocky, covering 44.6% of the region. This is followed by areas covered by colluvium (37.4 %), river-borne material (RBM) (15.8%), regolith (1.6%), glacio-fluvial material with a thin colluvial soil layer (3.5%), colluvium over RBM (1.4%), debris (0.2%), and scree (0.14%) (Fig. 4.9). The regolith, consisting of weathered rock fragments within in-situ soil, overlies the parent rock and is typically covered by dense vegetation. Scree, composed of rock fall fragments, accumulates primarily at the foot of slopes. Colluvium, a thin layer of

finer material, covers rocks, moraines, and debris. This heterogeneous mixture of fine mud/silt and highly angular rock fragments, more prevalent at the bottom of slopes, is referred to as colluvium. Debris is defined as a heterogeneous mixture of mud/silt and rock, with rock fragments randomly scattered within the mud. Loose debris and loose colluvium are differentiated when these materials are found to be non-cohesive. From Bela to Thattri, the area is mainly characterized by River-borne material (RBM) with a thin colluvial soil cover, making it vulnerable to debris slides. is present along the Rambhara River and its tributaries, while the area from Thattri to Lawa is rocky. The dominant slope-forming materials (SFM) in the study area are rocky slopes and soil, with colluvium cover primarily found from Ganpath bridge to Premnagar.

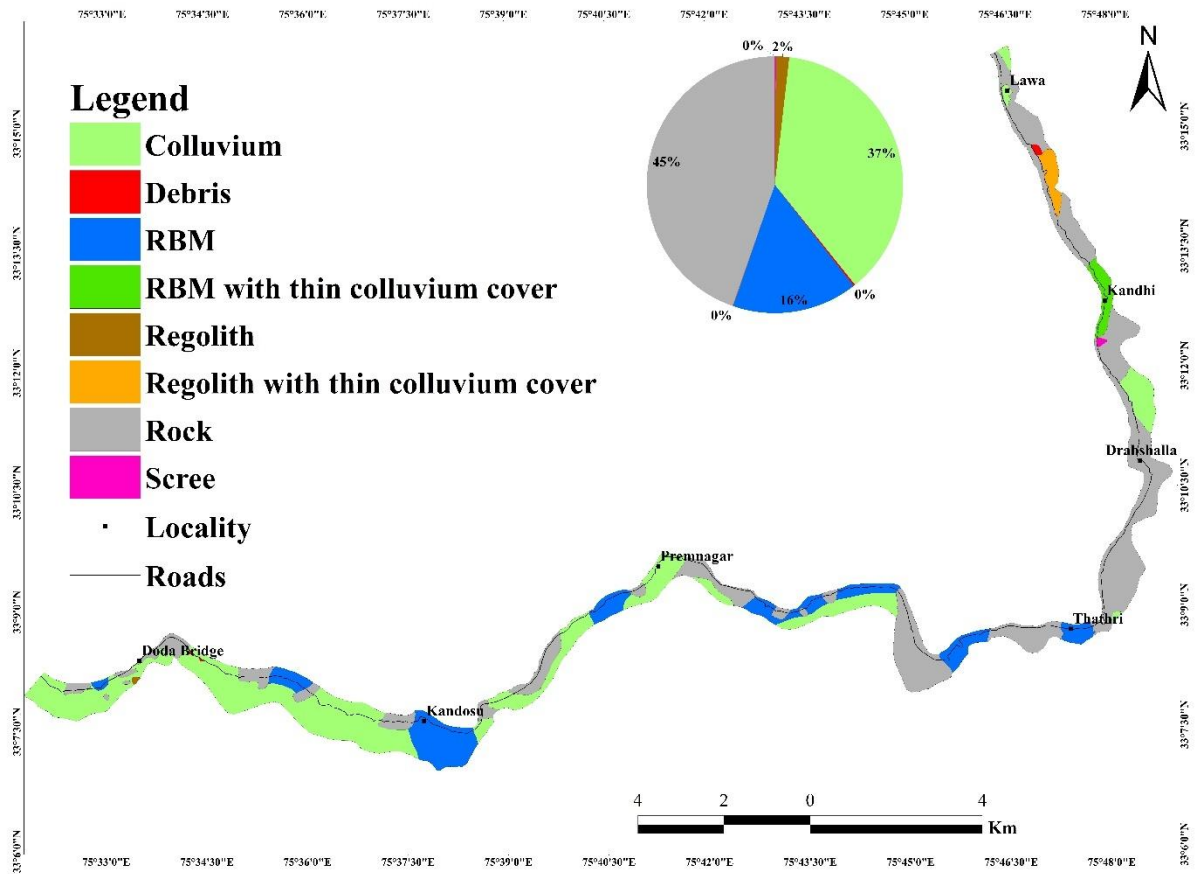


Fig.4.9 Slope forming material (SFM) map of the study area.

4.2.4. Slope forming material (SFM) thickness map: Facet based SFM thickness map was prepared along the cut slope failure domain along the road section of the study area. The thickness is varying from <1m to 20m thick (Fig.4.10.). The SW part of the study area is having higher thickness compared to other parts as here the slope forming material is more prone to

weathering due to freezing and thawing action also the presence of glaciers. The area is divided into various classes based on the classification by Anbalagan et al (2008). The thickness classes are defined as follows: nil to very thin (<1 m), thin (1-3 m), moderate (3-7 m), thick (7-10 m), very thick (10-15 m), and extremely thick (>15 m). Overburden thickness class of <1m covers 47.2% of the cut slope area followed by 1-3 m thin (23.5%), moderate (21%), extremely thick (4.74%), very thick (2.4%), and remaining 1% is having thick SFM cover.

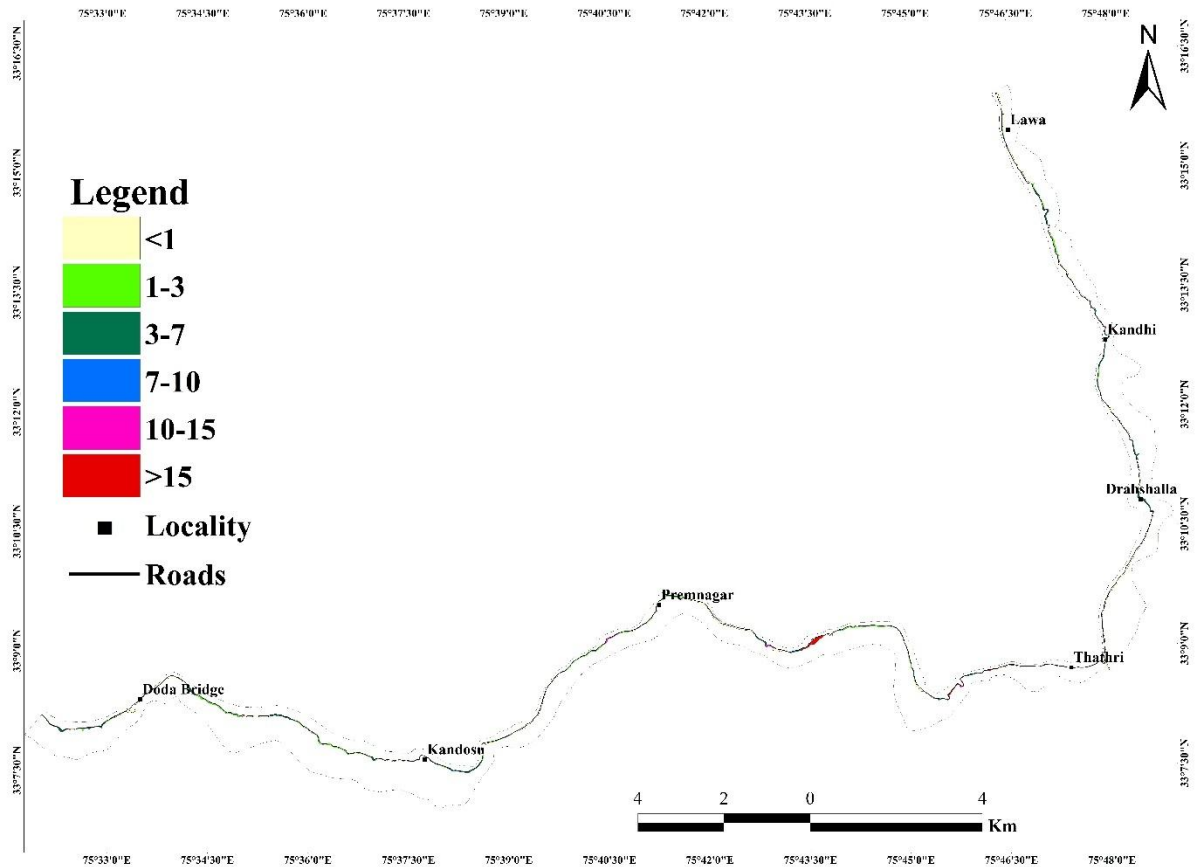


Fig. 4.10. Slope forming material (SFM) thickness map of the cut slope failure domain.

4.2.5. Hydrogeological map: The hydrogeological map was prepared based on the presence or absence of water seepages at shallow levels in the field area. Based on the hydrogeological condition, the area has been divided into five categories i.e. dry, damp, wet, dripping and flowing. Most of the area is having dry >damp >wet >dripping >flowing conditions respectively comprising 66.3%, 31.5%, 0.87%, 0.65% and 0.56% of the total area (Figs.4.11). Based on the water seepage observed in numbers of debris slides, shallow hydrogeological condition was considered as one of the causative factors for debris slides and therefore, was selected as an additional geo-factor map.

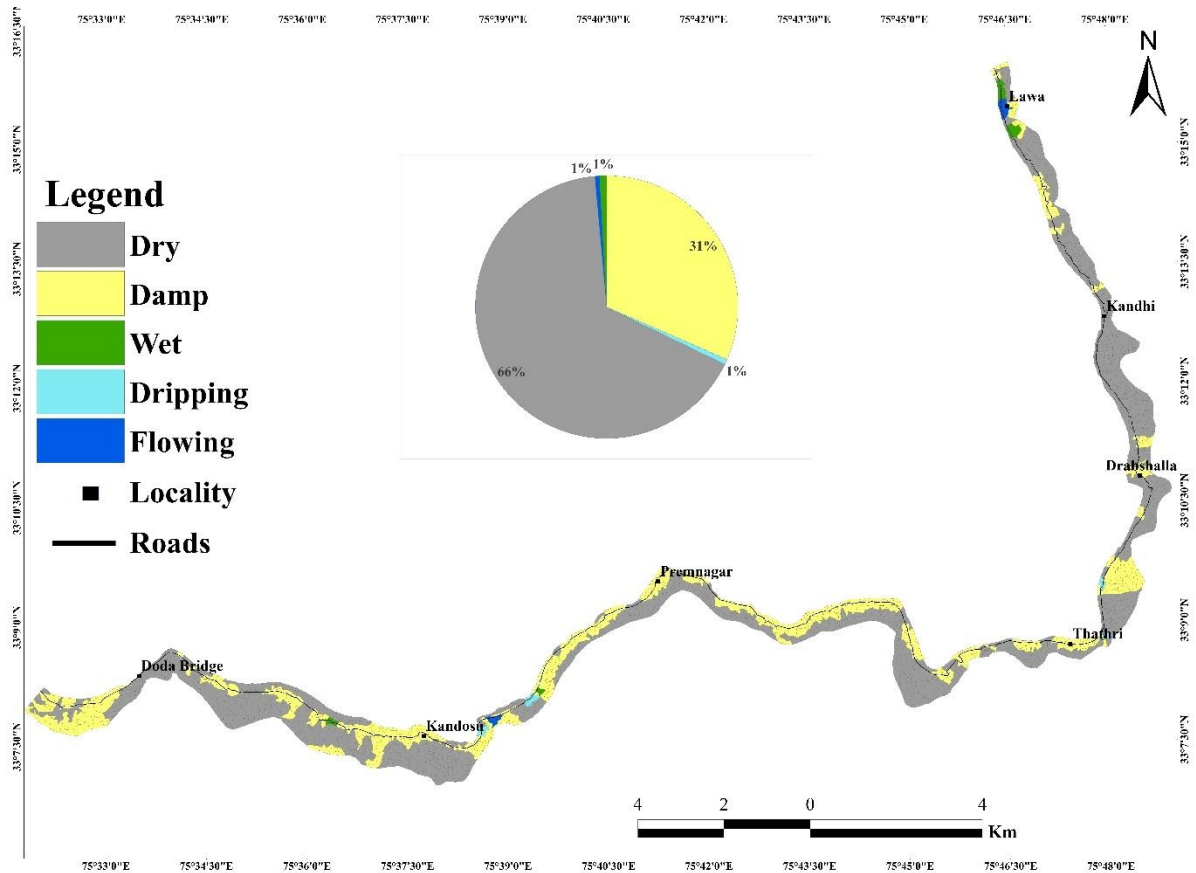


Fig. 4.11 Hydrogeological condition map of the study area.

4.2.6. Element at risk map:

Elements at risk encompass any object that might fail or be exposed to hazards during a disaster. This includes all valued attributes threatened by the hazard, in this case, landslides, and covers life, structures, land, resources, social and physical infrastructure, public service utilities, environmental qualities, economic activities, and physical and mental well-being. Some of these attributes can be quantified; some can be expressed in economic terms, while others resist easy quantification.

The NH-244 is a vital highway connecting Ramban district to Doda, Kishtwar and Anantnag playing a crucial role for the twin districts of Doda and Kishtwar as they connect with Jammu and Kashmir. As such, the Nh-244 itself is the most significant element at risk. Additionally, the settlements from Ganpath to Lawa are highly vulnerable along security establishment and few petrol pumps. Electric transmission towers are also considered elements at risk (Fig. 4.12). The risk elements of the area were evaluated into three important

categories:1. Essential and strategic covering roads, settlements,2. Public and administrative infrastructure in residential areas, 3. Agricultural and Forest land.

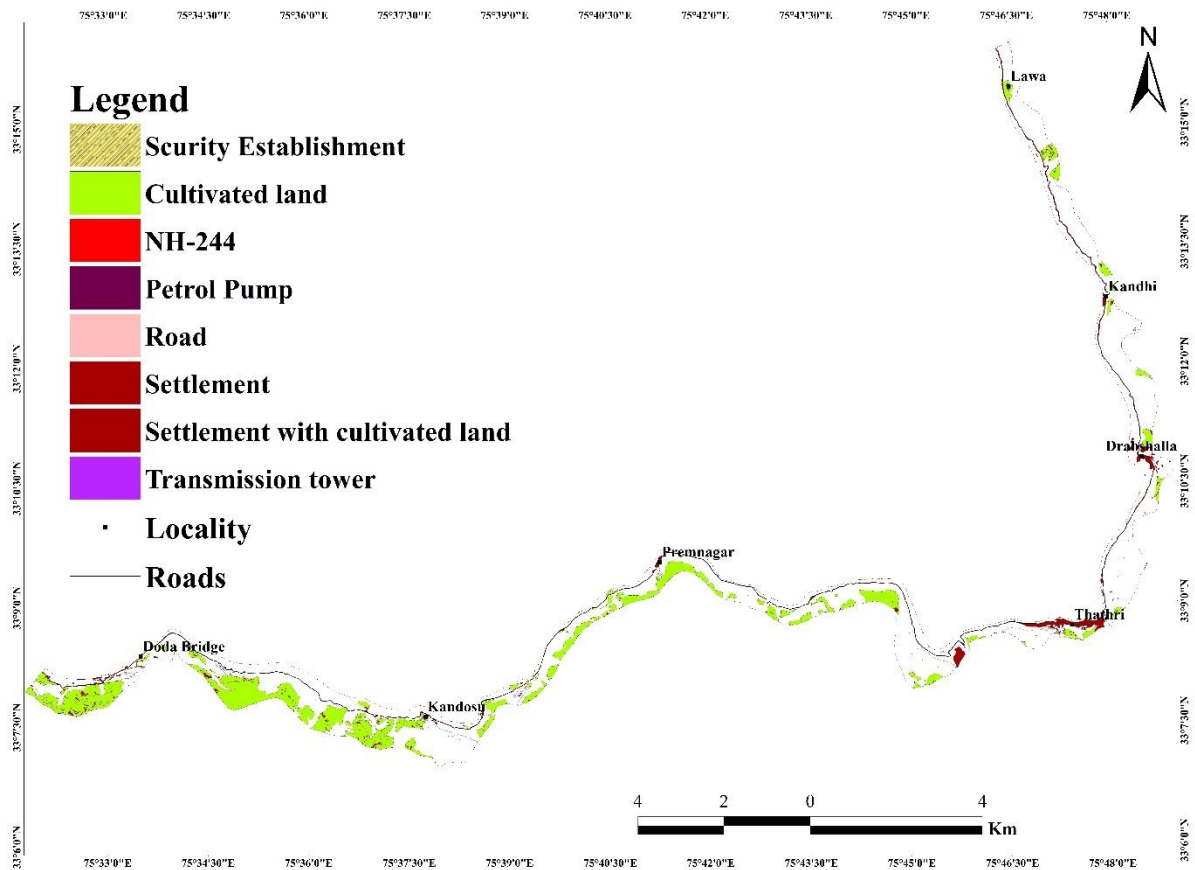


Fig. 4.12 Element at risk map of the study area.

4.2.7. Sample details:

Slope stability is predominantly influenced by the geotechnical parameters of rock masses and non-cohesive overburden materials. In the study area, overburden dominates the landscape, with limited rock exposures along the track route. The Slope Forming Material (SFM) map for the overburden-covered areas reveals a decreasing percentage of non-cohesive materials in the following order: regolith, colluvium, river-borne material (RBM), debris, soil, scree, and moraine deposits.

To assess the geotechnical parameters of both overburden materials and rocks, a total of fifteen undisturbed soil samples, each 12 inches long, and fifteen bedrock samples were collected (Fig. 6.13). These samples were sent to the Geotechnical Laboratory at the Geological Survey of India (Northern Region) in Lucknow for geotechnical analysis (Fig. 4.13). The soil

samples were taken from all major SFM classes, ensuring a representative collection across different overburden types. Similarly, to determine the strength of rock exposures, ten rock samples (each 10 cm³) were collected from the most competent rock types present in the area (Table 5.1).

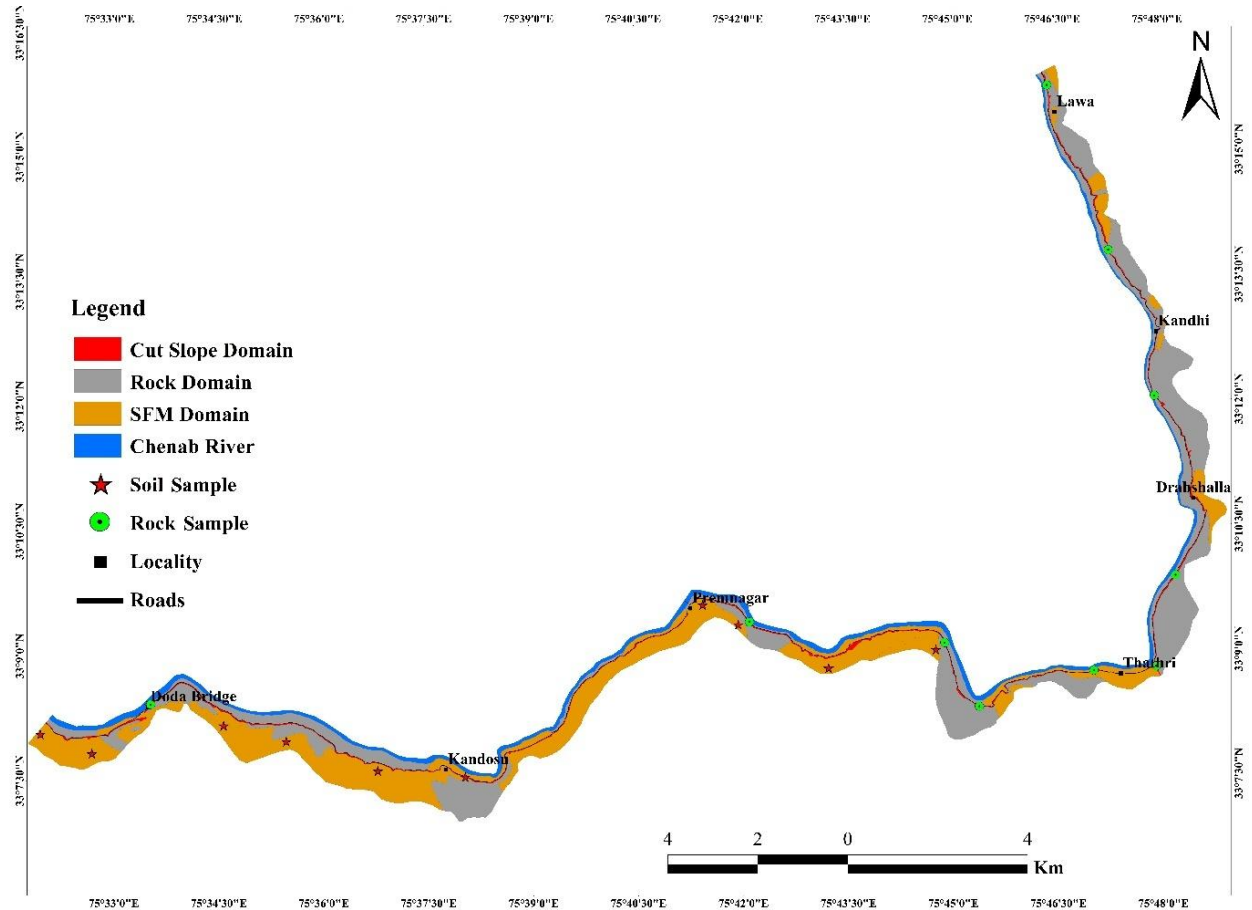


Fig. 4.13 Geotechnical map of the study area.

Table 4.1. Details of the GT samples collected

Rock Samples			Soil Samples		
Sample No	Sample Type	Type of Material	Sample No	Sample Type	Type of Material
DL/RS/01	UCS	Schist	DL/SS/01	SS	Clayey Soil
DL/RS/02	UCS	Quartzitic schist	DL/SS/02	SS	Silty soil
DL/RS/03	UCS	Marble	DL/SS/03	SS	Silty soil
DL/RS/04	UCS	Granite	DL/SS/04	SS	Clayey Soil
DL/RS/05	UCS	Marble	DL/SS/05	SS	Clayey Soil
DL/RS/06	UCS	Schist	DL/SS/06	SS	Clayey Soil
DL/RS/07	UCS	Quartzitic schist	DL/SS/07	SS	Clayey Soil
DL/RS/08	UCS	Pebbly Slate	DL/SS/08	SS	Silty soil
DL/RS/09	UCS	Slate	DL/SS/09	SS	Clayey Soil
DL/RS/10	UCS	Pebbly siltstone	DL/SS/10	SS	Clayey Soil

CHAPTER- 5

PREPARATION AND ANALYSIS OF LANDSLIDE SUSCEPTIBILITY MAPS

5.1. Selection of geo-factors and their LSER

The study rigorously examined the intrinsic factors contributing to slope instability across distinct slope domains, namely cut slope failure, debris slide, rock slide, and rock fall domains. This analysis involved the strategic weighting and selection of relevant causative geo-factor maps with respect to each domain. The weighting of these causative factors was determined based on their perceived significance relative to slope instability, aiming to ascertain the maximum Landslide Susceptibility Evaluation Rating (LSER) for each domain. The LSER, a heuristic metric, relies on established knowledge of the relationships between geo-factors and landslide occurrences, accommodating the specific conditions of different terrains. The methodology adopted for assigning LSER values to individual slope units or pixels entailed the selection and preparation of distinct causative factor maps aligned with the characteristics and recurrence patterns of landslide types within each domain. This approach adhered to standard procedures outlined in the Meso Scale Landslide Susceptibility Mapping (LSM) guidelines (GSI, 2021), with modifications informed by terrain-specific considerations as per the studies of Ghoshal et al. (2013) and Anbalagan et al. (2008).

The summation of the LSER values of all the thematic geofactor maps or/and overlay integration of LSER based raster maps, in each domain, gives the Total Estimated Susceptibility Value (TESV) for each slope unit/pixel and amounts maximum to 08 excluding the role of external factors.

Table 5.1.: Maximum LSER for selected causative Factors in Cut slope failure domain

Sr. No.	Factors		Maximum LSER Rating	
1	Slope angle		1	
2	Relative relief		2	
3	Slope Forming Material (SFM)	Geotechnical (SMR)	2	5
4	SFM Thickness		2	
5	Hydrogeology		1	
	Total Estimated Susceptibility Values (TESV)		8	

Table 5.2.: Maximum LSER for selected causative Factors in debris slide domain

Sr. No.	Factors	MaximumLSER Rating
1	Slope angle	1
2	Relative relief	1
3	Landuse Landcover (LULC)	1
4	Landform (Geomorphon)	1
5	Stream Power Index (SPI)	1
6	Slope Forming Material (SFM)	2
7	Hydrogeology	1
	Total Estimated Susceptibility Values (TESV)	8

Table 5.3: Maximum LSER for selected causative Factors in Rock slide and rock fall domains

Sr. No.	Factors	MaximumLSER
1	Slope Mass Rating (SMR)	5
2	Slope angle	1
3	Landuse Landcover (LULC)	1
4	Stream Power Index (SPI)	1
	Total Estimated Susceptibility Values (TESV)	8

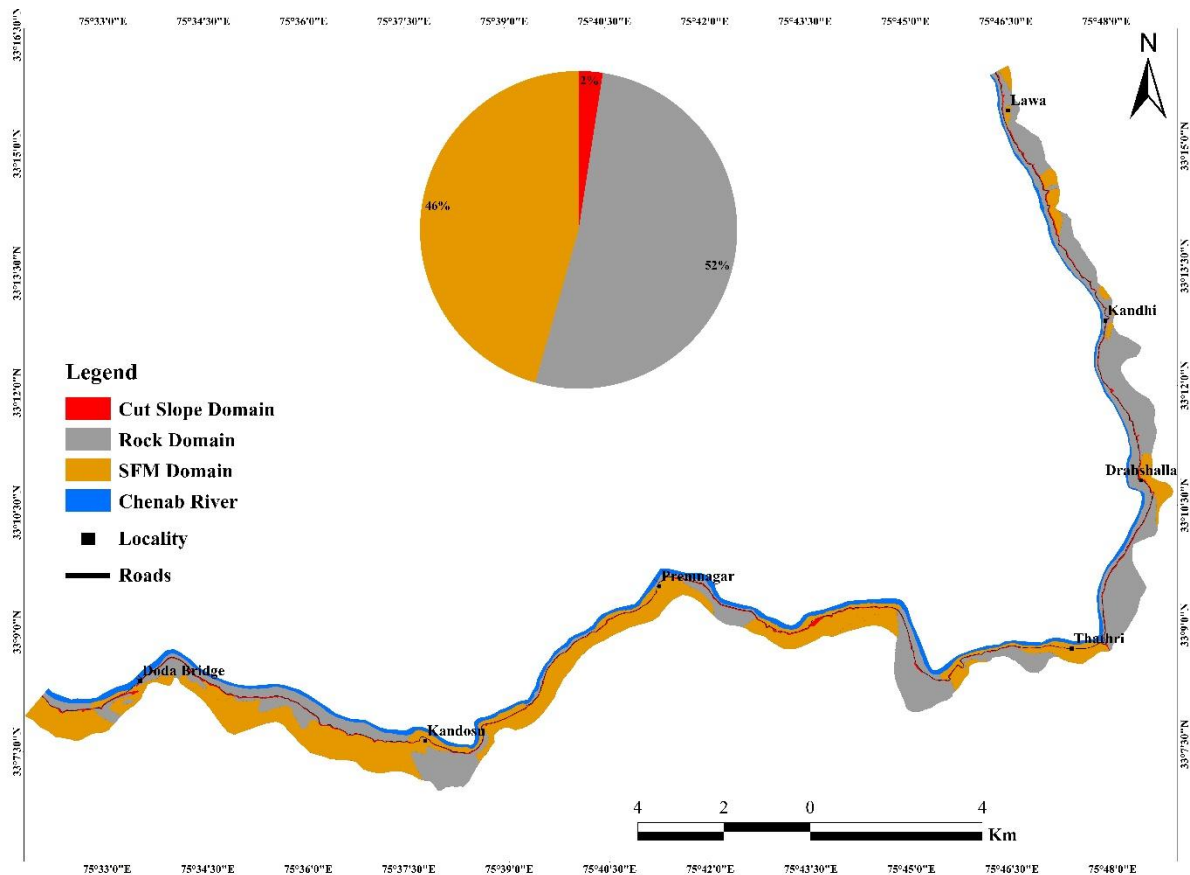


Fig. 5.1. Domain map showing cut slope, rock and overburden domains.

For any given slope unit, only one LSER can be assigned. Therefore, if more than two

classes of a theme are mapped in any given slope facet, instead of the centroid approach, the class with a larger proportion or higher ratings is considered for the particular facet of all categorical geo-factors (LULC, overburden, hydrogeology, structure). However, in geo-factors using continuous data (slope), either median or mean was considered for the class depending on the symmetry or central tendency of data (Tables 5.4 to 5.12). The soil samples show highly variable results and the Casagrande plasticity chart shows error in the results. Therefore, overburden map was rated on field-based classes. In case of rocky slopes, two approaches are available for LSER calculations: (i) direct SMR based rating approach and (ii) individual lithology (with weathering correction factor) + structure + hydrology-based rating approach. In case of later approach, the LSER for lithology are multiplied by weathering correction factor (Anbalagan et al (2008)). During the present work, the SMR based approach was adopted in rocky slopes. In case of thin weathered rock cover over massive rocky cliffs, LSER for weathered rock with necessary correction factor as proposed by Anbalagan et al, (2008) is adopted. The various causative factors and their ratings for each sub class are tabulated in tables 5.4 to 5.12. Relative relief for cut slope failure and debris slide domain has been separately weighed.

The effect of regional external triggering factors, like seismicity and rainfall which are dynamic in nature, cannot be assessed / studied on the slope facet (Paul and Kumar, 2022). The impact of these factors in the area are not incorporated for the LSER and final TESV map. Because a landslide susceptibility model is generally a static product, and is prepared by integrating rated and weighted pre-disposing geo-environmental and morphometric factors which are also static in nature, and represent a particular time period in sync with the time of generation of Land Use/Cover Map used in the exercise.

Table 5.4: LSER for Slope angle and Relative Relief

Slope angle			Relative Relief (m)		
Slope Angle	LSER	Class Name	Relative Relief	Debris slide domain	Cut slope failure domain
< 15	0.2	Very Low	<50	0.2	0.8
15-25	0.4	Low	50-100	0.4	1.4
25-35	0.6	Medium	100-150	0.6	1.6
35-45	0.8	High	150-200	0.8	1.8
> 45	1	Very High	> 200	1.0	2.0

Table 5.5: LSER for LULC in debris slide domain and rocky slopes

(A) LULC (Debris)	LSER	(B) LULC (Rock)	LSER
Dense Forest	0.2	Moderate Forest	0.3
Moderate Forest	0.4	Sparse Forest	0.6
Cultivated Land	0.4	Grass	0.7
Sparse Forest	0.8	Grass with isolated trees	0.8
Grass with isolated trees	0.8	Cultivated Land	0.4
Grass	0.9	Dense Forest	0.8
Barren	1	Barren	0.8
Settlement	1	Settlement	1
Transmission Tower	1	Army area	1
Army area	1	Transmission Tower	1
Road/NH-244	1	Road/NH-244	1
Petrol pump	1	Petrol pump	1

Table 5.6: LSER for landform in debris slide domain and hydrogeology in all facets

(A) Landform	LSER	(B) Hydrogeological	LSER
Valley	0.2	Dry	0.2
Summit	0.4	Damp	0.4
Ridge	0.4	Wet	0.6
Depression	0.4	Dripping	0.8
Foot slope	0.6	Flowing	1
Shoulder	0.6		
Spur	0.6		
Slope	0.6		
Hollow	1		

Table 5.7: LSER for SFM and SFM thickness in debris slide domain slopes

(A) SFM	LSER	(B) SFM Thickness	LSER
Colluvium	1	<1	0.6
Regolith	0.5	1 – 3	0.8
Colluvium over regolith	0.9	3 – 7	1.2
Colluvium over RBM	1.8	7 – 10	1.6
Scree	2	10 – 15	1.9
RBM	2	>15	2
Debris	2		

Table 5.8: LSER for Stream Power Index (SPI) map

SPI Rating		
Erosion condition	SPI	LSER
Least erosion	<0	0.1
Very Low Erosion	0.0-1.5	0.3
Low Erosion	1.5-3.0	0.5
Moderate Erosion	3.0-4.0	0.7
High Erosion	4.0-6.0	1.0

Table 5.9: LSER for (A) Lithology and (B) correction factor for weathering of lithology
(After Anbalagan et al. 2008)

(A) LSER for Lithology			(B) Correction factor for weathering of rock category	
Category	Lithology	LSER	Faintly weathered	Slightly (S) / Moderately (M) / Highly (H) weathered
Type – I	Granite	0.2	2	2.5 (S)
Type – I	Gneiss	0.2	2	2.5 (S)
Type – I	Quartzite	0.2	2	2.5 (S)
Type – I	Marble	0.2	2	2.5 (S)
Type – III	Schist	1.4	1	1.35 (M)
Type – III	Conglomerate	1.2	1	1.35 (M)
Type – III	Sheared quartzite	2.7	1	1.35 (H)

Table 5.10: LSER for relationship between slope and structural discontinuity for (A) parallelism in dip-slope direction and (B) angular difference in dip-slope amount

(A) Difference in angle of parallelism *		(B) Difference in amount of discontinuity and slope				Slope condition
		Planar / Wedge		Topple		
Angle	LSER	Difference	LSER	Difference	LSER	
>30°	0.20	>10	0.30	≤110	0.30	Very favourable
20.1-30°	0.25	0-10	0.50	110-130	0.50	Favourable
10.1-20°	0.30	0 - 0	0.70	130-140	0.70	Fair
5.1-10°	0.40	0 - (-10)	0.80	140-160	0.80	Unfavourable
≤5°	0.50	> -10	1.00	>160	1.00	Very unfavourable

Table 5.11: LSER for the amount of dip of structural discontinuity

Dip amount of discontinuity				Slope condition
Planar / Wedge		Topple		
Dip Amount	LSER	Dip Amount	LSER	
<15°	0.20	<50	0.20	Very favourable
15-25°	0.25	50-60	0.30	Favourable
25-35°	0.30	60-70	0.40	Fair
35-45°	0.40	70-80	0.45	Unfavourable
>45°	0.50	>80	0.50	Very unfavourable

Table 5.12: LSER for Geotechnical parameters

A. Shear Strength (Overburden material)				B. Slope Mass Rating (Rock mass)		
Cohesion	Angle of Internal Friction (Ø)			Class	SMR Value	LSER
	>30°	21-30°	11-20°			
5-15	1.4	1.8	2	Very Good	>80	1
16-30	0.8	1.4	1.8	Good	60-80	2
16-30	0.8	1.4	1.8	Normal	40-60	3
>30	0.6	0.8	1.4	Bad	20-40	5

5.2. Total Estimated Susceptibility Value (TESV) Determination:

The Total Estimated Susceptibility Value (TESV) is determined by summing the LSER values for each individual causative factor. The causative factors and their ratings for the study area were sourced from multiple references (Anbalagan et al., 2008; Ghoshal et al., 2013; GSI, 2021) and adjusted to reflect current terrain conditions as detailed in the preceding tables. Depending on whether "Additional Factors" are included, the maximum TESV can be either 8 or 10. The final TESV for each domain reflects the overall instability of the slopes. These TESV values are classified into three susceptibility categories as outlined in Table 5.13. Separate TESV maps were created for different domains, namely cut slope failure, debris slide, rock slide, and rock fall. These maps were produced by rasterizing various thematic maps and performing overlay analysis to generate pixel-based susceptibility maps. The comprehensive Mesoscale Landslide Susceptibility Map (LSM) was developed by integrating all four TESV domain maps.

Pixel based TESVM preparation was done by summation and rasterization of field based LSER of geo-factor maps followed by its overlay raster integration with DEM derivative LSER geo-factor maps, as calculated below.

Cut slope failure domain: Raster based overlay Integration (Weighted Sum): Slope Map + Relative Relief + $\sum$ LSER [Hydrogeological + SFM+ SFM Thickness] ----- Eq-1

Debris slide domain: Raster based overlay Integration (Weighted Sum): Slope Map + Relative Relief + Landform + SPI + $\sum$ LSER [LULC + Hydrogeological + SFM]----- Eq-2

*Rock slide domain: Raster based overlay Integration (Weighted Sum): Slope Map + SPI + $\sum$ LSER [LULC + SMR or weathered rock * weathering correction factor)] -----Eq-3*

*Rock fall domain: Raster based overlay Integration (Weighted Sum): Slope Map + SPI + $\sum$ LSER [LULC + SMR or weathered rock * weathering correction factor] -----Eq-4*

Final Mesoscale LSM: Mosaicking of raster layers of rock slide + rock fall + debris slide + cut slope failure domain susceptibility maps -----Eq-5

Based on TESV ranges, slope facets are categorized into three classes of relative hazard zones: low susceptibility, moderate susceptibility, and high landslide susceptibility (Table

5.13). Separate TESV maps will be created for cut slope failure domains, debris slide domains, and rock slide and rock fall domains, each following distinct LSER calculations. The resulting meso-scale Landslide Susceptibility map illustrates the spatial distribution of these zones, aiding planners in identifying safer areas and avoiding highly susceptible regions, as well as selecting appropriate measures to mitigate landslide hazards. While the model does not incorporate the effects of external factors like rainfall and earthquakes, their overall impact on the area is discussed.

Table.5.13: Total Landslide Susceptibility Class and TESV class categories

Hazard Zone	Susceptibility Category	TESV Class
I	Low	<3.8,
II	Moderate	3.8 – 4.6
III	High	>4.6
	Total	8

5.3. Landslide Susceptibility Map (LSM):

Paul and Kumar (2022) opined that the Total Estimated Susceptibility Values (TESV) calculations and subsequent Meso-scale Landslide Susceptibility Map (LSM) preparation through on pixel-based approach yielded relatively better results in part of Kashmir Himalaya than facet-based approach. Therefore, Pixel based approach was adopted during present study to prepare the TESV map. The same was validated with landslide inventory data by superimposing the landslide inventory polygons over TESV map, thereby cross verifying of the model accuracy. Pixel based Meso-scale LSM was prepared by combining and rasterization of LSER of field based thematic maps / causative factors and DEM derivative raster maps for each slope facet / pixel. These include Relative Relief, Land Use Land Cover (LULC), hydrogeological, Slope Forming Material (SFM) or slope mass rating (SMR), SFM thickness and DEM derivative raster maps of slope angle, Stream Power Index (SPI) and geomorphon. This was followed by overlay integration of rasterized LSER weighted field and DEM derivative maps separately for debris slide domain, rock slide and fall domain and cut slope failure domain slopes, thereby giving the TESV as per tables 5.1, 5.2 and 5.3.

5.4. Meso-scale (1:10000) LSM Map:

The TESV calculations were done separately for debris slide domain, rock slide and fall domain and cut slope failure domain and later combined together to prepare the final pixel based Meso-scale LSM for the Pul Doda to Lawa road along NH-244. The LSER for SFM and SMR maps respectively for overburden and rock covered areas was calculated using the field data, as lab based Geotechnical results for soil samples is erroneous and too limited for extrapolation. Individual Meso-scale LSM maps were prepared for planar, wedge and toppling rock failure slopes, which were mosaicked into a single susceptibility map for rocky slopes (Fig. 5.2-5.4).

The TESV calculations for debris slide domain and rocky slopes range from 3.4 to 7.4 (Figs. 5.2, to 5.4). The TESV calculations for final combined LSM map was prepared by mosaicking the individual domain-based susceptibility maps and have been categorized into low, moderate and high susceptibility zones respectively for <3.8, 3.8 – 4.6 and >4.6 TESV values (Figs. 7.4).

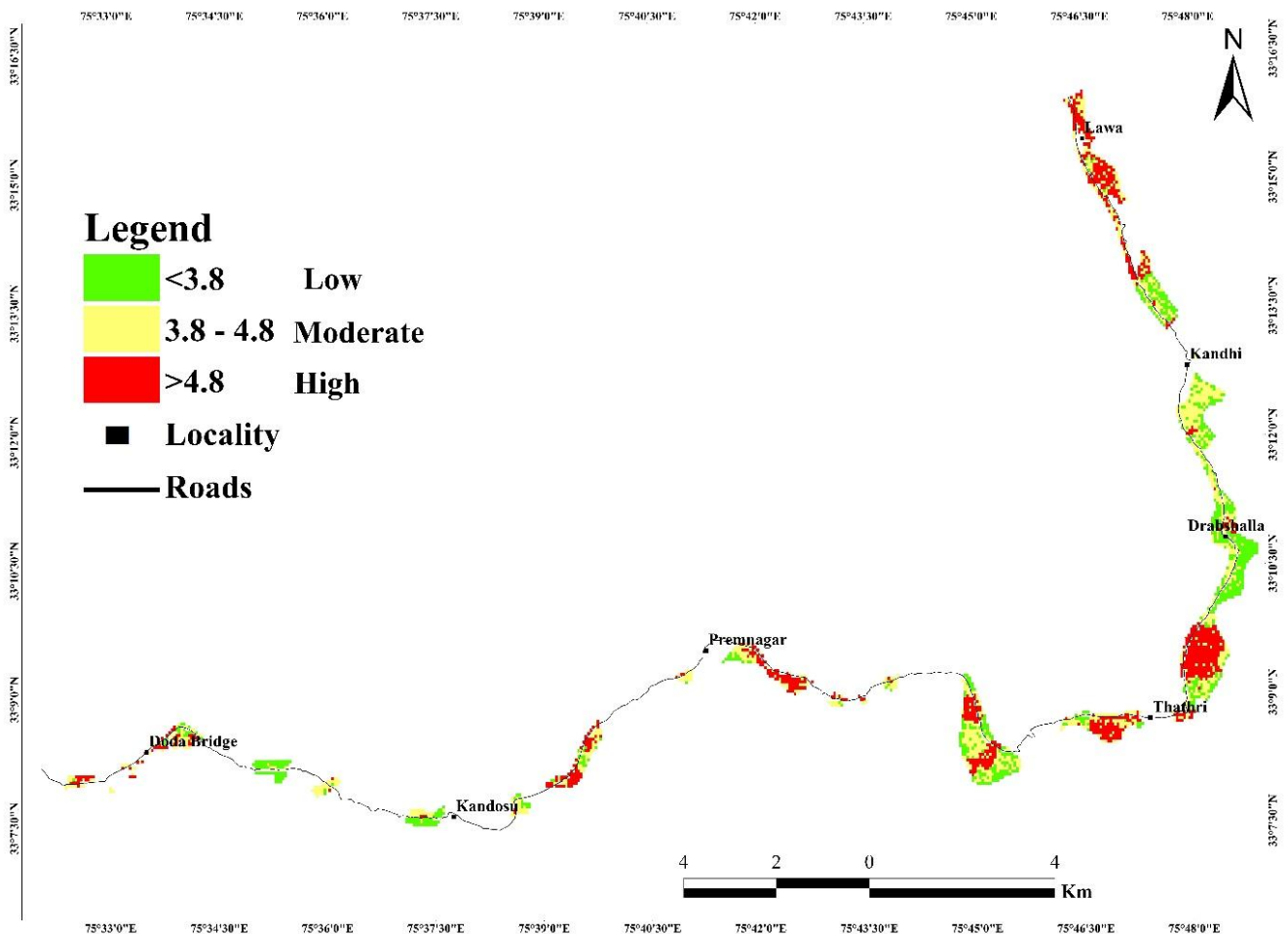


Fig.5.2 MLSM rock domain map of the study area

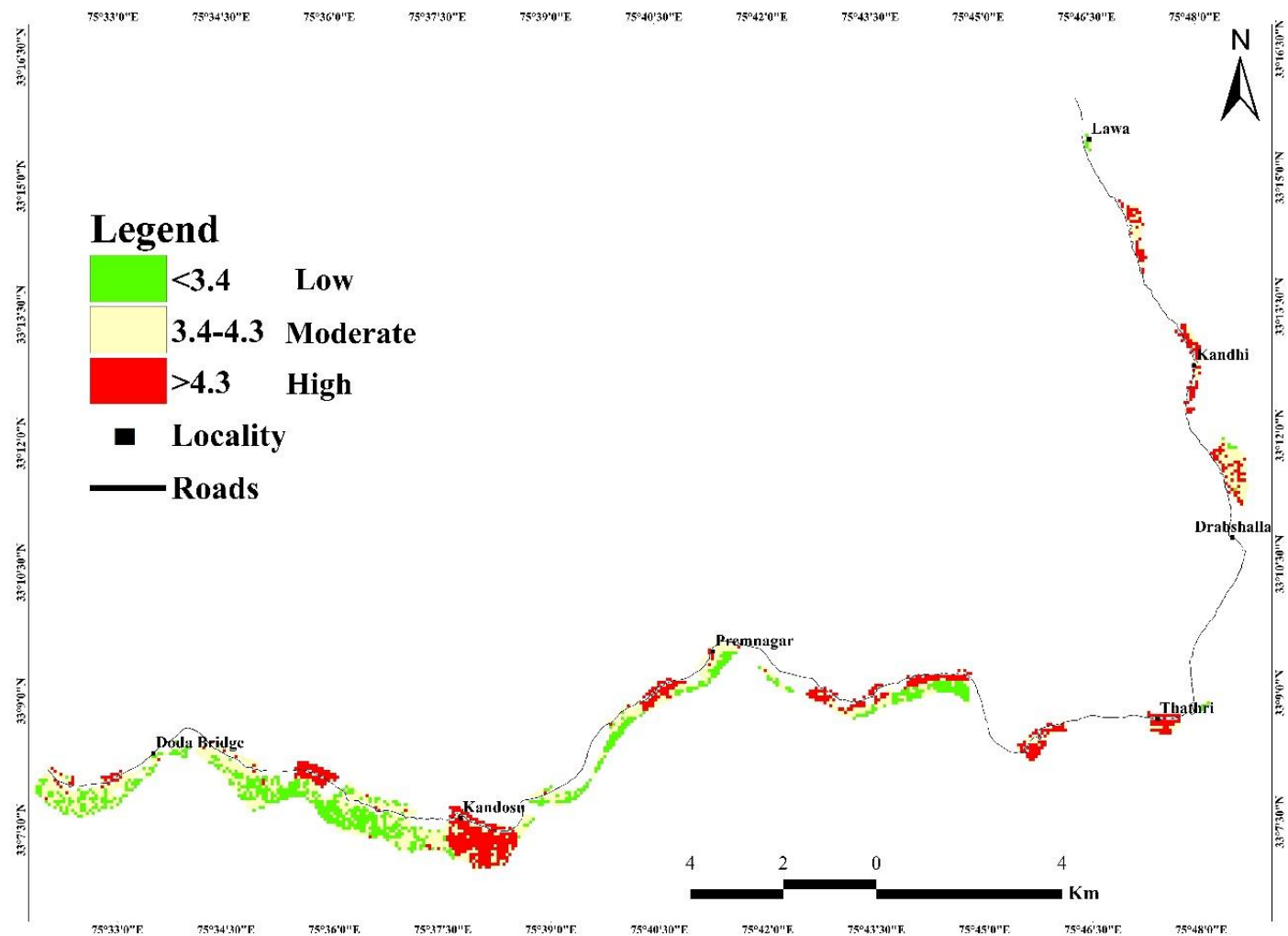


Fig. 5.3. MLSM debris domain map of the study area

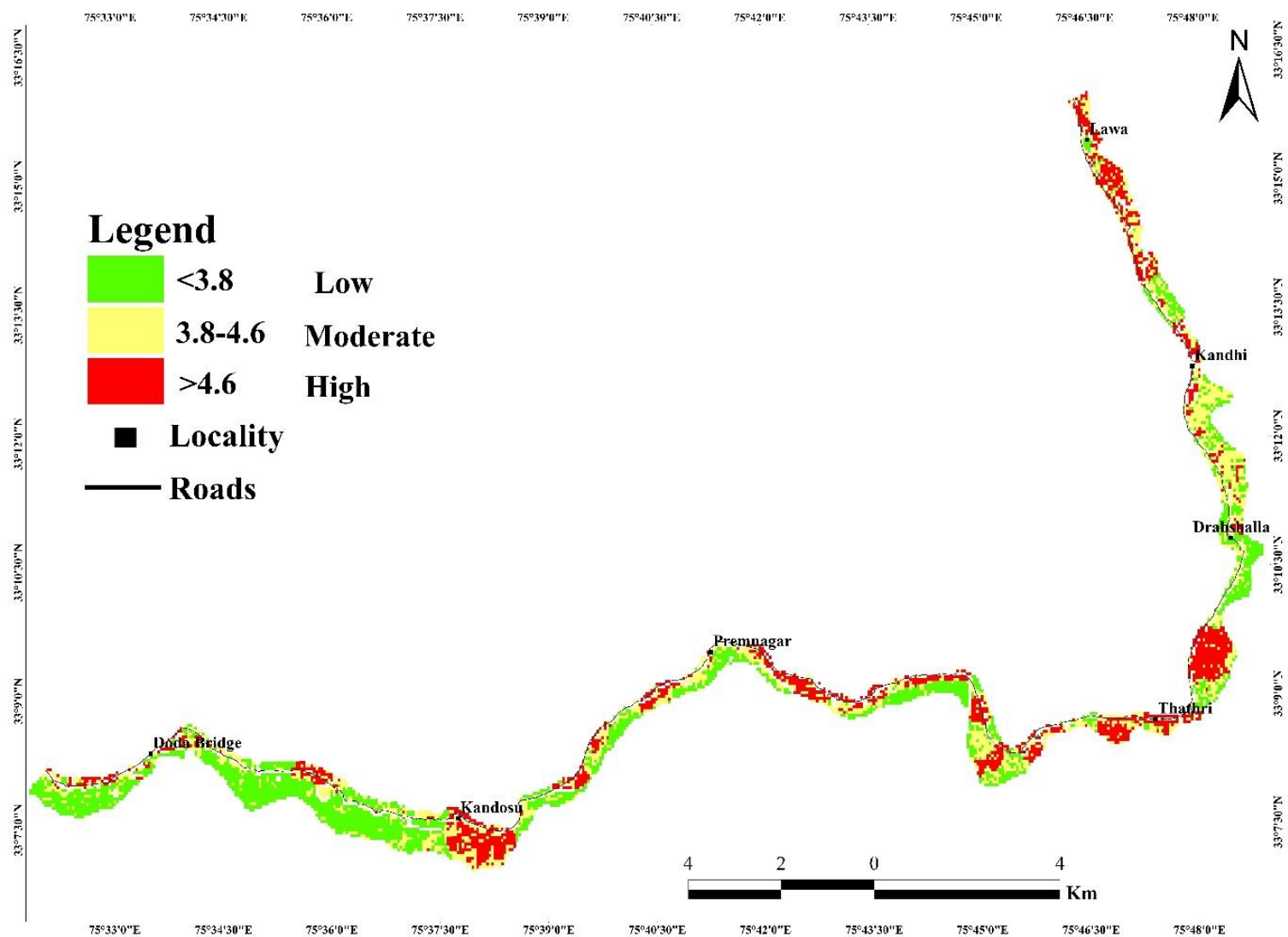


Fig.5.4 MLSM Map of the study area.

5.5. Validation of Meso-scale (1:10000) LSM Map:

The collection of landslide inventory data is essential to understand the causative factors of different slope failure domains and to validate the accuracy of Mesoscale LSM. The landslide and instability incidences were superimposed over pixel based Mesoscale LSM (Figs. 5.5 Table-5.14). The LSM of the area on validation shows 62.96% landslides in high susceptibility zone which covers only 26.87% area, 22.22% landslides in moderate susceptibility zone which covers 34.45% area and 14.81% landslides in low susceptibility zone which covers only 38.66% area. The cumulative percentage of landslides under high and moderate susceptibility classes in mesoscale LSM map is 85.18%.

Table 5.14: Landslide Susceptibility Classes of Mesoscale LSM for landslide-based validation.

Susceptibility Class	TESV	Landslide Susceptibility		Landslide Activity	
		Pixels	%	Nos.	%
Low Susceptibility	≤ 3.8	2444	38.66	4	14.81
Moderate Susceptibility	3.8- 4.6	2178	34.45	6	22.22
High Susceptibility	> 4.6	1699	26.87	17	62.96
Total		6321	100.0	27	100.0

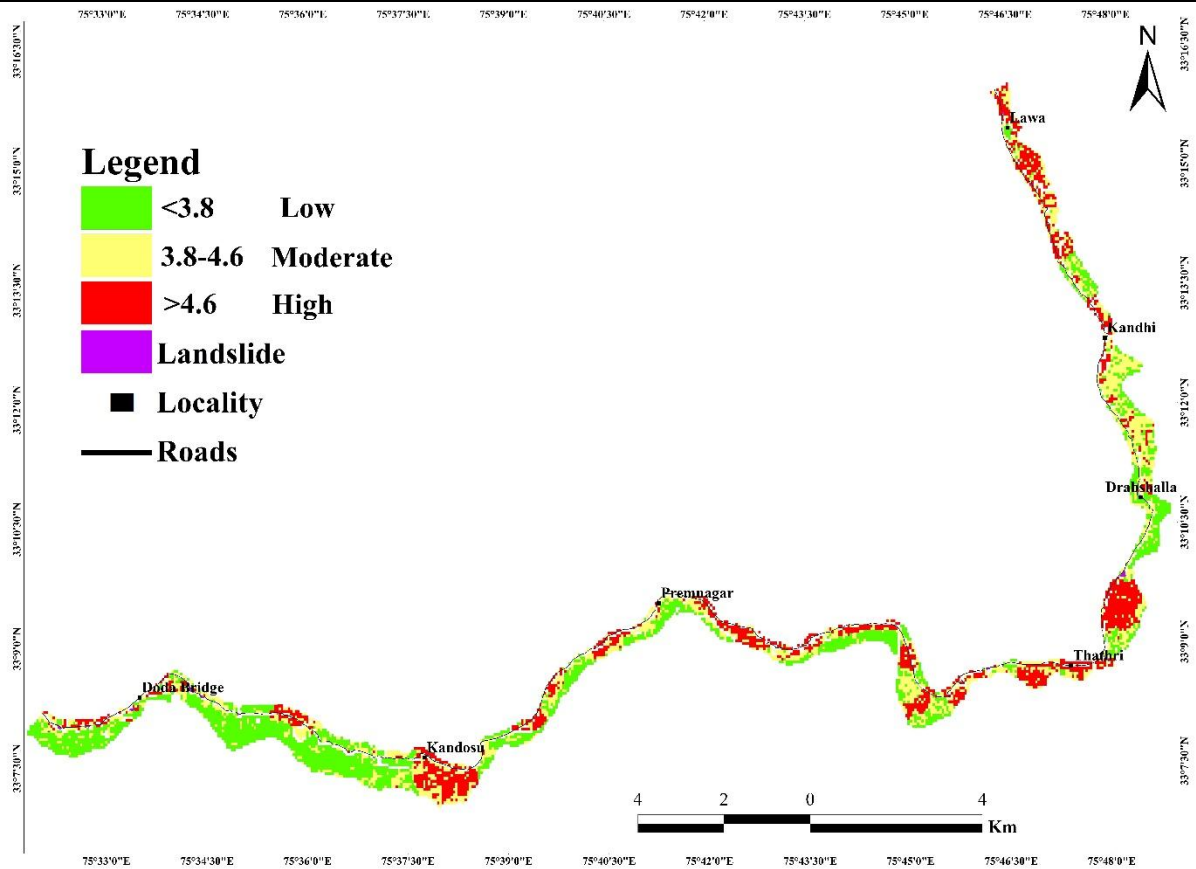


Fig 5.5. Overlay map of meso-scale LSM and landslide incidences.

5.6. Landslide Management Map of the study area:

Meso-scale LSM revealed low, moderate and high susceptibility zones having >90% validation. Based on Meso-scale LSM and Landslide inventory map, critical areas were identified for further necessary remedial measures to stabilize the slopes and reduce the landslide hazard risk in the area. The following slope stability observations have been used to plan for the relevant recommendations for the landslide management of the Ganpath bridge – Lawa along NH-244 road sector (Figs. 5.6), based on the present Meso-scale LSM.

A: The road section and Gadsu village are situated on unconsolidated colluvial deposits underlain by mica schist and interbedded quartzite bands. The area lies within a highly susceptible debris-slide zone and is presently affected by a large active debris slide. Given the geological and geomorphological conditions, the following mitigation measures are recommended: biotechnical slope stabilization using geofibre reinforcement, afforestation to enhance root cohesion and surface stability, drainage diversion in the crown area to minimize infiltration, construction of a protection wall at the toe to prevent undercutting and buttress support along the cut slope to improve stability. At present, the slide exhibits retrogressive movement, posing an immediate risk to structures located in the crown portion. Relocation of these houses is urgently required to ensure safety and prevent loss of life and property.

B: At Premnagar, the road passes through a densely built and narrow settlement area. To avoid disruption and accommodate safe traffic movement, the management plan recommends the construction of an 825 m viaduct to bypass the settlement.

C: The road section from Premnagar to New Thatri is predominantly constructed over Bimrock (Blocks-in-Matrix) material, comprising unconsolidated pebbles, cobbles, and boulders embedded within a sand and silty clay matrix. Due to its lithological composition and weak bonding, this stretch has been classified as a Debris Fall Zone. For slope stabilization and mitigation of rockfall hazards, it is recommended to install open-type drapery wire mesh at the base, supported by deep soil nails or anchors to ensure structural stability and containment of loose debris.

D: At Thatri, the road again traverses a narrow and congested settlement where widening is not feasible due to space constraints. A 1.4 km viaduct is therefore proposed in the management plan to reroute traffic and reduce congestion.

E: Near Darabshalla, a large rock slide zone characterized by planar failure has been identified. To bypass this unstable section, the recommended mitigation measure is a combination of 500 m of tunnelling along with a 70 m bridge.

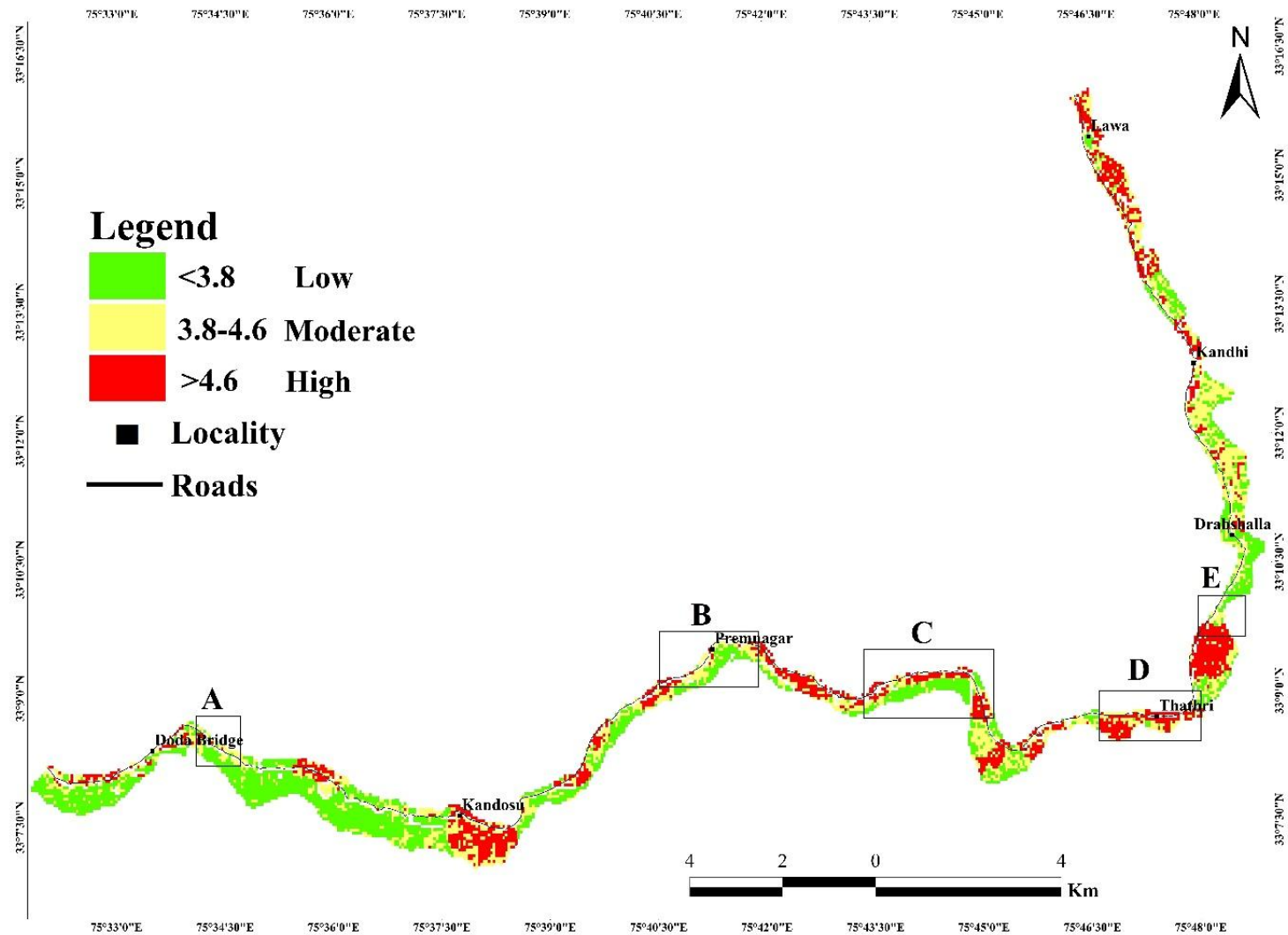


Fig. 5.6. Landslide Management Zones for landslide remedial measures along Doda – Lawa, NH-44 str

CHAPTER-6

SLOPE STABILITY ASSESSMENT AND LANDSLIDE RISK ANALYSIS

The slopes along NH-244, between Ganpath Bridge and Lawa, are predominantly covered with overburden material interspersed with rocky outcrops. These slopes have been categorized into distinct domains based on their characteristics: debris slide domain, rock slide and rock fall domain, and cut slope failure domain. To assess the strength of the overburden-covered slopes, fifteen soil samples representing various overburden classes were collected. Additionally, ten bedrock samples were obtained to determine key geotechnical parameters essential for evaluating rock mass strength. All laboratory testing and analyses were carried out at the Geotechnical Laboratory of the Geological Survey of India (GSI), Northern Region, Lucknow. Furthermore, slope stability assessments were conducted in the field at fifteen rock exposure sites using empirical approaches, including kinematic analysis (Markland test) and the Slope Mass Rating (SMR) system, following the methodologies proposed by Romana (1985) and Hoek and Bray (1981).

6.1. Geotechnical and geomechanical properties of undisturbed soil samples:

Ten undisturbed soil samples were collected to determine their geotechnical properties, ensuring representation of nearly all overburden classes. These samples underwent analysis for various parameters, including grain size, specific gravity, Atterberg limits (liquid limit, plastic limit, plastic index, shrinkage limit, shrinkage ratio), natural moisture content (NMC), wet and dry density, void ratio, porosity, cohesion (C), and friction angle. The descriptive statistical details of these properties are provided in Table 6.1 (see Annexure VI).

The particle (grain) size analysis of twenty undisturbed soil samples (USS) collected from the overburden-covered slopes was conducted using the sieving method. This method determined the percentage of particles within different size ranges. Sieve analysis for all samples was performed down to the 0.075 mm (silt fraction) sieve size. The samples were passed through a series of sieves with sizes of 12.5, 10, 6.3, 4.75, 2.36, 1.18, 0.6, 0.3, 0.15, and 0.075 mm (see Annexure VI). These sieve sizes correspond to gravel (75-4.75 mm), coarse sand (4.75-2.0 mm), medium sand (2.0-0.42 mm), fine sand (0.42-0.075 mm), and silt (<0.075 mm).

	Specific Gravity	Liquid Limit	Plastic Limit	Plastic index	Shrinkage limit	Shrinkage Ratio	NMC	Wet (gm/cc)	Dry	Void Ratio	Porosity	C (kg/cm ²)	Ø (°)
Mean	2.61	28.38	20.17	8.20	16.72	1.80	6.28	1.40	1.31	1.00	0.50	22.86	37.47
Median	2.60	28.99	19.60	7.54	16.15	1.81	5.47	1.37	1.29	1.02	0.50	20.00	37.30
Range	0.20	8.50	9.35	4.18	6.71	0.24	8.28	0.47	0.37	0.41	0.11	17.50	3.90
Minimum	2.51	24.70	17.33	6.52	14.67	1.63	2.74	1.24	1.17	0.76	0.43	17.50	35.60
Maximum	2.71	33.20	26.68	10.70	21.38	1.87	11.02	1.71	1.54	1.17	0.54	35.00	39.50
Std Deviation	0.08	3.02	3.09	1.52	2.17	0.08	2.73	0.13	0.12	0.14	0.04	5.85	1.30
Variance	0.01	9.14	9.55	2.30	4.70	0.01	7.46	0.02	0.01	0.02	0.00	34.23	1.68
Std Error	0.02	1.14	1.17	0.57	0.82	0.03	0.86	0.04	0.04	0.05	0.01	2.21	0.41
Kurtosis	-1.63	-0.83	4.20	-0.45	4.94	5.05	-0.88	3.04	0.07	-1.00	-0.80	3.68	-0.89
Skewness	-0.10	0.35	1.92	0.93	2.07	-2.07	0.36	1.55	0.83	-0.46	-0.55	1.84	0.30
95% conf. level	0.06	2.80	2.86	1.40	2.01	0.07	1.95	0.09	0.08	0.10	0.03	5.41	0.93
	Sieve opening mm (% finer)					Table 6.1. Descriptive statistics of Geotechnical parameters of the undisturbed soil samples. Total samples count is 10. But for atterberg limit, sample count is only 07 as result of the rest three samples are not available. Sieve sizes of 6.3, 2.36, 0.6, 0.15 and 0.075 respectively represent Gravel, Coarse, Medium, Fine and Silt fractions.							
	6.3	2.36	0.6	0.15	0.075								
Mean	90.13	88.39	85.64	75.73	71.94								
Median	99.61	97.32	96.56	88.39	84.03								
Range	93.70	97.42	98.88	99.04	98.51								
Minimum	6.30	2.36	0.60	0.15	0.08								
Maximum	100.00	99.78	99.48	99.19	98.58								
Std Deviation	27.87	28.63	28.68	29.77	30.49								
Variance	776.57	819.90	822.66	886.16	929.43								
Std Error	8.40	8.63	8.65	8.98	9.19								
Kurtosis	10.87	10.79	10.04	3.76	2.02								
Skewness	-3.29	-3.27	-3.12	-1.92	-1.54								
95% conf. level	18.72	19.24	19.27	20.00	20.48								

The analysis reveals 05 samples have $\geq 75\%$ finer particle content ($< 0.075\text{mm}$) and 03 samples have finer particle content between 50-75% and 02 samples have $\geq 40\%$ medium particle content (Fig 6.1). Therefore, 08 samples with $> 50\%$ finer particles than 0.075mm fall under the category of fine-grained soil. This indicates the higher percentage of fine particle content in samples with a predominance of silt fraction in the overburden.

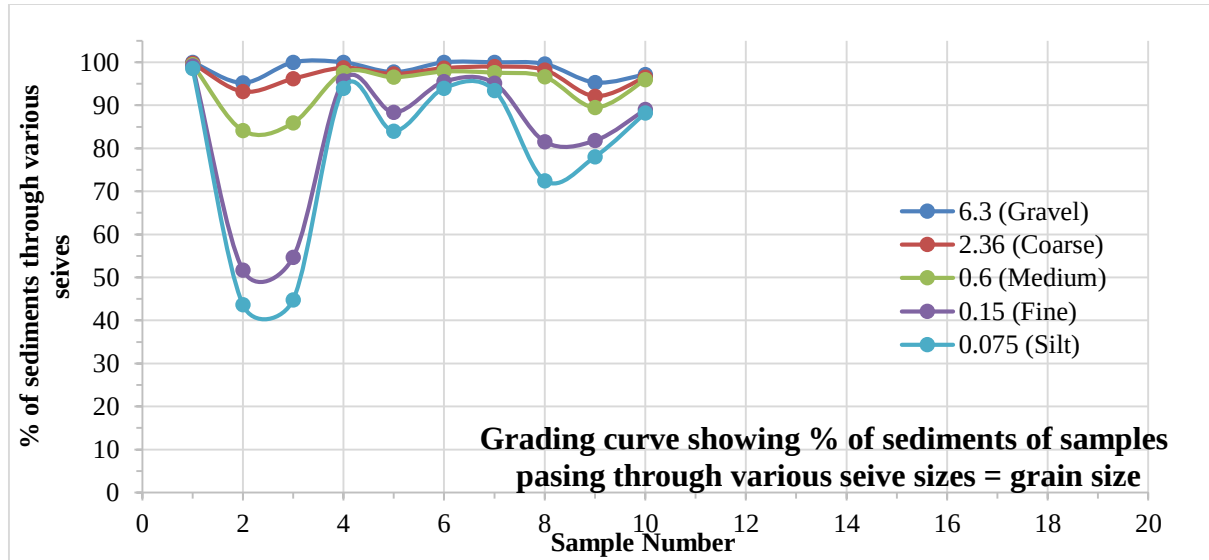


Fig.6.1. Grading curves show the percentage of sediments passed through different sieve sizes.

All the samples seem to be uniform in size, as $> 50\%$ quantity of $> 80\%$ samples pass through all the above coarser sieve sizes. six samples are highly uniform; two are moderately uniform and rest two shows some variations in size. The coefficient of uniformity (C_u) has not been calculated as none of the samples show $\leq 10\%$ sample quantity passing through any mesh size, necessary for its determination. Therefore, samples show overall uniformity. In view of the lack of clay fraction analysis for the samples, Casagrande's Unified Soil Classification System (USCS) is used to distinguish Clay from Silt. Casagrande's (1948) opinion that plasticity is the most important characteristic of fine-grained soils for engineering classification purpose rather than grain size and considered soils with Liquid limit (WL) > 50 as having high plasticity and vice versa. The present samples show $WL < 50$ in all the samples which reflect low plasticity in the samples. Ten samples are non-plastic and have not yielded any results for plasticity index. The USCS uses the laboratory-derived Atterberg Limits and the Casagrande plasticity chart to distinguish Clay from Silt, where the former generally plot above the plasticity chart's A-Line and later generally plot below it. Further U-Line in the Casagrande plasticity chart marks the upper bound for general soils. A U-Line is defined as the upper limit of the correlation between plasticity index and liquid

limit of the soil. The U-line is not used in classification but is an upper boundary of expected results for natural soils. Therefore, the values plotting above the U-line should be rechecked for errors. (After Casagrande 1948; Howard, 1977; Holtz and Kovacs, 1981). All the present samples are falling above the U-line and therefore showing erroneous results.

$$A\text{-line} = I_P = 0.73 (W_L - 20) \text{ ----- Eq-7}$$

$$U\text{-line} = I_P = 0.90 (W_L - 8) \text{ ----- Eq-8}$$

Where I_P = plasticity index

The void ratio (e) of the samples varies between 0.76 to 1.17 with the majority of the samples cluster around 1.02-1.17 indicating them to be of soft to stiff clay with a moderate void ratio (Das, 2011). The higher void ratio in three samples indicates a weaker or softer soil with greater deformability. The void ratio shows a strong positive correlation with the porosity of the samples and a strong negative correlation with wet and dry density (Fig 6.2b, Table 6.2). The samples show low porosity, ranging from 0.43 to 0.54, indicates the substantial presence of clayey soil (Charley et al., 1993). The porosity also shows strong negative correlation with density of the samples (Fig 6.2c). The dry density of the samples ranges from 1.17 to 1.54 gm/cc pointing to the clay material. The wet density is slightly higher.

The natural moisture content (NMC) of collected soil samples can be categorized into 15-20 % (dry) and 20-25% (Moderate) as stiff and 25-35 % (wet) as firm categories (Fig 6.2b). The relation of liquid limit (W_L), plastic limit (P_L) and NMC for all samples shows $W_L > NMC > P_L$, indicating soft to firm clay (Fig 7.2d). At the time of collection of undisturbed soil samples in the field, most of the sediments were moulded easily or strongly with fingers indicating them to fall in the soft or firm category. The results for the permeability have been determined for eleven samples. The samples show low permeability ranging between 0.00000369 to 0.0000273 cm/s indicating poor to very poor drainage properties. All the samples fall in less permeability of fine-grained soil ($<0.075\text{mm}$) with clay nature (CL). The grain size has a direct relation with permeability i.e. smaller the grain size, lower the permeability. Powers (1992) has given low permeability of 10^{-5} - 10^{-8}cm/s for low plasticity clay (CL) and 5×10^{-5} - 10^{-4}cm/s for low plasticity silt (ML) having poor to very poor drainage properties. The friction angle (ϕ°) of the samples ranges from 35.6° to 39.5° which are generally found in inorganic clay of low to slight plasticity. The cohesion values of the samples range from 0.0 to 35 (kg/cm²). The lower values of cohesion indicate poor

compaction of soil. The average cohesion value of the soil samples in the study area is 22.85 K pa showing it as inorganic clay.

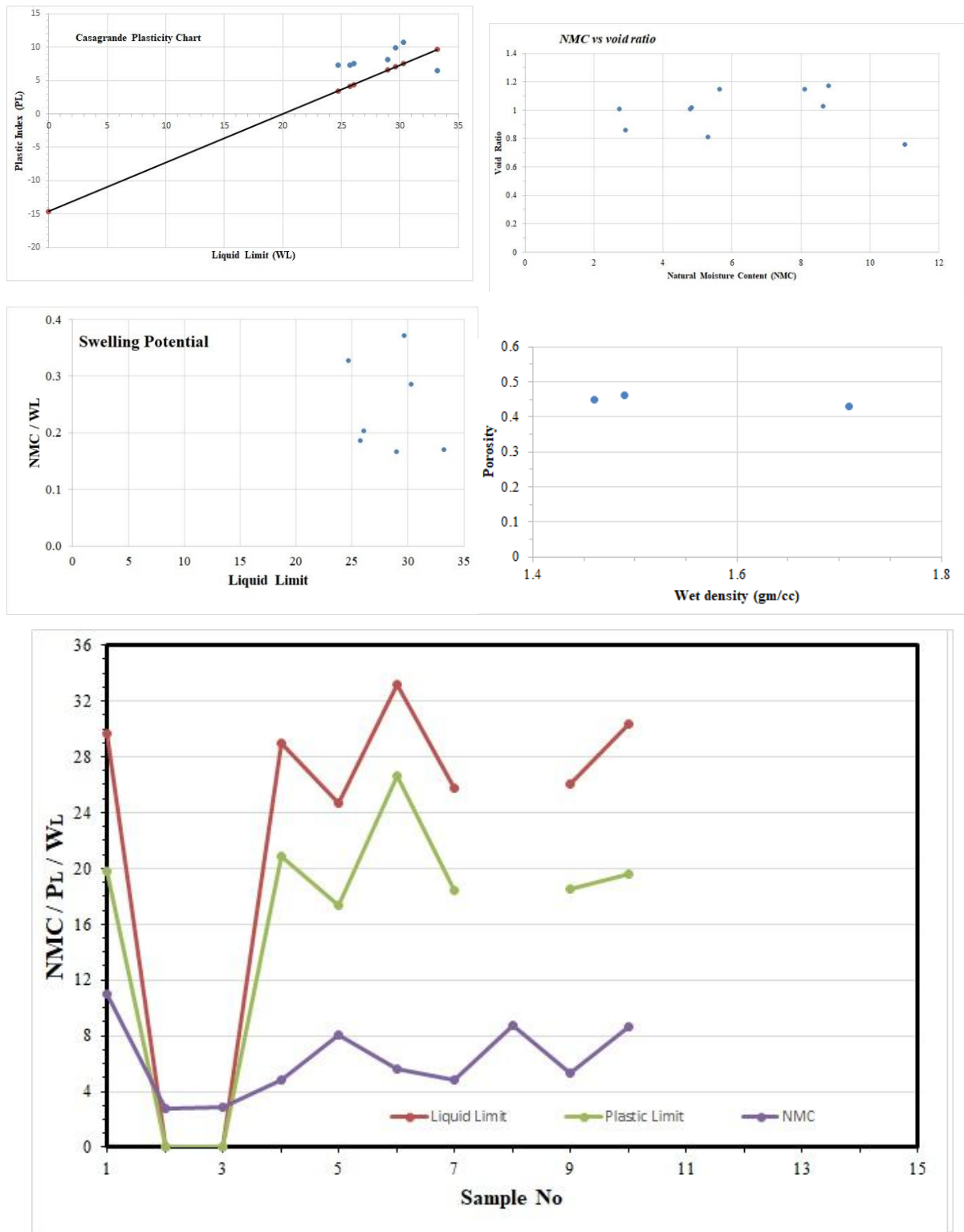


Fig.6.2. Bi-variate charts of (a) Casagrande Plasticity Chart showing nature of sediments, (b) NMC vs void ratio (c) Wet density vs Porosity, (d) comparative chart of Liquid limit, plastic limit and NMC.

Table 6.2. Correlation matrix of Geotechnical parameters of the undisturbed soil samples

	WL	PL	PI	Shrinkage limit	Shrinkage Ratio	NMC	Wet	Dry	Void Ratio	Porosity	C	Ø (°)	Fine sediment	SG
Liquid Limit (WL)	1													
Plastic Limit (PL)	0.88	1.00												
Plastic index (PI)	0.21	-0.29	1.00											
Shrinkage limit	0.82	0.98	-0.37	1.00										
Shrinkage Ratio	-0.61	-0.85	0.52	-0.91	1.00									
NMC	0.14	-0.21	0.71	-0.33	0.41	1.00								
Wet Density	-0.03	-0.33	0.62	-0.46	0.66	0.38	1.00							
Dry Density	-0.09	-0.35	0.54	-0.47	0.68	0.10	0.96	1.00						
Void Ratio	0.13	0.33	-0.43	0.40	-0.53	0.03	-0.86	-0.94	1.00					
Porosity	0.06	0.25	-0.39	0.32	-0.48	0.03	-0.87	-0.94	1.00	1.00				
Cohesion (C)	0.23	-0.23	0.93	-0.27	0.35	0.53	0.31	0.24	-0.16	-0.11	1.00			
Friction angle (Ø°)	-0.62	-0.60	-0.01	-0.51	0.21	-0.43	-0.23	-0.11	0.15	0.18	0.23	1.00		
Fine sediment	0.55	0.44	0.20	0.38	-0.03	0.60	0.07	-0.11	0.10	0.08	0.06	-0.88	1.00	
Specific Gravity	-0.05	-0.29	0.50	-0.45	0.66	0.28	0.61	0.57	-0.26	-0.26	0.32	0.10	-0.11	1.00

 Strongly correlated
 Moderately correlated

6.2. Geotechnical and geomechanical properties of rock Samples:

The strength of rocks was assessed in the field using indirect field-based method. Further bed rock samples were also taken for estimation of GT parameters to measure the strength of rocks. Fifteen rock samples of 10cm³ size were collected along the road and trekking route from various spots to determine the strength parameters of the rock masses. The samples were representative of the area and mostly collected from schist, quartzite, granite and marble. The GT parameters of Uniaxial Compressive Strength (UCS) through Schmidt Hammer, specific gravity and water absorption were assessed at GT Lab, GSI NR, Lucknow (Table 6.3). Result of density, void ratio and point load index was not received

Table 6.3. Geotechnical analysis of UCS rock samples

Field No.	Field No.	Rock Type	Water absorption (%)	Specific gravity	UCS (kg/cm ²)
1	RS-1	Schist	0.77	2.69	214
2	RS-2	Quartzitic schist	0.56	2.77	367
3	RS-3	Marble	0.33	2.79	326
4	RS-4	Granite	0.61	2.75	352
5	RS-5	Marble	0.32	2.81	321
6	RS-6	Schist	0.58	2.68	240
7	RS-7	Quartzitic schist	0.57	2.76	357
8	RS-8	Quartzite	0.65	2.73	336
9	RS-9	Schist	1.26	2.65	163
10	RS-10	Granite	0.65	2.74	336

The results show UCS values of such samples ranging from 163 to 367 kg/cm². Higher values are from quartzite schist and quartzite samples. Lowest values are from schist. The UCS values show strong to moderate positive correlation with specific gravity (Fig.6.3 and 6.4A; Table 6.3). Whereas, it shows moderate negative correlation with water absorption (Fig.6.3and 6.4B). The UCS shows positive correlation with RMR and SMR, which indicates lithological or rock mass control on the slope stability.

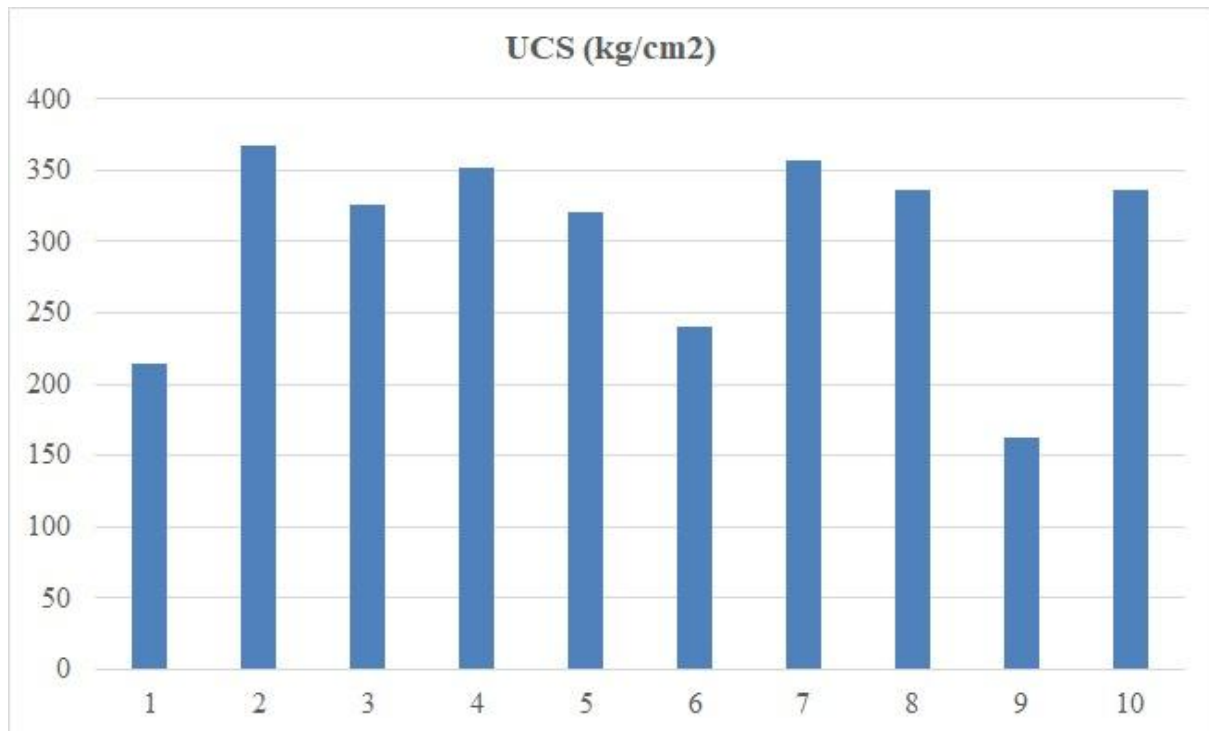


Fig. 6.3. Histogram showing UCS result of rock samples.

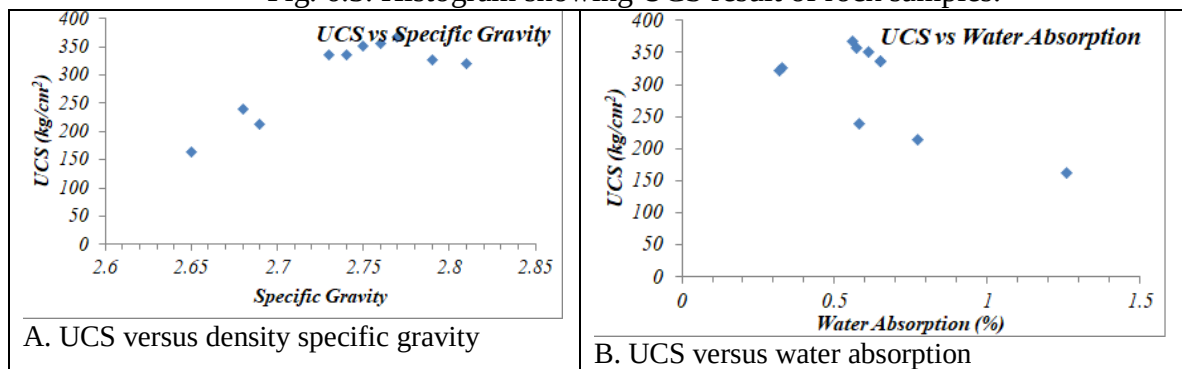


Fig.6.4. Bi-variate graphs showing Geotechnical parameters versus UCS of rock samples.

6.3. Slope stability assessment of Rock Slopes: Rock Mass Rating and Slope Mass Rating

Field based slope stability assessment of rocky slopes, mainly at cut slopes, was done at 118 locations / outcrops, using basic Rock Mass Rating (RMR_b) and slope mass rating (SMR) to identify the vulnerable slopes. Basic RMR is a commonly used geomechanical classification for the characterization of rock mass based on five geotechnical parameters (Bieniawski, 1976, 1979). RMR_b is calculated on the basis of field as well as laboratory inputs which include identification and collection of field data for parameters viz. number of joint sets (joint volume) for rock strength i.e. Rock Quality Designation (RQD), Spacing, Persistence, Aperture, Infilling, Roughness and Groundwater condition. Ten (10) representative rock samples were collected for determining the GT parameters from GT Lab, NR for RMR_b

calculations (Table 6.3). RMR_b calculations for rest of the rock exposures were based on field-based calculations (Annexure VII). Slope Mass Rating (SMR) is a rating system for rocky slopes where by adjustments is made in RMR_b for relation of structural discontinuities (dip direction and amount) with slope gradient (slope direction and angle) and for nature of excavation mechanism (Romana, 1985; Singh and Goel, 1999). The SMR classes are used to provide recommendations on the required slope reinforcement method. Slopes with very good stability will not need a reinforcement system, whereas those with lower levels of stability will require a variety of reinforcement types (Romana, 1993). The F_1 , F_2 and F_3 factors determine the existing relationship between discontinuities affecting the rocks and the slope, whereas, F_4 adjustment factor is used as per the slope excavation method. It is obtained using expression:

$$SMR = RMR_b + (F_1 \times F_2 \times F_3) + F_4 \text{-----} Eq-9$$

Where:

- RMR_b is the basic RMR index resulting from Bieniawski's rock mass classification;
- F_1 depends on the parallelism between discontinuity dip direction and slope dip,
- F_2 is related to the probability of discontinuity shear strength (Romana, 1993) and depends on the discontinuity dip,
- F_3 depends on the relationship between slope, and discontinuity dips,
- F_4 is a correction factor that depends on the excavation method.

The rocks are slight to moderately weathered. Based on the field based UCS values rock masses fall in weak to moderately strong classes. On the basis of field observations, the rock masses dominantly fall in fair to good category with RMR_b values ranging between avg. 33.6-79.3. Some out crops also fall in poor category with approximately 33.6 RMR_b . Field based RMR varies from Class IV Poor to Class II Good category. Most of the rock masses in the study area falls into Fair and Good categories, except three falling in Class IV Poor category. The SMR values, based on Romana (1985) for the same 10 outcrops, range from 54.6 to 79, categorizing them into Normal Partially Stable (III), and Good Stable (II) SMR classes. Specifically, there is no outcrop in Bad Unstable, five in Normal Partially Stable, five in Good Stable categories (Table 6.4).

The F_4 adjustment factor of 0.0 has been used as the excavation methods generally used for road cutting involve normal blasting and mechanical excavation because of jointed rocks. However, the technique of poor blasting will make the slopes more prone to sliding. The field based RMR_b and SMR assessment of representative outcrops reveals lithological control on slope instability, with fair to good RMR_b and falling under partially stable and stable SMR categories.

Table 6.4: Slope Mass Rating (SMR) calculation at UCS based Rocky outcrops

Sample No.	DL/RS/01	DL/RS/02	DL/RS/03	DL/RS/04	DL/RS/05	DL/RS/06	DL/RS/07	DL/RS/08	DL/RS/09	DL/RS/10
Location	285	286	287	288	289	290	291	292	293	294
Latitude	33.2631	33.23	33.2008	33.16483	33.147	33.146	33.139	33.1516	33.155957	33.14
Longitude	75.7736	75.788	75.79871	75.80347	75.799	75.784	75.756	75.7483	75.701807	75.559
Rock Name	Schist	Quartzitic schist	Marble	Granite	Marble	Schist	Quartzitic schist	Quartzite	Schist	Granite
RMRb	58.67	55.33	79.33	65.67	69.67	54.67	63.00	61.67	54.33	57.00
RMR Class	III	III	II	II	II	III	II	II	III	III
RMR Name	Fair	Fair	Good	Good	Good	Fair	Good	Good	Fair	Fair
Slope direction	245	245	220	320	250	10	20	75	55	315
Slope angle	80	80	80	80	80	80	80	80	80	80
F1	Romana 1985	0.15	0.15	0.40	0.15	1.00	0.15	0.15	0.15	0.15
F2		1.00	1.00	1.00	1.00	0.0	1.00	1.00	1.00	1.00
F3		-50.0	-50.0	-25.0	-6.0	-60	-60.0	-60.0	-60.0	-25.00
F4		0	0	0	0	0	0	0	0	0
SMR	51.2	47.8	69.3	64.8	69.7	45.7	54.0	52.7	50.6	48.0
Class	III	III	II	II	II	III	III	III	III	III
Class Name	Normal	Normal	Good	Good	Good	Normal	Normal	Normal	Normal	Normal
Stability	Partially stable	Partially stable	Stable	Stable	Stable	Partially stable	Partially stable	Partially stable	Partially stable	Partially stable
F1	Anbalgan et al. 2008	0.40	0.50	0.20	0.40	0.20	0.20	0.20	0.20	0.50
F2		0.40	0.50	0.50	0.50	0.40	0.50	0.50	0.50	0.50
F3		1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SMR	58.8	55.6	79.4	65.9	69.7	54.8	63.1	61.8	54.6	57.3
Class	III	III	II	II	II	III	II	II	III	III
Stability	Partially stable	Partially stable	Stable	Stable	Stable	Partially stable	Stable	Stable	Partially stable	Partially stable

The ten representatives outcrop samples are much less and limited for preparing the susceptibility and geotechnical maps of whole rocky slopes. Therefore, RMR_b and F_1 , F_2 , F_3 adjustment calculations / ratings as per Bieniawski, (1989) and Anbalagan et al. (2008) were made for all the 55 studied outcrops (Annexure VIII). These RMR_b and adjustment ratings were calculated on field-based data, representative for whole study area, all along the NH-244 from Pul Doda to Lawa bridge to prepare the SMR based landslide susceptibility and kinematic failure maps in rocky slopes. The field based RMR_b values of the 118 rocky slopes range from 32.16 to 79.33 representatives of Poor (IV), fair (III) and Good (II) RMR_b class (Annexure VIII). 45 slopes fall in good, 67 in fair and 06 in poor RMR_b category. The SMR values based on Anbalagan et al. (2008) for same 55 rocky slopes range from 33.26 to 79.43 falling in bad unstable (IV), normal partially stable (III) and good stable (II) SMR categories, respectively representing 6, 65 and 47 slopes (Annexure VIII).

The weightage of 5.0 was given to SMR map, which encompasses the ratings for lithology, joint characteristics, hydrogeology and relation with slope. The SMR map initially prepared from point data was projected to the similar lithologies in inaccessible areas. The SMR based susceptibility map was prepared by overlay analysis of relevant geo-factors maps. Separate SMR maps were prepared for rock slides (planar and wedge failure) and rock falls (Toppling). The highest pixel values (worst SMR) were selected for preparing combined susceptibility map.

6.4. Kinematic Analysis of structural discontinuities:

The study area is represented by 55.4% overburden cover slopes and 44.6% rock dominated slope for rocky slope, structure based kinematic analysis was carried out. The geological map of the area shows 48% mica schist with quartzite bands, 9% quartzite, 4.5% granite 0.11 of area is represented by marble band and 0.04% area represents by gneiss (Fig.6.9). The structural data of 381 discontinuity planes from five joint sets (J_1 , J_2 , J_3 and J_4) was collected from 118 spots and used for kinematic analysis. The data was analysed for all four joint sets for planar failure, wedge failure, flexure and direct toppling failure after analysing the mutual relations of dip/ intersection directions versus slope directions and assigning ratings as recommended by Anbalagan, et. al. 2008 (Annexure–IX–XII). Out of 381 discontinuity planes, 188 (49.34 %) planes at 116 spots show positive relation towards the slope direction and therefore, have the possibility of planar failure (Annexure–IX). The favourable relation is dominant in J_2 ($n=70$) and J_1 ($n=56$) planes which together account for 67%. The J_3 ($n=49$), and J_4 ($n=12$) planes comprising (33%) show positive favourable relation

for planar failure. Similarly, 188 planes (49.34%) from 116 spots show opposite relation with the slope direction and, therefore, have the possibility of toppling failure (Annexure–XI, XII). The opposite relation is dominant in J_3 ($n=67$) J_1 ($n=63$) and J_2 ($n=50$) representing 96 % of 188 joint planes. The J_4 contribute 4% ($n=7$) planes. Further, all the joint data ($n=381$) were analysed for wedge intersections formed by the intersection of different joint sets along various slope directions. Total 167 wedge intersections at 108 spots having positive favourable relation with slope direction were found (Annexure–X). The J_2 and J_3 fabrics are found to be the most crucial planes in the formation of such intersections and is involved in 37% ($n=61$) having favourable relation with slope direction. Various plane intersections involved in wedge formation are J_2 - J_3 ($n=49$), J_1 - J_3 ($n=33$), J_2 - J_4 ($n=08$), J_1 - J_4 ($n=08$), J_3 - J_4 ($n=08$) planes. The above synthesis reveals that J_1 and J_2 may show more planar and wedge failure than J_3 which mostly dips into the hill and may show more topple failure and their cumulative failure map were prepared (Fig Fig.6.5-6.8).

The geological map of the study area shows moderately deformed with widely spaced jointed rock masses of schist, quartzite, gneiss, granite and marble. The presence of such fractured and weak rock masses served as the base for selection of friction angle which was taken as 30° (After Vallejo and Ferrer, 2011; Anabalgan et. al., 2008). The dominant natural slope inclination in the study area is moderately steep to steep i.e. up to 45° . Further, the slope map prepared from the ALOS DEM reveals the maximum slope angle for the study area is 74° (Fig. 6.2). However, the study carried out along the road stretch where steepening of slopes due to cut slope is the predominant feature and forms sub vertical to vertical slope angles of $>80^\circ$. Therefore, slope inclination of 80° was considered for such cut slope road corridor for the calculations of F_3 to prepare the kinematic failure map (Fig.6.5-6.8). Similarly, it was found better to use slope direction as per the aspect class, as huge data was to be analysed stereographically for kinematic analysis purposes. Therefore, central slope direction of a particular aspect class was chosen for calculation of F_1 , e.g., in case of aspect class of 0 - 29° (NNE), slope direction of 15° was used for kinematic failure analysis. The aspect map has been classified into 12 directions from 0° to 360° with each class representing 30° . The joint discontinuity planes synthesised for planar, wedge and toppling analysis were plotted and presented stereographically to identify the probable failure joints (Annexure – IX-XII). The planar and flexure toppling stereographic projections were done on aspect class basis, whereas, the wedge and direct/block toppling stereographic projections were made on individual spot

basis. The stereographic maps were prepared at slope angles of 80° which represent the steep cut slope angles prevalent in the area.

Kinematic analysis reveals that the area shows 75% slope directions towards northern hemisphere with slight dominance towards eastern quarter (51%) than western (49%). Whereas the joint data shows pervasive J_1 and J_2 predominantly dip towards NW direction, moderately spaced J_3 fabric predominantly dipping towards SW direction, widely spaced J_4 uniformly dipping towards NW and SE directions. The opposite relation of slope with J_1 and positive relation of slope with J_2 strongly support the view that present topography of the area is controlled by D_2 deformation and is a mature topography. Further, involvement of J_3 fabric in wedge formation again supports its cross-cut relation with earlier fabric.

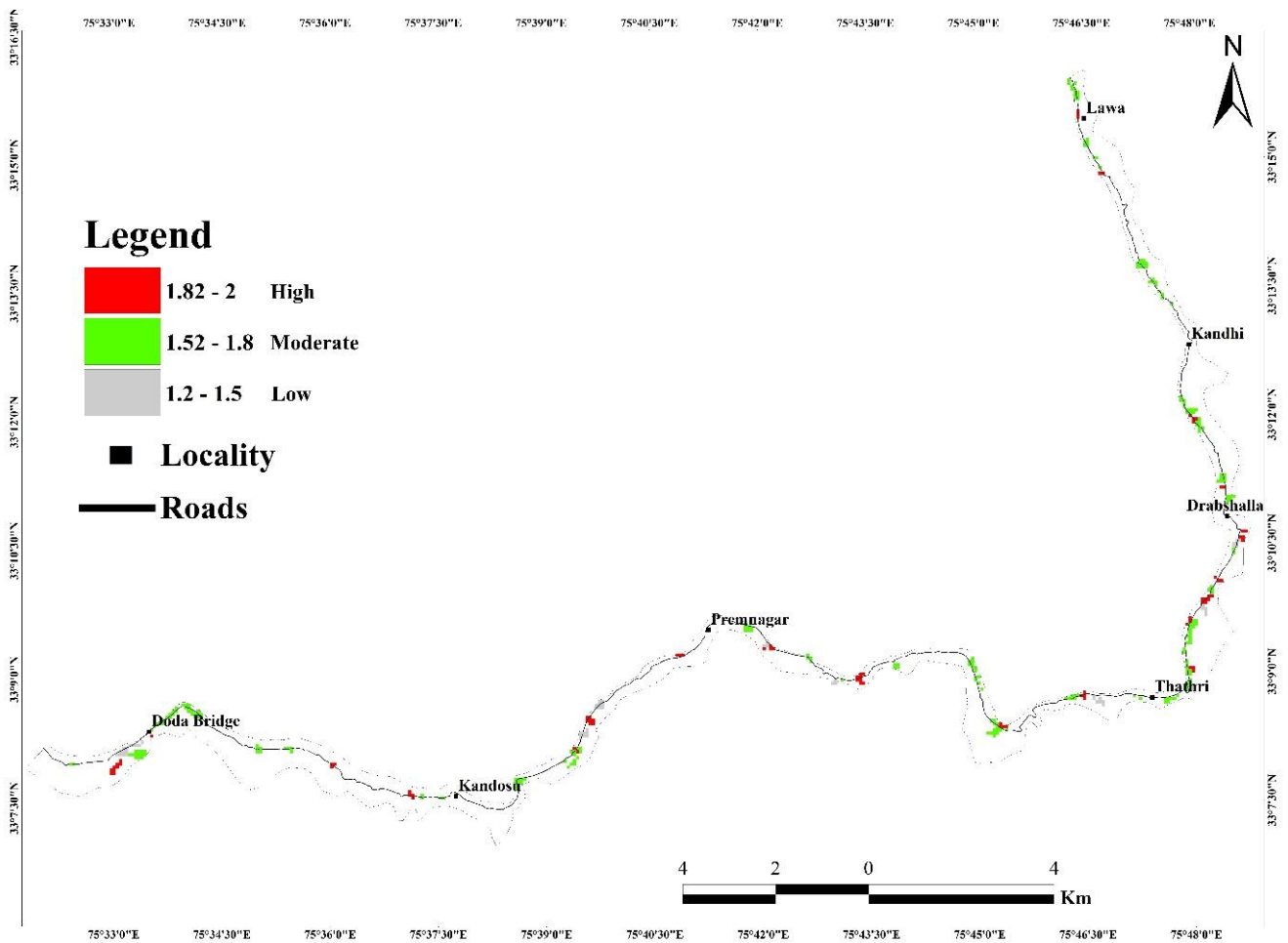


Fig. 6.5. Kinematic planar failure map of the rocky exposures of the study area.

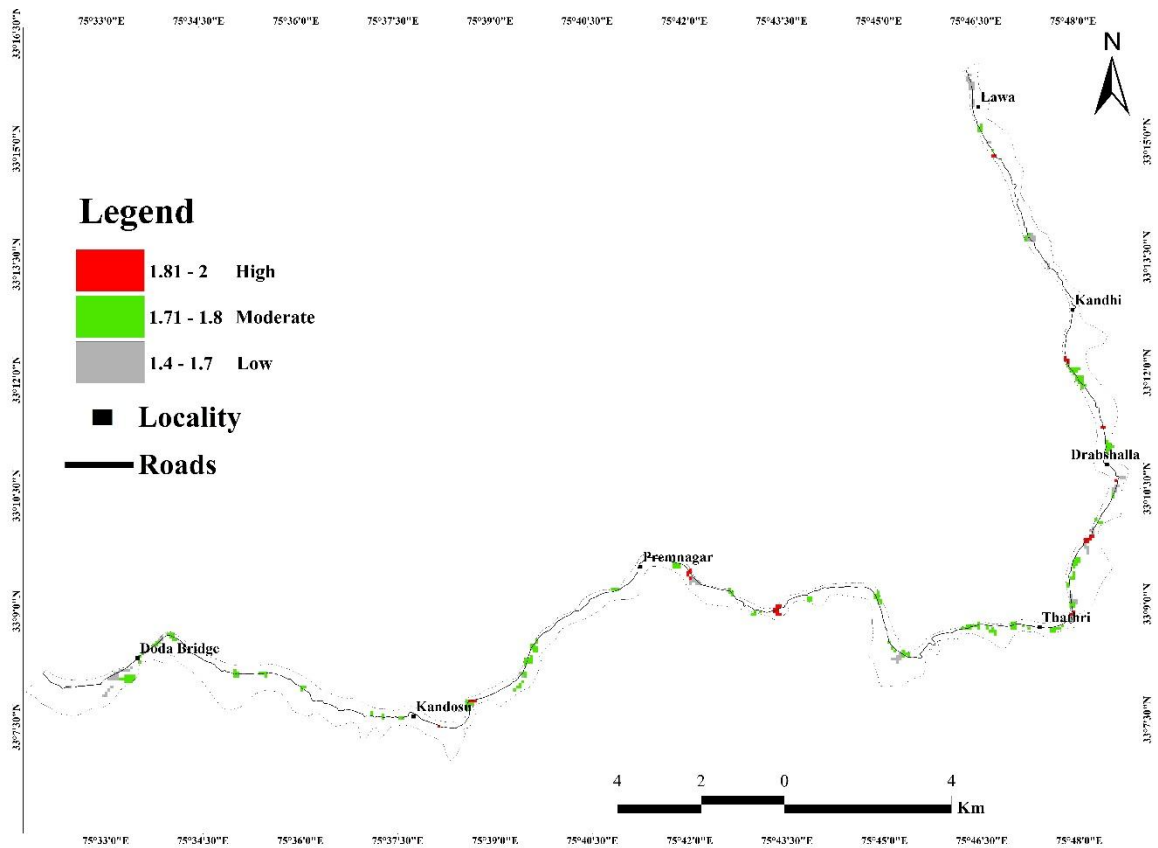


Fig. 6.6. Kinematic wedge failure map of the rocky exposures of the study area.

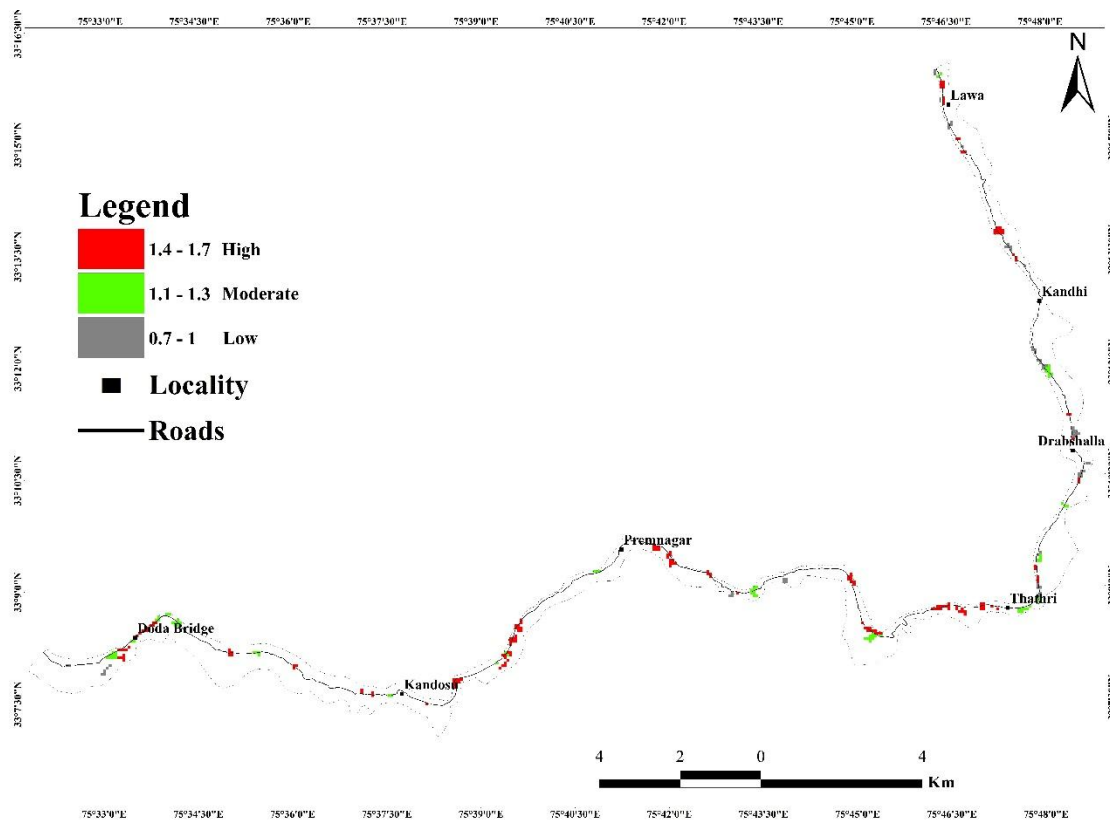


Fig. 6.7. Kinematic topple failure map of the rocky exposures of the study area.

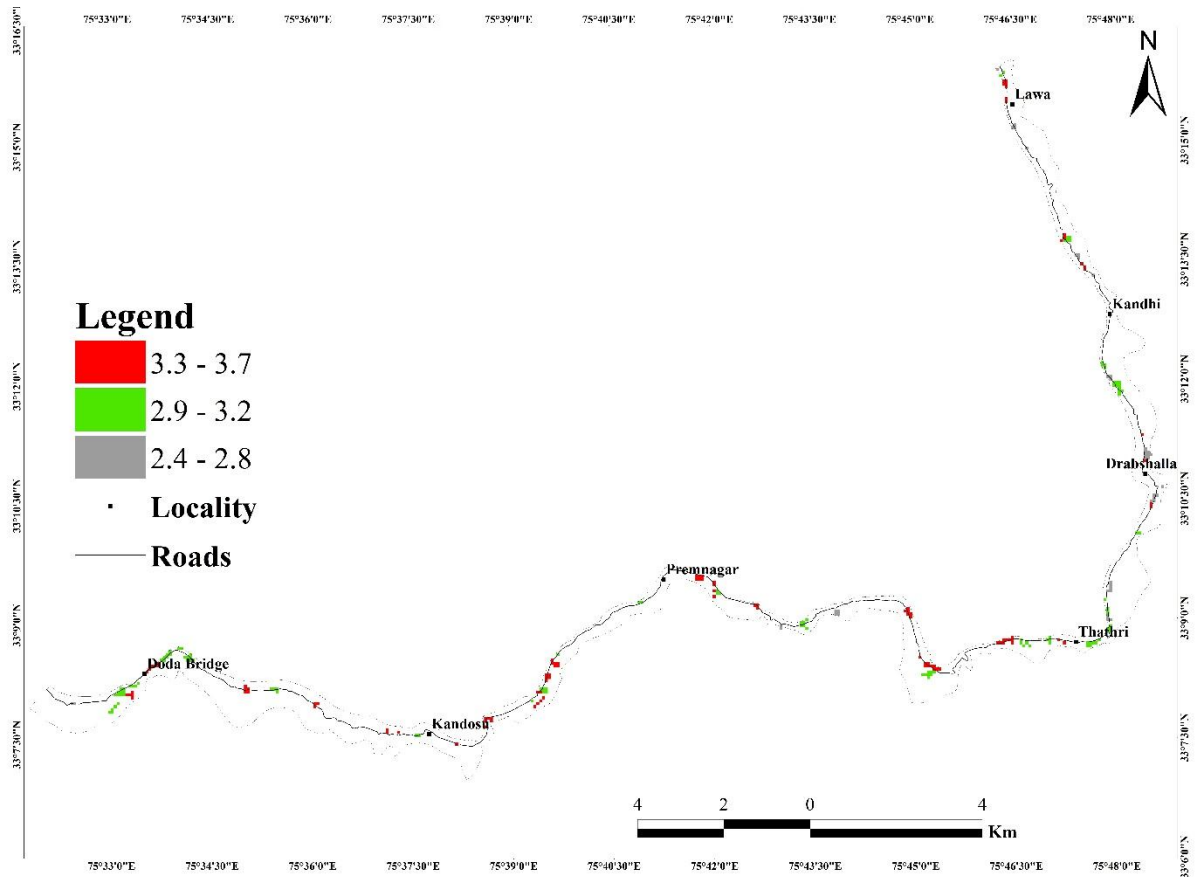


Fig. 6.8. Integrated kinematic failure map of the rocky exposures of the study area.

6.5. Stereographic analysis of structural discontinuities: To determine the potential failure mechanism and direction in jointed rock mass, kinematic analysis is graphically demonstrated using stereographic projection. In this way, the stability of rock mass is assessed based on the trend of joint planes, the orientation of slope faces and the angle of internal friction of the slope material. It helps to analyse the potential for the various modes of rock slope failures (plane, wedge, toppling failures), that occur due to the presence of unfavourably oriented discontinuities such as joints, faults, bedding planes, foliation, and shear zones that can potentially serve as failure planes. Kinematic analysis is based on Markland's test which is described in Hoek and Bray (1981).

According to the Markland's test, a plane failure is likely to occur when a discontinuity dips in the same direction (within 20°) as the slope face, at an angle gentler than the slope angle but greater than the friction angle along the failure plane (Hoek and Bray, 1981). A wedge failure may occur when the line of intersection of two discontinuities, forming the wedge-shaped block, plunges in the same direction as the slope face and the plunge angle is less than the slope angle but greater than the friction angle along the planes of failure (Hoek and Bray,

1981). A toppling failure may result when a steeply dipping discontinuity is parallel to the slope face (within 30°). In the study area kinematic analysis has been done by following the Hoek-Bray's Failure Criteria (Hoek and Bray, 1981) with the internal friction angle as 30° . The lateral limit buffer for planar and toppling failure stereographic projections was increased to 30° rather than 20° (as suggested by Romana, 1985), as it can give proper buffering to cover the whole aspect class of 30° each, for the extreme ends also (Paul and Kumar, 2022).

Wedge failure (Fig 6.7) analysis for the slope section has been done to determine the potential zone of failure. During the analysis the internal friction angle, slope angle and direction has been plotted in the stereonet analysis program DIPS of Rocscience. The area falling between slope and angle of internal friction constitute a potential zone. If the intersecting joints fall within the zone, the probability of wedge failure is high (Hoek and Bray, 1981 and Anbalagan, et al. 2008). Stereographic analysis for the planar and toppling failure (Fig.6.9 & 6.10) for structural discontinuities along 5 different slope directions were analysed (Fig.6.9 & 6.10). The wedge analysis was made individually for each spot including UCS sample spots (Fig. 6.11).

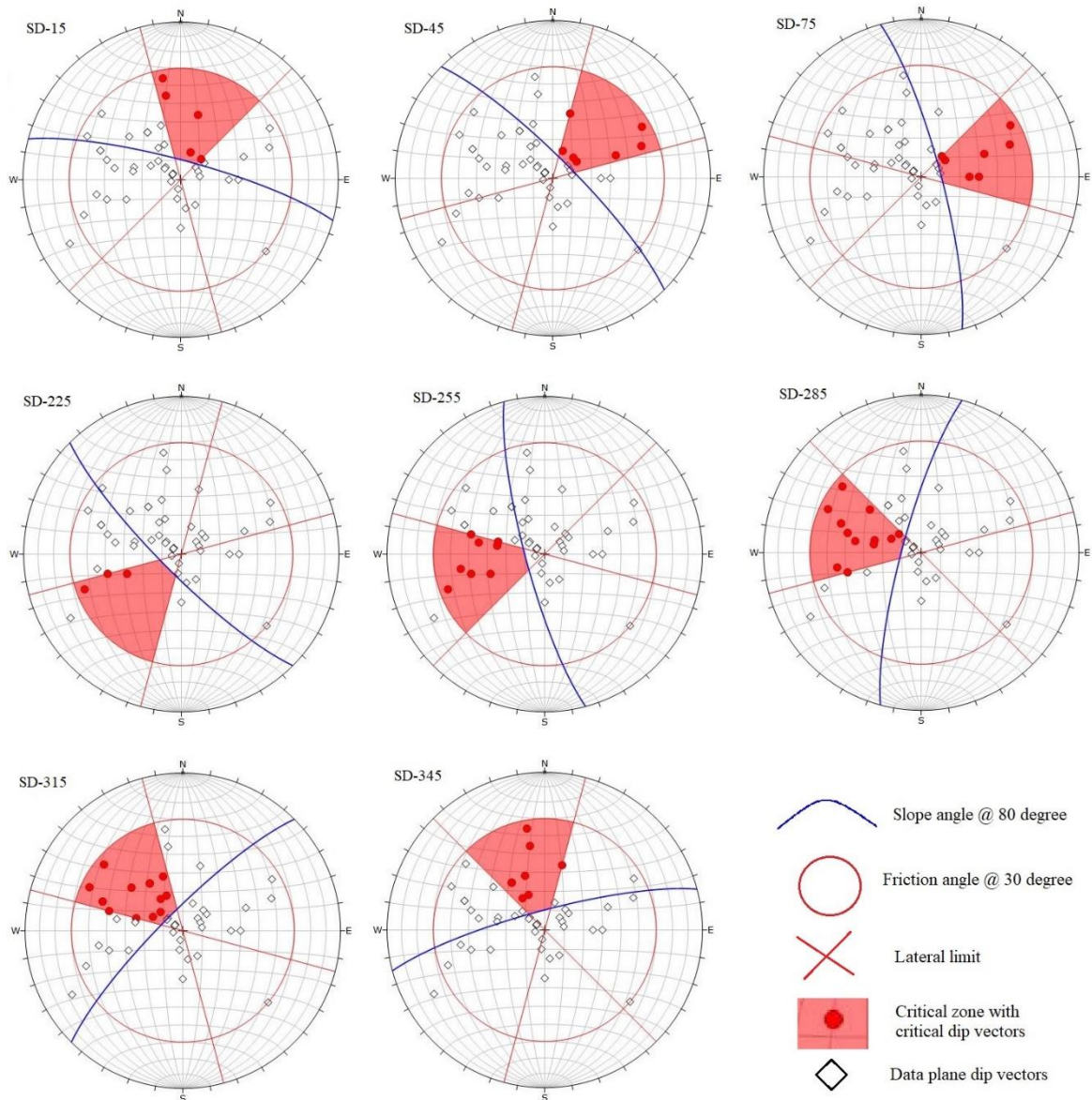


Fig.6.9. Kinematic analysis of rocky slopes on stereo plots showing dip vectors of the probable planar failure joints at various locations of the study area along different slope directions (SD) at internal friction angle of 30° .

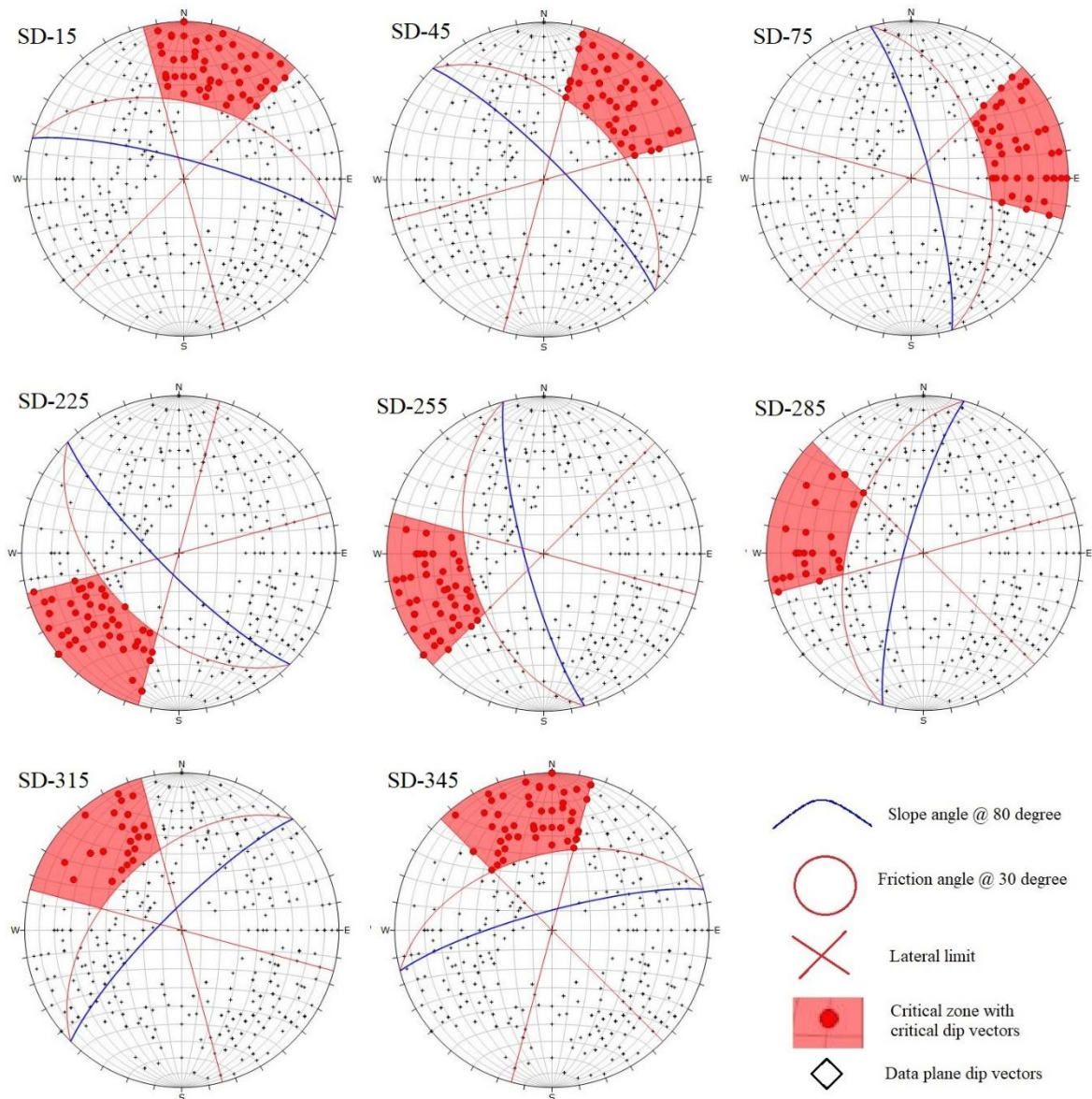


Fig.6.10. Kinematic analysis of rocky slopes on stereo plots showing dip vectors of the probable flexure topple failure joints at various locations of the study area along different slope directions (SD) at internal friction angle of 30° .

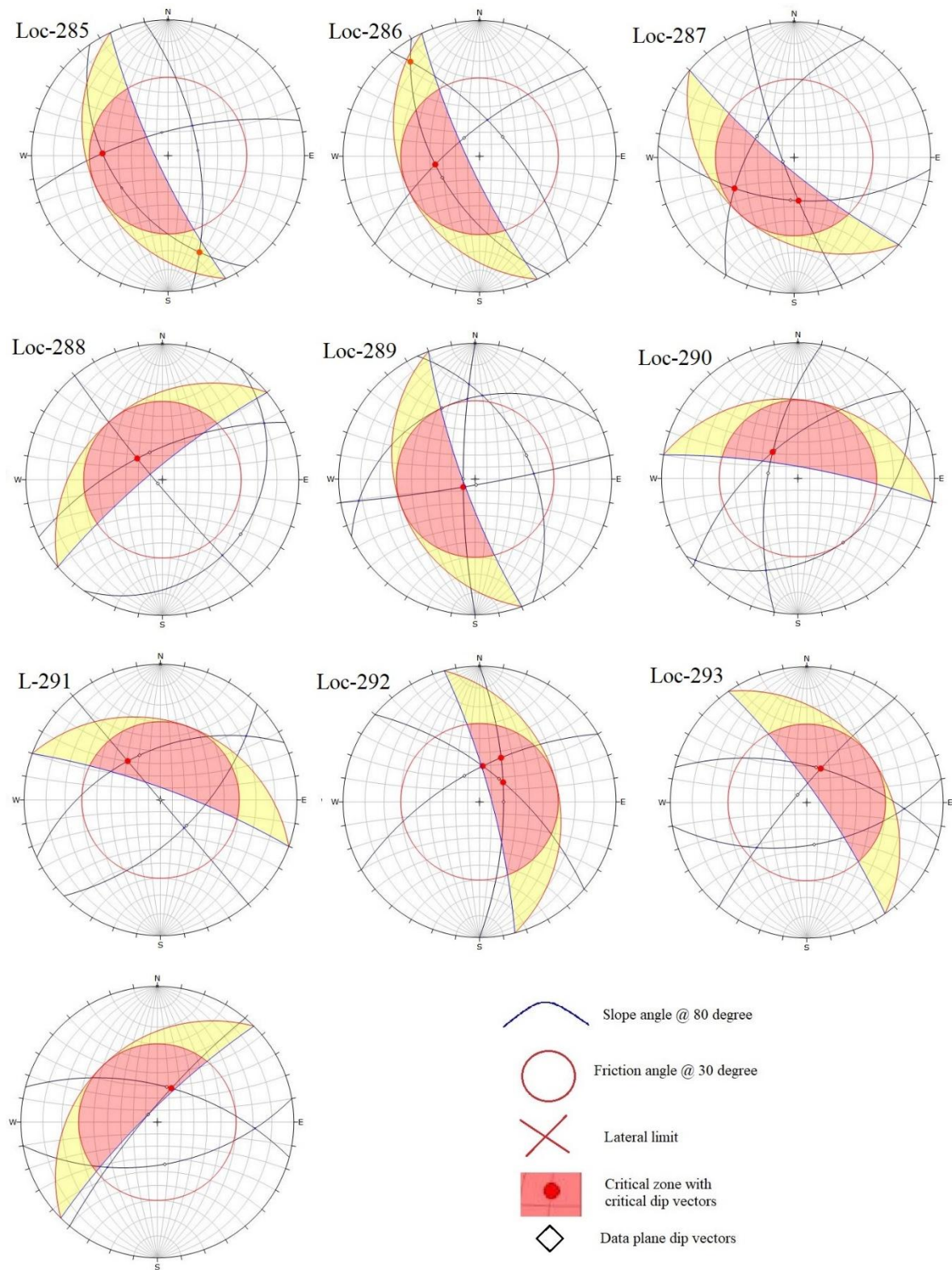


Fig.6.11. Kinematic analysis of rocky slopes on stereo plots showing dip vectors of the probable wedge failure joints at UCS sample locations of the study area at internal friction angle of 30°

6.6. Landslide hazard and risk analysis

The slope stability assessment along NH-244 between Ganpath Bridge and Lawa indicates that the corridor is characterized by variable lithological conditions, steep cut slopes, and high relief, which together control the occurrence of slope failures. To develop a comprehensive understanding of the landslide threat to the highway and associated infrastructure, the results of the geotechnical and kinematic analyses have been integrated into a Landslide Hazard and Risk Analysis framework. This analysis considers not only the mechanical stability of slopes but also the spatial probability of failure (hazard) and the potential consequences (risk) to road infrastructure, settlements, and other assets.

6.7 Methodology

The landslide risk analysis has been carried out following the integrated approach recommended by Varnes (1984), Anbalagan et al. (2008), and **NIDM (2017)**, wherein:

$$\text{Risk} = \text{Hazard} \times \text{Vulnerability} \times \text{Exposure}$$

The analysis was implemented through a GIS-based multi-criteria framework using thematic layers derived from field data, laboratory testing, and secondary datasets.

The following parameters were used:

Component	Factors / Inputs	Source
Hazard	SMR-based slope stability classes, slope angle (°), lithology, land use/land cover, drainage density, relative relief, and rainfall intensity	Field data, ALOS DEM,
Vulnerability	Structural condition of road cut slopes, proximity to habitations, building typology, and socioeconomic condition	Field survey, SOI toposheets
Exposure	Road alignment, bridges, habitations, powerlines, and other infrastructure	Road inventory and satellite imagery

Each factor was assigned a weight and rank according to its relative influence on slope failure, following Anbalagan (1992) and Van Westen (2004).

The weighted overlay analysis was then performed to produce a Landslide Hazard Zonation (LHZ) map, followed by Risk Zonation through integration with exposure and vulnerability data.

6.8 Landslide Hazard Zonation (LHZ)

The SMR-based stability ratings derived from the rock slope analysis formed the core of the hazard zonation. The field-based SMR values (range 33.26–79.43) and corresponding stability classes were reclassified into hazard categories as follows:

SMR Range	Stability Class	Hazard Level
<40	Poor / Unstable	High Hazard
40–55	Fair / Partially Stable	Moderate Hazard
>55	Good / Stable	Low Hazard

Incorporating additional slope and geomorphic factors refined this classification.

The resultant LHZ map delineates three major hazard zones along the NH-244 corridor:

- **High Hazard Zone (26%):** Concentrated near steep cut slopes in schistose and weathered quartzitic rocks between chainage 284–288 km and near Lawa bridge. These sections correspond to SMR Class III and IV, with active debris slides and rockfalls observed.
- **Moderate Hazard Zone (48%):** Representing moderately steep slopes in quartzite and granite, where stability is controlled by joint spacing and slope orientation.
- **Low Hazard Zone (26%):** Comprising gentle natural slopes or well-drained quartzite and marble exposures with limited overburden.

6.9 Vulnerability Assessment

Vulnerability was evaluated in both **physical** and **functional** terms:

- **Physical Vulnerability:** Degree of structural susceptibility of assets (road cut slopes, retaining walls, culverts, bridges) to landslide impacts. Ratings were assigned from 0.2 (very low) to 1.0 (very high).
- **Functional Vulnerability:** Importance of the asset for transport and connectivity. Critical lifelines such as NH-244 were assigned higher vulnerability scores.
- **Socioeconomic Vulnerability:** Derived from settlement proximity, population density, and dependence on the highway.

The assessment revealed that the road sections between Ganpath Bridge and Lawa exhibit moderate to high vulnerability, as repeated cut-slope failures directly impact the highway carriageway and adjoining habitations.

6.10 Exposure Mapping

Exposure analysis identified all built-up and infrastructure elements intersecting the hazard zones.

The following features were considered exposed:

- NH-244 alignment (entire corridor under study)
- 7 major bridges and 12 culverts
- 5 habitations within 100 m of the road (including Premnagar, Pul Doda, Ganpath, Thattri, and Lawa)
- Electric poles and roadside retaining structures

Approximately 42% of the total infrastructure length lies within moderate to high hazard zones, representing the primary exposed assets in the study area.

6.11 Risk Zonation

The spatial integration of the hazard, vulnerability, and exposure layers resulted in a Landslide Risk Zonation Map, which classifies the corridor into four risk categories:

Risk Level	Description	% of Corridor
Very High Risk	Frequent rockfall and debris slide zones threatening highway and settlements	14%
High Risk	Steep cut slopes, partially stable schist and quartzite zones	33%
Moderate Risk	Stable slopes with occasional block failures	38%
Low Risk	Gently dipping slopes with competent lithology	15%

The very high to high-risk sectors are primarily localized near Ganpath Bridge area and approach to Lawa area, corresponding to low SMR and RMR values, and areas of repeated slope re-activation.

CHAPTER – 7

DISCUSSION AND CONCLUSIONS

7.1. Summary:

- The NH-244 is a critical transportation corridor beginning near Khanabal and traversing Achabal, Kokernag, Daksum, and Sinthan Pass (Elevation: 3,748 m), continuing through Kishtwar, Thathri, and Doda, and terminating at NH-44 near Batote. This route is frequently disrupted by landslides due to extreme winter weather and heavy rainfall.
- Cumulative susceptibility values were determined for overburden, rock, and cut slope domains by overlaying the Landslide Susceptibility Evaluation Rating (LSER). The Total Estimated Susceptibility Value (TESV) was calculated for each domain. Various relevant maps used for DEM derivative analysis include Facet Map as base map, Slope Aspect, Slope Angle, Relative Relief, Drainage, Stream Power Index (SPI), Landform (Geomophon) and LanduseLand cover (LULC) maps. Various field based maps prepared are Lithological Map, Structural Map, Slope Forming Material (SFM), SFM thickness, Hydrogeological, LULC, Geotechnical (GT) and kinematic failure maps.
- The rock mass along the alignment is highly deformed, exhibiting three distinct stages of deformation. Combined with steep slope gradients, these structural weaknesses have degraded rock mass quality, ranging from good to bad conditions. Field mapping indicates the presence of three to four persistent joint sets, placing the rock mass predominantly in RMRb Classes II–III (Fair to Good), with average ratings between 32 and 79. Locally, some outcrops fall into Class IV (Poor), with RMRb values around 32. Weathering grades range from slight to moderate. Field-based UCS values classify the rock masses as weak to strong.
- The Total Estimated Susceptibility Value (TESV) for meso-scale landslide susceptibility mapping ranges from 1.4 to 7.8 and has been classified into three susceptibility classes—low, moderate, and high—using both equal interval and natural breaks classification methods. The results indicate that 38.6% of the area falls under the low susceptibility class, 34.4% under the moderate class, and 26.8% under the high susceptibility class. Validation against field data demonstrates over 85% knowledge-based accuracy for all three susceptibility zones. Landslide inventory-based validation further shows a 62.9% correspondence for high susceptibility zones (covering 26.8%

of the area) and over 85% correspondence when high and moderate susceptibility zones are considered together. During field investigations, a total of 27 landslides were identified and documented, with site-specific recommendations detailed in a 42-point proforma.

- The stretch from Premnagar to Thattri is composed predominantly of unconsolidated to semi-consolidated river-borne material (RBM) and glacio-fluvial deposits, making it highly susceptible to debris slides and debris falls due to the presence of weak pebble beds and steep cut slopes. Debris falls frequently occur as pebble shooting stones from over-steepened slopes, while sand interbeds within the sequence are easily eroded under such conditions. At Gadsu, an active colluvial debris slide exhibits translational failure and has progressed laterally, causing further destabilization of the slope and increasing the overall hazard potential in the area.

7.2. Recommendations:

- The road section and Gadsu village are located on unconsolidated colluvial deposits underlain by mica schist with interbedded quartzite bands. Given the prevailing geological conditions, the following mitigation measures are recommended: biotechnical slope stabilization using geofibre reinforcement, afforestation to improve root cohesion and enhance surface stability, drainage diversion in the crown area to reduce water infiltration, construction of a protection wall at the toe to prevent undercutting, and buttress support along the cut slope to increase overall stability. The slope currently exhibits retrogressive movement, presenting an immediate hazard to structures located in the crown portion. Urgent relocation of these houses is necessary to safeguard lives and property. Detailed Landslide Site specific Landslide study is recommended for future as the area is vulnerable to landslide.
- At Premnagar, the road passes through a densely built and narrow settlement, where widening would cause significant disruption. To maintain safe and uninterrupted traffic flow, the construction of a suitable viaduct in consultation with an experienced design engineer is suggested to bypass the settlement entirely.
- The road section from Premnagar to New Thattri is primarily constructed over river-borne material (RBM), consisting of unconsolidated pebbles, cobbles, and boulders within a sand and silty matrix. This composition makes the stretch highly susceptible to debris falls. For slope stabilization, the installation of an open-type drapery wire

mesh at the base, anchored with deep soil nails or rock anchors, is recommended to ensure structural stability and containment of loose debris in consultation with an experienced design engineer.

- At Thattri, the road again passes through a narrow and congested settlement where widening is not feasible due to severe space limitations. To address this, a 1.4 km viaduct is proposed to reroute traffic and ease congestion in consultation with an experienced design engineer.
- Near Darabshalla, a large rock slide zone exhibiting planar failure has been identified. To bypass this unstable area, a combination of 500 m of tunnelling and a 70 m bridge is recommended as part of the mitigation strategy.

Locality Index

S. No.	Name	Latitude	Longitude
1	Pul Doda	33°08'15.78"	75°33'26.89"
2	Ganpath Bridge	33°08'14.48"	75°32'02.16"
3	Prem Nagar	33°09'32.94"	75°41'15.08"
4	Thatri	33°08'42.85"	75°47'37.56"
5	Thelela	33°07'23.83"	75°38'58.76"
6	Darabshallah	33°10'47.64"	75°48'31.62"
7	Khandni	33°12'46.88"	75°47'56.71"
8	Lawa Bridge	33°15'53.53"	75°46'17.35"
9	Kandosu	33°07'35.98"	75°37'43.96"

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Annexure -I: Landslide Inventory of the study area (42 Point Proforma)

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	1 (L09)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/15
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33.26888
8	Longitude	75.773672
9	Length	30m
10	Width	25m
11	Height	25m
12	Area	750 sq m
13	Depth	1m
14	Volume	750 cu m
15	Run out distance	Nil
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	rapid
19	Activity	Active
20	Distribution	Confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	reoccurring
24	Geomorphology	Straight slope
25	Geology	Mica Schist
26	Structure	Jointed
27	Land use/ Land cover	Barren / grass

28	Hydrological condition	Damp
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	750m2 area
36	Geo-scientific Causes	Planar failure along J1 due to cut slope
37	Remedial measures	Rock bolting, wiremesh with shotcrete
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide has happened due to planar failure along J1 plane and due to cut slope.
41	Pdf	
42	Alert	II

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	2 (L10)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/15
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33.260787
8	Longitude	75.773958
9	Length	30m
10	Width	30m
11	Height	20m
12	Area	450 sq m
13	Depth	1m
14	Volume	450 cu m
15	Run out distance	2m
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	rapid
19	Activity	Active
20	Distribution	Retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Reoccurring
24	Geomorphology	Slope
25	Geology	Mica Schist
26	Structure	Jointed
27	Land use/ Land cover	Barren grass

28	Hydrological condition	Dry
29	Triggering Factor	Cut slope failure, rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	450m2 area
36	Geo-scientific Causes	Cut slope failure
37	Remedial measures	Rock bolting with wire mesh along with shotcrete for closing the joints so that water cannot percolate
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	3 (L12)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/15
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	32.26031
8	Longitude	75.774075
9	Length	100m
10	Width	200m
11	Height	90m
12	Area	20000sq m
13	Depth	1m
14	Volume	20000 cu m
15	Run out distance	1m
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	rapid
19	Activity	dormant
20	Distribution	confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Old slide
24	Geomorphology	Straight slope
25	Geology	Schist
26	Structure	Jointed

27	Land use/ Land cover	Barren
28	Hydrological condition	Dry
29	Triggering Factor	rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	20000m ² area
36	Geo-scientific Causes	Planar failure along J1 plane
37	Remedial measures	Retaining wall
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to cutting of toe for road widening
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	4(L13)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/15
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 15' 27.8784" N
8	Longitude	75° 46' 26.1696" E
9	Length	20m
10	Width	25m
11	Height	18m
12	Area	500sq m
13	Depth	1m
14	Volume	500 cu m
15	Run out distance	nil
16	Type of Material	Rock
17	Type of movement	fall
18	Rate of movement	Very rapid
19	Activity	active
20	Distribution	confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	flowing

24	Geomorphology	Straight slope
25	Geology	Schist
26	Structure	Jointed
27	Land use/ Land cover	Barren
28	Hydrological condition	Dry

29	Triggering Factor	Rainfall/cut slope
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	500m ² area
36	Geo-scientific Causes	Toppling failure due to cut slope , drainage water leakage along the slope
37	Remedial measures	Wiremesh with geojute and training of drainage
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to toppling failure along the water percolation from drainage
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	5 (L14)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/15
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 15' 24.7896" N
8	Longitude	33° 15' 24.7896" N
9	Length	15m
10	Width	10m
11	Height	10m
12	Area	150 sq m
13	Depth	1m
14	Volume	150 cu m
15	Run out distance	nil
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Reoccurring
24	Geomorphology	Slope
25	Geology	Schist
26	Structure	Jointed
27	Land use/ Land cover	Moderate vegetation

28	Hydrological condition	flowing
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	150m2 area
36	Geo-scientific Causes	Access water seepage into the slide zone has contributed for sliding.
37	Remedial measures	Channelization of the drainage, wiremesh with geojute along with erosion resistant plantation
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	. Access water seepage into the slide zone has contributed for sliding.
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	6 (L19)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/16
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 15' 1.2852" N
8	Longitude	75° 46' 39.1764" E
9	Length	70m
10	Width	100m
11	Height	60m
12	Area	7000sq m
13	Depth	1m
14	Volume	7000cu m
15	Run out distance	nil
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Reoccurring
24	Geomorphology	Straight steep slope
25	Geology	Garnetiferous mica schist
26	Structure	Jointed
27	Land use/ Land cover	barren
28	Hydrological	dry

	condition	
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	7000 ^{m2} area
36	Geo-scientific Causes	Planar failure along J1 plane due to toe cut.
37	Remedial measures	Rock bolting with wire mesh along shotcrete
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	. Planar failure along J1 plane due to toe cut.
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	7 (L20)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/16
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 14' 52.9836" N
8	Longitude	33° 14' 52.9836" N
9	Length	50m
10	Width	30m
11	Height	55m
12	Area	1500sq m
13	Depth	1m
14	Volume	1500cu m
15	Run out distance	nil
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight steep slope
25	Geology	Garnetiferous mica schist
26	Structure	Jointed
27	Land use/ Land cover	barren

28	Hydrological condition	dry
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1500 ^{m2} area
36	Geo-scientific Causes	Wedge failure
37	Remedial measures	Wire mesh with shotcrete
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to unscientific steep cut slope
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidence

No	Field	Description
1	Slide No (LS .No.)	8 (L20)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/16
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 14' 45.258" N
8	Longitude	33° 14' 45.258" N
9	Length	20m
10	Width	100m
11	Height	15m
12	Area	2000sq m
13	Depth	1m
14	Volume	2000cu m
15	Run out distance	nil
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Recurring
24	Geomorphology	Straight steep slope
25	Geology	Gneiss
26	Structure	Jointed
27	Land use/ Land cover	barren
28	Hydrological condition	dry

29	Triggering Factor	Cut slope due to road cutting and rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	2000 ^{m2} area
36	Geo-scientific Causes	Cut slope failure, toe cutting
37	Remedial measures	Retaining wall along wire mesh with geojute
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	.The slide is happened due to unscientific steep cut slope for road widening
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	9 (L28)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 1.8204" N
8	Longitude	75° 32' 17.9772" E
9	Length	40m
10	Width	60m
11	Height	35m
12	Area	2400sq m
13	Depth	1m
14	Volume	2400cu m
15	Run out distance	nil
16	Type of Material	Debris
17	Type of movement	Slide
18	Rate of movement	moderate
19	Activity	Active
20	Distribution	retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Recurring
24	Geomorphology	Straight slope
25	Geology	schist
26	Structure	Jointed
27	Land use/ Land cover	barren

28	Hydrological condition	damp
29	Triggering Factor	Cut slope due to road cutting and rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	2400 ^{m2} area
36	Geo-scientific Causes	Cut slope failure
37	Remedial measures	Retaining wall along wire mesh with geojute
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	. Slide has occurred due to cut slope failure
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	10(L32)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 2.1372" N
8	Longitude	75° 32' 42" E
9	Length	40m
10	Width	80m
11	Height	30m
12	Area	3200sq m
13	Depth	1m
14	Volume	3200cu m
15	Run out distance	nil
16	Type of Material	Debris
17	Type of movement	Slide
18	Rate of movement	slow
19	Activity	Active
20	Distribution	retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Recurring
24	Geomorphology	Straight slope
25	Geology	Mica schist
26	Structure	
27	Land use/ Land cover	grass

28	Hydrological condition	damp
29	Triggering Factor	Cut slope due to road cutting and rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	2000 ^{m2} area
36	Geo-scientific Causes	Toe cutting in loose unconsolidated material
37	Remedial measures	Benching along with retaining wall at toe
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to cutting of slope and toe cut
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	11(L37)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 14.2152" N
8	Longitude	75° 33' 21.6" E
9	Length	30m
10	Width	50m
11	Height	25m
12	Area	1500sq m
13	Depth	1m
14	Volume	1500cu m
15	Run out distance	nil
16	Type of Material	Debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	retrogressive
21	Style	multiple
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	schist
26	Structure	
27	Land use/ Land cover	barren

28	Hydrological condition	dry
29	Triggering Factor	Rainfall and toe cut
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1500 ^{m2} area
36	Geo-scientific Causes	Toe cutting due to road wideing has daylighted the joints
37	Remedial measures	Retaining wall, wire mesh with shotcrete along slope benching
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due toe cut
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	12(L44)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 37.9824" N
8	Longitude	75° 33' 50.4" E
9	Length	15m
10	Width	10m
11	Height	10m
12	Area	150sq m
13	Depth	1m
14	Volume	150cu m
15	Run out distance	10m
16	Type of Material	rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	retrogressive
21	Style	multiple
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	schist
26	Structure	joint
27	Land use/ Land cover	Sparse vegetation

28	Hydrological condition	dry
29	Triggering Factor	Rainfall/road cutting
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	150 ^{m2} area
36	Geo-scientific Causes	Toe cut, steep cut slope
37	Remedial measures	Rock bolt, wire mesh with shotcrete
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide occurred due to toe cutting by steep cut slope.
41	Pdf	
42	Alert	I

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	13(L48)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	Gadsu
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 30.03" N
8	Longitude	75° 34' 15.8376" E
9	Length	45m
10	Width	100m
11	Height	30m
12	Area	4500sq m
13	Depth	2m
14	Volume	9000cu m
15	Run out distance	nil
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	slow
19	Activity	Active
20	Distribution	retrogressive
21	Style	multiple
22	Failure mechanism	Rotational planer
23	History	Feb-23
24	Geomorphology	Straight slope
25	Geology	slate
26	Structure	fault
27	Land use/ Land cover	Sparse forest
28	Hydrological condition	damp

29	Triggering Factor	Steep toe cut
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1200 ^{m2} area
36	Geo-scientific Causes	Wedge failure making rotational planar failure
37	Remedial measures	Brust wall, wiremesh, drainage treatment and vegetation in the crown part
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	steep cut slope. Wedge failure making rotational planar failure
41	Pdf	
42	Alert	I

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	14(L49)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 24.1656" N
8	Longitude	75° 34' 16.9032" E
9	Length	40m
10	Width	50m
11	Height	30m
12	Area	2000sq m
13	Depth	1m
14	Volume	2000cu m
15	Run out distance	20
16	Type of Material	rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	retrogressive
21	Style	single
22	Failure mechanism	Shallow translational
23	History	
24	Geomorphology	Straight slope
25	Geology	Highly weathered mica schist
26	Structure	jointed

27	Land use/ Land cover	Sparse forest
28	Hydrological condition	damp
29	Triggering Factor	Rainfall, toe cut
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	2000 ^{m2} area
36	Geo-scientific Causes	Toe cutting due to cut slope destabilized the slope
37	Remedial measures	Toe cutting due to cut slope destabilized the slope
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to cut slope failure
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

	Field	Description
1	Slide No (LS .No.)	15(L84)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	Gadsu Slide
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 10.7304" N
8	Longitude	75° 34' 51.9816" E
9	Length	25m
10	Width	50m
11	Height	15m
12	Area	1250 sq m
13	Depth	1m
14	Volume	1250 cu m
15	Run out distance	15
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	enlarging
21	Style	multiple
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	Highly weathered Schist

26	Structure	Jointed
27	Land use/ Land cover	Sparse vegetation
28	Hydrological condition	Damp
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1250m2 area
36	Geo-scientific Causes	Toe cut due to road widening
37	Remedial measures	Toe support retaining wall, plantation
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	Un scientific cut slope in the loose material.
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	16 (L59)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 10.5684" N
8	Longitude	75° 35' 33.5364" E
9	Length	20m
10	Width	40m
11	Height	15m
12	Area	800sq m
13	Depth	1m
14	Volume	800 cu m
15	Run out distance	15
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	enlarging
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Reoccurring

24	Geomorphology	Straight slope
25	Geology	
26	Structure	
27	Land use/ Land cover	Moderate fast

28	Hydrological condition	Damp
29	Triggering Factor	rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	800m2 area
36	Geo-scientific Causes	Cut slope failure due to road cutting
37	Remedial measures	Retaining wall for toe support, wire mesh with geojute, plantation
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	

40	Summary/Abstract	Cut slope failure due to road cutting
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	17(L69)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 7' 41.4192" N
8	Longitude	75° 36' 37.7748" E
9	Length	20m
10	Width	50m
11	Height	17m
12	Area	1000sq m
13	Depth	1m
14	Volume	1000 cu m
15	Run out distance	nil
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	active
20	Distribution	retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Recurring
24	Geomorphology	Straight slope
25	Geology	Highly weathered Schist
26	Structure	

27	Land use/ Land cover	Moderate vegetation
28	Hydrological condition	Damp
29	Triggering Factor	Rainfall/toe cut
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	2400m ² area
36	Geo-scientific Causes	Cut slope failure due to unscientific slope cutting
37	Remedial measures	Retaining wall for toe support, wire mesh with shotcrete
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	Cut slope failure due to unscientific slope cutting
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	18(L107)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 9' 13.3128" N
8	Longitude	75° 42' 29.4444" E
9	Length	15m
10	Width	10m
11	Height	12m
12	Area	150sq m
13	Depth	1m
14	Volume	150 cu m
15	Run out distance	5
16	Type of Material	Rock
17	Type of movement	slide
18	Rate of movement	fast
19	Activity	active
20	Distribution	confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	flowing

24	Geomorphology	Steep slope
25	Geology	Highly weathered Schist
26	Structure	Jointed

27	Land use/ Land cover	Dense vegetation
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42-point detailed Geo-parametric attributes of landslide incidences

28	Hydrological condition	Damp
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	150m ² area
36	Geo-scientific Causes	Cut slope failure
37	Remedial measures	Wire mesh ,buttress support
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	Cut slope failure occurred due to steep cut for road widening
41	Pdf	
42	Alert	I / II / III

No	Field	Description
1	Slide No (LS .No.)	19 (L152)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 40.9524" N
8	Longitude	75° 46' 26.976" E
9	Length	25m
10	Width	50m
11	Height	22m
12	Area	1250 sq m
13	Depth	1m
14	Volume	1250 cu m
15	Run out distance	15
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	slow
19	Activity	Active
20	Distribution	enlarging
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight Slope
25	Geology	RBM
26	Structure	
27	Land use/ Land cover	Grass
28	Hydrological condition	Damp

29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1250m2 area
36	Geo-scientific Causes	Cut slope failure due to steep cut slope
37	Remedial measures	Retaining wall , wire mesh with geojute, erosion resistant plantation
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to cut slope failure by road cutting and erosion by drainage
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	20 (L155)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 43.4112" N
8	Longitude	75° 46' 23.484" E
9	Length	25m
10	Width	30m
11	Height	30m
12	Area	750sq m
13	Depth	1m
14	Volume	750cu m
15	Run out distance	20m
16	Type of Material	Debris
17	Type of movement	Slide
18	Rate of movement	Moderate
19	Activity	Active
20	Distribution	Retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	Reoccurring
24	Geomorphology	Straight slope
25	Geology	Mica schist with quartzite bands
26	Structure	Jointed
27	Land use/ Land cover	Grass
28	Hydrological condition	dry

29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	750 ^{m2} area
36	Geo-scientific Causes	Cut slope failure due to road cutting and erosion by drainage.
37	Remedial measures	Channelizing the drainage, retaining wall to support toe along with erosion resistant plantation
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to cutting of road and failed due to wedge failure
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	21 (L161)
2	State	JK
3	District	Doda
4	Toposheet	43O/16
5	Name of the slide	Thatri
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 45.024" N
8	Longitude	75° 47' 2.9472" E
9	Length	15m
10	Width	20m
11	Height	10m
12	Area	300sq m
13	Depth	2
14	Volume	600cu m
15	Run out distance	10m
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	mica schist
26	Structure	Jointed
27	Land use/ Land cover	Settlement

28	Hydrological condition	dry
29	Triggering Factor	Rainfall / cut slope
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	1 house
35	Agriculture/Forest/Barren	300 ^{m2} area
36	Geo-scientific Causes	Wedge failure
37	Remedial measures	Retaining wall, dislodging of the debris from the slide
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	cutting of road and failed
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	22 (L202)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 7' 27.426" N
8	Longitude	75° 38' 12.066" E
9	Length	70m
10	Width	50m
11	Height	60m
12	Area	3500sq m
13	Depth	1m
14	Volume	3500cu m
15	Run out distance	55
16	Type of Material	debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	enlarging
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	Quaternary deposit
26	Structure	Jointed
27	Land use/ Land cover	Sparse vegetation

28	Hydrological condition	dry
29	Triggering Factor	rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	3500 ^{m2} area
36	Geo-scientific Causes	Low cohesion due to the presence of boulder with limited matrix
37	Remedial measures	Retaining wall for toe support, drapery wire mesh
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to Low cohesion due to the presence of boulder with limited matrix
41	Pdf	
42	Alert	I

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	23(L213)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 7' 57.7992" N
8	Longitude	75° 32' 8.25" E
9	Length	20m
10	Width	50m
11	Height	15m
12	Area	1000sq m
13	Depth	3m
14	Volume	1000cu m
15	Run out distance	15
16	Type of Material	Debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Enlarging
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	Quaternary
26	Structure	Jointed
27	Land use/ Land cover	moderate

28	Hydrological condition	flowing
29	Triggering Factor	Unscientific steep cut slope in the unconsolidated overburden/ rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1000 ^{m2} area
36	Geo-scientific Causes	Unscientific steep cut slope in the unconsolidated overburdenwith,
37	Remedial measures	Retaining wall, wire mesh with geojute, erosion resistant
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to Unscientific steep cut slope in the unconsolidated overburdenwith, saturation due to rainfall.
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	24(L261)
2	State	JK
3	District	Kishtwar
4	Toposheet	43O/16
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 13' 33.5208" N
8	Longitude	75° 47' 27.5352" E
9	Length	40m
10	Width	50m
11	Height	30m
12	Area	2000sq m
13	Depth	2m
14	Volume	4000cu m
15	Run out distance	15m
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Enlarging
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	gneiss
26	Structure	Jointed
27	Land use/ Land cover	barren

28	Hydrological condition	dry
29	Triggering Factor	Rainfall/cut slope
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Bare land	2000 ^{m2} area
36	Geo-scientific Causes	Wedge failure and planar failure
37	Remedial measures	Retaining wall for toe support, drapery wire mesh
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to Wedge failure and planar failure
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	25(L263)
2	State	JK
3	District	kishtwar
4	Toposheet	43O/16
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 13' 43.6548" N
8	Longitude	75° 47' 20.292" E
9	Length	20m
10	Width	50m
11	Height	15m
12	Area	1000sq m
13	Depth	1m
14	Volume	1000cu m
15	Run out distance	15m
16	Type of Material	Rock
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	confined
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	gneiss
26	Structure	Jointed
27	Land use/ Land cover	moderate
28	Hydrological condition	dry

29	Triggering Factor	Rainfall/cut slope
30	Death of persons	Nil
31	People affected	Nil
32	Livestock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1000 ^{m2} area
36	Geo-scientific Causes	Planar failure along J1 plane
37	Remedial measures	Retaining wall for toe support, drapery wire mesh
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to planar failure along J1 plane
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	26(L295)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 9' 14.0364" N
8	Longitude	75° 40' 55.3188" E
9	Length	20m
10	Width	50m
11	Height	18m
12	Area	1000sq m
13	Depth	1m
14	Volume	1000cu m
15	Run out distance	10
16	Type of Material	Rock
17	Type of movement	Slide/fall
18	Rate of movement	fast
19	Activity	Active
20	Distribution	Retrogressive
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	Schist
26	Structure	
27	Land use/ Land cover	Moderate vegetation
28	Hydrological condition	dry
29	Triggering Factor	Rainfall
30	Death of persons	Nil

31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	1000 ^{m2} area
36	Geo-scientific Causes	Toppling failure and planar failure along J1 plane
37	Remedial measures	Retaining wall for toe support, drapery wire mesh
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	The slide is happened due to planar failure along J1 plane
41	Pdf	
42	Alert	I / II / III

42-point detailed Geo-parametric attributes of landslide incidences

No	Field	Description
1	Slide No (LS .No.)	27(L2)
2	State	JK
3	District	Doda
4	Toposheet	43O/12
5	Name of the slide	
6	NH/SH/Locality	NH-244
7	Latitude	33° 8' 4.1028" N
8	Longitude	75° 32' 6.0072" E
9	Length	70m
10	Width	50m
11	Height	60m
12	Area	3500sq m
13	Depth	1m
14	Volume	3500cu m
15	Run out distance	55
16	Type of Material	Debris
17	Type of movement	Slide
18	Rate of movement	fast
19	Activity	Active
20	Distribution	enlarging
21	Style	Single
22	Failure mechanism	Translational Shallow
23	History	
24	Geomorphology	Straight slope
25	Geology	Quaternary deposit
26	Structure	Jointed
27	Land use/ Land cover	Sparse vegetation

28	Hydrological condition	dry
29	Triggering Factor	Rainfall
30	Death of persons	Nil
31	People affected	Nil
32	Live stock loss	Nil
33	Communication	NH-244
34	Infrastructure	Nil
35	Agriculture/Forest/Barren	3500 ^{m2} area
36	Geo-scientific Causes	Low cohesion due to the presence of boulder with limited matrix
37	Remedial measures	Retaining wall for toe support, drapery wire mesh
38	Remarks, if any	Road is vulnerable
39	Photos. Sketch of Plan & section of the slide	
40	Summary/Abstract	Low cohesion due to the presence of boulder with limited matrix
41	Pdf	
42	Alert	I

Annexure VI: GT parameters of Undisturbed soil samples.

Sl. No	Lab No.	Field No.	Sieve opening mm (% finer)										Specific Gravity
			Classification of Soil										
			Gravel				Coarse Sand	Medium Sand		Fine Sand		Silt+ Clay	
			12.5	10.0	6.3	4.75	2.36	1.18	0.60	0.30	0.15	0.075	
1	9922-S	DL/SS/01	100	100	100	100	99.78	99.57	99.48	99.14	99.19	98.58	2.71
2	9923-S	DL/SS/02	96.64	96.64	95.26	94.93	93.20	90.32	84.14	74.42	51.71	43.66	2.59
3	9924-S	DL/SS/03	100	100	100	98.74	96.19	92.63	85.92	68.76	54.68	44.79	2.69
4	9925-S	DL/SS/04	100	100	100	99.12	98.75	98.24	97.65	96.73	95.72	94.00	2.68
5	9926-S	DL/SS/05	100	98.27	97.74	97.58	97.32	97.03	96.56	95.17	88.39	84.03	2.67
6	9927-S	DL/SS/06	100	100	100	98.94	98.67	98.41	97.89	97.45	95.53	93.96	2.51
7	9928-S	DL/SS/07	100	100	100	99.82	99.02	98.49	97.61	96.60	95.17	93.44	2.51
8	9929-S	DL/SS/08	100	100	99.61	98.91	98.24	97.71	96.66	93.21	81.56	72.49	2.60
9	9930-S	DL/SS/09	100	98.46	95.32	93.68	92.11	90.85	89.53	86.97	81.82	78.04	2.52
10	9931-S	DL/SS/10	100	98.91	97.21	96.93	96.61	96.39	96.01	95.30	89.07	88.24	2.60

Annexure VI: GT parameters of Undisturbed soil samples(continue)

Lab No.	Moisture Content %	Wet Density Gms/cc	Dry Density Gms/cc	Porosity %	Void Ratio	Atterberg's Limit					Direct Shear Test		Permeability Test Cm/sec
						Liquid Limit %	Plastic Limit %	Shrinkage Limit %	Shrinkage Ratio	Plastic Index	'C' (Kpa)	'Ø' degree	
9922-S	11.02	1.71	1.54	0.43	0.76	29.66	19.78	15.84	1.87	9.88	25.00	35.6	Impermeable
9923-S	2.74	1.33	1.29	0.50	1.01	Non- Plastic Soil					NP	38.8	0.0000273
9924-S	2.92	1.49	1.45	0.46	0.86	Non- Plastic Soil					NP	39.5	0.0000149
9925-S	4.83	1.39	1.33	0.50	1.02	28.99	20.90	16.94	1.83	8.09	22.5	36.7	0.00000523
9926-S	8.10	1.34	1.24	0.54	1.15	24.70	17.33	14.67	1.83	7.37	20.0	37.4	0.00000817
9927-S	5.64	1.24	1.17	0.53	1.15	33.20	26.68	21.38	1.63	6.52	17.50	36.0	0.00000369
9928-S	4.79	1.31	1.25	0.50	1.01	25.74	18.41	16.15	1.81	7.33	20.00	37.0	0.00000426
9929-S	8.79	1.31	1.20	0.54	1.17	Non- Plastic Soil					NP	39.1	0.00000833
9930-S	5.30	1.46	1.39	0.45	0.81	26.06	18.52	15.71	1.80	7.54	20.00	37.2	0.00000539
9931-S	8.64	1.39	1.28	0.51	1.03	30.3	19.60	16.38	1.80	10.70	35.00	37.4	Impermeable

Annexure III: Rock Mass Rating (RMR) of rocky slopes (Bieniawski, 1979)											
	Sample No.	DL/RS/01	DL/RS/02	DL/RS/03	DL/RS/04	DL/RS/05	DL/RS/06	DL/RS/07	DL/RS/08	DL/RS/09	DL/RS/10
	Latitude	33.26312	33.23005	33.200797	33.164832	33.14661	33.145845	33.138787	33.151578	33.155957	33.140167
	Longitude	75.773583	75.787977	75.79871	75.803473	75.798697	75.783833	75.75647	75.74831	75.701807	75.558975
	Jv	10	25	10	15	10	20	15	15	20	20
UCS	Kg/cm2	214	367	326	352	321	240	357	336	163	336
	Mpa	21	36	32	34.5	31.5	23.5	35	33	16	33
	Rat.	2	4	4	4	4	2	4	4	2	4
RQD	Value	82	-3.8	-3.8	-3.8	-3.8	49	49	65.5	49	49
	Rat.	5	5	5	5	5	8	8	23	8	8
Spacing (mm)	J1	Value	<60	200-600	>2m	200-600	200-600	60-200	<60	60-200	<60
		Rat.	5	10	20	10	10	8	5	8	5
	J2	Value	200-600	200-600	0.6-2m	0.6-2m	0.6-2m	0.6-2m	200-600	200-600	200-600
		Rat.	10	10	15	15	15	15	10	10	8
	J3	Value	200-600	0.6-2m	>2m	>2m	>2m	0.6-2m	0.6-2m	200-600	200-600
		Rat.	10	15	20	20	20	15	15	10	10
	Mean Rating		8.33	11.67	18.33	15.00	15.00	12.67	10.00	9.33	8.33
Persistence (m)	J1	Value	3-10	3-10	3-10	10-20	10-20	3-10	10-20	10-20	3-10
		Rat.	2	2	2	1	1	2	1	1	2
	J2	Value	1-3	1-3	1-3	3-10	3-10	1-3	1-3	1-3	1-3
		Rat.	4	4	4	2	2	4	4	4	4
	J3	Value	1-3	1-3	1-3	3-10	3-10	1-3	1-3	1-3	1-3
		Rat.	4	4	4	2	2	4	4	4	4
	Mean Rating		3.3	3.3	3.3	1.7	1.7	3.3	3.0	3.0	3.3
Aperture / Separation (mm)	J1	Value	1-5	>5	None	>5	>5	>5	>5	None	1-5
		Rat.	1	0	6	0	0	0	0	6	1
	J2	Value	1-5	>5	>5	None	None	1-5	None	None	1-5
		Rat.	1	0	0	6	6	1	6	6	1
	J3	Value	1-5	1-5	None	>5	>5	>5	None	None	None
		Rat.	1	1	6	0	0	0	6	6	6
	Mean Rating		1.0	0.3	4.0	2.0	2.0	0.3	4.0	4.0	2.7

Annexure III: Rock Mass Rating (RMR) of rocky slopes (Bieniawski, 1979)													
	Sample No.		DL/RS/01	DL/RS/02	DL/RS/03	DL/RS/04	DL/RS/05	DL/RS/06	DL/RS/07	DL/RS/08	DL/RS/09	DL/RS/10	
Roughness	J1	Value	Rough	Rough	Sl Rough	Rough	Rough	Sl Rough	Sl Rough	Rough	Sl Rough	Rough	
		Rat.	5	5	5	5	5	3	3	5	3	5	
	J2	Value	Rough	Rough	Rough	Rough	Rough	Rough	Rough	Rough	Rough	Sl Rough	Rough
		Rat.	5	5	5	5	5	5	5	5	5	3	5
	J3	Value	Rough	Rough	Rough	Smooth	Smooth	Rough	Rough	Rough	Rough	Rough	Rough
		Rat.	5	5	5	1	1	5	5	5	5	5	5
	Mean Rating		5.0	5.0	5.0	3.7	3.7	4.3	4.3	5.0	3.7	5.0	
Infilling	J1	Value	S Clay	S Clay	None	S Clay	S Clay	S Clay	S Clay	Calcite	None	S Clay	
		Rat.	2	2	6	2	2	2	2	4	6	2	
	J2	Value	S Clay	S Clay	S Clay	None	None	S Clay	None	None	S Clay	S Clay	
		Rat.	2	2	2	6	6	2	6	6	2	2	
	J3	Value	S Clay	S Clay	None	S Clay	S Clay	S Clay	None	None	None	None	
		Rat.	2	2	6	2	2	2	6	6	6	6	
	Mean Rating		2.0	2.0	4.7	3.3	3.3	2.0	4.7	5.3	4.7	3.3	
Weathering	Value	Moderate	Moderate	Slightly	Slightly	Slightly	Slightly	Slightly	Slightly	Moderate	Moderate	Slightly	
	Rat.	3	3	5	5	5	5	5	5	3	3	5	
Hydrogeological Condition	Value	Dry	Dry	Dry	Dry	Dry	Dry	Dry	Dry	Dry	Dry	Dry	
	Rat.	15	15	15	15	15	15	15	15	15	15	15	
RMRb	Σ Rat		44.67	49.33	64.33	54.67	54.67	52.67	58.00	71.67	52.33	54.00	
RMR Class			III	III	II	III	III	III	III	II	III	III	
RMR Name			Fair	Fair	Good	Fair	Fair	Fair	Fair	Good	Fair	Fair	

**Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes
based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)**

	Location No.		2	4	6A	7	8	9
	Latitude		33.144183	33.154086	33.280441	33.265113	33.264356	33.26888
	Longitude		75.661317	75.681106	75.767318	75.772847	75.773128	75.773672
		Jv	10	15	7	9	10	10
UCS		Value	20	35	75	5	5	45
		Rat.	2	4	7	1	1	4
RQD		Value	82	65.5	91.9	85.3	82	82
		Rat.	17	13	20	17	17	18
Spacing (mm)	J1	Value	<60	<60	200-600	60-200	<60	60-200
		Rat.	5	5	10	8	5	8
	J2	Value	60-200	600-2000	200-600	60-200	60-200	60-200
		Rat.	8	15	10	8	8	8
	J3	Value	60-200	200-600	>2m	0.6-2m	0.6-2m	200-600
		Rat.	8	10	20	15	15	10
	J4	Value						
		Rat.						
	Mean Rating		7	10.00	13.3	10.333333	9.333333	8.666667
Persisten ce (m)	J1	Value	10-20	3-10	3-10	3-10	3-10	3-10
		Rat.	1	2	2	2	2	2
	J2	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	2	4
	J3	Value	1-3	1-3	1-3	3-10	3-10	3-10
		Rat.	1	4	4	2	2	2
	J4	Value						
		Rat.						
	Mean Rating		2	2.5	3.3	2.666667	2	2.7
Aperture / Separatio n (mm)	J1	Value	>5	>5	>5	1-5	None	1-5
		Rat.	0	0	0	1	6	1
	J2	Value	>5	1-5	1-5	0	>5	1-5
		Rat.	0	1	1	0	0	1
	J3	Value	<1	1-5	1-5	>5	1-5	1-5
		Rat.	4	1	1	0	1	1
	J4	Value						
		Rat.						
	Mean Rating		1.333333	0.666667	0.67	0.333333	2.333333	1.00
Roughnes s	J1	Value	V Rough	Sl Rough	Sl Rough	V Rough	Smooth	Rough
		Rat.	6	3	3	6	1	5
	J2	Value	V Rough	Rough	Rough	V Rough	Rough	Rough
		Rat.	4	5	5	6	5	5
	J3	Value	Sl Rough	Rough	Rough	Rough	Rough	Rough
		Rat.	3	5	5	5	5	5
	J4	Value						
		Rat.						

	Mean Rating		4.3333333	4.3333333	4.3333333	5.6666667	3.6666667	5
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	None	S Clay
		Rat.	2	0	0	2	6	2
	J2	Value	S Clay	S Clay	S Clay	None	S Clay	S Clay
		Rat.	0	2	2	6	0	2
	J3	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J4	Value						
		Rat.						
	Mean Rating		1.3333333	1.3333333	1.3	3.3333333	2.6666667	2
Weathering		Value	Slight	Slight	Slight	Moderate	Slight	Moderate
		Rat.	5	5	5	4	5	3
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Damp	Wet
		Rat.	15	15	15	15	10	7
RMRb			55	55.8	70.0	59.333333	53	51.333333
RMR Class			III	III	II	III	III	III
RMR Class			FAIR	FAIR	GOOD	FAIR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008		F1	0.40	0.20	0.20	0.20	0.20	0.20
		F2	0.50	0.50	0.50	0.50	0.50	0.50
		F3	0.80	1.00	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			55.2	55.9	70.1	59.4	53.1	51.4
SMR Class			III	III	III	III	III	III
SMR Name			Normal	Normal	Normal	Normal	Normal	Normal
Stability			Partially Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was calculated. The RMR varies from Poor to Good class. Most of the area falls in categories.

**Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes
based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)**

	Location No.		10	11	17	18	19	20
	Latitude		33.260787	33.258659	33.253358	33.25224	33.250357	33.248048
	Longitude		75.773958	75.774086	75.775706	75.776247	75.777549	75.77926
		Jv	10	15	14	8	15	15
UCS		Value	45	35	5	35	35	30
		Rat.	4	4	1	4	4	8
RQD		Value	82	65.5	68.8	88.6	65.5	65.5
		Rat.	18	13	13	17	13	13
Spacing (mm)	J1	Value	60-200	<60	<60	<60	<60	<60
		Rat.	8	5	5	5	5	5
	J2	Value	60-200	60-200	60-200	>2m	60-200	>2
		Rat.	8	8	8	20	8	10
	J3	Value	200-600	0.6-2m	200-600		60-200	200-600
		Rat.	10	15	10		8	10
	J4	Value						>2m
		Rat.						20
	Mean Rating		8.6666667	9.3333333	7.7	12.5	7	11.25
Persisten ce (m)	J1	Value	3-10	3-10	<1	10-20	10-20	1-3
		Rat.	2	2	6	1	1	4
	J2	Value	1-3	1-3	1-3	3-10	3-10	1-3
		Rat.	4	4	4	2	2	4
	J3	Value	3-10	1-3	3-10		1-3	1-3
		Rat.	2	4	2		4	4
	J4	Value						3-10
		Rat.						2
	Mean Rating		2.7	3.3333333	4	1.5	2.3	3.5
Aperture / Separatio n (mm)	J1	Value	1-5	0.1	1-5	None	>5	>5
		Rat.	1	1	1	6	0	0
	J2	Value	1-5	1-5	>5	>5	>5	>5
		Rat.	1	1	0	0	0	0
	J3	Value	1-5	None	>5		>5	>5
		Rat.	1	6	0		0	0
	J4	Value						1-5
		Rat.						1
	Mean Rating		1.00	2.6666667	0.33	3	0.0	0.3
Roughnes s	J1	Value	Rough	V Rough	Smooth	Sl Rough	Sl Rough	Smooth
		Rat.	5	6	1	3	3	1
	J2	Value	Rough	V Rough	V Rough	Rough	Sl Rough	Sl rough
		Rat.	5	6	6	5	3	3
	J3	Value	Rough	V Rough	V Rough		Rough	Rough
		Rat.	5	6	6		5	5
	J4	Value						V Rough
		Rat.						6

	Mean Rating		5	6.0	4.3	4	3.7	3.8			
Infilling	J1	Value	S Clay	S Clay	S Clay	None	S Clay	S Clay			
		Rat.	2	0	2	6	0	2			
	J2	Value	S Clay	S Clay	S Clay	S Clay	None	S Clay			
		Rat.	2	0	2	2	6	2			
	J3	Value	S Clay	None	S Clay		S Clay	S Clay			
		Rat.	2	6	2		0	2			
	J4	Value						S Clay			
		Rat.						2			
Mean Rating		2	2	2	4	2.0	2.0				
Weathering		Value	Moderate	Moderate	Moderate	Slight	Moderate				
		Rat.	3	3	3	5	3	3			
Groundwater Condition		Value	Wet	Flowing	Dry	Dry	Dry	Dry			
		Rat.	7	0	15	15	15	15			
RMRb			51.333333	43.3	50.3	66	50	59.75			
RMR Class			III	III	III	II	III	III			
RMR Class			FAIR	FAIR	FAIR	GOOD	FAIR	FAIR			
Adjustment: Anbalgan et al. 2008		F1	0.20	0.4	0.20	0.50	0.50	0.20			
		F2		0.5					0.50	0.50	0.50
		F3		1.00					1	1.00	1.00
		F4	0	0	0	0	0	0			
SMR			51.4	43.5	50.4	66.3	50.2	59.8			
SMR Class			III	III	III	II	III	III			
SMR Name			Normal	Normal	Normal	Good	Normal	Normal			
Stability			Partially Stable	Partially Stable	Partially Stable	Stable	Partially Stable	Partially Stable			

Field based RMR was into Fair category, some falls in Poor and Good

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		21	30	36	37	39	41
	Latitude		33.247208	33.133601	33.136723	33.137282	33.139351	33.141353
	Longitude		75.77983	75.541399	75.553572	75.556153	75.55834	75.561288
		Jv	25	30	20	26	20	20
UCS		Value	75	35	35	35	75	75
		Rat.	17	4	4	4	7	7
RQD		Value	32.5	16	49	29.2	49	49
		Rat.	8	5	8	8	8	8
Spacing (mm)	J1	Value	60-200	<60	<60	<60	<60	<60
		Rat.	8	5	5	5	5	5
	J2	Value	60-200	60-200	<60	<60	60-200	60-200
		Rat.	8	8	5	5	8	8
	J3	Value	60-200	200-600	200-600	200-600	200-600	60-200
		Rat.	8	10	10	10	10	8
	J4	Value	200-600					
		Rat.	10					
	Mean Rating		8.50	7.67	6.67	6.67	7.67	7.00
Persisten ce (m)	J1	Value	3-10	3-10	3-10	3-10	3-10	10-20
		Rat.	2	2	2	2	2	1
	J2	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J3	Value	3-10	1-3	1-3	1-3	1-3	1-3
		Rat.	2	4	4	4	4	4
	J4	Value	03-Jan					
		Rat.	4					
	Mean Rating		3.0	3.3	3.3	3.3	3.3	3.0
Aperture / Separatio n (mm)	J1	Value	None	1-5	1-5	1-5	1-5	>5
		Rat.	6	1	1	1	1	0
	J2	Value	None	>5	None	1-5	1-2	>5
		Rat.	6	0	6	1	1	0
	J3	Value	1-5	>5	>5	1-5	None	None
		Rat.	1	0	0	1	6	6
	J4	Value	>5					
		Rat.	0					
	Mean Rating		3.3	0.3	2.3	1.0	2.7	2.0
Roughnes s	J1	Value	Rough	Rough	Smooth	Sl Rough	Rough	Sl Rough
		Rat.	1	5	1	3	5	3
	J2	Value	V Rough	Rough	Rough	Rough	Rough	Sl Rough
		Rat.	6	5	5	5	5	3
	J3	Value	Rough	Rough	Sl Rough	Rough	Rough	V Rough
		Rat.	5	5	3	5	5	6
	J4	Value	Rough					
		Rat.	5					

		Mean Rating	4.3	5.0	3.0	4.3	5.0	4.0
Infilling	J1	Value	None	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	6	2	2	2	2	2
	J2	Value	None	S Clay	None	S Clay	S Clay	S Clay
		Rat.	6	2	6	2	2	2
	J3	Value	S Clay	S Clay	S Clay	S Clay	None	None
		Rat.	2	2	2	2	6	6
	J4	Value	S Clay					
		Rat.	2					
		Mean Rating	4.0	2.0	3.3	2.0	3.3	3.3
Weathering		Value	Slight	Slight	Moderate	Slight	Moderate	Slight
		Rat.	5	3	5	3	5	5
Groundwater Condition		Value	Dry	Damp	Dry	Dry	Dry	Dry
		Rat.	15	10	15	15	15	15
RMRb			68.00	40.33	50.67	47.33	57.00	54.33
RMR Class			III	III	III	III	III	III
RMR Class			GOOD	FAIR	FAIR	FAIR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.40	0.25	0.50	0.25	0.50	0.25
	F2		0.50	0.50	0.50	0.50	0.50	0.50
	F3		1.00	0.50	0.80	0.80	0.50	0.50
		F4	0	0	0	0	0	0
SMR			68.2	40.4	50.9	47.4	57.1	54.4
SMR Class			II	III	III	III	III	III
SMR Name			Good	Normal	Normal	Normal	Normal	Normal
Stability			Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was c

**Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes
based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)**

	Location No.		42	43	45	55	57	63
	Latitude		33.14234	33.143387	33.144455	33.135982	33.136076	33.132858
	Longitude		75.56294	75.56434	75.567036	75.583426	75.589186	75.600713
		Jv	8	15	10	15	15	15
UCS		Value	35	35	35	35	35	35
		Rat.	4	4	4	4	4	4
RQD		Value	88.6	65.5	82	65.5	65.5	65.5
		Rat.	17	13	17	13	13	13
Spacing (mm)	J1	Value	<60	60-200	<60	<60	<60	60-200
		Rat.	5	8	5	5	5	8
	J2	Value	>2m	200-600	60-200	200-600	0.6-2m	200-600
		Rat.	20	10	8	10	15	10
	J3	Value	60-200	200-600	200-600	>2m	200-600	>2m
		Rat.	8	10	10	20	10	20
	J4	Value						
		Rat.						
	Mean Rating		11.00	9.33	7.67	11.67	10.00	12.67
Persisten ce (m)	J1	Value	10-20	10-20	3-10	3-10	3-10	3-10
		Rat.	1	1	2	2	2	2
	J2	Value	1-3	3-10	1-3	3-10	3-10	1-3
		Rat.	4	2	4	2	2	4
	J3	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J4	Value						
		Rat.						
	Mean Rating		3.0	2.3	3.3	2.7	2.7	3.3
Aperture / Separatio n (mm)	J1	Value	>5	1-5	>5	>5	None	>5
		Rat.	0	1	0	0	6	0
	J2	Value	None	>5	1-5	>5	>5	1-5
		Rat.	6	0	1	0	0	1
	J3	Value	1-5	None	None	None	1-5	1-5
		Rat.	1	6	6	6	1	1
	J4	Value						
		Rat.						
	Mean Rating		2.3	2.3	2.3	2.0	2.3	0.7
Roughnes s	J1	Value	SI Rough	SI Rough	Rough	Rough	Rough	Rough
		Rat.	3	3	5	5	5	5
	J2	Value	Rough	Rough	Rough	Rough	Rough	V Rough
		Rat.	5	5	5	5	5	6
	J3	Value	Rough	V Rough	Rough	Rough	V Rough	SI Rough
		Rat.	5	6	5	5	6	3
	J4	Value						
		Rat.						

		Mean Rating	4.3	4.7	5.0	5.0	5.3	4.7
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	None	S Clay
		Rat.	2	2	2	2	6	2
	J2	Value	None	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	6	2	2	2	2	2
	J3	Value	S Clay	None	None	None	S Clay	S Clay
		Rat.	2	6	6	6	2	2
	J4	Value						
		Rat.						
		Mean Rating	3.3	3.3	3.3	3.3	3.3	2.0
Weathering		Value	Slight	Slight	Slight	Moderate	Slight	Slight
		Rat.	5	5	3	5	5	3
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			65.00	59.00	60.67	61.67	60.67	58.33
RMR Class			II	III	II	II	III	III
RMR Class			GOOD	FAIR	GOOD	GOOD	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.30	0.30	0.25	0.20	0.20	0.50
	F2		0.50	0.50	0.50	0.50	0.40	0.40
	F3		0.50	0.50	1.00	0.30	1.00	1.00
		F4	0	0	0	0	0	0
SMR			65.1	59.1	60.8	61.7	60.7	58.5
SMR Class			II	III	II	II	II	III
SMR Name			Good	Normal	Good	Good	Good	Normal
Stability			Stable	Partially Stable	Stable	Stable	Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		101	102	103	104	105	108
	Latitude		33.159566	33.159348	33.157771	33.157206	33.155529	33.153402
	Longitude		75.696095	75.697988	75.700157	75.70045	75.702246	75.710224
		Jv	12	30	25	26	20	20
UCS		Value	75	35	5	35	25	35
		Rat.	7	4	2	4	4	4
RQD		Value	75.4	16	32.5	29.2	49	49
		Rat.	17	5	8	8	8	8
Spacing (mm)	J1	Value	200-600	60-200	<60	<60	<60	60-200
		Rat.	10	8	5	5	5	8
	J2	Value	200-600	60-200	0.6-2	200-600	200-600	200-600
		Rat.	10	8	15	10	10	10
	J3	Value	0.6-2m	60-200	>2m	0.6-2m	200-600	200-600
		Rat.	15	8	20	15	10	10
	J4	Value			>2m	0.6-2m		
		Rat.			20	15		
	Mean Rating		11.67	8.00	15.00	11.25	8.33	9.33
Persisten ce (m)	J1	Value	10-20	<1	10-20	10-20	3-10	3-10
		Rat.	1	6	1	1	2	2
	J2	Value	3-10	1-3	1-3	3-10	1-3	1-3
		Rat.	2	4	4	2	4	4
	J3	Value	1-3	<1	1-3	3-10	1-3	1-3
		Rat.	4	6	4	2	4	4
	J4	Value			3-10	3-10		
		Rat.			2	2		
	Mean Rating		2.3	5.3	2.8	1.8	3.3	3.3
Aperture / Separatio n (mm)	J1	Value	>5	1-5	>5	>5	None	><5
		Rat.	0	1	0	0	6	0
	J2	Value	1-5	1-5	None	None	1-5	1-5
		Rat.	1	1	6	6	1	1
	J3	Value	1-5	>5	None	None	1-5	None
		Rat.	1	0	6	6	1	6
	J4	Value			>5	>5		
		Rat.			0	0		
	Mean Rating		0.7	0.7	3.0	3.0	2.7	2.3
Roughnes s	J1	Value	Rough	SI Rough	Rough	SI Rough	SI Rough	Rough
		Rat.	5	3	5	3	3	5
	J2	Value	SI Rough	V Rough	Rough	Smooth	SI Rough	SI Rough
		Rat.	3	6	5	1	3	3
	J3	Value	Smooth	Rough	Rough	Rough	Rough	V Rough
		Rat.	1	5	5	5	5	6
	J4	Value			Rough	Rough		
		Rat.			5	5		

		Mean Rating	3.0	4.7	5.0	3.5	3.7	4.7
Infilling	J1	Value	S Clay	S Clay	Hard calcite	Hard filling	None	S Clay
		Rat.	2	2	2	2	6	2
	J2	Value	S Clay	None	None	None	S Clay	S Clay
		Rat.	2	6	6	6	2	2
	J3	Value	S Clay	None	None	None	None	None
		Rat.	2	6	6	6	6	6
	J4	Value			S Clay	S Clay		
		Rat.			2	2		
		Mean Rating	2.0	4.7	4.0	4.0	4.7	3.3
Weathering		Value	Moderate	Slight	Slight	Moderate	Moderate	Moderate
		Rat.	5	5	3	3	3	3
Groundwater Condition		Value	Damp	Dry	Damp	Dry	Damp	Damp
		Rat.	10	15	10	15	10	10
RMRb			58.67	52.33	52.75	53.50	47.67	48.00
RMR Class			III	III	III	III	III	III
RMR Class			FAIR	FAIR	FAIR	FAIR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.20	0.20	0.30	0.20	0.50	0.20
	F2		0.40	0.50	0.40	0.50	0.50	0.50
	F3		1.00	0.50	1.00	1.00	0.80	1.00
	F4		0	0	0	0	0	0
SMR			58.7	52.4	52.9	53.6	47.9	48.1
SMR Class			III	III	III	III	III	III
SMR Name			Normal	Normal	Normal	Normal	Normal	Normal
Stability			Partially Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		109	113	114	121	131	132
	Latitude		33.152577	33.114917	33.148959	33.152158	33.150923	33.14903
	Longitude		75.711449	75.716763	75.71873	75.730446	75.758523	75.749322
		Jv	20	10	22	15	25	25
UCS		Value	40	35	100	35	75	40
		Rat.	4	4	12	4	7	4
RQD		Value	49	82	42.4	65.5	32.5	32.5
		Rat.	8	17	8	13	8	8
Spacing (mm)	J1	Value	<60	60-200	200-600	<60	60-200	<60
		Rat.	5	8	10	5	8	5
	J2	Value	200-600	200-600	60-200	60-200	200-600	60-200
		Rat.	10	10	8	8	10	8
	J3	Value	200-600	0.6-2m	200-600	>2m	200-600	200-600
		Rat.	10	15	10	20	10	10
	J4	Value	>2m		>2m			
		Rat.	20		20			
	Mean Rating		11.25	11.00	12.00	11.00	9.33	7.67
Persistence (m)	J1	Value	3-10	3-10	3-10	3-10	10-20	3-10
		Rat.	2	2	2	2	1	2
	J2	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J3	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J4	Value	1-3		1-3			
		Rat.	4		4			
	Mean Rating		3.5	2.5	3.5	3.3	3.0	3.3
Aperture / Separation (mm)	J1	Value	>5	1-5	>5	>5	>5	1-5
		Rat.	0	1	0	0	0	1
	J2	Value	1-5	None	1-5	None	None	None
		Rat.	1	6	1	6	6	6
	J3	Value	None	None	>5	None	None	None
		Rat.	6	6	0	6	6	6
	J4	Value	1-5		1-5			
		Rat.	1		1			
	Mean Rating		2.0	4.3	0.5	4.0	4.0	4.3
Roughness	J1	Value	Sl Rough	Rough	Smooth	Sl Rough	Rough	Sl Rough
		Rat.	3	5	1	3	5	3
	J2	Value	Rough	Rough	Rough	Rough	Rough	Rough
		Rat.	5	5	5	5	5	5
	J3	Value	Sl Rough	V Rough	Rough	V Rough	Rough	V Rough
		Rat.	3	6	5	6	6	6
	J4	Value	Smooth		Silken side			
		Rat.	1		0			

		Mean Rating	3.0	5.3	2.8	4.7	5.3	4.7
Infilling	J1	Value	S Clay	S Clay	Hard fill	S Clay	Hard fill	S Clay
		Rat.	2	2	4	2	4	2
	J2	Value	S Clay	None	S Clay	None	None	None
		Rat.	2	6	2	6	6	6
	J3	Value	None	None	None	None	None	None
		Rat.	6	6	6	6	6	6
	J4	Value	S Clay		None			
		Rat.	2		6			
		Mean Rating	3.0	4.7	4.5	4.7	5.3	4.7
Weathering		Value	Moderate	Slight	Moderate	Slight	Slight	Moderate
		Rat.	5	3	5	5	3	3
Groundwater Condition		Value	Damp	Damp	Damp	Damp	Dry	Dry
		Rat.	10	10	10	10	15	15
RMRb			49.75	61.83	58.25	59.67	60.00	54.67
RMR Class			III	II	III	III	III	III
RMR Class			FAIR	GOOD	FAIR	FAIR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.20	0.20	0.25	0.30	0.20	0.20
	F2		0.50	0.50	0.50	0.50	0.50	0.50
	F3		1.00	1.00	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			49.9	61.9	58.4	59.8	60.1	54.8
SMR Class			III	II	III	III	II	III
SMR Name			Normal	Good	Normal	Normal	Good	Normal
Stability			Partially Stable	Stable	Partially Stable	Partially Stable	Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		138	139	140	141	142	153
	Latitude		33.141364	33.140342	33.139797	33.139309	33.138819	33.144899
	Longitude		75.752038	75.753379	75.754368	75.7552	75.756441	75.771077
		Jv	10	25	20	22	15	30
UCS		Value	40	40	35	40	35	35
		Rat.	4	4	4	4	4	4
RQD		Value	82	32.5	49	42.4	65.5	16
		Rat.	17	8	8	8	13	5
Spacing (mm)	J1	Value	60-200	<60	<60	<60	<60	<60
		Rat.	8	5	5	5	5	5
	J2	Value	60-200	60-200	60-200	60-200	200-600	200-600
		Rat.	8	8	8	8	10	10
	J3	Value	>2m	>2m	0.6-2m	200-600	0.6-2m	200-600
		Rat.	20	20	15	10	15	10
	J4	Value						
		Rat.						
	Mean Rating		12.00	11.00	9.33	7.67	10.00	8.33
Persistence (m)	J1	Value	3-10	3-10	10-20	10-20	10-20	10-20
		Rat.	2	2	1	1	1	1
	J2	Value	1-3	3-10	1-3	1-3	1-3	1-3
		Rat.	4	2	4	4	4	4
	J3	Value	1-3	3-10	1-3	1-3	1-3	1-3
		Rat.	4	2	4	4	4	4
	J4	Value						
		Rat.						
	Mean Rating		3.3	2.0	3.0	3.0	3.0	3.0
Aperture / Separation (mm)	J1	Value	1-5	>5	>5	>5	1-5	>5
		Rat.	1	0	0	0	1	0
	J2	Value	>5	1-5	None	1-5	None	>5
		Rat.	0	1	6	1	6	0
	J3	Value	None	>5	1-5	None	None	None
		Rat.	6	0	1	6	6	6
	J4	Value						
		Rat.						
	Mean Rating		2.3	0.3	2.3	2.3	4.3	2.0
Roughness	J1	Value	Rough	Sl Rough	Sl Rough	Sl Rough	Sl Rough	Sl Rough
		Rat.	5	3	3	3	3	3
	J2	Value	Rough	Rough	Rough	Rough	Rough	Rough
		Rat.	5	5	5	5	5	5
	J3	Value	Rough	V Rough	Sl Rough	Rough	Rough	Rough
		Rat.	5	6	3	5	5	5
	J4	Value						
		Rat.						

		Mean Rating	5.0	4.7	3.7	4.3	4.3	4.3
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J2	Value	S Clay	S Clay	None	None	None	S Clay
		Rat.	2	2	6	6	6	2
	J3	Value	None	S Clay	S Clay	None	None	None
		Rat.	6	2	2	6	6	6
	J4	Value						
		Rat.						
		Mean Rating	3.3	2.0	3.3	4.7	4.7	3.3
Weathering		Value	Moderate	Slight	Moderate	Slight	Slight	Slight
		Rat.	5	3	5	5	5	3
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			67.00	50.00	53.67	54.00	63.33	48.00
RMR Class			II	III	III	III	II	III
RMR Class			GOOD	FAIR	FAIR	FAIR	GOOD	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.25	0.30	0.50	0.30	0.25	0.20
	F2		0.50	0.50	0.50	0.50	0.50	0.50
	F3		1.00	1.00	0.80	0.80	0.50	0.70
		F4	0	0	0	0	0	0
SMR			67.1	50.2	53.9	54.1	63.4	48.1
SMR Class			II	III	III	III	II	III
SMR Name			Good	Normal	Normal	Normal	Good	Normal
Stability			Stable	Partially Stable	Partially Stable	Partially Stable	Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		154	156	157	161	162	165
	Latitude		33.145329	33.145793	33.145724	33.14584	33.145436	33.145775
	Longitude		75.772722	75.774793	75.77636	75.784152	75.786789	75.796005
		Jv	25	15	25	25	18	20
UCS		Value	35	35	35	35	35	75
		Rat.	4	4	4	4	4	7
RQD		Value	32.5	65.5	32.5	32.5	55.6	49
		Rat.	8	13	8	8	13	8
Spacing (mm)	J1	Value	<60	200-600	60-200	60-200	<60	<60
		Rat.	5	10	8	8	5	5
	J2	Value	0.6-2m	0.6-2m	200-600	0.6-2m	200-600	60-200
		Rat.	15	15	10	15	10	8
	J3	Value	>2m	0.6-2m	>2m	0.6-2m	0.6-2m	200-600
		Rat.	20	15	20	15	15	10
	J4	Value			200-600			
		Rat.			10			
	Mean Rating		13.33	13.33	12.00	12.67	10.00	7.67
Persisten ce (m)	J1	Value	10-20	3-10	3-10	3-10	3-10	3-10
		Rat.	1	2	2	2	2	2
	J2	Value	3-10	1-3	1-3	1-3	1-3	3-10
		Rat.	2	4	4	4	4	2
	J3	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J4	Value			1-3			
		Rat.			4			
	Mean Rating		2.3	2.5	3.5	3.3	3.3	2.7
Aperture / Separatio n (mm)	J1	Value	>5	>5	>5	>5	>5	1-5
		Rat.	0	0	0	0	0	1
	J2	Value	None	>5	None	1-5	None	>5
		Rat.	6	0	6	1	6	0
	J3	Value	None	None	>5	>5	1-5	1-5
		Rat.	6	6	0	0	1	1
	J4	Value			None			
		Rat.			6			
	Mean Rating		4.0	2.0	3.0	0.3	2.3	0.7
Roughnes s	J1	Value	Sl Rough	Sl Rough	Sl Rough	Sl Rough	Sl Rough	Smooth
		Rat.	3	3	3	3	3	1
	J2	Value	Sl Rough	Rough	Sl Rough	Rough	Rough	Sl Rough
		Rat.	3	5	3	5	5	3
	J3	Value	Rough	Rough	Sl Rough	Rough	Rough	Rough
		Rat.	5	5	3	5	5	5
	J4	Value			Rough			
		Rat.			5			

		Mean Rating	3.7	4.3	3.5	4.3	4.3	3.0
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J2	Value	None	S Clay	None	None	None	S Clay
		Rat.	6	2	6	6	6	2
	J3	Value	None	None	S Clay	S Clay	S Clay	S Clay
		Rat.	6	6	2	2	2	2
	J4	Value			None			
		Rat.			6			
		Mean Rating	4.7	3.3	4.0	3.3	3.3	2.0
Weathering		Value	Moderate	Slight	Moderate	Slight	Slight	Moderate
		Rat.	5	3	5	5	3	5
Groundwater Condition		Value	Dry	Dry	Damp	Dry	Damp	Damp
		Rat.	15	15	10	15	10	10
RMRb			60.00	60.50	53.00	56.00	53.33	46.00
RMR Class			III	II	III	IV	III	III
RMR Class			FAIR	GOOD	FAIR	POOR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.20	0.50	0.20	0.20	0.20	0.40
	F2		0.40	0.50	0.50	0.50	0.50	0.40
	F3		1.00	0.70	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			60.1	60.7	53.1	56.1	53.4	46.2
SMR Class			II	II	III	III	III	III
SMR Name			Good	Good	Normal	Normal	Normal	Normal
Stability			Stable	Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		167	168	169	170	171	173
	Latitude		33.146523	33.148597	33.149917	33.151113	33.152203	33.154328
	Longitude		75.798642	75.798573	75.798645	75.798456	75.798281	75.798177
		Jv	10	9	10	10	30	33
UCS		Value	75	75	35	35	10	35
		Rat.	7	7	4	4	2	4
RQD		Value	82	85.3	82	82	16	6.1
		Rat.	17	17	17	17	5	6
Spacing (mm)	J1	Value	60-200	200-600	<60	60-200	60-200	60-200
		Rat.	8	10	5	8	8	8
	J2	Value	200-600	0.6-2m	0.6-2m	200-600	60-200	200-600
		Rat.	10	15	15	10	8	10
	J3	Value	0.6-2m	200-600	>2m	>2m	200-600	200-600
		Rat.	15	10	20	20	10	10
	J4	Value	>2m	>2m				
		Rat.	20	20				
	Mean Rating		13.25	13.75	13.33	12.67	8.67	9.33
Persistence (m)	J1	Value	10-20	10-20	3-10	3-10	1-3	10-20
		Rat.	1	1	2	2	4	1
	J2	Value	1-3	1-3	3-10	1-3	1-3	3-10
		Rat.	4	4	2	4	4	2
	J3	Value	3-10	1-3	3-10	1-3	<1	1-3
		Rat.	2	4	2	4	6	4
	J4	Value	3-10	3-10				
		Rat.	2	2				
	Mean Rating		2.3	2.8	2.0	3.3	4.7	2.3
Aperture / Separation (mm)	J1	Value	>5	>5	>5	1-5	>5	>5
		Rat.	0	0	0	1	0	0
	J2	Value	None	None	None	None	1-5	None
		Rat.	6	6	6	6	1	6
	J3	Value	1-5	1-5	>5	None	None	1-5
		Rat.	1	1	0	6	6	1
	J4	Value	1-5	>5				
		Rat.	1	0				
	Mean Rating		2.0	1.8	2.0	4.3	2.3	2.3
Roughness	J1	Value	SI Rough	SI Rough	Rough	SI Rough	SI Rough	SI Rough
		Rat.	3	3	5	3	3	3
	J2	Value	Rough	Rough	SI Rough	Rough	Rough	Rough
		Rat.	5	5	3	5	5	5
	J3	Value	Rough	Rough	Smooth	Rough	Rough	Rough
		Rat.	5	5	1	5	5	5
	J4	Value	SI Rough	SI Rough				
		Rat.	3	3				

		Mean Rating	4.0	4.0	3.0	4.3	4.3	4.3
Infilling	J1	Value	Hard filling	Hard filling	S Clay	S Clay	S Clay	S Clay
		Rat.	4	4	2	2	2	2
	J2	Value	None	None	None	None	None	None
		Rat.	6	6	6	6	6	6
	J3	Value	S Clay	S Clay	S Clay	None	None	S Clay
		Rat.	2	2	2	6	6	2
	J4	Value	S Clay	S Clay				
		Rat.	2	2				
		Mean Rating	3.5	3.5	3.3	4.7	4.7	3.3
Weathering		Value	Slight	Moderate	Slight	Moderate	Moderate	Moderate
		Rat.	3	5	3	3	3	3
Groundwater Condition		Value	Dripping	Dry	Dry	Dry	Dry	Dry
		Rat.	4	15	15	15	15	15
RMRb			56.00	69.75	62.67	68.33	49.67	49.67
RMR Class			III	II	II	II	III	III
RMR Class			FAIR	GOOD	GOOD	GOOD	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.30	0.20	0.20	0.50	0.20	0.20
	F2		0.50	0.50	0.50	0.50	0.50	0.50
	F3		0.70	1.00	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			56.1	69.9	62.8	68.6	49.8	49.8
SMR Class			III	II	II	II	III	III
SMR Name			Normal	Good	Good	Good	Normal	Normal
Stability			Partially Stable	Stable	Stable	Stable	Partially Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		174	176	177	178	181	182
	Latitude		33.155484	33.157951	33.159011	33.160223	33.162745	33.163724
	Longitude		75.79793	75.798662	75.798824	75.799552	75.80161	75.80234
		Jv	30	28	20	23	29	20
UCS		Value	35	35	35	35	75	75
		Rat.	4	4	4	4	7	7
RQD		Value	16	22.6	49	39.1	19.3	49
		Rat.	6	6	8	8	6	8
Spacing (mm)	J1	Value	60-200	<60	60-200	<60	<60	60-200
		Rat.	8	5	8	5	5	8
	J2	Value	200-600	200-600	200-600	200-600	0.6-2m	0.6-2m
		Rat.	10	10	10	10	15	15
	J3	Value	0.6-2m	>2m	200-600	>2m	0.6-2m	>2m
		Rat.	15	20	10	20	15	20
	J4	Value						
		Rat.						
	Mean Rating		11.00	11.67	9.33	11.67	11.67	14.33
Persistence (m)	J1	Value	10-20	10-20	3-10	10-20	3-10	3-10
		Rat.	1	1	2	1	2	2
	J2	Value	3-10	1-3	3-10	1-3	1-3	1-3
		Rat.	2	4	2	4	4	4
	J3	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J4	Value						
		Rat.						
	Mean Rating		2.3	3.0	2.7	3.0	3.3	3.3
Aperture / Separation (mm)	J1	Value	>5	>5	>5	>5	>5	>5
		Rat.	0	0	0	0	0	0
	J2	Value	1-5	1-5	None	None	>5	None
		Rat.	1	1	6	6	0	6
	J3	Value	1-5	1-5	1-5	>5	>5	>5
		Rat.	1	1	1	0	0	0
	J4	Value						
		Rat.						
	Mean Rating		0.7	0.7	2.3	2.0	0.0	2.0
Roughness	J1	Value	SI Rough	SI Rough	Rough	SI Rough	Rough	Rough
		Rat.	3	3	5	3	5	5
	J2	Value	SI Rough	Rough	Smooth	Rough	Rough	Rough
		Rat.	3	5	1	5	5	5
	J3	Value	Rough	SI Rough	Rough	SI Rough	SI Rough	Smooth
		Rat.	5	3	5	3	5	1
	J4	Value						
		Rat.						

		Mean Rating	3.7	3.7	3.7	3.7	5.0	3.7
Infilling	J1	Value	S Clay	S Clay	Hard filling	S Clay	S Clay	S Clay
		Rat.	2	2	4	2	2	2
	J2	Value	S Clay	S Clay	None	None	S Clay	None
		Rat.	2	2	6	6	2	6
	J3	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J4	Value						
		Rat.						
		Mean Rating	2.0	2.0	4.0	3.3	2.0	3.3
Weathering		Value	Moderate	Moderate	Moderate	Slight	Moderate	Slight
		Rat.	3	3	5	3	5	5
Groundwater Condition		Value	Dry	Dripping	Dry	Dry	Damp	Dry
		Rat.	15	4	15	15	10	15
RMRb			47.67	38.00	54.00	53.67	50.00	61.67
RMR Class			III	IV	III	III	III	II
RMR Class			FAIR	POOR	FAIR	FAIR	FAIR	GOOD
Adjustment: Anbalgan et al. 2008	F1		0.20	0.20	0.40	0.30	0.40	0.50
	F2		0.50	0.50	0.50	0.40	0.50	0.50
	F3		1.00	1.00	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			47.8	38.1	54.2	53.8	50.2	61.9
SMR Class			III	III	III	III	III	II
SMR Name			Normal	Normal	Normal	Normal	Normal	Good
Stability			Partially Stable	Partially Stable	Partially Stable	Partially Stable	Partially Stable	Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		183	184	185	186	191	192
	Latitude		33.164369	33.165337	33.166219	33.167513	33.173814	33.174742
	Longitude		75.803146	75.80395	75.804149	75.805248	75.809621	75.810071
		Jv	15	15	15	20	15	15
UCS		Value	75	35	75	75	35	35
		Rat.	7	4	7	7	4	4
RQD		Value	65.5	65.5	65.5	49	65.5	65.5
		Rat.	13	13	13	8	13	13
Spacing (mm)	J1	Value	200-600	60-200	60-200	200-600	200-600	60-200
		Rat.	10	8	8	8	8	8
	J2	Value	0.6-2m	0.6-2m	0.6-2m	>2m	0.6-2m	200-600
		Rat.	15	15	15	20	15	10
	J3	Value	>2m	>2m	>2m	>2m	>2m	>2m
		Rat.	20	20	20	20	20	20
	J4	Value						
		Rat.						
	Mean Rating		15.00	14.33	14.33	16.00	14.33	12.67
Persistence (m)	J1	Value	10-20	10-20	10-20	3-10	3-10	3-10
		Rat.	1	1	1	2	2	2
	J2	Value	0.6-2m	1-3	1-3	1-3	1-3	1-3
		Rat.	15	4	4	4	4	4
	J3	Value	>2m	1-3	3-10	1-3	1-3	1-3
		Rat.	20	4	2	4	4	4
	J4	Value						
		Rat.						
	Mean Rating		12.0	3.0	2.3	3.3	3.3	3.3
Aperture / Separation (mm)	J1	Value	>5	>5	>5	>5	None	1-5
		Rat.	0	0	0	1	6	1
	J2	Value	None	None	None	None	>5	1-5
		Rat.	6	6	6	6	0	1
	J3	Value	>5	>5	>5	>5	None	>5
		Rat.	0	0	0	0	6	0
	J4	Value						
		Rat.						
	Mean Rating		2.0	2.0	2.0	2.3	4.0	0.7
Roughness	J1	Value	Rough	V Rough	Rough	Rough	Rough	Rough
		Rat.	5	6	5	5	5	5
	J2	Value	Rough	Rough	Rough	Rough	V Rough	Rough
		Rat.	5	5	5	5	6	5
	J3	Value	Smooth	Sl Rough	Smooth	Smooth	Rough	Rough
		Rat.	1	3	1	1	5	5
	J4	Value						
		Rat.						

		Mean Rating	3.7	3.5	2.8	2.8	4.0	5.0
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	None	S Clay
		Rat.	2	2	2	2	6	2
	J2	Value	None	None	None	None	S Clay	S Clay
		Rat.	6	6	6	6	2	2
	J3	Value	S Clay	S Clay	S Clay	S Clay	None	S Clay
		Rat.	2	2	2	2	6	2
	J4	Value						
		Rat.						
		Mean Rating	3.3	3.3	3.3	3.3	4.7	2.0
Weathering		Value	Slight	Slight	Moderate	Slight	Moderate	Moderate
		Rat.	5	3	5	3	3	5
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			76.00	61.17	64.75	60.75	65.33	60.67
RMR Class			II	II	II	II	II	II
RMR Class			GOOD	GOOD	GOOD	GOOD	GOOD	GOOD
Adjustment: Anbalgan et al. 2008	F1		0.30	0.25	0.30	0.50	0.50	0.25
	F2		0.50	0.50	0.50	0.50	0.50	0.50
	F3		1.00	1.00	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			76.2	61.3	64.9	61.0	65.6	60.8
SMR Class			II	II	II	II	II	II
SMR Name			Good	Good	Good	Good	Good	Good
Stability			Stable	Stable	Stable	Stable	Stable	Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		193	194	196	198	200	201
	Latitude		33.17725	33.17725	33.12659	33.126467	33.126412	33.124325
	Longitude		75.810596	75.810985	75.619085	75.621455	75.625133	75.636263
		Jv	10	15	18	25	10	18
UCS		Value	75	35	35	75	75	75
		Rat.	7	4	4	7	7	7
RQD		Value	82	65.5	55.6	32.5	82	55.6
		Rat.	17	13	13	8	17	13
Spacing (mm)	J1	Value	60-200	60-200	0.6-2m	60-200	60-200	60-200
		Rat.	8	8	15	8	8	8
	J2	Value	60-200	>2m	200-600	200-600	200-600	200-600
		Rat.	8	20	10	10	10	10
	J3	Value	>2m	>2m	0.6-2m	200-600	0.6-2m	200-600
		Rat.	20	20	15	10	15	10
	J4	Value				200-600	0.6-2m	0.6-2m
		Rat.				10	15	15
	Mean Rating		12.00	16.00	13.33	9.50	12.00	10.75
Persistence (m)	J1	Value	10-20	10-20	3-10	3-10	10-20	3-10
		Rat.	1	1	2	2	1	2
	J2	Value	3-10	3-10	3-10	3-10	1-3	1-3
		Rat.	2	2	2	2	4	4
	J3	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J4	Value				<1	1-3	1-3
		Rat.				6	4	4
	Mean Rating		2.3	2.3	2.7	3.5	3.3	3.5
Aperture / Separation (mm)	J1	Value	>5	>5	1-5	1-5	1-5	1-5
		Rat.	0	0	1	1	1	1
	J2	Value	1-5	>5	1-5	1-5	None	1-5
		Rat.	1	0	1	1	6	1
	J3	Value	>5	1-5	>5	>5	>5	>5
		Rat.	0	1	0	0	0	0
	J4	Value				None	1-5	>5
		Rat.				6	1	0
	Mean Rating		0.3	0.3	0.7	2.0	2.0	0.5
Roughness	J1	Value	Sl Rough	Rough	Rough	Sl Rough	Sl Rough	Rough
		Rat.	3	3	5	3	3	5
	J2	Value	Smooth	Sl Rough	Sl Rough	Smooth	Rough	Sl Rough
		Rat.	1	3	3	1	5	3
	J3	Value	Rough	Rough	Rough	Sl Rough	Sl Rough	Rough
		Rat.	5	5	5	3	3	5
	J4	Value				Rough	Rough	Rough
		Rat.				5	5	5

		Mean Rating	3.0	3.7	4.3	3.0	4.0	4.5
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J2	Value	None	S Clay	S Clay	S Clay	None	S Clay
		Rat.	6	2	2	2	6	2
	J3	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J4	Value				None	S Clay	S Clay
		Rat.				6	2	2
		Mean Rating	3.3	2.0	2.0	3.0	3.0	2.0
Weathering		Value	Slight	Slight	Slight	Moderate	Slight	Moderate
		Rat.	5	5	3	5	3	3
Groundwater Condition		Value	Dry	Dry	Damp	Damp	Damp	Dry
		Rat.	15	15	10	10	10	15
RMRb			65.00	61.33	53.00	51.00	61.25	59.25
RMR Class			II	II	III	III	II	III
RMR Class			GOOD	GOOD	FAIR	FAIR	GOOD	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.40	0.40	0.40	0.30	0.50	0.20
	F2		0.50	0.50	0.50	0.50	0.50	0.50
	F3		1.00	0.80	1.00	1.00	1.00	1.00
	F4		0	0	0	0	0	0
SMR			65.2	61.5	53.2	51.2	61.5	59.4
SMR Class			II	II	III	III	II	III
SMR Name			Good	Good	Normal	Normal	Good	Normal
Stability			Stable	Stable	Partially Stable	Partially Stable	Stable	Partially Stable

Field based RMR was c

**Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes
based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)**

	Location No.		224	227	228	231	232	233
	Latitude		33.131925	33.1351	33.1353	33.181962	33.183963	33.18551
	Longitude		75.55004	75.554247	75.556028	75.807401	75.807797	75.807523
		Jv	18	15	16	36	20	15
UCS		Value	35	75	75	5	75	75
		Rat.	4	7	7	2	7	7
RQD		Value	55.6	65.5	62.2	-3.8	49	65.5
		Rat.	13	13	13	5	8	13
Spacing (mm)	J1	Value	60-200	0.6-2m	0.6-2m	<60	<60	<60
		Rat.	8	15	15	5	5	5
	J2	Value	200-600	0.6-2m	60-200	60-200	60-200	0.6-2m
		Rat.	10	15	8	8	8	15
	J3	Value	0.6-2m	200-600	200-600	60-200	200-600	0.6-2m
		Rat.	15	10	10	8	10	15
	J4	Value						
		Rat.						
	Mean Rating		11.00	13.33	11.00	7.00	7.67	11.67
Persisten ce (m)	J1	Value	3-10	3-10	10-20	>5	3-10	10-20
		Rat.	2	2	1	0	2	1
	J2	Value	1-3	1-3	3-10	<1	1-3	3-10
		Rat.	4	4	2	6	4	2
	J3	Value	<1	1-3	1-3	<1	1-3	1--3
		Rat.	6	4	4	6	4	4
	J4	Value						
		Rat.						
	Mean Rating		4.0	3.3	2.3	4.0	3.3	2.3
Aperture / Separatio n (mm)	J1	Value	1-5	1-5	None	>5	>5	None
		Rat.	1	1	6	0	0	6
	J2	Value	None	1-5	None	1-5	>5	>5
		Rat.	6	1	6	1	0	0
	J3	Value	None	>5	>5	0.1-1	>5	>5
		Rat.	6	0	0	4	0	0
	J4	Value						
		Rat.						
	Mean Rating		4.3	0.7	4.0	1.7	0.0	2.0
Roughnes s	J1	Value	Sl Rough	Rough	Rough	Smooth	Sl Rough	Sl Rough
		Rat.	3	5	5	1	3	3
	J2	Value	Sl Rough	Rough	Sl Rough	Rough	Rough	Smooth
		Rat.	3	5	3	5	5	1
	J3	Value	Rough	Rough	Rough	Rough	Rough	Rough
		Rat.	5	5	5	5	5	5
	J4	Value						
		Rat.						

	Mean Rating		3.7	5.0	4.3	3.7	4.3	3.0
Infilling	J1	Value	S Clay	S Clay	None	S Clay	S Clay	None
		Rat.	2	2	6	2	2	6
	J2	Value	None	Hard filling	None	None	S Clay	S Clay
		Rat.	6	4	6	6	2	2
	J3	Value	None	S Clay	S Clay	S Clay	S Clay	None
		Rat.	6	2	2	2	2	6
	J4	Value						
		Rat.						
Mean Rating		4.7	2.7	4.7	3.3	2.0	4.7	
Weathering		Value	Moderate	Slight	Slightly	Slightly	Moderate	Moderate
		Rat.	5	5	5	3	3	3
Groundwater Condition		Value	Dry	Dry	Dry	Dripping	Dry	Dry
		Rat.	15	15	15	4	15	15
RMRb			64.67	65.00	66.33	33.67	50.33	61.67
RMR Class			II	II	II	IV	III	II
RMR Class			GOOD	GOOD	GOOD	POOR	FAIR	GOOD
Adjustment: Anbalgan et al. 2008		F1	0.40	0.4	0.40	0.20	0.30	0.50
		F2	0.50	0.5	0.50	0.50	0.50	0.40
		F3	1.00	1	1.00	1.00	1.00	1.00
		F4	0	0	0	0	0	0
SMR			64.9	65.2	66.5	33.8	50.5	61.9
SMR Class			III	III	II	IV	III	II
SMR Name			Normal	Normal	Good	Bad	Normal	Good
Stability			Partially Stable	Partially Stab	Stable	Unstable	Partially Stable	Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		235	238	239	240	241	242
	Latitude		33.187785	33.197417	33.197938	33.198627	33.200047	33.200483
	Longitude		75.807167	75.80223	75.80125	75.800468	75.799583	75.799057
		Jv	10	18	10	18	26	10
UCS		Value	75	35	75	75	75	75
		Rat.	7	4	7	7	7	7
RQD		Value	82	55.6	82	55.6	29.2	82
		Rat.	17	13	17	13	8	17
Spacing (mm)	J1	Value	<60	<60	60-200	60-200	<60	>2m
		Rat.	5	5	8	8	5	20
	J2	Value	0.6-2m	>2m	0.6-2m	200-600	60-200	0.6-2m
		Rat.	15	20	15	10	8	15
	J3	Value	0.6-2m	200-600	200-600	200-600	0.6-2m	0.6-2m
		Rat.	15	10	10	10	15	15
	J4	Value	>2m			>2m		
		Rat.	20			20		
	Mean Rating		13.75	11.67	11.00	12.00	9.33	16.67
Persistence (m)	J1	Value	3-10	1-3	10-20	10-20	10-20	3-10
		Rat.	2	4	1	1	1	2
	J2	Value	3-10	<1	3-10	3-10	3-10	1-3
		Rat.	2	6	2	2	2	4
	J3	Value	1-3	1-3	10-20	1-3	1-3	1-3
		Rat.	4	4	1	4	4	4
	J4	Value	1-3			<1		
		Rat.	4			6		
	Mean Rating		3.0	4.7	1.3	3.3	2.3	3.3
Aperture / Separation (mm)	J1	Value	None	1-5	>5	>5	>5	None
		Rat.	6	1	0	0	0	6
	J2	Value	>5	None	1-5	>5	>5	>5
		Rat.	0	6	1	0	0	0
	J3	Value	1-5	None	1-5	1-5	1-5	None
		Rat.	1	6	1	1	1	6
	J4	Value	1-5			None		
		Rat.	1			6		
	Mean Rating		2.0	4.3	0.7	1.8	0.3	4.0
Roughness	J1	Value	Rough	Sl Rough	Sl Rough	Sl Rough	Sl Rough	Rough
		Rat.	5	3	3	3	3	5
	J2	Value	Smooth	Sl Rough	Sl Rough	Smooth	Smooth	Rough
		Rat.	1	3	3	1	1	5
	J3	Value	Smooth	Rough	Rough	Rough	Rough	Rough
		Rat.	1	5	5	5	5	5
	J4	Value	Smooth			V Rough		
		Rat.	1			6		

	Mean Rating		2.0	3.7	3.7	3.8	3.0	5.0
Infilling	J1	Value	None	S Clay	S Clay	S Clay	S Clay	None
		Rat.	6	2	2	2	2	6
	J2	Value	S Clay	None	S Clay	S Clay	S Clay	S Clay
		Rat.	2	6	2	2	2	2
	J3	Value	None	None	S Clay	S Clay	S Clay	None
		Rat.	6	6	2	2	2	6
	J4	Value	None			None		
		Rat.	6			6		
Mean Rating		5.0	4.7	2.0	3.0	2.0	4.7	
Weathering		Value	Moderate	Slight	Moderate	Slight	Moderate	Slightly
		Rat.	5	3	5	3	5	5
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			69.75	64.00	62.67	61.75	52.00	77.67
RMR Class			II	II	II	II	III	II
RMR Class			GOOD	GOOD	GOOD	GOOD	FAIR	GOOD
Adjustment: Anbalgan et al. 2008		F1	0.30	0.20	0.5	0.20	0.30	0.50
		F2	0.50	0.50	0.3	0.40	0.50	0.50
		F3	1.00	1.0	1	1.0	1.0	0.7
		F4	0	0	0	0	0	0
SMR			69.9	64.1	62.8	61.8	52.2	77.8
SMR Class			II	III	II	III	II	III
SMR Name			Good	Normal	Good	Normal	Good	Normal
Stability			Stable	artially Stabl	Stable	artially Stabl	Stable	artially Stabl

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		244	245	257	259	260	263
	Latitude		33.202068	33.20293	33.221722	33.224042	33.225127	33.228793
	Longitude		75.797967	75.797857	75.795562	75.792543	75.791738	75.788897
		Jv	10	10	15	15	20	15
UCS		Value	75	75	75	75	75	75
		Rat.	7	7	7	7	7	7
RQD		Value	82	82	65.5	65.5	49	65.5
		Rat.	17	17	13	13	8	13
Spacing (mm)	J1	Value	200-600	0.6-2m	<60	200-600	<60	0.6-2m
		Rat.	10	15	5	10	5	15
	J2	Value	0.6-2m	>2m	>2m	>2m	60-200	200-600
		Rat.	15	20	20	20	8	10
	J3	Value	>2m	0.6-2m	200-600	0.6-2m	200-600	200-600
		Rat.	20	15	10	15	10	10
	J4	Value						
		Rat.						
	Mean Rating		15.00	16.67	11.67	15.00	7.67	11.67
Persistence (m)	J1	Value	10-20	10-20	3-10	1-3	3-10	3-10
		Rat.	1	1	2	4	2	2
	J2	Value	10-20	1-3	3-10	1-3	1-3	1-3
		Rat.	1	4	2	4	4	4
	J3	Value	<1	1-3	1-3	1-3	1-3	1-3
		Rat.	6	4	4	4	4	4
	J4	Value						
		Rat.						
	Mean Rating		2.7	3.0	2.7	4.0	3.3	3.3
Aperture / Separation (mm)	J1	Value	None	1-5	>5	None	>5	>5
		Rat.	6	1	0	6	0	0
	J2	Value	>5	>5	>5	>5	1-5	1-5
		Rat.	0	0	0	0	1	1
	J3	Value	None	1-5	1-5	>5	1-5	>5
		Rat.	6	1	1	0	1	0
	J4	Value						
		Rat.						
	Mean Rating		4.0	0.7	0.3	2.0	0.7	0.3
Roughness	J1	Value	Rough	SI Rough	Rough	V Rough	Rough	Rough
		Rat.	5	3	5	6	5	5
	J2	Value	Smooth	Rough	Smooth	Smooth	Rough	Rough
		Rat.	1	5	1	1	5	5
	J3	Value	Rough	Rough	V Rough	Rough	Rough	Rough
		Rat.	5	5	6	5	5	5
	J4	Value						
		Rat.						

		Mean Rating	3.7	4.3	4.0	4.0	5.0	5.0
Infilling	J1	Value	None	S Clay	S Clay	None	S Clay	S Clay
		Rat.	6	2	2	6	2	2
	J2	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J3	Value	Rough	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	5	2	2	2	2	2
	J4	Value						
		Rat.						
		Mean Rating	4.3	2.0	2.0	3.3	2.0	2.0
Weathering		Value	Slight	Moderate	Moderate	Moderate	Slightly	Slightly
		Rat.	3	3	3	5	5	3
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			71.67	68.67	58.67	68.33	53.67	60.33
RMR Class			II	II	III	II	III	II
RMR Class			GOOD	GOOD	FAIR	GOOD	FAIR	GOOD
Adjustment: Anbalgan et al. 2008	F1		0.5	0.20	0.20	0.5	0.20	0.20
	F2		0.5	0.50	0.50	0.5	0.50	0.50
	F3		1	1.0	1.0	1	1.0	1.0
	F4		0	0	0	0	0	0
SMR			71.9	68.8	58.8	68.6	53.8	60.4
SMR Class			III	IV	IV	III	III	III
SMR Name			Normal	Bad	Bad	Normal	Normal	Normal
Stability			Partially Stable	Unstable	Unstable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was c

**Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes
based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)**

	Location No.		264	285	286	287	288	289
	Latitude		33.229868	33.26312	33.23005	33.200797	33.164832	33.14661
	Longitude		75.787828	75.773583	75.787977	75.79871	75.803473	75.798697
		Jv	25	10	25	10	15	10
UCS		Value	75	35	75	75	75	75
		Rat.	7	4	7	7	7	7
RQD		Value	32.5	82	32.5	82	65.5	82
		Rat.	8	17	8	17	13	17
Spacing (mm)	J1	Value	200-600	<60	200-600	>2m	200-600	200-600
		Rat.	10	5	10	20	10	10
	J2	Value	200-600	200-600	200-600	0.6-2m	0.6-2m	0.6-2m
		Rat.	10	10	10	15	15	15
	J3	Value	600-2000	200-600	0.6-2m	>2m	>2m	>2m
		Rat.	15	10	15	20	20	20
	J4	Value						
		Rat.						
	Mean Rating		8.75	8.33	11.67	18.33	15.00	15.00
Persistence (m)	J1	Value	3-10	3-10	3-10	3-10	10-20	10-20
		Rat.	2	2	2	2	1	1
	J2	Value	1-3	1-3	1-3	1-3	3-10	3-10
		Rat.	4	4	4	4	2	2
	J3	Value	1-3	1-3	1-3	1-3	3-10	3-10
		Rat.	4	4	4	4	2	2
	J4	Value						
		Rat.						
	Mean Rating		3.3	3.3	3.3	3.3	1.7	1.7
Aperture / Separation (mm)	J1	Value	>5	1-5	>5	None	>5	>5
		Rat.	0	1	0	6	0	0
	J2	Value	>5	1-5	>5	>5	None	None
		Rat.	0	1	0	0	6	6
	J3	Value	>5	1-5	1-5	None	>5	>5
		Rat.	0	1	1	6	0	0
	J4	Value						
		Rat.						
	Mean Rating		0.0	1.0	0.3	4.0	2.0	2.0
Roughness	J1	Value	Rough	Rough	Rough	Sl Rough	Rough	Rough
		Rat.	5	5	5	5	5	5
	J2	Value	Rough	Rough	Rough	Rough	Rough	Rough
		Rat.	5	5	5	5	5	5
	J3	Value	Rough	Rough	Rough	Rough	Smooth	Smooth
		Rat.	5	5	5	5	1	1
	J4	Value						
		Rat.						

		Mean Rating	5.0	5.0	5.0	5.0	3.7	3.7
Infilling	J1	Value	S Clay	S Clay	S Clay	None	S Clay	S Clay
		Rat.	2	2	2	6	2	2
	J2	Value	S Clay	S Clay	S Clay	S Clay	None	None
		Rat.	2	2	2	2	6	6
	J3	Value	S Clay	S Clay	S Clay	None	S Clay	S Clay
		Rat.	2	2	2	6	2	2
	J4	Value						
		Rat.						
		Mean Rating	2.0	2.0	2.0	4.7	3.3	3.3
Weathering		Value	Moderate	Moderate	Moderate	Moderate	Slightly	Slightly
		Rat.	3	3	3	5	5	5
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			52.08	58.67	55.33	79.33	65.67	69.67
RMR Class			III	III	III	II	II	II
RMR Class			FAIR	FAIR	FAIR	GOOD	GOOD	GOOD
Adjustment: Anbalgan et al. 2008	F1		0.20	0.25	0.20	0.20	0.4	0.20
	F2		0.50	0.50	0.50	0.50	0.5	0.50
	F3		1.0	1.0	1.0	1.0	0.8	1.00
	F4		0	0	0	0	0	0
SMR			52.2	58.8	55.4	79.4	65.8	69.8
SMR Class			III	III	III	II	II	II
SMR Name			Normal	Normal	Normal	Good	Good	Good
Stability			Partially Stable	Partially Stable	Partially Stable	Stable	Stable	Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		290	291	292	293	294	295
	Latitude		33.145845	33.138787	33.151578	33.155957	33.140167	33.153898
	Longitude		75.783833	75.75647	75.74831	75.701807	75.558975	75.682033
		Jv	20	15	15	20	20	25
UCS		Value	35	35	75	35	75	35
		Rat.	4	4	4	4	7	4
RQD		Value	49	65.5	65.5	49	49	32.5
		Rat.	8	13	13	8	8	8
Spacing (mm)	J1	Value	60-200	<60	60-200	<60	<60	<60
		Rat.	8	5	8	5	5	5
	J2	Value	0.6-2m	200-600	200-600	200-600	60-200	60-200
		Rat.	15	10	10	10	8	8
	J3	Value	0.6-2m	0.6-2m	200-600	200-600	200-600	0.6-2m
		Rat.	15	15	10	10	10	15
	J4	Value						200-600
		Rat.						10
	Mean Rating		12.67	10.00	9.33	8.33	7.67	9.50
Persistence (m)	J1	Value	3-10	10-20	10-20	3-10	3-10	10-20
		Rat.	2	1	1	2	2	1
	J2	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J3	Value	1-3	1-3	1-3	1-3	1-3	1-3
		Rat.	4	4	4	4	4	4
	J4	Value						1-3
		Rat.						4
	Mean Rating		3.3	3.0	3.0	3.3	3.3	3.3
Aperture / Separation (mm)	J1	Value	>5	>5	>5	None	1-5	>5
		Rat.	0	0	0	6	1	0
	J2	Value	1-5	None	None	1-5	1-5	1-5
		Rat.	1	6	6	1	1	1
	J3	Value	>5	None	None	None	None	>5
		Rat.	0	6	6	6	6	0
	J4	Value						1-5
		Rat.						1
	Mean Rating		0.3	4.0	4.0	4.3	2.7	0.5
Roughness	J1	Value	SI Rough	SI Rough	Rough	SI Rough	Rough	SI Rough
		Rat.	3	3	5	3	5	3
	J2	Value	Rough	Rough	Rough	SI Rough	Rough	Rough
		Rat.	5	5	5	3	5	5
	J3	Value	Rough	Rough	Rough	Rough	Rough	Smooth
		Rat.	5	5	5	5	5	1
	J4	Value						Rough
		Rat.						5

		Mean Rating	4.3	4.3	5.0	3.7	5.0	3.5
Infilling	J1	Value	S Clay	S Clay	Calcite	None	S Clay	S Clay
		Rat.	2	2	4	6	2	2
	J2	Value	S Clay	None	None	S Clay	S Clay	S Clay
		Rat.	2	6	6	2	2	2
	J3	Value	S Clay	None	None	None	None	S Clay
		Rat.	2	6	6	6	6	2
	J4	Value						S Clay
		Rat.						2
		Mean Rating	2.0	4.7	5.3	4.7	3.3	2.0
Weathering		Value	Slightly	Slightly	Slightly	Moderate	Moderate	Slightly
		Rat.	5	5	3	3	5	3
Groundwater Condition		Value	Dry	Dry	Dry	Dry	Dry	Dry
		Rat.	15	15	15	15	15	15
RMRb			54.67	63.00	61.67	54.33	57.00	48.75
RMR Class			III	II	II	III	III	III
RMR Class			FAIR	GOOD	GOOD	FAIR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.30	0.3	0.30	0.5	0.20	0.50
	F2		0.50	0.5	0.50	0.4	0.50	0.50
	F3		1.00	1	1.00	0.9	1.0	0.8
	F4		0	0	0	0	0	0
SMR			54.8	63.2	61.8	54.5	57.1	49.0
SMR Class			III	II	II	III	III	III
SMR Name			Normal	Good	Good	Normal	Normal	Normal
Stability			Partially Stable	Stable	Stable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		305	307	308	309	312	313
	Latitude		33.132483	33.127823	33.129343	33.130203	33.132308	33.133633
	Longitude		75.655035	75.643645	75.643285	75.644507	75.651418	75.654475
		Jv	15	15	36	14	20	30
UCS		Value	75	75	35	75	5	75
		Rat.	7	7	4	7	2	7
RQD		Value	65.5	65.5	-3.8	68.8	49	16
		Rat.	13	13	5	13	8	5
Spacing (mm)	J1	Value	60-200	60-200	<60	200-600	<60	60-200
		Rat.	8	8	5	10	5	8
	J2	Value	200-600	200-600	<60	200-600	200-600	0.6-2m
		Rat.	10	10	5	10	10	15
	J3	Value	0.6-2m	0.6-2m	60-200	60-200	0.6-2m	0.6-2m
		Rat.	15	15	8	8	15	15
	J4	Value			60-200	>2m		200-600
		Rat.			8	20		10
	Mean Rating		11.00	11.00	8.67	16.00	7.50	12.00
Persisten ce (m)	J1	Value	10-20	10-20	10-20	3-10	3-10	10-20
		Rat.	1	1	1	2	2	1
	J2	Value	3-10	3-10	3-10	3-10	3-10	3-10
		Rat.	2	2	2	2	2	2
	J3	Value	3-10	1-3	1-3	3-10	1-3	3-10
		Rat.	2	4	4	2	4	2
	J4	Value			1-3	1-3		1-3
		Rat.			4	4		4
	Mean Rating		1.7	2.3	2.8	2.5	2.7	2.3
Aperture / Separatio n (mm)	J1	Value	>5	>5	>5	>5	>5	>5
		Rat.	0	1	0	0	0	0
	J2	Value	>5	1-5	>5	>5	1-5	None
		Rat.	0	1	0	0	1	6
	J3	Value	>5	>5	None	>5	>5	None
		Rat.	0	0	6	0	0	6
	J4	Value			1-5	None		>5
		Rat.			1	6		0
	Mean Rating		0.0	0.7	1.8	1.5	0.3	3.0
Roughnes s	J1	Value	Sl Rough	Rough	Rough	Rough	Sl Rough	Rough
		Rat.	3	5	5	5	3	5
	J2	Value	Rough	Rough	Sl Rough	Sl Rough	Rough	Smooth
		Rat.	5	5	3	3	5	1
	J3	Value	Rough	Sl Rough	Rough	Smooth	Sl Rough	Rough
		Rat.	5	3	5	1	3	5
	J4	Value			Rough	Rough		Rough
		Rat.			5	5		5

		Mean Rating	4.3	4.3	4.5	3.5	3.7	4.0
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J2	Value	S Clay	S Clay	S Clay	S Clay	S Clay	None
		Rat.	2	2	2	2	2	6
	J3	Value	S Clay	S Clay	None	S Clay	S Clay	None
		Rat.	2	2	6	2	2	6
	J4	Value			S Clay	None		S Clay
		Rat.			2	6		2
		Mean Rating	2.0	2.0	4.0	4.0	2.0	4.0
Weathering		Value	Moderate	Moderate	Slight	Moderate	Moderate	Moderate
		Rat.	3	5	3	3	3	3
Groundwater Condition		Value	Dry	Dripping	Dry	Damp	Damp	Dripping
		Rat.	15	4	15	10	10	4
RMRb			57.00	49.33	48.67	60.50	39.17	44.25
RMR Class			III	III	III	II	IV	III
RMR Class			FAIR	FAIR	FAIR	GOOD	POOR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.30	0.20	0.5	0.5	0.30	0.20
	F2		0.50	0.40	0.5	0.45	0.50	0.50
	F3		1.00	1.0	1	1	1.0	1.0
	F4		0	0	0	0	0	0
SMR			57.2	49.4	48.9	60.7	39.3	44.4
SMR Class			III	III	III	IV	III	III
SMR Name			Normal	Normal	Normal	Bad	Normal	Normal
Stability			Partially Stable	Partially Stable	Partially Stable	Unstable	Partially Stable	Partially Stable

Field based RMR was c

Annexure:IV Basic Rock Mass Rating (RMRb) and Slope Mass Rating (SMR) of rocky slopes based on field parameters (Bieniawski, 1979; Anbalagan et al. 2008)

	Location No.		314	315	316	317	318	319
	Latitude		33.13438	33.135058	33.135492	33.136372	33.137852	33.13899
	Longitude		75.65566	75.65626	75.65678	75.657562	75.658207	75.658658
		Jv	30	28	32	25	15	15
UCS		Value	35	50	75	25	75	75
		Rat.	4	7	7	4	7	7
RQD		Value	16	22.6	9.4	32.5	65.5	65.5
		Rat.	5	5	5	8	13	13
Spacing (mm)	J1	Value	<60	<60	<60	<60	<60	<60
		Rat.	5	5	5	5	5	5
	J2	Value	0.6-2m	60-200	60-200	60-200	0.6-2m	0.6-2m
		Rat.	15	8	8	10	15	15
	J3	Value	200-600	60-200	60-200	200-600	>2m	>2m
		Rat.	10	8	8	10	20	20
	J4	Value						
		Rat.						
	Mean Rating		7.50	7.00	7.00	8.33	13.33	13.33
Persistence (m)	J1	Value	10-20	10-20	10-20	10-20	10-20	10-20
		Rat.	1	1	1	1	1	1
	J2	Value	3-10	1-3	1-3	1-3	1-3	1-3
		Rat.	2	4	4	4	4	4
	J3	Value	1-3	<1	1-3	1-3	<1	<1
		Rat.	4	6	4	4	6	6
	J4	Value						
		Rat.						
	Mean Rating		2.3	3.7	3.0	3.0	3.7	3.7
Aperture / Separation (mm)	J1	Value	>5	>5	>5	>5	1-5	1-5
		Rat.	0	0	0	0	1	1
	J2	Value	>5	>5	>5	>5	None	None
		Rat.	0	0	0	0	6	6
	J3	Value	>5	>	>5	>5	>5	>5
		Rat.	0	0	0	0	0	0
	J4	Value						
		Rat.						
	Mean Rating		0.0	0.0	0.0	0.0	2.3	2.3
Roughness	J1	Value	Sl Rough	Sl Rough	Sl Rough	Rough	Rough	Rough
		Rat.	3	3	3	5	5	5
	J2	Value	Rough	Rough	Rough	Rough	V Rough	V Rough
		Rat.	5	5	5	5	6	6
	J3	Value	Rough	Rough	Rough	Rough	V Rough	V Rough
		Rat.	5	5	5	5	6	6
	J4	Value						
		Rat.						

		Mean Rating	4.3	4.3	4.3	5.0	5.7	5.7
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J2	Value	S Clay	S Clay	S Clay	S Clay	None	None
		Rat.	2	2	2	2	6	6
	J3	Value	S Clay	S Clay	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2	2	2
	J4	Value						
		Rat.						
		Mean Rating	2.0	2.0	2.0	2.0	3.3	3.3
Weathering		Value	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
		Rat.	3	3	3	3	3	3
Groundwater Condition		Value	Dripping	Dripping	Damp	Wet	Damp	Damp
		Rat.	4	4	10	7	10	10
RMRb			32.17	36.00	41.33	40.33	61.33	61.33
RMR Class			IV	IV	III	III	II	II
RMR Class			POOR	POOR	FAIR	FAIR	GOOD	GOOD
Adjustment: Anbalgan et al. 2008	F1		0.20	0.30	0.20	0.40	0.20	0.30
	F2		0.50	0.50	0.50	0.45	0.45	0.50
	F3		1.0	1.00	1.0	1.00	1.00	1.0
	F4		0	0	0	0	0	0
SMR			32.3	36.2	41.4	40.5	61.4	61.5
SMR Class			III	III	II	III	III	III
SMR Name			Normal	Normal	Good	Normal	Good	Normal
Stability			Partially Stable	Partially Stable	Stable	Partially Stable	Stable	Partially Stable

Field based RMR was c

Cont.....

	Location No.		320	321	323	326
	Latitude		33.140572	33.141988	33.149162	33.14862
	Longitude		75.659328	75.660102	75.72308	75.72373
		Jv	15	30	17	17
UCS		Value	35	35	75	75
		Rat.	4	4	7	7
RQD		Value	65.5	16	58.9	58.9
		Rat.	13	5	13	13
Spacing (mm)	J1	Value	<60	<60	60-200	60-200
		Rat.	5	5	8	8
	J2	Value	60-200	200-600	200-600	200-600
		Rat.	8	10	10	10
	J3	Value	200-600		60-200	60-200
		Rat.	10		8	8
	J4	Value			0.6-2m	0.6-2m
		Rat.			15	15
	Mean Rating		7.67	7.50	10.25	10.25
Persisten ce (m)	J1	Value	10-20	10-20	10-20	10-20
		Rat.	1	1	1	1
	J2	Value	3-10	1-3	3-10	3-10
		Rat.	2	4	2	2
	J3	Value	1-3		1-3	1-3
		Rat.	4		4	4
	J4	Value			1-3	1-3
		Rat.			4	4
	Mean Rating		2.3	2.5	2.8	2.8
Aperture / Separatio n (mm)	J1	Value	>5	>5	>5	>5
		Rat.	0	0	0	0
	J2	Value	>5	1-5	>5	>5
		Rat.	0	1	0	0
	J3	Value	>5		>5	>5
		Rat.	0		0	0
	J4	Value			>5	>5
		Rat.			0	0
	Mean Rating		0.0	0.5	0.0	0.0
Roughnes s	J1	Value	Rough	Rough	Sl Rough	Sl Rough
		Rat.	5	5	3	3
	J2	Value	Rough	Rough	Smooth	Smooth
		Rat.	5	5	1	1
	J3	Value	Rough		Rough	Rough
		Rat.	5		5	5
	J4	Value			Rough	Rough
		Rat.			5	5

		Mean Rating	5.0	5.0	3.5	3.5
Infilling	J1	Value	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2
	J2	Value	S Clay	S Clay	S Clay	S Clay
		Rat.	2	2	2	2
	J3	Value	S Clay		S Clay	S Clay
		Rat.	2		2	2
	J4	Value			S Clay	S Clay
		Rat.			2	2
		Mean Rating	2.0	2.0	2.0	2.0
Weathering		Value	Moderate	Moderate	Moderate	Slight
		Rat.	3	3	5	5
Groundwater Condition		Value	Damp	Damp	Damp	Damp
		Rat.	10	10	10	10
RMRb			47.00	39.50	53.50	53.50
RMR Class			III	IV	III	III
RMR Class			FAIR	POOR	FAIR	FAIR
Adjustment: Anbalgan et al. 2008	F1		0.30	0.40	0.20	0.20
	F2		0.50	0.45	0.50	0.50
	F3		1.0	1.00	1.0	1.0
	F4		0	0	0	0
SMR			47.2	39.7	53.6	53.6
SMR Class			III	III	III	III
SMR Name			Normal	Normal	Normal	Normal
Stability			Partially Stable	Partially Stable	Partially Stable	Partially Stable

Field based RMR was c

Annexue-V: Kinematic Analysis for planar failure

Latitude	Longitude	Loc	Strike	DA	DD	Name	SDA	SDD	Aj-As	Bj	Bj-Bs	F1	F2	F3	Rating	RMRb	RMR Class	RMR Class
33.144183	75.661317	L2		55	280	J3	80	314	34	55	-25	0.20	0.50	0.50	1.20	55.00	III	1.8
33.154086	75.681106	L4		70	335	J3	80	345	10	70	-10	0.40	0.50	1.00	1.90	55.83	III	1.8
33.265113	75.772547	L7		35	310	J1	80	230	-80	35	-45	0.20	0.30	1.00	1.50	59.33	III	1.8
33.265113	75.772547	L7		65	180	J2	80	230	50	65	-15	0.20	0.50	1.00	1.70	59.33	III	1.8
33.264356	75.773128	L8		50	285	J1	80	255	-30	50	-30	0.25	0.50	1.00	1.75	53.00	III	1.8
33.262888	75.773672	L9		45	290	J1	80	265	-25	45	-35	0.25	0.50	1.00	1.75	51.33	III	1.8
33.262888	75.773672	L10		45	290	J1	80	260	-30	45	-35	0.25	0.50	1.00	1.75	51.33	III	1.8
33.258659	75.774086	L11		35	295	J1	80	295	0	35	-45	0.50	0.40	1.00	1.90	43.33	III	1.8
33.253358	75.775706	L17		65	285	J1	80	250	-35	65	-15	0.20	0.50	1.00	1.70	50.33	III	1.8
33.253358	75.775706	L17		75	170	J2	80	250	80	75	-5	0.20	0.50	1.00	1.70	50.33	III	1.8
33.25224	75.776247	L18		50	255	J1	80	255	0	50	-30	0.50	0.50	1.00	2.00	66.00	II	1.2
33.25224	75.776247	L18		80	190	J2	80	255	65	80	0	0.20	0.50	1.00	1.70	66.00	II	1.2
33.250357	75.777549	L19		45	260	J1	80	255	-5	45	-35	0.50	0.50	0.80	1.80	50.00	III	1.8
33.248048	75.77926	L20		65	280	J1	80	240	-40	65	-15	0.20	0.50	0.70	1.40	59.75	III	1.8
33.248048	75.77926	L20		85	195	J3	80	240	45	85	5	0.20	0.50	1.00	1.70	59.75	III	1.8
33.247208	75.77983	L21		75	310	J1	80	240	-70	75	-5	0.20	0.50	1.00	1.70	68.00	II	1.2
33.247208	75.77983	L21		60	250	J3	80	240	-10	60	-20	0.40	0.50	1.00	1.90	68.00	II	1.2
33.247208	75.77983	L21		75	150	J4	80	240	90	75	-5	0.20	0.50	1.00	1.70	68.00	II	1.2
33.247208	75.77983	L21		20	240	J5	80	240	0	20	-60	0.50	0.25	1.00	1.75	68.00	II	1.2
33.1336	75.541399	L30		80	50	J3	80	20	-30	80	0	0.25	0.50	1.00	1.75	40.33	III	1.8
33.136723	75.553572	L36		45	350	J2	80	300	-50	45	-35	0.20	0.50	0.50	1.20	50.67	III	1.8
33.136723	75.553572	L36		73	295	J3	80	300	5	73	-7	0.50	0.50	1.00	2.00	50.67	III	1.8
33.137282	75.556153	L37		70	325	J2	80	300	-25	70	-10	0.25	0.50	0.80	1.55	47.33	III	1.8
33.139351	75.55834	L39		80	310	J3	80	315	5	80	0	0.50	0.50	1.00	2.00	57.00	III	1.8
33.141353	75.561288	L41		85	310	J3	80	335	25	85	5	0.25	0.50	1.00	1.75	54.33	III	1.8
33.1424	75.56294	L42		85	300	J3	80	320	20	85	5	0.30	0.50	0.80	1.60	65.00	II	1.2
33.143387	75.56434	L43		85	305	J3	80	320	15	85	5	0.30	0.50	1.00	1.80	59.00	III	1.8
33.143884	75.564991	L44		85	305	J3	80	320	15	85	5	0.30	0.50	0.50	1.30	59.00	III	1.8
33.144455	75.567036	L45		60	325	J2	80	350	25	60	-20	0.25	0.50	1.00	1.75	60.67	II	1.2
33.142534	75.570268	L47		60	325	J2	80	350	25	60	-20	0.25	0.50	1.00	1.75	60.67	II	1.2
33.135982	75.583426	L55		55	310	J1	25	350	40	55	30	0.20	0.50	1.00	1.70	61.67	II	1.2
33.136076	75.589186	L57		35	250	J1	80	340	90	35	-45	0.20	0.40	1.00	1.60	60.67	II	1.2

33.136076	75.589186	L57		80	305	J3	80	340	-35	80	0	0.20	0.50	1.00	1.70	60.67	II	1.2
33.132858	75.600713	L63		35	350	J1	80	350	0	35	-45	0.50	0.40	1.00	1.90	58.33	III	1.8
33.159566	75.696095	L101		35	60	J2	80	25	-35	35	-45	0.20	0.40	1.00	1.60	58.67	III	1.8
33.159348	75.697988	L102		80	65	J3	80	20	-45	80	0	0.20	0.50	1.00	1.70	52.33	III	1.8
33.157771	75.00157	L103		40	70	J2	80	50	-20	40	-40	0.30	0.40	1.00	1.70	52.75	III	1.8
33.157771	75.00157	L103		80	80	J3	80	50	-30	80	0	0.25	0.50	0.50	1.25	52.75	III	1.8
33.157771	75.00157	L103		30	130	J4	80	50	-80	30	-50	0.20	0.30	1.00	1.50	52.75	III	1.8
33.155529	75.702246	L105		75	55	J3	80	55	0	75	-5	0.50	0.50	1.00	2.00	47.67	III	1.8
33.153402	75.710224	L108		55	70	J2	80	30	-40	55	-25	0.20	0.50	1.00	1.70	48.00	III	1.8
33.152577	75.711449	L109		65	90	J2	80	55	-35	65	-15	0.20	0.50	1.00	1.70	49.75	III	1.8
33.152577	75.711449	L109		75	45	J3	80	55	10	75	-5	0.40	0.50	0.50	1.40	49.75	III	1.8
33.149171	75.716763	L113		75	20	J2	80	35	15	75	-5	0.30	0.50	0.50	1.30	61.83	II	1.2
33.149171	75.716763	L113		60	90	J3	80	35	-55	60	-20	0.20	0.50	0.80	1.50	61.83	II	1.2
33.148959	75.71873	L114		55	15	J2	80	40	25	55	-25	0.25	0.50	1.00	1.75	58.25	III	1.8
33.152158	75.730446	L121		60	340	J1	80	320	-20	60	-20	0.30	0.50	1.00	1.80	59.67	III	1.8
33.152158	75.730446	L121		85	260	J3	80	320	60	85	5	0.20	0.50	1.00	1.70	59.67	III	1.8
33.15201	75.748057	L130		90		J1	80	75	75	90	10	0.20	0.50	0.50	1.20	60.00	II	1.2
33.15201	75.748057	L130		90		J2	80	75	75	90	10	0.20	0.50	1.00	1.70	60.00	II	1.2
33.150923	75.758523	L131		70	90	J2	80	75	-15	70	-10	0.30	0.50	0.70	1.50	60.00	II	1.2
33.150923	75.758523	L131		65	40	J3	80	75	35	65	-15	0.20	0.50	0.50	1.20	60.00	II	1.2
33.14903	75.749322	L132		65	100	J1	80	60	-40	65	-15	0.20	0.50	1.00	1.70	54.67	III	1.8
33.14903	75.749322	L132		45	140	J2	80	60	-80	45	-35	0.20	0.50	0.50	1.20	54.67	III	1.8
33.14903	75.749322	L132		85	80	J3	80	60	-20	85	5	0.30	0.50	1.00	1.80	54.67	III	1.8
33.141364	75.752038	L138		55	20	J2	80	50	30	55	-25	0.25	0.50	0.80	1.55	67.00	II	1.2
33.140342	75.753379	L139		60	55	J2	80	40	-15	60	-20	0.30	0.50	1.00	1.80	50.00	III	1.8
33.139797	75.754368	L140		75	50	J2	80	45	-5	75	-5	0.50	0.50	1.00	2.00	53.67	III	1.8
33.139309	75.7552	L141		75	50	J3	80	30	-20	75	-5	0.30	0.50	1.00	1.80	54.00	III	1.8
33.138819	75.756441	L142		90	50	J3	80	20	-30	90	10	0.25	0.50	1.00	1.75	63.33	II	1.2
33.144899	75.771077	L153		80	70	J3	80	35	-35	80	0	0.20	0.50	1.00	1.70	48.00	III	1.8
33.145329	75.772722	L154		40	70	J2	80	15	-55	40	-40	0.20	0.40	1.00	1.60	60.00	II	1.2
33.145793	75.774793	L156		80	15	J3	80	15	0	80	0	0.50	0.50	1.00	2.00	60.50	II	1.2
33.145724	75.77636	L157		50	45	J2	80	10	-35	50	-30	0.20	0.50	0.50	1.20	53.00	III	1.8
33.145724	75.77636	L157		80	55	J4	80	10	-45	80	0	0.20	0.50	0.80	1.50	53.00	III	1.8
33.145436	75.786789	L162		65	65	J2	80	20	-45	65	-15	0.20	0.50	1.00	1.70	53.33	III	1.8

33.145775	75.796005	L165		40	320	J2	80	330	10	40	-40	0.40	0.40	1.00	1.80	46.00	III	1.8
33.146523	75.798642	L167		85	170	J1	80	250	80	85	5	0.20	0.50	1.00	1.70	56.00	III	1.8
33.146523	75.798642	L167		30	335	J2	80	250	-85	30	-50	0.20	0.30	1.00	1.50	56.00	III	1.8
33.146523	75.798642	L167		80	270	J3	80	250	-20	80	0	0.30	0.50	1.00	1.80	56.00	III	1.8
33.148597	75.798573	L168		60	345	J1	80	280	-65	60	-20	0.20	0.50	0.80	1.50	69.75	II	1.2
33.148597	75.798573	L168		70	210	J2	80	280	70	70	-10	0.20	0.50	1.00	1.70	69.75	II	1.2
33.148597	75.798573	L168		85	270	J3	80	280	10	85	5	0.40	0.50	0.70	1.60	69.75	II	1.2
33.149917	75.798645	L169		65	335	J1	80	260	-75	65	-15	0.20	0.50	0.70	1.40	62.67	II	1.2
33.149917	75.798645	L169		80	225	J2	80	260	35	80	0	0.20	0.50	1.00	1.70	62.67	II	1.2
33.151113	75.798456	L170		65	335	J1	80	245	-90	65	-15	0.20	0.50	1.00	1.70	68.33	II	1.2
33.151113	75.798456	L170		40	250	J2	80	245	-5	40	-40	0.50	0.40	1.00	1.90	68.33	II	1.2
33.151113	75.798456	L170		80	260	J3	80	245	-15	80	0	0.30	0.50	0.80	1.60	68.33	II	1.2
33.152203	75.798281	L171		60	350	J1	80	265	-85	60	-20	0.20	0.50	1.00	1.70	49.67	II	1.2
33.152203	75.798281	L171		45	230	J2	80	265	35	45	-35	0.20	0.50	1.00	1.70	49.67	II	1.2
33.154328	75.798177	L173		60	340	J1	80	265	-75	60	-20	0.20	0.50	0.50	1.20	49.67	II	1.2
33.154328	75.798177	L173		75	250	J2	80	265	15	75	-5	0.30	0.50	1.00	1.80	49.67	II	1.2
33.155484	75.79793	L174		60	325	J1	80	255	-70	60	-20	0.20	0.50	1.00	1.70	47.67	II	1.2
33.155484	75.79793	L174		55	215	J2	80	255	40	55	-25	0.20	0.50	1.00	1.70	47.67	II	1.2
33.157951	75.798662	L176	70	70	340	J1	80	255	-85	70	-10	0.20	0.50	1.00	1.70	38.00	IV	2.4
33.157951	75.798662	L176	290	60	190	J2	80	255	65	60	-20	0.20	0.50	1.00	1.70	38.00	IV	2.4
33.159042	75.798824	L177	50	65	320	J1	80	310	-10	65	-15	0.40	0.50	1.00	1.90	54.00	III	1.8
33.159042	75.798824	L177	320	65	220	J2	80	310	90	65	-15	0.20	0.50	1.00	1.70	54.00	III	1.8
33.160223	75.799552	L178	50	45	325	J1	80	305	-20	45	-35	0.30	0.40	1.00	1.70	53.67	III	1.8
33.160223	75.799552	L178	310	70	225	J2	80	305	80	70	-10	0.20	0.50	1.00	1.70	53.67	III	1.8
33.162745	75.80161	L181	55	60	320	J1	80	310	-10	60	-20	0.40	0.50	0.50	1.40	50.00	III	1.8
33.162745	75.80161	L181	305	55	220	J2	80	310	90	55	-25	0.20	0.50	0.80	1.50	50.00	III	1.8
33.163724	75.80234	L182	45	60	320	J1	80	315	-5	60	-20	0.50	0.50	1.00	2.00	61.67	II	1.2
33.164369	75.803146	L183	65	65	335	J1	80	315	-20	65	-15	0.30	0.50	0.80	1.60	76.00	II	1.2
33.164369	75.803146	L183	325	85	230	J2	80	315	85	85	5	0.20	0.50	1.00	1.70	76.00	II	1.2
33.165377	75.80395	L184	55	57	335	J1	80	310	-25	57	-23	0.25	0.50	1.00	1.75	61.17	II	1.2
33.166219	75.804149	L185	60	55	330	J1	80	310	-20	55	-25	0.30	0.50	1.00	1.80	64.75	II	1.2
33.167513	75.805248	L186	40	60	305	J1	80	300	-5	60	-20	0.50	0.50	1.00	2.00	60.75	II	1.2
33.167513	75.805248	L186	305	80	210	J2	80	300	90	80	0	0.20	0.50	1.00	1.70	60.75	II	1.2
33.173814	75.809621	L191	40	65	305	J1	80	310	5	65	-15	0.50	0.50	0.70	1.70	65.33	II	1.2

33.174742	75.810071	L192	40	60	315	J1	80	290	-25	60	-20	0.25	0.50	0.80	1.55	60.67	II	1.2
33.176175	75.810596	L193	10	65	280	J1	80	290	10	65	-15	0.40	0.50	1.00	1.90	65.00	II	1.2
33.176175	75.810596	L193	300	45	210	J2	80	290	80	45	-35	0.20	0.50	0.50	1.20	65.00	II	1.2
33.176175	75.810596	L193	140	80	230	J3	80	290	60	80	0	0.20	0.50	1.00	1.70	65.00	II	1.2
33.17725	75.810985	L194	40	75	310	J1	80	300	-10	75	-5	0.40	0.50	1.00	1.90	61.33	II	1.2
33.17725	75.810985	L194	325	75	235	J2	80	300	65	75	-5	0.20	0.50	1.00	1.70	61.33	II	1.2
33.126513	75.619017	L196	300	60	30	J2	80	20	-10	60	-20	0.40	0.50	1.00	1.90	53.00	III	1.8
33.126467	75.621455	L198	320	50	55	J2	80	25	-30	50	-30	0.20	0.50	0.80	1.50	51.00	III	1.8
33.126467	75.621455	L198	100	40	10	J4	80	25	15	40	-40	0.30	0.50	1.00	1.80	51.00	III	1.8
33.126467	75.621455	L199	100	40	10	J1	80	25	15	40	-40	0.30	0.50	1.00	1.80	51.00	III	1.8
33.126467	75.621455	L199	100	40	10	J2	80	25	15	40	-40	0.30	0.50	0.80	1.60	51.00	III	1.8
33.126467	75.621455	L199	100	40	10	J3	80	25	15	40	-40	0.30	0.50	1.00	1.80	51.00	III	1.8
33.126412	75.625133	L200	285	80	15	J2	80	20	5	80	0	0.50	0.50	0.80	1.80	61.25	II	1.2
33.126412	75.625133	L200	295	45	25	J4	80	20	-5	45	-35	0.50	0.50	0.70	1.70	61.25	II	1.2
33.124325	75.636263	L201	330	55	60	J2	80	20	-40	55	-25	0.20	0.50	0.80	1.50	59.25	III	1.8
33.131925	75.55004	L224	230	50	330	J2	80	320	-10	50	-30	0.40	0.50	1.00	1.90	64.67	II	1.2
33.131925	75.55004	L224	30	75	280	J3	80	320	40	75	-5	0.20	0.50	1.00	1.70	64.67	II	1.2
33.1353	75.556028	L228	270	40	360	J2	80	350	-10	40	-40	0.40	0.50	0.50	1.40	66.33	II	1.2
33.1353	75.556028	L228	360	89	270	J3	80	350	80	89	9	0.20	0.50	1.00	1.70	66.33	II	1.2
33.181962	75.807401	L231	325	40	240	J2	80	290	50	40	-40	0.20	0.40	1.00	1.60	33.67	IV	2.4
33.183963	75.807797	L232	210	60	305	J2	80	290	-15	60	-20	0.30	0.50	1.00	1.80	50.33	III	1.8
33.183963	75.807797	L232	290	80	200	J3	80	290	90	80	0	0.20	0.50	0.70	1.40	50.33	III	1.8
33.18551	75.807523	L233	360	43	270	J1	80	270	0	43	-37	0.50	0.40	1.00	1.90	61.67	II	1.2
33.18551	75.807523	L233	295	45	300	J3	80	270	-30	45	-35	0.25	0.40	1.00	1.65	61.67	II	1.2
33.187785	75.807167	L235	340	50	250	J1	80	270	20	50	-30	0.30	0.50	1.00	1.80	69.75	II	1.2
33.187785	75.807167	L235	250	70	340	J4	80	270	-70	70	-10	0.20	0.50	0.80	1.50	69.75	II	1.2
33.197417	75.80223	L238	360	35	270	J1	80	240	-30	35	-45	0.25	0.30	1.00	1.55	64.00	II	1.2
33.197938	75.80125	L239	20	53	290	J1	80	210	-80	53	-27	0.20	0.50	1.00	1.70	62.67	II	1.2
33.197938	75.80125	L239	275	80	175	J3	80	210	35	80	0	0.20	0.50	0.70	1.40	62.67	II	1.2
33.198627	75.800468	L240	360	50	270	J1	80	260	-10	50	-30	0.40	0.50	1.00	1.90	61.75	II	1.2
33.198627	75.800468	L240	95	67	185	J3	80	260	75	67	-13	0.20	0.50	1.00	1.70	61.75	II	1.2
33.198627	75.800468	L240	115	85	195	J4	80	260	65	85	5	0.20	0.50	0.50	1.20	61.75	II	1.2
33.200047	75.799583	L241	360	45	270	J1	80	235	-35	45	-35	0.20	0.40	1.00	1.60	52.00	III	1.8
33.200047	75.799583	L241	305	70	210	J3	80	235	25	70	-10	0.25	0.50	0.80	1.55	52.00	III	1.8

33.200483	75.799057	L242	30	55	300	J1	80	220	-80	55	-25	0.20	0.50	1.00	1.70	77.67	II	1.2
33.200483	75.799057	L242	275	55	185	J2	80	220	35	55	-25	0.20	0.50	1.00	1.70	77.67	II	1.2
33.200483	75.799057	L242	155	80	250	J3	80	220	-30	80	0	0.25	0.50	0.70	1.45	77.67	II	1.2
33.202068	75.797967	L244	70	50	290	J1	80	260	-30	50	-30	0.25	0.50	1.00	1.75	71.67	II	1.2
33.202068	75.797967	L244	120	60	210	J3	80	260	50	60	-20	0.20	0.50	1.00	1.70	71.67	II	1.2
33.20293	75.797857	L245	20	55	290	J1	80	250	-40	55	-25	0.20	0.50	1.00	1.70	68.67	II	1.2
33.20293	75.797857	L245	105	75	195	J3	80	250	55	75	-5	0.20	0.50	0.80	1.50	68.67	II	1.2
33.221722	75.795562	L257	85	55	175	J1	80	260	85	55	-25	0.20	0.50	1.00	1.70	58.67	III	1.8
33.221722	75.795562	L257	200	55	290	J3	80	260	-30	55	-25	0.25	0.50	1.00	1.75	58.67	III	1.8
33.224042	75.792543	L259	140	70	220	J2	80	240	20	70	-10	0.30	0.50	0.80	1.60	68.33	II	1.2
33.225127	75.791738	L260	25	46	305	J1	80	240	-65	46	-34	0.20	0.50	1.00	1.70	53.67	III	1.8
33.225127	75.791738	L260	260	53	160	J2	80	240	80	53	-27	0.20	0.50	1.00	1.70	53.67	III	1.8
33.225978	75.790982	L261	25	46	305	J1	80	240	-65	46	-34	0.20	0.50	1.00	1.70	53.67	III	1.8
33.225978	75.790982	L261	260	53	160	J2	80	240	80	53	-27	0.20	0.50	1.00	1.70	53.67	III	1.8
33.228793	75.788897	L263	360	40	270	J1	80	220	-50	40	-40	0.20	0.40	1.00	1.60	60.33	II	1.2
33.228793	75.788897	L263	70	80	160	J2	80	220	60	80	0	0.20	0.50	0.70	1.40	60.33	II	1.2
33.229868	75.787828	L264	35	70	320	J1	80	300	-20	70	-10	0.30	0.50	0.80	1.60	52.08	III	1.8
33.229868	75.787828	L264	140	55	240	J2	80	300	60	55	-25	0.20	0.50	1.00	1.70	52.08	III	1.8
33.26312	75.773583	L285	345	45	235	J1	80	245	10	45	-35	0.40	0.40	1.00	1.80	58.67	III	1.8
33.23005	75.787977	L286	215	70	320	J1	80	245	-75	70	-10	0.20	0.50	0.80	1.50	55.33	III	1.8
33.23005	75.787977	L286	140	55	240	J2	80	245	5	55	-25	0.50	0.50	1.00	2.00	55.33	III	1.8
33.200797	75.79871	L287	210	55	300	J1	80	220	-80	55	-25	0.20	0.50	1.00	1.70	79.33	II	1.2
33.200797	75.79871	L287	275	55	185	J2	80	220	35	55	-25	0.20	0.50	1.00	1.70	79.33	II	1.2
33.200797	75.79871	L287	155	80	250	J3	80	220	-30	80	0	0.25	0.50	0.70	1.45	79.33	II	1.2
33.164832	75.803473	L288	245	65	335	J1	80	320	-15	65	-15	0.30	0.50	1.00	1.80	65.67	II	1.2
33.164832	75.803473	L288	325	85	230	J2	80	320	90	85	5	0.20	0.50	0.50	1.20	65.67	II	1.2
33.14661	75.798697	L289		85	170	J1	80	250	80	85	5	0.20	0.50	0.50	1.20	69.67	II	1.2
33.14661	75.798697	L289		30	335	J2	80	250	-85	30	-50	0.20	0.30	1.00	1.50	69.67	II	1.2
33.14661	75.798697	L289		80	270	J3	80	250	-20	80	0	0.30	0.50	0.70	1.50	69.67	II	1.2
33.138787	75.75647	L291		90	50	J3	80	15	-50	90	10	0.20	0.50	0.50	1.20	63.00	II	1.2
33.151578	75.74831	L292		70	90	J2	80	75	-90	70	-10	0.20	0.50	0.80	1.50	61.67	II	1.2
33.151578	75.74831	L292		65	40	J3	80	75	-40	65	-15	0.20	0.50	1.00	1.70	61.67	II	1.2
33.155957	75.701807	L293		75	55	J3	80	45	-55	75	-5	0.20	0.50	0.80	1.50	54.33	III	1.8
33.140167	75.558975	L294		60	15	J2	80	315	-15	60	-20	0.30	0.50	1.00	1.80	57.00	III	1.8

33.153898	75.682033	L295	80	65	350	J2	80	330	-20	65	-15	0.30	0.50	1.00	1.80	48.75	III	1.8
33.153898	75.682033	L295	35	55	310	J4	80	330	20	55	-25	0.30	0.50	1.00	1.80	48.75	III	1.8
33.132483	75.655035	L305	30	50	310	J3	80	330	20	50	-30	0.30	0.50	1.00	1.80	57.00	III	1.8
33.127823	75.643645	L307	220	68	315	J2	80	300	-15	68	-12	0.30	0.50	1.00	1.80	49.33	III	1.8
33.129343	75.643285	L308	360	65	270	J3	80	300	30	65	-15	0.25	0.50	1.00	1.75	48.67	III	1.8
33.129343	75.643285	L308	345	60	255	J4	80	300	45	60	-20	0.20	0.50	1.00	1.70	48.67	III	1.8
33.130203	75.644507	L309	50	54	315	J2	80	345	30	54	-26	0.25	0.50	1.00	1.75	60.50	II	1.2
33.130203	75.644507	L309	30	60	300	J4	80	345	45	60	-20	0.20	0.50	1.00	1.70	60.50	II	1.2
33.132308	75.651418	L312	20	80	285	J2	80	340	55	80	0	0.20	0.50	0.70	1.40	39.17	IV	2.4
33.133633	75.654475	L313	360	57	270	J2	80	315	45	57	-23	0.20	0.50	1.00	1.70	44.25	III	1.8
33.135058	75.65626	L315	345	65	255	J2	80	330	75	65	-15	0.20	0.50	1.00	1.70	36.00	IV	2.4
33.135492	75.65678	L316	35	65	310	J2	80	310	0	65	-15	0.50	0.50	1.00	2.00	41.33	III	1.8
33.136372	75.657562	L317	30	75	300	J2	80	300	0	75	-5	0.50	0.50	0.80	1.80	40.33	III	1.8
33.137852	75.658207	L318	180	75	270	J2	80	295	25	75	-5	0.25	0.50	0.80	1.55	61.33	II	1.2
33.13899	75.658658	L319	180	75	270	J2	80	295	25	75	-5	0.25	0.50	0.80	1.55	61.33	II	1.2
33.141988	75.660102	L321	30	55	310	J2	80	300	-10	55	-25	0.40	0.50	1.00	1.90	39.50	IV	2.4
33.149162	75.72308	L323	25	85	305	J3	80	330	25	85	5	0.25	0.50	0.50	1.25	53.50	III	1.8
33.149162	75.72308	L323	55	55	325	J4	80	330	5	55	-25	0.50	0.50	1.00	2.00	53.50	III	1.8
33.148618	75.723726	L326	25	85	305	J3	80	330	25	85	5	0.25	0.50	0.50	1.25	54.50	III	1.8
33.148618	75.723726	L326	55	55	325	J4	80	330	5	55	-25	0.50	0.50	1.00	2.00	54.50	III	1.8

Annexue-VI: Kinematic Analysis for Wedge failure

Latitude	Longitude	Loc	Name	As	Bs	Ai	Bi	Ai-As	Bi	Bi-Bs	F1	F2	F3	Rating	RMRb	RMR Class	
33.1442	75.66132	L2	J1-J3	314	80	249	50	-65	50	-30	0.20	0.50	1.00	1.70	55.00	III	1.8
33.1541	75.68111	L4	J1-J3	345	80	265	43	-80	43	-37	0.20	0.40	1.00	1.60	55.83	III	1.8
33.1541	75.68111	L4	J2-J3	345	80	28	58	43	58	-22	0.20	0.50	1.00	1.70	55.83	III	1.8
33.2644	75.77313	L8	J1-J3	255	80	337	35	82	35	-45	0.20	0.30	1.00	1.50	53.00	III	1.8
33.2644	75.77313	L8	J1-J2	255	80	224	31	-31	31	-49	0.20	0.30	1.00	1.50	53.00	III	1.8
33.2629	75.77367	L9	J1-J2	265	80	234	28	-31	28	-52	0.20	0.30	1.00	1.50	51.33	III	1.8
33.2629	75.77367	L9	J1-J3	265	80	341	31	76	31	-49	0.20	0.30	1.00	1.50	51.33	III	1.8
33.2629	75.77367	L10	J1-J2	260	80	234	28	-26	28	-52	0.25	0.30	1.00	1.55	51.33	III	1.8
33.2629	75.77367	L10	J1-J3	260	80	341	31	81	31	-49	0.20	0.30	1.00	1.50	51.33	III	1.8
33.2587	75.77409	L11	J1-J3	295	80	2	14	67	14	-66	0.20	0.20	1.00	1.40	43.33	III	1.8
33.2587	75.77409	L11	J1-J2	295	80	24	0	89	0	-80	0.20	0.20	1.00	1.40	43.33	III	1.8
33.2534	75.77571	L17	J1-J2	250	80	236	55	-14	55	-25	0.30	0.50	1.00	1.80	50.33	III	1.8
33.2522	75.77625	L18	J1-J2	255	80	269	49	14	49	-31	0.30	0.50	1.00	1.80	66.00	II	1.2
33.2504	75.77755	L19	J1-J3	255	80	235	42	-20	42	-38	0.25	0.40	1.00	1.65	50.00	III	1.8
33.248	75.77926	L20	J1-J3	240	80	275	65	35	65	-15	0.20	0.50	1.00	1.70	59.75	III	1.8
33.2472	75.77983	L21	J1-J3	240	80	246	60	6	60	-20	0.40	0.50	1.00	1.90	68.00	II	1.2
33.2472	75.77983	L21	J1-J4	240	80	216	56	-24	56	-24	0.25	0.50	1.00	1.75	68.00	II	1.2
33.2472	75.77983	L21	J3-J4	240	80	230	33	-10	33	-47	0.30	0.30	1.00	1.60	68.00	II	1.2
33.1336	75.5414	L30	J2-J3	20	80	326	32	-54	32	-48	0.20	0.30	1.00	1.50	40.33	III	1.8
33.1367	75.55357	L36	J3-J2	300	80	8	42	68	42	-38	0.20	0.40	1.00	1.60	50.67	III	1.8
33.1367	75.55357	L36	J1-J3	300	80	221	41	-79	41	-39	0.20	0.40	1.00	1.60	50.67	III	1.8
33.1373	75.55615	L37	J1-J2	300	80	248	31	-52	31	-49	0.20	0.30	1.00	1.50	47.33	III	1.8
33.1394	75.55834	L39	J2-J3	315	80	22	59	67	59	-21	0.20	0.50	1.00	1.70	57.00	III	1.8
33.1424	75.56294	L42	J3-J2	320	80	21	54	61	54	-26	0.20	0.50	1.00	1.70	65.00	II	1.2
33.1434	75.56434	L43	J2-J3	320	80	31	29	71	29	-51	0.20	0.30	1.00	1.50	59.00	III	1.8
33.1439	75.56499	L44	J2-J3	320	80	32	29	72	29	-51	0.20	0.30	1.00	1.50	60.67	II	1.2
33.1445	75.56704	L45	J2-J3	350	80	7	51	17	51	-29	0.30	0.50	1.00	1.80	60.67	II	1.2
33.136	75.58343	L55	J1-J2	350	80	302	54	-48	54	-26	0.20	0.50	1.00	1.70	61.67	II	1.2
33.136	75.58343	L55	J2-J3	350	80	43	71	53	71	-9	0.20	0.50	0.80	1.50	61.67	II	1.2
33.136	75.58343	L55	J1-J3	350	80	3	40	13	40	-40	0.40	0.40	1.00	1.80	61.67	II	1.2
33.1361	75.58919	L57	J2-J3	340	80	19	58	39	58	-22	0.20	0.50	1.00	1.70	60.67	II	1.2

33.1329	75.60071	L63	J1-J2	350	80	262	1	-88	1	-79	0.20	0.20	1.00	1.40	58.33	III	1.8
33.1329	75.60071	L63	J1-J3	350	80	348	34	-2	34	-46	0.50	0.30	1.00	1.80	58.33	III	1.8
33.1596	75.6961	L101	J1-J2	25	80	56	34	31	34	-46	0.20	0.30	1.00	1.50	58.67	III	1.8
33.1593	75.69799	L102	J2-J3	20	80	350	53	-30	53	-27	0.25	0.50	1.00	1.75	52.33	III	1.8
33.1578	75.00157	L103	J1-J2	50	80	75	39	25	39	-41	0.25	0.40	1.00	1.65	52.75	III	1.8
33.1578	75.00157	L103	J2-J4	50	80	117	29	67	29	-51	0.20	0.30	1.00	1.50	52.75	III	1.8
33.1578	75.00157	L103	J1-J3	50	80	58	79	8	79	-1	0.40	0.50	0.80	1.70	59.85	III	1.8
33.1572	75.70045	L104	J1-J2	55	80	65	54	10	54	-26	0.40	0.50	1.00	1.90	59.85	III	1.8
33.1572	75.70045	L104	J1-J3	55	80	60	74	5	74	-6	0.40	0.50	0.80	1.70	59.85	III	1.8
33.1572	75.70045	L104	J2-J3	55	80	343	11	-72	11	-69	0.20	0.20	1.00	1.40	59.85	III	1.8
33.1555	75.70225	L105	J1-J2	55	80	44	74	-11	74	-6	0.30	0.50	0.80	1.60	47.67	III	1.8
33.1555	75.70225	L105	J3-J2	55	80	131	38	76	38	-42	0.20	0.40	1.00	1.60	47.67	III	1.8
33.1534	75.71022	L108	J2-J3	30	80	59	53	29	53	-27	0.25	0.50	1.00	1.75	48.00	III	1.8
33.1526	75.71145	L109	J1-J2	55	80	39	53	-16	53	-27	0.30	0.50	1.00	1.80	49.75	III	1.8
33.1526	75.71145	L109	J2-J3	55	80	91	64	36	64	-16	0.20	0.50	1.00	1.70	49.75	III	1.8
33.1526	75.71145	L109	J1-J3	55	80	358	68	-57	68	-12	0.20	0.50	1.00	1.70	49.75	III	1.8
33.1492	75.71676	L113	J2-J3	55	80	83	59	28	59	-21	0.25	0.50	1.00	1.75	61.83	II	1.2
33.149	75.71873	L114	J1-J2	40	80	334	47	-66	47	-33	0.20	0.50	1.00	1.70	58.25	III	1.8
33.149	75.71873	L114	J2-J3	40	80	77	33	37	33	-47	0.20	0.30	1.00	1.50	58.25	III	1.8
33.1522	75.73045	L121	J1-J3	320	80	341	58	21	58	-22	0.25	0.50	1.00	1.75	59.67	III	1.8
33.1509	75.75852	L131	J1-J2	75	80	24	49	-51	49	-31	0.20	0.50	1.00	1.70	60.00	II	1.2
33.1509	75.75852	L131	J2-J3	75	80	49	64	-26	64	-16	0.25	0.50	1.00	1.75	60.00	II	1.2
33.1509	75.75852	L131	J3-J1	75	80	6	60	-69	60	-20	0.20	0.50	1.00	1.70	60.00	II	1.2
33.1414	75.75204	L138	J1-J2	50	80	32	54	-18	54	-26	0.30	0.50	1.00	1.80	67.00	IV	2.4
33.1403	75.75338	L139	J1-J2	40	80	67	59	27	59	-21	0.25	0.50	1.00	1.75	50.00	III	1.8
33.1398	75.75437	L140	J1-J2	45	80	114	57	69	57	-23	0.20	0.50	1.00	1.70	53.67	III	1.8
33.1393	75.7552	L141	J2-J3	30	80	332	39	-58	39	-41	0.20	0.40	1.00	1.60	54.00	III	1.8
33.1388	75.75644	L142	J2-J3	20	80	320	49	-60	49	-31	0.20	0.50	1.00	1.70	63.33	II	1.2
33.1449	75.77108	L153	J2-J3	35	80	357	58	-38	58	-22	0.20	0.50	1.00	1.70	48.00	III	1.8
33.1453	75.77272	L154	J1-J2	15	80	51	37	36	37	-43	0.20	0.40	1.00	1.60	60.00	II	1.2
33.1458	75.77479	L156	J2-J3	15	80	314	70	-61	70	-10	0.20	0.50	1.00	1.70	60.50	II	1.2
33.1457	75.77636	L157	J1-J2	10	80	41	50	31	50	-30	0.20	0.50	1.00	1.70	53.00	III	1.8
33.1457	75.77636	L157	J1-J4	10	80	1	73	-9	73	-7	0.40	0.50	0.80	1.70	53.00	III	1.8
33.1458	75.78415	L161	J2-J3	10	80	317	59	-53	59	-21	0.20	0.50	1.00	1.70	56.00	III	1.8

33.1454	75.78679	L162	J1-J2	20	80	44	62	24	62	-18	0.25	0.50	1.00	1.75	53.33	III	1.8
33.1458	75.79601	L165	J2-J3	330	80	346	36	16	36	-44	0.30	0.40	1.00	1.70	46.00	III	1.8
33.1465	75.79864	L167	J1-J3	250	80	236	78	-14	78	-2	0.30	0.50	0.80	1.60	56.00	III	1.8
33.1486	75.79857	L168	J1-J2	280	80	283	38	3	38	-42	0.50	0.40	1.00	1.90	69.75	II	1.2
33.1486	75.79857	L168	J1-J3	280	80	351	59	71	59	-21	0.20	0.50	1.00	1.70	69.75	II	1.2
33.1499	75.79865	L169	J1-J2	260	80	297	59	37	59	-21	0.20	0.50	1.00	1.70	62.67	II	1.2
33.1511	75.79846	L170	J1-J2	245	80	267	38	22	38	-42	0.25	0.40	1.00	1.65	68.33	II	1.2
33.1511	75.79846	L170	J2-J3	245	80	171	9	-74	9	-71	0.20	0.20	1.00	1.40	74.00	II	1.2
33.1522	75.79828	L171	J1-J2	265	80	280	32	15	32	-48	0.30	0.30	1.00	1.60	49.67	III	1.8
33.1543	75.79818	L173	J1-J2	265	80	315	57	50	57	-23	0.20	0.50	1.00	1.70	49.67	III	1.8
33.1555	75.79793	L174	J1-J2	255	80	266	42	11	42	-38	0.30	0.40	1.00	1.70	47.67	III	1.8
33.159	75.79882	L177	J1-J2	310	80	270	54	-40	54	-26	0.20	0.50	1.00	1.70	54.00	III	1.8
33.1602	75.79955	L178	J1-J2	305	80	296	41	-9	41	-39	0.40	0.40	1.00	1.80	53.67	III	1.8
33.1627	75.80161	L181	J1-J2	310	80	266	44	-44	44	-36	0.20	0.40	1.00	1.60	50.00	III	1.8
33.1637	75.80234	L182	J1-J2	315	80	301	59	-14	59	-21	0.30	0.50	1.00	1.80	61.67	II	1.2
33.1644	75.80315	L183	J1-J2	315	80	309	63	-6	63	-17	0.40	0.50	1.00	1.90	76.00	II	1.2
33.1654	75.80395	L184	J1-J2	310	80	264	27	-46	27	-53	0.20	0.30	1.00	1.50	61.17	II	1.2
33.1662	75.80415	L185	J1-J2	310	80	273	38	-37	38	-42	0.20	0.40	1.00	1.60	64.75	II	1.2
33.1675	75.80525	L186	J1-J2	300	80	284	58	-16	58	-22	0.30	0.50	1.00	1.80	60.75	II	1.2
33.1738	75.80962	L191	J1-J2	310	80	294	64	-16	64	-16	0.30	0.50	1.00	1.80	65.33	II	1.2
33.1747	75.81007	L192	J1-J2	290	80	241	25	-49	25	25	0.20	0.25	1.00	1.45	60.67	II	1.2
33.1762	75.8106	L193	J1-J2	290	80	217	45	-73	45	-35	0.20	0.40	1.00	1.60	65.00	II	1.2
33.1762	75.8106	L193	J1-J3	290	80	300	63	10	63	-17	0.40	0.50	1.00	1.90	65.00	II	1.2
33.1773	75.81099	L194	J1-J2	300	80	273	70	-27	70	-10	0.25	0.50	0.80	1.55	61.33	II	1.2
33.1265	75.61902	L196	J2-J3	20	80	354	54	-26	54	-26	0.25	0.50	1.00	1.75	53.00	III	1.8
33.1265	75.62146	L198	J2-J3	25	80	48	49	23	49	-31	0.25	0.50	1.00	1.75	51.00	III	1.8
33.1265	75.62146	L198	J2-J4	25	80	53	31	28	31	-49	0.25	0.30	1.00	1.55	51.00	III	1.8
33.1265	75.62146	L198	J3-J4	25	80	10	39	-15	39	-41	0.30	0.40	1.00	1.70	51.00	III	1.8
33.1264	75.62513	L200	J1-J2-J4	20	80	102	11	82	11	-69	0.20	0.20	1.00	1.40	61.25	II	1.8
33.1264	75.62513	L200	J3-J4	20	80	3	42	-17	42	-38	0.30	0.40	1.00	1.70	61.25	II	1.8
33.1243	75.63626	L201	J2-J3	20	80	10	42	-10	42	-38	0.40	0.40	1.00	1.80	59.25	III	1.8
33.1243	75.63626	L201	J2-J4	20	80	21	47	1	47	-33	0.50	0.50	1.00	2.00	59.25	III	1.8
33.1243	75.63626	L201	J3-J4	20	80	8	50	-12	50	-30	0.40	0.50	1.00	1.90	59.25	III	1.8
33.1319	75.55004	L224	J1-J2	320	80	263	23	-57	23	-57	0.20	0.25	1.00	1.45	64.67	II	1.2

33.1319	75.55004	L224	J2-J3	320	80	352	43	32	43	-37	0.20	0.40	1.00	1.60	64.67	II	1.2
33.1351	75.55425	L227	J2-J3	20	80	10	44	-10	44	-36	0.40	0.40	1.00	1.80	65.00	II	1.2
33.1353	75.55603	L228	J2-J3	350	80	359	39	9	39	-41	0.40	0.40	1.00	1.80	66.33	II	1.2
33.182	75.8074	L231	J2-J3	290	80	273	34	-17	34	-46	0.30	0.30	1.00	1.60	33.67	IV	2.4
33.184	75.8078	L232	J2-J3	290	80	325	47	35	47	-33	0.20	0.50	1.00	1.70	50.33	III	1.8
33.1855	75.80752	L233	J1-J3	270	80	277	42	7	42	-38	0.40	0.40	1.00	1.80	61.67	II	1.2
33.1878	75.80717	L235	J1-J3	270	80	239	49	-31	49	-31	0.20	0.50	1.00	1.70	69.75	II	1.2
33.1878	75.80717	L235	J1-J4	270	80	273	47	3	47	-33	0.50	0.50	1.00	2.00	69.75	II	1.2
33.1974	75.80223	L238	J1-J2	240	80	320	23	80	23	-57	0.20	0.25	1.00	1.45	64.00	II	1.2
33.1979	75.80125	L239	J1-J2	210	80	254	47	44	47	-33	0.20	0.50	1.00	1.70	62.67	II	1.2
33.1986	75.80047	L240	J1-J3	260	80	248	47	-12	47	-33	0.30	0.50	1.00	1.80	61.75	II	1.2
33.1986	75.80047	L240	J3-J4	260	80	279	49	19	49	-31	0.30	0.50	1.00	1.80	61.75	II	1.2
33.2	75.79958	L241	J1-J3	235	80	278	44	43	44	-36	0.20	0.50	1.00	1.70	52.00	III	1.8
33.2005	75.79906	L242	J1-J2	220	80	242	37	22	37	-43	0.25	0.40	1.00	1.65	77.67	II	1.2
33.2005	75.79906	L242	J2-J3	220	80	174	54	-46	54	-26	0.20	0.50	1.00	1.70	77.67	II	1.2
33.2021	75.79797	L244	J1-J3	260	80	262	45	2	45	-35	0.50	0.40	1.00	1.90	71.67	II	1.2
33.2029	75.79786	L245	J1-J3	250	80	265	52	15	52	-28	0.30	0.50	1.00	1.80	68.67	II	1.2
33.2217	75.79556	L257	J1-J3	260	80	232	37	-28	37	-43	0.25	0.40	1.00	1.65	58.67	III	1.8
33.224	75.79254	L259	J2-J3	240	80	217	70	-23	70	-10	0.25	0.50	0.80	1.55	68.33	II	1.2
33.2288	75.7889	L263	J1-J2	220	80	242	36	22	36	-44	0.25	0.40	1.00	1.65	60.33	II	1.2
33.2299	75.78783	L264	J1-J2	300	80	261	53	-39	53	-27	0.20	0.50	1.00	1.70	52.08	III	1.8
33.2299	75.78783	L264	J1-J3	300	80	10	59	70	59	-21	0.20	0.50	1.00	1.70	52.08	III	1.8
33.2631	75.77358	L285	J1-J2	245	80	162	16	-83	16	-64	0.20	0.20	1.00	1.40	58.67	III	1.8
33.2631	75.77358	L285	J2-J3	245	80	272	38	27	38	-42	0.25	0.40	1.00	1.65	58.67	III	1.8
33.2301	75.78798	L286	J1-J2	245	80	263	53	18	53	-27	0.30	0.50	1.00	1.80	55.33	III	1.8
33.2301	75.78798	L286	J2-J3	245	80	324	8	79	8	-72	0.20	0.20	1.00	1.40	55.33	III	1.8
33.2008	75.79871	L287	J1-J2	220	80	242	37	22	37	-43	0.25	0.40	1.00	1.65	65.67	II	1.2
33.2008	75.79871	L287	J2-J3	220	80	174	54	-46	54	-26	0.20	0.50	1.00	1.70	65.67	II	1.2
33.1648	75.80347	L288	J1-J2	320	80	311	62	-9	62	-18	0.40	0.50	1.00	1.90	69.67	II	1.2
33.1466	75.7987	L289	J1-J3	250	80	234	78	-16	78	-2	0.30	0.50	0.80	1.60	69.67	II	1.2
33.1458	75.78383	L290	J2-J3	10	80	314	59	-56	59	-21	0.20	0.50	1.00	1.70	54.67	III	1.8
33.1388	75.75647	L291	J2-J3	20	80	320	48	-60	48	-32	0.20	0.50	1.00	1.70	63.00	II	1.2
33.1516	75.74831	L292	J1-J2	75	80	27	50	-48	50	-30	0.20	0.50	1.00	1.70	61.67	II	1.2
33.1516	75.74831	L292	J2-J3	75	80	5	59	-70	59	-21	0.20	0.50	1.00	1.70	61.67	II	1.2

33.1516	75.74831	L292	J1-J3	75	80	51	64	-24	64	-16	0.25	0.50	1.00	1.75	61.67	II	1.2
33.156	75.70181	L293	J1-J2	55	80	131	39	76	39	-41	0.20	0.40	1.00	1.60	54.33	III	1.8
33.156	75.70181	L293	J1-J3	55	80	40	75	-15	75	-5	0.30	0.50	0.80	1.60	54.33	III	1.8
33.1402	75.55898	L294	J2-J3	315	80	22	59	67	59	-21	0.20	0.50	1.00	1.70	57.00	III	1.8
33.1539	75.68203	L295	J1-J2	330	80	272	23	-58	23	-57	0.20	0.25	1.00	1.45	48.75	III	1.8
33.1539	75.68203	L295	J2-J4	330	80	300	54	-30	54	-26	0.25	0.50	1.00	1.75	48.75	III	1.8
33.1539	75.68203	L295	J1-J4	330	80	255	39	-75	39	-41	0.20	0.40	1.00	1.60	48.75	III	1.8
33.1325	75.65504	L305	J1-J3	330	80	249	30	-81	30	-50	0.20	0.30	1.00	1.50	57.00	III	1.8
33.1325	75.65504	L305	J2-J3	330	80	313	50	-17	50	-30	0.30	0.50	1.00	1.80	57.00	III	1.8
33.1278	75.64365	L307	J1-J2	300	80	251	47	-49	47	-33	0.20	0.50	1.00	1.70	49.33	III	1.8
33.1293	75.64329	L308	J1-J3	300	80	229	58	-71	58	-22	0.20	0.50	1.00	1.70	48.67	III	1.8
33.1293	75.64329	L308	J1-J4	300	80	231	57	-69	57	-23	0.20	0.50	1.00	1.70	48.67	III	1.8
33.1293	75.64329	L308	J2-J3	300	80	336	40	36	40	-40	0.20	0.40	1.00	1.60	48.67	III	1.8
33.1293	75.64329	L308	J2-J4	300	80	324	32	24	32	-48	0.25	0.30	1.00	1.55	48.67	III	1.8
33.1302	75.64451	L309	J1-J2	345	80	285	49	-60	49	-31	0.20	0.50	1.00	1.70	60.50	II	1.2
33.1302	75.64451	L309	J1-J4	345	80	273	56	-72	56	-24	0.20	0.50	1.00	1.70	60.50	II	1.2
33.1302	75.64451	L309	J3-J2	345	80	4	36	3	36	-32	0.50	0.40	1.00	1.90	60.50	II	1.2
33.1302	75.64451	L309	J4-J3	345	80	9	39	24	39	-41	0.25	0.40	1.00	1.65	60.50	II	1.2
33.1302	75.64451	L309	J2-J4	345	80	348	48	3	48	-32	0.50	0.50	1.00	2.00	60.50	II	1.2
33.1355	75.65678	L316	J1-J2	310	80	240	36	-70	36	-44	0.20	0.40	1.00	1.60	41.33	III	1.8
33.1355	75.65678	L316	J2-J3	310	80	38	2	88	2	-78	0.20	0.20	1.00	1.40	41.33	III	1.8
33.1364	75.65756	L317	J1-J2	300	80	236	59	-64	59	-21	0.20	0.50	1.00	1.70	40.33	III	1.8
33.1364	75.65756	L317	J2-J3	300	80	27	9	87	9	-71	0.20	0.20	1.00	1.40	40.33	III	1.8
33.1379	75.65821	L318	J1-J2	295	80	269	75	-26	75	-5	0.50	0.50	0.80	1.80	61.33	II	1.2
33.139	75.65866	L319	J1-J2	295	80	269	75	-26	75	-5	0.50	0.50	0.80	1.80	61.33	II	1.2
33.142	75.6601	L321	J1-J2	300	80	261	42	-39	42	-38	0.40	0.40	1.00	1.80	39.50	IV	2.4
33.1492	75.72308	L323	J1-J4	330	80	271	39	-59	39	-41	0.40	0.40	1.00	1.80	53.50	III	1.8
33.1492	75.72308	L323	J2-J3	330	80	29	49	59	49	-31	0.50	0.50	1.00	2.00	53.50	III	1.8
33.1492	75.72308	L323	J2-J4	330	80	9	45	39	45	-35	0.40	0.40	1.00	1.80	53.50	III	1.8
33.1492	75.72308	L323	J3-J4	330	80	32	28	62	28	-52	0.30	0.30	1.00	1.60	53.50	III	1.8
33.1486	75.72373	L326	J1-J4	330	80	271	39	-59	39	-41	0.40	0.40	1.00	1.80	53.50	III	1.8
33.1486	75.72373	L326	J2-J3	330	80	29	49	59	49	-31	0.50	0.50	1.00	2.00	53.50	III	1.8
33.1486	75.72373	L326	J2-J4	330	80	9	45	39	45	-35	0.40	0.40	1.00	1.80	53.50	III	1.8
33.1486	75.72373	L326	J3-J4	330	80	32	28	62	28	-52	0.30	0.30	1.00	1.60	53.50	III	1.8

Annexue-VII: Kinematic Analysis for Flexure Topple failure

Latitude	Longitude	Loc	Strike	DA	DD	Name	SDA	SDD	Aj-As+180	Bj	Bj+Bs	F1	F2	F3	Rating	RMRb	RMR Class	RMR Class
33.14418	75.66132	L2		70	185	J1	80	314	129	70	150	0.20	0.40	0.90	1.50	55.00	III	1.8
33.14418	75.66132	L2		35	60	J2	80	314	254	35	115	0.20	0.20	0.50	0.90	55.00	III	1.8
33.15409	75.68111	L4		45	245	J1	80	345	100	45	125	0.20	0.20	0.50	0.90	55.83	III	1.8
33.15409	75.68111	L4		60	50	J2	80	345	295	60	140	0.20	0.30	0.70	1.20	55.83	III	1.8
33.28044	75.76732	L6A		50	240	J1	80	345	105	50	130	0.20	0.20	0.50	0.90	70.00	III	1.8
33.28044	75.76732	L6A		58	75	J2	80	314	239	58	138	0.20	0.30	0.70	1.20	70.00	III	1.8
33.28044	75.76732	L6A		65	130	J3	80	314	184	65	145	0.20	0.40	0.90	1.50	70.00	II	1.2
33.28044	75.76732	L6A		58	40	J4	80	314	274	58	138	0.20	0.30	0.70	1.20	70.00	II	1.2
33.26511	75.77255	L7		50	100	J3	80	230	130	50	130	0.20	0.20	0.50	0.90	59.33	II	1.2
33.26436	75.77313	L8		60	155	J2	80	255	100	60	140	0.20	0.30	0.70	1.20	53.00	II	1.2
33.26436	75.77313	L8		45	20	J3	80	255	235	45	125	0.20	0.20	0.50	0.90	53.00	II	1.2
33.26289	75.77367	L9		70	155	J2	80	265	110	70	150	0.20	0.40	0.90	1.50	51.33	II	1.2
33.26289	75.77367	L9		80	65	J3	80	265	200	80	160	0.20	0.45	0.90	1.55	51.33	II	1.2
33.26289	75.77367	L10		70	155	J2	80	260	105	70	150	0.20	0.40	0.90	1.50	51.33	II	1.2
33.26289	75.77367	L10		80	65	J3	80	260	195	80	160	0.20	0.45	0.90	1.55	51.33	II	1.2
33.25866	75.77409	L11		65	115	J2	80	295	180	65	145	0.20	0.40	0.90	1.50	43.33	II	1.2
33.25866	75.77409	L11		35	70	J3	80	295	225	35	115	0.20	0.20	0.50	0.90	43.33	II	1.2
33.25336	75.77571	L17		25	80	J3	80	250	170	25	105	0.20	0.20	0.30	0.70	50.33	II	1.2
33.25036	75.77755	L19		30	70	J2	80	255	185	30	110	0.20	0.20	0.30	0.70	50.33	II	1.2
33.25036	75.77755	L19		80	155	J3	80	255	100	80	160	0.20	0.45	0.90	1.55	50.33	II	1.2
33.24805	75.77926	L20		35	65	J2	80	240	175	35	115	0.20	0.20	0.50	0.90	59.75	II	1.2
33.24721	75.77983	L21		60	50	J2	80	240	190	60	140	0.20	0.30	0.70	1.20	68.00	II	1.2
33.24721	75.77983	L21		75	150	J4	80	240	90	75	155	0.20	0.45	0.90	1.55	68.00	II	1.2
33.1336	75.5414	L30		30	160	J1	80	20	-140	30	110	0.20	0.20	0.30	0.70	40.33	II	1.2
33.1336	75.5414	L30		35	350	J2	80	20	-330	35	115	0.20	0.20	0.50	0.90	40.33	II	1.2
33.13672	75.55357	L36		55	170	J1	80	300	130	55	135	0.20	0.30	0.70	1.20	50.67	III	1.8
33.13728	75.55615	L37		65	175	J1	80	300	125	65	145	0.20	0.40	0.90	1.50	47.33	III	1.8
33.13728	75.55615	L37		35	15	J3	80	300	285	35	115	0.20	0.20	0.50	0.90	47.33	III	1.8
33.13935	75.55834	L39		55	170	J1	80	315	145	55	135	0.20	0.30	0.70	1.20	57.00	II	1.2
33.13935	75.55834	L39		60	15	J2	80	315	300	60	140	0.20	0.30	0.70	1.20	57.00	II	1.2
33.14035	75.55947	L40		85	60	FP	80	335	275	85	165	0.20	0.50	1.00	1.70	57.00	II	1.2
33.14135	75.56129	L41		75	205	J1	80	335	130	75	155	0.20	0.45	0.90	1.55	54.33	II	1.2

33.14135	75.56129	L41		35	100	J2	80	335	235	35	115	0.20	0.20	0.50	0.90	54.33	II	1.2
33.1424	75.56294	L42		70	210	J1	80	320	110	70	150	0.20	0.40	0.70	1.30	65.00	II	1.2
33.1424	75.56294	L42		55	25	J2	80	320	295	55	135	0.20	0.30	0.70	1.20	65.00	II	1.2
33.14339	75.56434	L43		60	210	J1	80	320	110	60	140	0.20	0.30	0.70	1.20	59.00	II	1.2
33.14339	75.56434	L43		30	20	J2	80	320	300	30	110	0.20	0.20	0.30	0.70	59.00	III	1.8
33.14388	75.56499	L44		60	210	J1	80	320	110	60	140	0.20	0.30	0.70	1.20	59.00	III	1.8
33.14388	75.56499	L44		30	20	J2	80	320	300	30	110	0.20	0.20	0.30	0.70	59.00	II	1.2
33.14446	75.56704	L45		45	125	J1	80	350	225	45	125	0.20	0.20	0.50	0.90	60.66	II	1.2
33.14446	75.56704	L45		55	35	J3	80	350	315	55	135	0.20	0.30	0.70	1.20	59.83	III	1.8
33.14253	75.57027	L47		45	125	J1	80	350	225	45	125	0.20	0.20	0.50	0.90	59.83	III	1.8
33.14253	75.57027	L47		55	35	J3	80	350	315	55	135	0.20	0.30	0.70	1.20	59.83	III	1.8
33.13598	75.58343	L55		75	80	J2	80	350	270	75	155	0.20	0.45	0.90	1.55	61.67	II	1.2
33.13598	75.58343	L55		75	10	J3	80	350	340	75	155	0.20	0.45	0.90	1.55	61.67	II	1.2
33.13608	75.58919	L57		60	30	J2	80	340	310	60	140	0.20	0.30	0.70	1.20	60.67	II	1.2
33.13286	75.60071	L63		35	175	J2	80	350	175	35	115	0.20	0.20	0.50	0.90	58.33	III	1.8
33.13286	75.60071	L63		85	75	J3	80	350	275	85	165	0.20	0.45	1.00	1.65	54.83	III	1.8
33.15957	75.6961	L101		85	330	J1	80	25	-305	85	165	0.20	0.50	1.00	1.70	58.67	III	1.8
33.15957	75.6961	L101		60	230	J3	80	25	-205	60	140	0.20	0.30	0.70	1.20	58.67	II	1.2
33.15935	75.69799	L102		50	145	J1	80	20	-125	50	130	0.20	0.20	0.50	0.90	52.33	III	1.8
33.15935	75.69799	L102		55	330	J2	80	20	-310	55	135	0.20	0.30	0.70	1.20	52.33	III	1.8
33.15777	75.00157	L103		85	350	J1	80	50	-300	85	165	0.20	0.50	1.00	1.70	52.75	III	1.8
33.15721	75.70045	L104		87	340	J1	80	55	-285	87	167	0.20	0.50	1.00	1.70	53.50	III	1.8
33.15721	75.70045	L104		40	210	J4	80	55	-155	40	120	0.20	0.20	0.30	0.70	53.50	III	1.8
33.15553	75.70225	L105		80	355	J1	80	55	-300	80	160	0.20	0.45	0.90	1.55	47.67	III	1.8
33.15553	75.70225	L105		40	150	J2	80	55	-95	40	120	0.20	0.20	0.50	0.90	47.67	III	1.8
33.1534	75.71022	L108		30	230	J1	80	30	-200	30	110	0.20	0.20	0.30	0.70	48.00	III	1.8
33.1534	75.71022	L108		70	360	J3	80	30	-330	70	150	0.20	0.40	0.90	1.50	48.00	II	1.2
33.15258	75.71145	L109		70	340	J1	80	55	-285	70	150	0.20	0.40	0.90	1.50	49.75	III	1.8
33.14917	75.71676	L113		50	210	J1	80	35	-175	50	130	0.20	0.20	0.50	0.90	61.83	III	1.8
33.14896	75.71873	L114		55	295	J1	80	40	-255	55	135	0.20	0.30	0.70	1.20	58.25	III	1.8
33.14896	75.71873	L114		65	150	J3	80	40	-110	65	145	0.20	0.40	0.90	1.50	58.25	III	1.8
33.15216	75.73045	L121		25	170	J2	80	320	150	25	105	0.20	0.20	0.30	0.70	59.67	III	1.8
33.15092	75.75852	L131		65	330	J1	80	75	-255	65	145	0.20	0.30	0.90	1.40	60.00	II	1.2
33.14136	75.75204	L138		80	315	J1	80	50	-265	80	160	0.20	0.45	0.90	1.55	67.00	II	1.2

33.14136	75.75204	L138		40	215	J3	80	50	-165	40	120	0.20	0.20	0.50	0.90	67.00	II	1.2
33.14034	75.75338	L139		85	345	J1	80	40	-305	85	165	0.20	0.50	1.00	1.70	50.00	III	1.8
33.14034	75.75338	L139		20	225	J3	80	40	-185	20	100	0.20	0.20	0.30	0.70	50.00	III	1.8
33.1398	75.75437	L140		70	170	J1	80	45	-125	70	150	0.20	0.40	0.90	1.50	53.67	III	1.8
33.1398	75.75437	L140		35	235	J3	80	45	-190	35	115	0.20	0.20	0.50	0.90	53.67	III	1.8
33.13931	75.7552	L141		60	160	J1	80	30	-130	60	140	0.20	0.30	0.70	1.20	54.00	III	1.8
33.13931	75.7552	L141		40	340	J2	80	30	-310	40	120	0.20	0.20	0.50	0.90	54.00	III	1.8
33.13882	75.75644	L142		60	135	J1	80	20	-115	60	140	0.20	0.30	0.70	1.20	63.33	II	1.2
33.13882	75.75644	L142		50	335	J2	80	20	-315	50	130	0.20	0.20	0.50	0.90	63.33	II	1.2
33.1449	75.77108	L153		45	145	J1	80	35	-110	45	125	0.20	0.20	0.50	0.90	48.00	III	1.8
33.1449	75.77108	L153		60	335	J2	80	35	-300	60	140	0.20	0.30	0.70	1.20	48.00	III	1.8
33.14533	75.77272	L154		85	325	J1	80	15	-310	85	165	0.20	0.50	1.00	1.70	60.00	II	1.2
33.14533	75.77272	L154		65	220	J3	80	15	-205	65	145	0.20	0.40	0.90	1.50	60.00	II	1.2
33.14579	75.77479	L156		30	170	J1	80	15	-155	30	110	0.20	0.20	0.30	0.70	60.50	II	1.2
33.14579	75.77479	L156		70	310	J2	80	15	-295	70	150	0.20	0.40	0.90	1.50	60.50	II	1.2
33.14572	75.77636	L157		75	330	J1	80	10	-320	75	155	0.20	0.45	0.90	1.55	53.00	III	1.8
33.14572	75.77636	L157		55	225	J3	80	10	-215	55	135	0.20	0.30	0.70	1.20	53.00	III	1.8
33.14584	75.78415	L161		30	145	J1	80	10	-135	30	110	0.20	0.20	0.30	0.70	56.00	III	1.8
33.14584	75.78415	L161		60	320	J2	80	10	-310	60	140	0.20	0.30	0.70	1.20	56.00	III	1.8
33.14584	75.78415	L161		65	280	J3	80	10	-270	65	145	0.20	0.40	0.90	1.50	56.00	III	1.8
33.14544	75.78679	L162		80	335	J1	80	20	-315	80	160	0.20	0.45	0.90	1.55	53.33	III	1.8
33.14544	75.78679	L162		50	230	J3	80	20	-210	50	130	0.20	0.20	0.50	0.90	53.33	III	1.8
33.14578	75.79601	L165		50	150	J1	80	330	180	50	130	0.20	0.20	0.50	0.90	46.00	III	1.8
33.14578	75.79601	L165		55	45	J3	80	330	285	55	135	0.20	0.30	0.70	1.20	46.00	III	1.8
33.14652	75.79864	L167		45	65	J4	80	250	185	45	125	0.20	0.20	0.70	1.10	56.00	III	1.8
33.1486	75.79857	L168		15	95	J4	80	280	185	15	95	0.20	0.20	0.50	0.90	69.75	II	1.2
33.14992	75.79865	L169		35	95	J3	80	260	165	35	115	0.20	0.20	0.50	0.90	62.67	II	1.2
33.1522	75.79828	L171		65	85	J3	80	265	180	65	145	0.20	0.40	0.90	1.50	49.67	II	1.2
33.15433	75.79818	L173		65	80	J3	80	265	185	65	145	0.20	0.40	0.70	1.30	49.67	II	1.2
33.15548	75.79793	L174		55	80	J3	80	255	175	55	135	0.20	0.30	0.50	1.00	47.67	II	1.2
33.15795	75.79866	L176	160	30	85	J3	80	255	170	30	110	0.20	0.20	0.30	0.70	38.00	II	1.2
33.15904	75.79882	L177	220	25	135	J3	80	310	175	25	105	0.20	0.20	0.30	0.70	54.00	III	1.8
33.16022	75.79955	L178	210	75	120	J3	80	305	185	75	155	0.20	0.45	0.90	1.55	53.67	III	1.8
33.16275	75.80161	L181	210	25	120	J3	80	310	190	25	105	0.20	0.20	0.30	0.70	50.00	III	1.8

33.16372	75.80234	L182	195	20	135	J3	80	315	180	20	100	0.20	0.20	0.30	0.70	61.67	II	1.2
33.16437	75.80315	L183	210	20	125	J3	80	315	190	20	100	0.20	0.20	0.30	0.70	76.00	II	1.2
33.16538	75.80395	L184	290	55	195	J2	80	310	115	55	135	0.20	0.30	0.70	1.20	61.17	II	1.2
33.16538	75.80395	L184	195	20	110	J3	80	310	200	20	100	0.20	0.20	0.30	0.70	61.17	II	1.2
33.16622	75.80415	L185	295	65	205	J2	80	310	105	65	145	0.20	0.40	0.90	1.50	64.75	II	1.2
33.16622	75.80415	L185	195	30	110	J3	80	310	200	30	110	0.20	0.20	0.30	0.70	64.75	II	1.2
33.16751	75.80525	L186	40	40	130	J3	80	300	170	40	120	0.20	0.20	0.70	1.10	60.75	II	1.2
33.17381	75.80962	L191	310	85	215	J2	80	310	95	85	165	0.20	0.50	1.00	1.70	65.33	II	1.2
33.17381	75.80962	L191	50	45	135	J3	80	310	175	45	125	0.20	0.20	0.50	0.90	65.33	II	1.2
33.17474	75.81007	L192	270	45	180	J2	80	290	110	45	125	0.20	0.20	0.50	0.90	60.67	II	1.2
33.17474	75.81007	L192	175	45	85	J3	80	290	205	45	125	0.20	0.20	0.50	0.90	60.67	II	1.2
33.17725	75.81099	L194	45	25	125	J3	80	300	175	25	105	0.20	0.20	0.30	0.70	61.33	II	1.2
33.12651	75.61902	L196	100	48	190	J1	80	20	-170	48	128	0.20	0.20	0.50	0.90	53.00	III	1.8
33.12651	75.61902	L196	20	70	295	J3	80	20	-275	70	150	0.20	0.40	0.90	1.50	53.00	III	1.8
33.12647	75.62146	L198	115	50	195	J1	80	25	-170	50	130	0.20	0.20	0.50	0.90	51.00	III	1.8
33.12647	75.62146	L198	40	80	330	J3	80	25	-305	80	160	0.20	0.45	0.90	1.55	51.00	III	1.8
33.12641	75.62513	L200	65	25	165	J1	80	20	-145	25	105	0.20	0.20	0.30	0.70	61.25	II	1.2
33.12641	75.62513	L200	50	60	305	J3	80	20	-285	60	140	0.20	0.30	0.70	1.20	61.25	II	1.2
33.12433	75.63626	L201	80	50	170	J1	80	20	-150	50	130	0.20	0.20	0.50	0.90	59.25	III	1.8
33.12433	75.63626	L201	20	80	290	J3	80	20	-270	80	160	0.20	0.45	0.90	1.55	59.25	III	1.8
33.12433	75.63626	L201	90	50	360	J4	80	20	-340	50	130	0.20	0.20	0.50	0.90	59.25	III	1.8
33.13193	75.55004	L224	105	50	195	J1	80	320	125	50	130	0.20	0.20	0.50	0.90	64.67	II	1.2
33.1351	75.55425	L227	80	57	170	J1	80	20	-150	57	137	0.20	0.30	0.70	1.20	65.00	II	1.2
33.1351	75.55425	L227	275	45	355	J2	80	20	-335	45	125	0.20	0.20	0.50	0.90	65.00	II	1.2
33.1351	75.55425	L227	20	75	295	J3	80	20	-275	75	155	0.20	0.45	0.90	1.55	65.00	II	1.2
33.1353	75.55603	L228	90	65	180	J1	80	350	170	65	145	0.20	0.40	0.90	1.50	66.33	II	1.2
33.18196	75.8074	L231	10	78	100	J1	80	290	190	78	158	0.20	0.45	0.90	1.55	33.67	IV	2.4
33.18196	75.8074	L231	100	80	190	J3	80	290	100	80	160	0.20	0.45	0.90	1.55	33.67	IV	2.4
33.18396	75.8078	L232	75	38	160	J1	80	290	130	38	118	0.20	0.20	0.50	0.90	50.33	III	1.8
33.18551	75.80752	L233	180	38	90	J2	80	270	180	38	118	0.20	0.20	0.50	0.90	61.67	II	1.2
33.18779	75.80717	L235	190	40	80	J2	80	270	190	40	120	0.20	0.20	0.50	0.90	69.75	II	1.2
33.18779	75.80717	L235	65	85	155	J3	80	270	115	85	165	0.20	0.50	1.00	1.70	69.75	II	1.2
33.19742	75.80223	L238	305	45	25	J2	80	240	215	45	125	0.20	0.20	0.50	0.90	64.00	II	1.2
33.19742	75.80223	L238	230	65	130	J3	80	240	110	65	145	0.20	0.40	0.90	1.50	64.00	II	1.2

33.19794	75.80125	L239	335	57	70	J2	80	210	140	57	137	0.20	0.30	0.70	1.20	62.67	II	1.2
33.19863	75.80047	L240	345	40	65	J2	80	260	195	40	120	0.20	0.20	0.50	0.90	61.75	II	1.2
33.20005	75.79958	L241	30	35	125	J2	80	235	110	35	115	0.20	0.20	0.50	0.90	52.00	III	1.8
33.20207	75.79797	L244	210	35	120	J2	80	260	140	35	115	0.20	0.20	0.50	0.90	71.67	II	1.2
33.20293	75.79786	L245	360	48	90	J2	80	250	160	48	128	0.20	0.20	0.50	0.90	68.67	II	1.2
33.22172	75.79556	L257	340	50	70	J2	80	260	190	50	130	0.20	0.20	0.50	0.90	58.67	II	1.2
33.22404	75.79254	L259	325	60	55	J1	80	240	185	60	140	0.20	0.30	0.70	1.20	68.33	II	1.2
33.22404	75.79254	L259	230	85	140	J3	80	240	100	85	165	0.20	0.50	1.00	1.70	68.33	II	1.2
33.22513	75.79174	L260	150	45	60	J3	80	240	180	45	125	0.20	0.20	0.50	0.90	53.67	III	1.8
33.22598	75.79098	L261	150	45	60	J3	80	240	180	45	125	0.20	0.20	0.50	0.90	53.67	III	1.8
33.22879	75.7889	L263	340	65	60	J3	80	220	160	65	145	0.20	0.40	0.90	1.50	60.33	II	1.2
33.22987	75.78783	L264	310	65	50	J3	80	300	250	65	145	0.20	0.40	0.90	1.50	52.08	III	1.8
33.26312	75.77358	L285	60	70	345	J2	80	245	-100	70	150	0.20	0.40	0.90	1.50	58.67	III	1.8
33.26312	75.77358	L285	170	65	80	J3	80	245	165	65	145	0.20	0.40	0.90	1.50	55.33	III	1.8
33.23005	75.78798	L286	310	65	50	J3	80	245	195	65	145	0.20	0.40	0.90	1.50	65.67	II	1.2
33.16483	75.80347	L288	210	20	125	J3	80	320	195	20	100	0.20	0.20	0.30	0.70	69.67	II	1.2
33.14661	75.7987	L289		45	65	J4	80	250	185	45	125	0.20	0.20	0.50	0.90	54.67	III	1.8
33.14585	75.78383	L290	55	30	145	J1	80	10	-135	30	110	0.20	0.20	0.30	0.70	54.67	III	1.8
33.14585	75.78383	L290		60	320	J2	80	10	-310	60	140	0.20	0.30	0.70	1.20	63.00	II	1.2
33.14585	75.78383	L290		65	280	J3	80	10	-270	65	145	0.20	0.40	0.90	1.50	63.00	II	1.2
33.13879	75.75647	L291		60	135	J1	80	15	-120	60	140	0.20	0.30	0.70	1.20	63.00	II	1.2
33.13879	75.75647	L291		50	335	J2	80	15	-320	50	130	0.20	0.30	0.50	1.00	61.67	II	1.2
33.15158	75.74831	L292	80	65	330	J1	80	75	-255	65	145	0.20	0.40	0.90	1.50	54.33	III	1.8
33.15596	75.70181	L293		80	355	J1	80	45	-310	80	160	0.20	0.45	0.90	1.55	54.33	III	1.8
33.15596	75.70181	L293		40	150	J2	80	45	-105	40	120	0.20	0.20	0.50	0.90	54.33	III	1.8
33.14017	75.55898	L294		55	170	J1	80	315	145	55	135	0.20	0.30	0.70	1.20	57.00	III	1.8
33.14017	75.55898	L294	35	80	310	J3	80	315	-310	80	160	0.20	0.45	0.90	1.55	57.00	III	1.8
33.1539	75.68203	L295	290	55	200	J1	80	330	130	55	135	0.20	0.30	0.70	1.20	48.75	III	1.8
33.1539	75.68203	L295	180	20	90	J3	80	330	240	20	100	0.20	0.20	0.30	0.70	48.75	III	1.8
33.13248	75.65504	L305	105	45	195	J1	80	330	135	45	125	0.20	0.20	0.50	0.90	57.00	III	1.8
33.13248	75.65504	L305	100	65	10	J2	80	330	320	65	145	0.20	0.40	0.90	1.50	57.00	III	1.8
33.12782	75.64365	L307	95	70	185	J1	80	300	115	70	150	0.20	0.40	0.90	1.50	49.33	III	1.8
33.12782	75.64365	L307	315	40	45	J3	80	300	255	40	120	0.20	0.20	0.50	0.90	49.33	III	1.8
33.12934	75.64329	L308	95	65	190	J1	80	300	110	65	145	0.20	0.40	0.90	1.50	48.67	III	1.8

33.12934	75.64329	L308	295	52	25	J2	80	300	275	52	132	0.20	0.30	0.70	1.20	48.67	III	1.8
33.1302	75.64451	L309	35	64	230	J1	80	345	115	64	144	0.20	0.40	0.90	1.50	60.50	II	1.2
33.1302	75.64451	L309	120	47	50	J3	80	345	295	47	127	0.20	0.20	0.5	0.90	60.50	II	1.2
33.13231	75.65142	L312	95	77	185	J1	80	340	155	77	157	0.20	0.45	0.90	1.55	39.17	IV	2.4
33.13231	75.65142	L312	350	45	80	J3	80	340	260	45	125	0.20	0.20	0.50	0.90	39.17	IV	2.4
33.13363	75.65448	L313	95	54	195	J1	80	315	120	54	134	0.20	0.30	0.70	1.20	44.25	III	1.8
33.13363	75.65448	L313	360	60	90	J3	80	315	225	60	140	0.20	0.30	0.70	1.20	44.25	III	1.8
33.13363	75.65448	L313	55	55	155	J4	80	315	160	55	135	0.20	0.30	0.70	1.20	44.25	III	1.8
33.13506	75.65626	L315	85	81	175	J1	80	330	155	81	161	0.20	0.50	1.00	1.70	36.00	IV	2.4
33.13506	75.65626	L315	360	68	90	J3	80	330	240	68	148	0.20	0.40	0.90	1.50	36.00	IV	2.4
33.13549	75.65678	L316	90	55	180	J1	80	310	130	55	135	0.20	0.30	0.70	1.20	41.33	III	1.8
33.13549	75.65678	L316	30	20	120	J3	80	310	190	20	100	0.20	0.20	0.30	0.70	41.33	III	1.8
33.13637	75.65756	L317	110	65	200	J1	80	300	100	65	145	0.20	0.40	0.90	1.50	40.33	III	1.8
33.13637	75.65756	L317	360	20	90	J3	80	300	210	20	100	0.20	0.20	0.30	0.70	40.33	III	1.8
33.13785	75.65821	L318	105	90		J1	80	295	295	90	170	0.20	0.50	1.00	1.70	61.33	II	1.2
33.13785	75.65821	L318	70	55	150	J3	80	295	145	55	135	0.20	0.30	0.70	1.20	61.33	II	1.2
33.13899	75.65866	L319	105	90		J1	80	295	295	90	170	0.20	0.50	1.00	1.70	61.33	II	1.2
33.13899	75.65866	L319	70	55	150	J3	80	295	145	55	135	0.20	0.30	0.70	1.20	61.33	II	1.2
33.14199	75.6601	L321	90	80	180	J1	80	300	120	80	160	0.20	0.45	0.90	1.55	39.50	IV	2.4
33.14916	75.72308	L323	140	60	210	J1	80	330	120	60	140	0.20	0.30	0.70	1.20	53.50	III	1.8
33.14916	75.72308	L323	300	50	40	J2	80	330	290	50	130	0.20	0.20	0.50	0.90	53.50	III	1.8
33.14862	75.72373	L326	140	60	210	J1	80	330	120	60	140	0.20	0.30	0.70	1.20	53.50	III	1.8
33.14862	75.72373	L326	300	50	40	J2	80	330	290	50	130	0.20	0.20	0.50	0.90	53.50	III	1.8



Annexure-VIII

Geological Survey of India

Data sheet on report movement in connection with its scrutiny

Division: LSSM/EGG Division

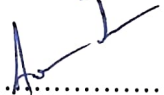
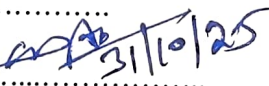
Item No.: M4LSSM/NC/NR/SU-JK/2024/50064

Field Season: 2024-2025

Mission: IV

Title: Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir,

Name of author(s) Arun Dev, Director Mohd. Ibrahim, Senior Geologist	Date of submission of report to the Supervisory Officer: Signature of the senior most author 30.06.2025
Name of Supervisory Officer Arun Dev, Director	Date of receipt of report: 30.06.2025 Date of completion of scrutiny: 16.07.2025 Date of handing over report to the author after scrutiny: 17.07.2025 Signature
Name of author(s) Mohd. Ibrahim, Senior Geologist To whom the report has been handed over by the Supervisory Officer	Date of receipt after scrutiny 17.07.2025 Signature..... Date of resubmission 11.08.2025 Signature.....
Supervisory Officer Arun Dev, Director	Date of receipt after modification 11.08.2025 Date of handing over report to the Regional Mission Head...14.08.2025 Signature.....
Name of Regional Mission Head Shri Sanjay Kumar DDG & RMH IV	Date of receipt of report...14.8.2025 Date of handing over the report after peer review to the Project Director...12.09.2025 Signature.....

Name of Supervisory Officer Arun Dev, Director	Date of receipt from the Regional Mission Head after peer review ...12.09.2025..... Date of handing over to author(s) ...15.09.2025..... Signature..... 
Name of author(s) Mohd. Ibrahim, Senior Geologist	Date of receipt of report from Supervisory Officer ...15.09.2025..... Date of handing over report after modification to Supervisory Officer ...10.10.2025..... Signature..... 
Name of Supervisory Officer Arun Dev, Director	Date of receipt of report after modification from the author(s) ...10.10.2025..... Date of handing over report to Regional Mission Head ...17.10.2025..... Signature..... 
Name of Regional Mission Head Shri Sanjiv Kumar ADG & RMHIV	Date of receipt after modification from the Project Director....24.10.2025 Date of handing over the report to the HoD of the Region for approval27.10.2025..... Signature..... 
Name of the HoD of the Region Shri. Rajinder Kumar ADG & HODNR	Date of receipt of report for approval27.10.2025..... Date of approval...31.10.2025 Date of sending back report to Project Director...03.11.2025 Signature of the HoD..... 
Name of the Project Director Shri. Arun Dev (Director)	Date of receipt of report after approval of HoD....30.11.2025. Date of circulation of report...3/12/2025 Signature..... 



Geological Survey of India

Certificate of Quality of Report

Project/Division: Engineering Geology & Landslide,

FSP No.: - M4LSSM/NC/NR/SU-JK/2024/50064,

Mission: IVA

Title of the report: Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir.

Part-A

This is to certify that the following officers have carried out the above mentioned FSP, and are responsible for collection of field data, carrying out the required laboratory analysis, compilation and synthesis of field data, coalition of the data and writing of the report as per the approved guidelines. The duration of field stay of each officer(s) is as against his name

Name(s)/Designation of the field officer	Total field stay during the FSP (no. of days)	Signature with date
1. Arun Dev, Director	86 days	<i>[Signature]</i> 16/10/25
2. Mohd. Ibrahim, Sr. Geologist	83 days	<i>[Signature]</i> 16/10/25

Part-B

This is to certify that drilling has been carried out and completed as per the table given below:

Drilling unit no.	Date of commencement of drilling	Date of completion of drilling	Name and signature of Director Drilling
NA			

Part-C

This is to certify that geophysical component of the item has been carried out and completed as per the table given below:

Name(s)/Designation of the Geophysist	Date of commencement of work	Date of completion of work	Signature with date of Geophysist
NA			
Name and signature with date of the Director (Geophysics)			

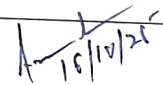
Part-D

This is to certify that the chemical/geotechnical analysis of the geological samples has been carried out as per the table given below:

Name of the Director in charge of the chemical/geotechnical lab	Date of receipt of samples from the field officer/party	Date of providing complete analytical results to the field officer	Signature with date of the Director Chemistry / Geotechnical

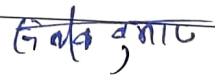
Part-E

This is to certify that the below mentioned Supervisory Officer(s) have supervised the above item and have provided guidance to the Field Officer (s). The number of inspection visits and the total duration of field visits is as per the below table. It is also certified that the work carried out under the project is of high standards, the norms of the field work, lab work etc have been followed. It is also certified that thorough scrutiny of the report for maintaining the scientific contents and its quality has been overseen.

Name/Designation of the Supervisory Officer	No. of inspection visit made to the field area	Total no. of days spent on inspection visit	Signature with date
Arun Dev, Director	02	10	 15/10/25

Part-F

This is to certify that the report has been thoroughly scrutinized at the Regional Mission Head Level. The scientific contents and the quality of the report are certified. The report has been peer reviewed and the needful corrections have been attended to.

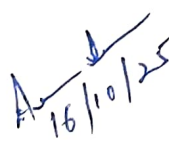
Name(s)/Designation of the Regional Mission Head	Signature with date
Dr. Sanjay Kumar, DDG & RMH-IV	 31/10/25

Part-G

This is to certify that the above report has been scrutinized at the various levels and has also been peer reviewed. On being satisfied regarding the scientific contents and the quality of report, the undersigned has accorded approval for its circulation

Name(s)/Designation of the Regional HOD	Signature with date
Shri. Rajinder Kumar AOG & HOD/NR	 31/10/25

FIVE POINT PERFORMA (Annexure-X)

Title of the report	Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir.
Name(s) of the author(s)	Arun Dev, Director Mohd. Ibrahim, Senior Geologist
Background of the investigation at whose instance or with what particular objective in view the investigation was undertaken	The meso scale study is taken up along along NH-244, Doda-Lawa, so that the output is used by stakeholders in developmental planning at local scale. Hence, in addition to the landslide susceptibility map, an user friendly theme based qualitative Landslide mitigation and management map.
Whether the item of investigation was included in the approved programme for the year concerned. If so, particulars, such as item No., page reference etc. may be given.	Yes Item Code: M4LSSM/NC/NR/SU-JK/2024/50064 Field Season: 2024-25
Date of commencement of the investigation with month and year, and exact number of days spent by the officer(s) in the field in connection with the investigation.	FS: 2024-25 Date of commencement of field work: 25-06-2024. Arun Dev, Director- 86 days Mohd. Ibrahim, Senior Geologist - 83 days Arun Dev, Director - 10 days
Brief comments by Director / Supervisory officer on the conclusion and recommendation by the author(s) with suggestion for implementing the same.	The various remedial measures suggested by the authors will help in identifying the slope stability. The work has been completed to the satisfaction of the undersigned.
Please state specifically how the results of the investigation were achieved compared with the objective for which it was undertaken.	The results of the investigation commensurate with the objective.
Signature of the Project Director: 	Name: Arun Dev Designation: Director Project/ Division: EG & LSSM State Unit/ Region: UT : Jammu & Kashmir and Ladakh, NR
Place: Jammu Date: 16/10/25	Jammu

Certificate of Supervisory officer

It is certified that the Report pertaining to FS 2024-25 entitled 'Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir" (Item Code: M4LSSM/NC/NR/SU-JK/2024/50064) authored by Sh. Mohd Ibrahim Sr. Geologist has been modified as per the suggestions of the peer reviewer. This is verified by the undersigned.

Signature

Name: ARUN DEV

Designation: Director

Place: Jammu

Date: 16/10/25

INFORMATION SHEET FOR BIBLIOGRAPHY DATABASE IN UNPUBLISHED REPORTS

Title	Meso Scale (1:10000) Landslide susceptibility corridor mapping along NH-244, Doda-Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir.
Authors	Arun Dev, Director and Mohd. Ibrahim, Senior Geologist
Field Season	2024-25
Year (s)	One year assignment
FSP ID	M4LSSM/NC/NR/SU-JK/2024/50064
Theme	Landslide susceptibility mapping
Sub Theme	Meso Scale (1:10000) Landslide susceptibility mapping
Status	Category-B, Open File Report
State (s)/UT	Jammu and Kashmir
District (s)	Doda and Kishtwar
Toposheets	43O/12 ,15 & 16.
Key Locations	Doda, Kishtwar,
Short Abstract	The Doda-Lawa section, forming a vital part of National Highway NH-244, serves as the primary lifeline for the Doda and Kishtwar districts. Due to the recurrent occurrence of landslides along the stretch between Doda Bridge and Lawa, a meso-scale (1:10,000) landslide susceptibility mapping was undertaken, covering a total length of approximately 45 line-kilometres. The study area falls in the Survey of India (SOI) toposheet nos. 43O/12 ,15 & 16. The cumulative susceptibility values are calculated by integrating the LSER which gives different Total Estimated Susceptibility Values (TESV) for various domains of the area. The domain based TESSV was calculated for cut slope, overburden and rock domains. Based on field observations and satellite imagery, different landslide types were identified in the study area, including debris slides, minor debris flows, and rock slides. A heuristic cum deterministic model was used to categorize the area into various landslide susceptibility classes, and the resulting map showed that 28% of the area had low susceptibility, 40% had moderate susceptibility, and 32% had high susceptibility. Validation of the map using landslide inventory data indicated that 60.3% of landslides fell within the high susceptibility indicated that 60.3% of landslides fell within the high susceptibility zone

Plate-I

Mesoscale Landslide Susceptibility Map of NH-244, Doda- Lawa, Doda and Kishtwar Districts, UT: Jammu and Kashmir

Legend

